Supplemental Material to "A dictionary between finite spectral triples and Kaluza-Klein data"

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This is the supplement of the paper *A dictionary between finite spectral triples and Kaluza-Klein data*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

Parts contained. Lemma parts: 05, 16, 22, 23, 24, 25, 28, 29, 30, 31, 32, 40, 35, 36, 37, 38, 41, 42, 43.

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

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Part 05

The geometric identification exists: the minimal grading-preserving spectral truncation of a suitable chiral interval is exactly Connes–Lott, with

$$|m| = \frac{\pi}{2l}, \qquad d_{\text{trunc}} = \frac{2l}{\pi}.$$

The truncated endpoint states are smeared states on the original interval. Moreover, this interval has no zero mode, which limits the physical interpretation of both extremes.

Part 1. Deconstructed chain

Statement. There is a natural family with opposite endpoint chiralities, an exact Connes–Lott member at $N = 2$, and endpoint distance converging to l. The answer depends on the representation of the site algebra; the familiar lattice-distance anomaly is not universal.

Hypotheses. We consider complex even spectral triples, without imposing additional real-structure or KO-dimension axioms. Let $N \geq 2$, $n = N - 1$, and

$$A_N = \mathbb{C}^N, \qquad H_N = \text{span}\{L_1, \dots, L_{N-1}, R_2, \dots, R_N\}.$$

Thus interior sites carry both components, site 1 only L, and site N only R. Define

$$\pi_N(a) = \text{diag}(a_1, \dots, a_{N-1}, a_2, \dots, a_N), \qquad \Gamma_N = \text{diag}(I_n, -I_n).$$

If S_n is the lower shift, set

$$B_N = h_N(I_n - S_n), \qquad D_N = \begin{pmatrix} 0 & B_N \\ B_N^* & 0 \end{pmatrix}.$$

These are forward/backward differences:

$$(D_N\psi)_{L_i} = h_N(R_{i+1} - R_i), \qquad (D_N\psi)_{R_j} = h_N(L_{j-1} - L_j),$$

with the missing components interpreted as

$$R_1 = 0, \qquad L_N = 0.$$

There are nearest-neighbour couplings and onsite L-R terms at interior sites.

For compatibility with the spectral truncation in Part 3, choose

$$\theta_N = \frac{\pi}{4N - 2}, \qquad h_N = \frac{\pi}{4l \sin\theta_N} \sim \frac{N}{l}. \tag{1}$$

This fixes the smallest positive eigenvalue to $\pi/(2l)$ at every N.

Proof.

The representation is faithful, $[\Gamma_N, \pi_N(a)] = 0$, and $\Gamma_N D_N = -D_N\Gamma_N$. At $N = 2$,

$$A_2 = \mathbb{C}^2, \quad H_2 = \mathbb{C}^2, \quad D_2 = \begin{pmatrix} 0 & h_2 \\ h_2 & 0 \end{pmatrix}, \quad h_2 = \frac{\pi}{2l}.$$

This is exactly the stated two-point triple; a diagonal unitary supplies any desired phase of m.

Ordering the basis as

$$L_1, R_2, L_2, R_3, \dots, L_{N-1}, R_N$$

makes D_N a path adjacency matrix with alternating signs. A diagonal unitary removes those signs. The usual sine eigenvectors of a path therefore give

$$\operatorname{spec}(D_N) = \left\{ 2h_N \cos\frac{j\pi}{2N-1} : 1 \le j \le 2N-2 \right\} = \{\pm\lambda_{N,r} : 1 \le r \le N-1\},$$

where

$$\boxed{\lambda_{N,r} = 2h_N \sin((2r-1)\theta_N).} \tag{2}$$

Consequently,

$$\boxed{K_N(t) = \operatorname{Tr} e^{-tD_N^2} = 2\sum_{r=1}^{N-1} e^{-4th_N^2 \sin^2((2r-1)\theta_N)}.} \tag{3}$$

For a real site function a, the onsite terms commute with $\pi_N(a)$, while

$$B_N\pi_R(a) - \pi_L(a)B_N = h_N \operatorname{diag}(a_2 - a_1, \dots, a_N - a_{N-1}).$$

Hence

$$\|[D_N, \pi_N(a)]\| = h_N \max_i |a_{i+1} - a_i|.$$

The triangle inequality bounds $|a_j - a_i|$ by $|j-i|/h_N$; the function $a_i = (i-1)/h_N$ attains this bound. Thus

$$\boxed{d_N(i,j) = \frac{|i-j|}{h_N}.} \tag{4}$$

In particular,

$$\boxed{d_N(1,N) = \frac{4l(N-1)}{\pi}\sin\frac{\pi}{4N-2} \longrightarrow l.} \tag{5}$$

One can instead choose the geometric normalization

$$h_N = \frac{N-1}{l}. \tag{6}$$

Then $d_N(1,N) = l$ exactly for every N, with the same continuum limit. However, its $N = 2$ mass is $1/l$, rather than the continuum lowest eigenvalue $\pi/(2l)$. Equations (1) and (6) distinguish spectral calibration from metric calibration at finite resolution.

Two other natural choices clarify the literature:

- **One-component staggered path.** Take $A = \mathbb{C}^M$, $H = \mathbb{C}^M$, alternating grading, and $D = h$ times path adjacency. Opposite endpoint chiralities require even M. Its spectrum is $2h\cos(j\pi/(M+1))$. For $k = |i-j|$,

$$\boxed{d(i,j) = \begin{cases} (k+1)/(2h), & k \text{ odd}, \\ \sqrt{k(k+2)}/(2h), & k \text{ even}. \end{cases}} \tag{7}$$

  In particular $d(1,M) = M/(2h)$. Thus $h = M/(2l)$ gives endpoint distance l, whereas $h = M/l$ gives $l/2$.

  Here is a direct proof of (7). Restrict the commutator to the connecting subpath and extend the function constantly outside it. After removing phases, its norm is that of weighted adjacency, with

dimensionless edge weights $w_i = h|\Delta a_i|$. If its two bipartition sizes are p, q, the off-diagonal block C satisfies

$$\sum_i w_i = \mathbf{1}_p^T C \mathbf{1}_q \le \sqrt{pq}\,\|C\|.$$

For $p = q$, alternating weights $1, 0, 1, \dots, 1$ attain the bound. For $p = q + 1$, the weights

$$\frac{(q, 1, q-1, 2, \dots, 1, q)}{\sqrt{q(q+1)}}$$

attain it: the normalized constant vectors on the two bipartitions are singular vectors with singular value 1, and positivity gives the norm 1. This proves (7).

These are the nonadditive distances studied by Bimonte–Lizzi–Sparano, §5 and given explicitly by Atzmon, Eq. (30). Reduction to the connecting path also agrees with Iochum–Krajewski–Martinetti, Proposition 4.

- **Vertex-edge incidence Dirac.** Let $\partial : \mathbb{C}^N \to \mathbb{C}^{N-1}$ be the path difference matrix and use

$$D_{\mathrm{inc}} = h \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}.$$

Its spectrum and heat trace are

$$\{0\} \cup \left\{ \pm 2h \sin \frac{k\pi}{2N} : 1 \le k < N \right\},$$

$$K_{\mathrm{inc}}(t) = 1 + 2 \sum_{k=1}^{N-1} e^{-4th^2 \sin^2(k\pi/(2N))}.$$

These follow by substituting the cosine eigenvectors into the path Laplacian $\partial^*\partial$. Representing a site function on each edge by its value at the tail gives distance $|i-j|/h$. But H has dimension $2N-1$, has a zero mode, and both endpoint vertex components have the same grading. At $N = 2$, its spectrum is $0, \pm\sqrt{2}h$, so it is not Connes–Lott.

Thus the quoted integer deconstruction tower belongs to the incidence construction. Opposite chiral boundary conditions produce the shifted tower (2).

Status: PROVED

Part 2. Continuum limit

Statement. The chain above converges to an interval Dirac operator in three precise senses: ordered low eigenvalues, heat traces locally uniformly for $t > 0$, and pure-state metric spaces in Gromov–Hausdorff distance. It also has a controlled one-dimensional heat window.

Hypotheses. On $H_l = L^2([0, l], \mathbb{C}^2)$, define

$$D_l = \begin{pmatrix} 0 & \partial_x \\ -\partial_x & 0 \end{pmatrix}, \qquad \mathrm{Dom}\, D_l = \{(L, R) \in H^1 : R(0) = 0,\ L(l) = 0\}, \tag{8}$$

with scalar multiplication by $C^\infty([0, l])$ and grading $\Gamma = \mathrm{diag}(1, -1)$.

These are opposite local chiral conditions. Reflection parity is Γ at 0 and $-\Gamma$ at l; their composition gives antiperiodicity on the doubled interval.

Proof.

Integration by parts gives the boundary form

$$[\overline{L}\,\widetilde{R} - \overline{R}\,\widetilde{L}]_0^l.$$

The conditions in (8) form a maximal isotropic boundary subspace, so D_l is self-adjoint. Its resolvent is compact by the compact embedding $H^1 \hookrightarrow L^2$.

Solving $D_l\psi = \lambda\psi$ gives

$$\psi_\lambda(x) = c(\cos\lambda x, \sin\lambda x), \qquad \cos(\lambda l) = 0.$$

Therefore

$$\boxed{\operatorname{spec}(D_l) = \left\{ \pm\frac{(r-\frac{1}{2})\pi}{l} : r = 1, 2, ... \right\}.} \tag{9}$$

The kernel is zero. Squaring gives mixed Neumann–Dirichlet conditions on L, and Dirichlet–Neumann conditions on R.

For scalar f,

$$\|[D_l, f]\| = \|f'\|_\infty, \qquad d_l(x,y) = |x-y|. \tag{10}$$

Under normalization (1), with $m_1 = \pi/(2l)$,

$$\lambda_{N,r} = m_1 \frac{\sin((2r-1)\theta_N)}{\sin\theta_N} \longrightarrow (2r-1)m_1. \tag{11}$$

This proves low-eigenvalue convergence. Since

$$\sin y \geq \frac{2y}{\pi} \quad (0 \leq y \leq \pi/2),$$

we also have

$$\lambda_{N,r} \geq \frac{2r-1}{l}. \tag{12}$$

A summable Gaussian majorant now proves, uniformly on every $t \in [\varepsilon, T]$,

$$K_N(t) \longrightarrow K_l(t) = 2\sum_{r=1}^{\infty} e^{-t((r-\frac{1}{2})\pi/l)^2}. \tag{13}$$

Poisson summation gives

$$\boxed{K_l(t) = \frac{l}{\sqrt{\pi t}}\left(1 + 2\sum_{q=1}^{\infty}(-1)^q e^{-q^2l^2/t}\right).} \tag{14}$$

In particular, $K_l(t) \sim l/\sqrt{\pi t}$ as $t/l^2 \to 0$.

The finite-chain heat window has the following rigorous meaning. For every sequence $t_N > 0$ satisfying

$$\frac{t_N}{l^2} \to 0, \qquad \frac{N^2 t_N}{l^2} \to \infty, \tag{15}$$

one has

$$\boxed{K_N(t_N) \sim \frac{l}{\sqrt{\pi t_N}}, \qquad -2t_N\frac{K_N'(t_N)}{K_N(t_N)} \to 1.} \tag{16}$$

To prove this, put $u_r = \pi\sqrt{t_N}(r-\frac{1}{2})/l$. On bounded u_r-intervals, the sine argument in (11) tends uniformly to zero, so $\sqrt{t_N}\lambda_{N,r}/u_r \to 1$. The mesh tends to zero and its upper endpoint tends to infinity. Bound (12) controls the tails. Gaussian Riemann sums, and then the same sums weighted by $t_N\lambda_{N,r}^2$, prove both assertions.

Finally, identify site i with $x_i = (i-1)l/(N-1)$. Equation (4) gives

$$d_N(i,j) = \rho_N|x_i - x_j|, \qquad \rho_N = \frac{N-1}{lh_N} = 1 + O(N^{-1}).$$

The metric distortion and the grid mesh are both $O(l/N)$, proving Gromov–Hausdorff convergence to $[0, l]$.

The endpoint chirality here concerns allowed boundary components. It does not supply localized chiral zero modes.

Status: PROVED

Part 3. Minimal spectral truncation

Statement. The minimal nonzero grading-preserving spectral truncation of (8) is unitarily equivalent to Connes–Lott, with

$$\boxed{|m| = m_1 = \frac{\pi}{2l}, \qquad c = \frac{\pi}{2}.}$$

Its natural truncated endpoint states have distance $1/m_1 = 2l/\pi$. Literal endpoint evaluations do not descend through compression.

Hypotheses. Let

$$P = \mathbf{1}_{[0,E]}(|D_l|), \qquad m_1 \le E < 3m_1,$$

and give $S = P\,C([0, l])\,P$ its inherited operator-system structure. This is the compression construction described in Connes–van Suijlekom, §1.

Proof.

An orthonormal chiral basis of PH_l is

$$u_L(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(\pi x/(2l)) \\ 0 \end{pmatrix}, \qquad u_R(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(\pi x/(2l)) \end{pmatrix}.$$

In this basis,

$$PD_lP = \begin{pmatrix} 0 & m_1 \\ m_1 & 0 \end{pmatrix}, \qquad P\Gamma P = \mathrm{diag}(1, -1). \tag{17}$$

Compression of multiplication is

$$Q(f) = \begin{pmatrix} \dfrac{2}{l}\displaystyle\int_0^l f(x)\cos^2\frac{\pi x}{2l}\,dx & 0 \\ 0 & \dfrac{2}{l}\displaystyle\int_0^l f(x)\sin^2\frac{\pi x}{2l}\,dx \end{pmatrix}. \tag{18}$$

Since

$$Q(1) = I, \qquad Q(\cos(\pi x/l)) = \tfrac{1}{2}\,\mathrm{diag}(1, -1),$$

its range is exactly the full diagonal algebra:

$$\boxed{S = \mathbb{C} \oplus \mathbb{C}.} \tag{19}$$

Here the compressed operator system happens to be an algebra, although Q is not multiplicative.

Evaluation on the finite-dimensional space PH_l is bounded. At 0, only u_L has a nonzero boundary value; at l, only u_R does. The normalized evaluation vectors therefore define

$$\omega_0(\mathrm{diag}(a, b)) = a, \qquad \omega_l(\mathrm{diag}(a, b)) = b.$$

Using (17),

$$\|[PD_lP, \mathrm{diag}(a, b)]\| = m_1|a - b|,$$

and hence

$$\boxed{d_{\mathrm{trunc}}(\omega_0, \omega_l) = \frac{1}{m_1} = \frac{2l}{\pi} < l.} \tag{20}$$

Minimality follows because there is no zero mode and the grading exchanges the eigenspaces at $+m_1$ and $-m_1$. A nonzero grading-preserving spectral subspace must retain both.

There is an essential distinction between these states and literal evaluations. For

$$g(x) = \cos(3\pi x/l),$$

equation (18) gives $Q(g) = 0$, while $g(0) = 1$ and $g(l) = -1$. Thus neither endpoint evaluation factors through Q.

The pullbacks of the truncated endpoint states are instead the probability measures

$$d\mu_L = \frac{2}{l}\cos^2\frac{\pi x}{2l}\,dx, \qquad d\mu_R = \frac{2}{l}\sin^2\frac{\pi x}{2l}\,dx.$$

Their distance using the **original** continuum seminorm is

$$d_{D_l}(\mu_L, \mu_R) = \int_0^l \left| \int_0^x (d\mu_L - d\mu_R) \right| dx = \frac{4l}{\pi^2}. \tag{21}$$

The equality follows by integration by parts and is attained by $f = -x$. Equations (10), (20), and (21) concern three different state/metric comparisons.

Status: PROVED

Part 4. The two limits

Amendment (Terra 6.1, 3 Nov 2026; report). The statements of this part are correct for the stated bare chain, with one precision: the condition N/l -> infinity gives uniform convergence of the dispersion on bounded energy windows and of normalized heat traces and counting densities, and does not by itself give convergence of the unnormalized difference of heat traces (for N = floor(l^(5/4)) it diverges). Escapes for the empty light sector: a tuned bulk mass or a scaled boundary coupling.

Statement. At fixed finite rank, $m \to 0$ produces infinitely separated finite geometry, with heat dimension zero. A resolved interval requires a growing tower. As $m \to \infty$, the metric collapses, but the light Hilbert space is zero.

Hypotheses. Identify

$$m = m_1 = \frac{\pi}{2l},$$

use normalization (1), and hold physical energies or heat times fixed unless stated otherwise.

Proof.

For fixed N, $D_N = mT_N$ with T_N independent of m. Therefore, as $m \to 0$,

$$D_N \to 0, \qquad K_N(t) \to 2(N-1), \qquad d_N(i,j) \to \infty \quad (i \neq j).$$

For the minimal truncation,

$$K_2(t) = 2e^{-tm^2}.$$

Every finite N has ultraviolet heat dimension zero. Making its finitely many modes light does not change that conclusion.

For the full interval, a fixed energy E is in the resolved one-dimensional regime when

$$\frac{1}{l} \ll E.$$

For its finite-chain approximation, one additionally needs

$$\boxed{\frac{1}{l} \ll E \ll \frac{N}{l}.} \tag{22}$$

Thus, as $l \to \infty$, an exact continuum limit throughout a fixed positive energy window follows if $N/l \to \infty$. Equivalently, as $m \to 0$, one needs $Nm \to \infty$.

The continuum counting function and heat density make the emergence explicit:

$$\#\{|\lambda| \le E\} = 2\left\lfloor \frac{lE}{\pi} + \frac{1}{2} \right\rfloor, \qquad \frac{K_l(t)}{l} \longrightarrow \frac{1}{\sqrt{\pi t}} \quad (l \to \infty). \tag{23}$$

Based at an endpoint, the metric limit is a half-line; the opposite endpoint recedes to infinity.

The order of limits matters even for the normalized heat trace:

$$\lim_{l\to\infty}\lim_{N\to\infty} \frac{K_N(t)}{l} = \frac{1}{\sqrt{\pi t}}, \qquad \lim_{N\to\infty}\lim_{l\to\infty} \frac{K_N(t)}{l} = 0. \tag{24}$$

Likewise, taking $t \to 0$ at fixed N gives dimension zero, whereas taking $N \to \infty$ first gives the interval's dimension one.

As $m \to \infty$, all distances tend to zero. At fixed N, the algebra elements with uniformly bounded commutators reduce to constants. But every eigenvalue has magnitude at least m, so

$$\|(D_N - i)^{-1}\| \le \frac{1}{\sqrt{m^2+1}} \longrightarrow 0. \tag{25}$$

The same holds for D_l, after identifying the varying intervals by dilation. Its heat trace tends to zero at every fixed $t > 0$.

Thus the metric limit is a point, while the embedded resolvent limit has

$$H_{\text{light}} = \ker T_N = 0.$$

There is no finite operator limit on the original nonzero Hilbert space, and no surviving fermion pair at fixed energy.

Finally, the two-state spectral truncation exists for

$$m_1 \le E < 3m_1.$$

A genuine spectral cutoff $E \ll 1/l$ retains no modes. The ratio between the first two positive levels is always 3, so the minimal pair has no parametrically large separation from the next level.

Status: PROVED

Verdict on hypothesis H (theorem T1)

Supported for T1's geometric assertion. Connes–Lott is exactly the minimal nonzero grading-preserving spectral compression of the opposite-chirality interval above, with

$$l = \frac{\pi}{2|m|}.$$

The chain recovers its spectrum, heat window, and metric.

The precise replacement for the stronger interpretation of H is:

> The two sheets are truncated endpoint evaluation states, whose pullbacks are smeared over the interval. Refinement restores the interval. Decompactification produces a one-dimensional heat regime only when the number of retained modes grows sufficiently rapidly. Collapse identifies the metric points but removes all fermionic modes of this particular interval at fixed energy.

Accordingly, a nonempty two-state sector at $E \ll 1/l$, or surviving light fermions after collapse, is disproved by this model. The construction supplies no KO-6 or Standard Model identification.

Independent numerical check (Claude, 2 Nov 2026)

For l = 1 and N = 2, 3, 5, 9 we checked the following. The spectrum of D_N agrees with +/- 2 h_N sin((2r-1) theta_N) to 1e-14. The lowest positive eigenvalue equals pi/(2l) for every N. For random real site functions, ||[D_N, pi_N(a)]|| equals h_N max abs(a_{i+1} - a_i), which is the identity behind d_N(i,j) = abs(i-j)/h_N. The endpoint distance d_N(1,N) = 0.6366, 0.7869, 0.8844, 0.9398 tends to l. For the minimal truncation, Q(cos(3 pi x/l)) = 0, and the smeared-state distance 0.405290 matches 4l/pi^2. The truncated distance is 2l/pi = 0.6366, identical to d_2(1,2).

Literature (../LITERATURE.md, section A). The Kaluza-Klein reading of the two-point space is in Connes-Lott (sections I and III: "two copies of Z separated by a distance L"); the two points as points of a fifth dimension are in Viet-Wali 1996; the two-point model as an approximation of a manifold, to be related to dimensional reduction, is in Dimakis-Muller-Hoissen 1994 (section III, Example 4, and section VII). Connes-van Suijlekom (section 1) note that compressions P A P are operator systems, of which the chiral interval is an exception. The truncation theorem with |m| = pi/(2l) was not found in print.

Part 16

Noncommutative finite real triples from commutative data: truncating the coarse correspondence

Proved by Claude Opus 5.5 without Astra or Fable, in response to the challenge that the T0 note overread lemma 15 and the Farsi-Latrémolière limits. Signs and the level-two algebra were checked with numpy on random data (section "Numerical check"). The term "coarse correspondence" is that of Connes, *Noncommutative Geometry* (1994), chapter V, for the bimodule of Hilbert-Schmidt operators, and the form D = Delta + J Delta J^-1 is in Krajewski (arXiv:hep-th/9701081) and Paschke-Sitarz (arXiv:q-alg/9612029); all checked against the sources. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): the examples and Corollary 16.4 are correct; amendments to Proposition 16.1, Theorem 16.2, Corollary 16.3 and the caveats are incorporated below; Remark 16.5 and Example 16.6 come from the referee's assessment of what the lemma achieves. The two-sheet section was refereed separately (Terra 6.1, 3 Nov 2026; report), with one error in Corollary 16.8 and one in Example 16.9, corrected below.

Outcome. Compressing functions on a connected commutative space to a fixed spectral window produces a noncommutative finite algebra with no limit taken. With the standard reality J a J^-1 = a*, the inherited structure fails order zero exactly when that algebra is noncommutative (Proposition 16.1). With the parent Hilbert space doubled to H_0 (x) conj(H_0), order zero and order one hold for every such algebra (Theorem 16.2, Corollary 16.3); these two conditions are formal consequences of the tensor square, available for any finite even representation (Remark 16.5). The KO-6 signs and the count dim H_F = 2 k^2 are reached by a product with a Clifford point, which is an input and fails finite orientability (Corollary 16.4), and more faithfully by a parent with two sheets, which gives the form of the Chamseddine-Connes classification (Theorem 16.7, Corollary 16.8, Example 16.9). This selects no particular algebra and gives no dynamics.

Setting

Let (A_0, H_0, D_0, gamma_0) be an even spectral triple, A_0 commuting with gamma_0, with D_0 of compact resolvent. Let Sigma = -Sigma be a finite subset of R and P_0 = 1_Sigma(D_0), of rank k, so that P_0 commutes with gamma_0. Put V = P_0 H_0, graded by gamma_V, with d = D_0 restricted to V, and

```
Q(a) = P_0 a P_0 on V,     O_F = Q(A_0),     A_F = C*(O_F) in B(V).
```

O_F is an operator system, in general not an algebra, and A_F consists of even operators. Compression is standard: for functions it is the Toeplitz compression, and its use in truncating spectral triples is due

to D'Andrea-Lizzi-Martinetti and Connes-van Suijlekom. The latter note that the C*-algebra generated by PAP is in many examples a noninformative full matrix algebra. Since Q(1) = 1, A_F is the bicommutant of O_F, so its structure is set by the operators commuting with all compressed functions, such as the grading in example (b) and j in Example 16.9. That is where the information lies, and the fact that A_F is noncommutative is modest.

Examples of A_F.

(a) *Chiral interval, lowest level.* For condition (c) with M = 0 and Sigma = {+/- pi/(2l)}, O_F = C + C = A_F (lemma 05).

(b) *Chiral interval, next level.* For Sigma = {+/- pi/(2l), +/- 3 pi/(2l)}, V has basis (c_1, 0), (c_2, 0), (0, s_1), (0, s_2), with c_r = (2/l)^{1/2} cos((r - 1/2) pi x/l) and s_r = (2/l)^{1/2} sin((r - 1/2) pi x/l). With C_n = (1/l) int f cos(n pi x/l),

Q(f) = (Q_L, Q_R), Q_L = [[C_0 + C_1, C_1 + C_2], [C_1 + C_2, C_0 + C_3]], Q_R = [[C_0 - C_1, C_1 - C_2], [C_1 - C_2, C_0 - C_3]].

The values C_0, C_1, C_2, C_3 are independent and determine Q(f), so O_F has dimension 4. Each of Q_L and Q_R alone ranges over all symmetric 2 x 2 matrices, which generate M_2(C), so A_F projects onto each summand. For f_1 = 1/2 + cos(pi x/l) + cos(2 pi x/l) + cos(3 pi x/l) and f_2 = 1/2 - cos(pi x/l) + cos(2 pi x/l) - cos(3 pi x/l),

```
Q(f_1) = ([[1, 1], [1, 1]], 0),     Q(f_2) = (0, [[1, -1], [-1, 1]]).
```

M_2(C) is simple, so multiplying Q(f_1) and Q(f_2) by elements of A_F gives M_2(C) + 0 and 0 + M_2(C). Hence A_F = M_2(C) + M_2(C), of dimension 8. The finite algebra of the interval itself is noncommutative at the second level.

(c) *Twistor space.* For the level-one Toeplitz compression on CP^3, Q(x_a) = Gamma_a/5 (Proposition 10.1), and these generate A_F = M_4(C) on V = C^4. Here V is the kernel of the Dolbeault-Dirac operator twisted by O(1), so d = 0 on V.

The standard correspondence

Proposition 16.1 (order zero fails with inherited reality). Let J_0 be a real structure on H_0 (antiunitary, J_0^2 = +/-1, with the grading sign), with J_0 a J_0^-1 = a* for all a in A_0 and J_0 D_0 = eps' D_0 J_0, and let J_F = J_0 restricted to V. Then J_F Q(a)* J_F^-1 = Q(a), and the data (A_F, V, d, J_F) satisfy order zero if and only if A_F is commutative. The same holds for a product with a fixed finite factor that itself satisfies order zero, with the compatible product real structure.

Proof. J_0 preserves V because Sigma = -Sigma. Then J_0 P_0 a* P_0 J_0^-1 = P_0 a P_0. Order zero would make Q(a) commute with Q(b)° = Q(b) for all a, b, so O_F and hence A_F would be commutative; the converse is immediate, since J_F A_F J_F^-1 = A_F. For the product, take the unit in the finite factor: [1 (x) Q(a), 1 (x) Q(b)] = 1 (x) [Q(a), Q(b)]. ■

Together with Amendment 2 of lemma 07 (first order fails for C + C with inherited J_2), this is the correct form of the obstruction. It concerns the standard correspondence L^2(A_0) with the pointwise real structure.

The coarse correspondence

Let conj(H_0) be the conjugate Hilbert space, with canonical antiunitary xi -> xi^c, and with gamma^c, D^c the conjugate operators, gamma^c xi^c = (gamma_0 xi)^c and D^c xi^c = (D_0 xi)^c. On H = H_0 (x) conj(H_0), which is the coarse correspondence of A_0, equivalently the Hilbert-Schmidt operators on H_0, define for homogeneous xi, eta

```
pi(a) = a (x) 1,
DD = D_0 (x) 1 + gamma_0 (x) D^c,
Gamma = gamma_0 (x) gamma^c,
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c.
```

Theorem 16.2. (A_0, H, DD, J, Gamma) is a real even spectral triple with signs J^2 = 1, J DD = DD J, J Gamma = Gamma J (KO-dimension 0). For even a, b with bounded commutators,

```
a° := J pi(a)\* J^-1 = 1 (x) (a\*)^c,     [pi(a), b°] = 0,     [[DD, pi(a)], b°] = 0,
```

and norm([DD, pi(a)]) = norm([D_0, a]). So if the original triple is metric, the new triple is metric with the same Connes distance on A_0. If the Weyl exponent p = lim log N_0(R)/log R of D_0 exists, that of DD is 2 p, since N_0(R/sqrt(2))^2 <= N_DD(R) <= N_0(R)^2.

Proof. J is antilinear, and J^2 = 1 because the sign is applied twice with the degrees exchanged. Gamma has eigenvalue (-1)^{|xi| + |eta|}, symmetric under the exchange. For the Dirac operator,

```
J DD (xi (x) eta^c) = (-1)^{(|xi| + 1) |eta|} eta (x) (D_0 xi)^c + (-1)^{|xi|} (-1)^{|xi| (|eta| +
↪  1)} D_0 eta (x) xi^c
                    = (-1)^{|xi| |eta|} [D_0 eta (x) xi^c + (-1)^{|eta|} eta (x) (D_0 xi)^c] = DD J
                     ↪  (xi (x) eta^c).
```

For even a, a* preserves degree, so J pi(a)* J^-1 (eta (x) xi^c) = eta (x) (a* xi)^c, which gives the formula for a° and order zero. Since a is even, [DD, pi(a)] = [D_0, a] (x) 1, which commutes with 1 (x) (b*)^c. Finally DD^2 = D_0^2 (x) 1 + 1 (x) (D^c)^2, because gamma_0 anticommutes with D_0. An eigenbasis of D_0 decomposes DD into finite-dimensional invariant blocks, so DD is essentially self-adjoint on the algebraic tensor product, with compact resolvent, and the spectrum is +/- (lambda_i^2 + lambda_j^2)^{1/2}. ■

Geometrically, when H_0 = L^2(S_X) on an even-dimensional spin manifold X, a charge conjugation commuting with D_X identifies conj(H_0) with H_0, H is L^2(S_{X x X}) and DD is the ordinary product Dirac operator. (The grading may acquire an overall sign under this identification.) So the data are a product-manifold Dirac operator on X x X with a restricted observable algebra C(X) (x) 1, and a J covering the flip of the two factors, which differs from the pointwise spin real structure. They count as commutative Kaluza-Klein data only with that restriction and that J: they are neither the ordinary spin triple of X x X with its full function algebra nor a finite-rank twist over X. The restriction is essential: on all of C(X x X), the compressions generate A_F (x) conj(A_F), and order zero generally fails again.

Corollary 16.3 (truncation). Let P = P_0 (x) conj(P_0). It commutes with DD, Gamma and J. It is a joint cutoff in the two factors, and need not be a spectral projection of DD: for absolute eigenvalues 1, 5, 7, sqrt(73) of D_0 with P_0 selecting the first three, the selected pair (5, 7) and the excluded pair (1, sqrt(73)) both give DD^2 = 74. The compression of pi(A_0) is O_F (x) 1; passing to the generated algebra gives the finite real even triple

```
T_F = (A_F, V (x) conj(V), D_F, J_F, Gamma_F),     D_F = d (x) 1 + gamma_V (x) d^c = Delta + J_F
↪  Delta J_F^-1,     Delta = d (x) 1,
```

with A_F acting on the left. It has KO-dimension 0, and satisfies order zero and order one for every A_F, noncommutative or not, with dim H_F = k^2. The family is exact at each truncation; no limit is taken.

Proof. The proof of Theorem 16.2 applies verbatim on V with arbitrary even a, b in B(V), since it uses only the grading and the tensor structure. For J_F Delta J_F^-1, apply J to (d (x) 1) J (xi (x) eta^c): the signs combine to (-1)^{|xi|}, giving gamma_V (x) d^c. ■

So H_F has the bimodule form of Connes' finite geometries, with A_F on the left and its opposite on the right, and D_F has the first-order form of Krajewski and Paschke-Sitarz.

Example (a) gives (C + C, C^2 (x) C^2, m (sigma_1 (x) 1 + sigma_3 (x) sigma_1)). There [[D_F, e], e°] = 0 for e = diag(1, 0), which repairs the first-order failure of Counterexample 08.4 at the price of doubling. Example (b) gives (M_2(C) + M_2(C), C^16, D_F).

Corollary 16.4 (KO-6 by a Clifford point). The product of T_F with (C, C^2, 0, sigma_1 K, sigma_3) is

```
(A_F, V (x) conj(V) (x) C^2, D_F (x) 1, J_F (x) sigma_1 K, Gamma_F (x) sigma_3),
```

with signs (+, +, -), order zero, order one and dim H = 2 k^2. It fails the usual zero-dimensional orientability axiom: every zero-cycle built from the left and opposite actions acts as T (x) 1_2, while the grading is Gamma_F (x) sigma_3.

Proof. (sigma_1 K)^2 = 1 and sigma_1 K sigma_3 = -sigma_3 sigma_1 K, while the D term is untouched. a° is a° of T_F tensored with 1. ■

Remark 16.5 (the order conditions are formal). For every finite even representation (A, V, d, gamma), the map V -> V (x) conj(V), d -> d (x) 1 + gamma (x) d^c gives a real bimodule with order zero and order one, using no prior finite real structure. So these two conditions follow from choosing the tensor square, and say nothing specific about A_F. The construction does not recover an arbitrary prescribed finite (H, D, J). What is specific to T0 is that A_F is generated by compressed functions, that the tensor square is itself the truncation of a commutative parent in the restricted sense above, and families such as the following.

Example 16.6 (a tower-decoupling family, from the referee). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with V = ker D_0 = C^4, and let DD_l = DD/l on the doubled parent. Its kernel is V (x) conj(V), and its gap grows like 1/l. Here P is a spectral projection. Compression followed by algebra generation gives the constant finite data A_F = M_4(C), dim H_F = 16, D_F = 0, and the Clifford point gives dim H = 32 with the KO-6 signs. Order one is automatic, since D_F = 0. The whole triple converges to (C, C^16, 0) by Theorem 15.3, consistently with lemma 15: the two statements concern the whole algebra and the truncation.

Two sheets: the form of the classification

Chamseddine and Connes (*Why the Standard Model*, arXiv:0706.3688, sections 2-3) classify irreducible pairs (A, J) with order zero and a separating vector: the centre of the complexified algebra is C, with A_C = M_k(C) on itself and J x = x*, or C + C, with A_C = M_k(C) + M_k(C) on itself and J(x, y) = (y*, x*). With centre C, J gamma = -gamma J is impossible. The pair of Corollary 16.4 is therefore reducible: the projection 1 (x) (1 + sigma_1)/2 commutes with A_F and J. The irreducible form comes from a parent with two sheets. At the finite level the bimodule (W (x) conj(V)) (+) (V (x) conj(W)) with its adjoint-flip J is already stated in Remark 2.9 of that paper; what we add is its realization from parents and compressed functions.

Theorem 16.7 (two-sheet coarse parent). Let (A_X, H_X, D_X, gamma_X) and (A_Y, H_Y, D_Y, gamma_Y) be even spectral triples. On H = (H_X (x) conj(H_Y)) (+) (H_Y (x) conj(H_X)), for even a in A_X and b in A_Y, put

```
pi(a, b) = (a (x) 1) (+) (b (x) 1),
DD = (D_X (x) 1 + gamma_X (x) D_Y^c) (+) (D_Y (x) 1 + gamma_Y (x) D_X^c),
Gamma = (gamma_X (x) gamma_Y^c) (+) (-gamma_Y (x) gamma_X^c),
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c,
```

where J maps each summand to the other. Then J^2 = 1, J DD = DD J and J Gamma = -Gamma J, the KO-6 signs. Moreover

```
(a, b)° = (1 (x) (b\*)^c) (+) (1 (x) (a\*)^c),
```

so the first summand is an (A_X, A_Y)-bimodule and the second an (A_Y, A_X)-bimodule, and order zero and order one hold.

Proof. The computation of Theorem 16.2 applies with D_Y in place of the second factor: on xi (x) eta^c in the first summand, J DD and DD J both give (-1)^{|xi| |eta|} [D_Y eta (x) xi^c + (-1)^{|eta|} eta (x) (D_X xi)^c]. The sign in Gamma on the second summand gives J Gamma = -Gamma J. For a°, apply J, then pi(a, b)*, then J, as in Theorem 16.2: an element of the first summand goes to the second, where b* acts on the left, and returns with b* on the right factor. Since a and b are even, [DD, pi(a, b)] = ([D_X, a] (x) 1) (+) ([D_Y, b] (x) 1), which commutes with the opposite actions. ■

Corollary 16.8 (truncation; the form of the classification). For symmetric windows with V_X, V_Y of dimensions k_X, k_Y, and even subalgebras A_{F,X} in B(V_X), A_{F,Y} in B(V_Y), the compression

```
T_F = (A_{F,X} + A_{F,Y}, (V_X (x) conj(V_Y)) (+) (V_Y (x) conj(V_X)), D_F, J_F, Gamma_F)
```

has the KO-6 signs, order zero and order one, and dim H_F = 2 k_X k_Y. If gamma_{V_X} lies in A_{F,X} and gamma_{V_Y} in A_{F,Y}, the real orientability relation holds: with c = (gamma_{V_X}, gamma_{V_Y}) and z = (1, -1), Gamma_F = pi(z c) c°, a zero-cycle with coefficients in the algebra and its opposite. This is weaker than Gamma_F in pi(A_F). If V_X and V_Y are ungraded with k_X = k_Y = k and the generated algebras are full, then (A_F, H_F, J_F) is exactly the irreducible pair with centre C + C: identify both summands with M_k(C), so that J_F(x, y) = (y*, x*); an invertible matrix in each summand is a separating vector, and a projection commuting with both full matrix actions is scalar on each summand, with J equating the scalars. In all cases D_F commutes with the two sheet projections, so the sheets are not connected. It need not commute with the whole centre: for V_X = C^2, gamma_X = sigma_3, d_X = sigma_1, A_{F,X} = C + C acting diagonally and V_Y = C, the central element (diag(1, 0), 0) does not commute with D_F.

Proof. Theorem 16.7 applies on the truncation with arbitrary even operators. For orientability, pi(c) c° = (gamma_{V_X} (x) gamma_{V_Y}^c) (+) (gamma_{V_Y} (x) gamma_{V_X}^c), and z flips the sign of the second summand. The last statements are direct. ■

Example 16.9 (twistor level one, and M_2(H) + M_4(C)). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with kernel H^0(CP^3, O(1)) = C^4, and let j be the antiunitary with j^2 = -1 commuting with the Gamma_a of Proposition 10.1, which makes C^4 a quaternionic space H^2. Use this space twice, as W (with the functions of S^4 pulled back along the twistor projection) and as V (with the functions of CP^3).

1. A real function f on S^4 pulls back to a function invariant under w -> j w. So j Q(f) j^-1 = Q(f), and Q(f) is a Hermitian element of End_H(W). Since Q(x_a) = Gamma_a/5, the real algebra generated is End_H(W) = M_2(H), of real dimension 16.
2. For real functions on CP^3, Q(w* B w / w* w) = (B + tr(B) 1)/5 for Hermitian B, so Q is onto the Hermitian matrices, and the real algebra generated is End_C(V) = M_4(C), of real dimension 32.

Hence Corollary 16.8, with the parent gradings, which are trivial on the kernels, gives the real algebra A = M_2(H) + M_4(C) on H_F = Hom(V, W) (+) Hom(W, V) = E (-) E* of dimension 32, with J(x, y) = (y*, x*). These are the algebra, bimodule and real structure of Chamseddine and Connes. The generation is of real algebras from real functions: complex generation on the S^4 side would give M_4(C).

The nontrivial grading of W in that construction is not obtained from this parent. An even parent has [P gamma P, P a P] = 0 for every compressed function, while Gamma_5 = 5 Q(x_5) anticommutes with Q(x_1). So the grading has to be added after truncation, as abstract finite data, fixing an axis. The full algebra is then a graded prealgebra rather than the algebra of an even triple, with even part H + H + M_4(C), and Gamma_F(xi, eta) = (Gamma_5 xi, -eta Gamma_5), which is Proposition 3.2 of that paper; the real orientability relation holds on the even part. Similarly d = mu n^a Gamma_a with n orthogonal to the axis is the compression of a bounded real multiplication operator, but adding that operator to the parent Dirac operator keeps neither the kernel as a spectral projection nor the parent grading as an odd perturbation, so d too is abstract data. It commutes with the two sheet projections and with the centre R + C of the full prealgebra, and not with the centre of the even algebra.

This bears on the step left open in canonical section 4, where M_2(H) in M_4(C) is a nested pair on one four-space, with "the independent E, F module roles" to be derived. The example closes that abstract finite-module problem after adding a second sheet: the nested actions on one four-space become independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the Dirac operator connecting the two pieces that the selection of A_F = C + H + M_3(C) requires. The fibre-rank problem remains open.

The inputs are the second sheet, the restriction of one algebra to base functions, real algebra generation, the four-dimensional window, the grading axis and d. The content beyond the formal bimodule is the concrete compression calculation in items 1 and 2, already in the corpus, and its explicit match with the classified data.

Caveats

These results should be read as narrowly as the note they correct.

1. **Metric.** T_F is usually not a metric spectral triple. For A_F = B(V_+) + B(V_-) with V_+ of dimension at least 2 and d odd, the commutant of d in A_F contains the pairs (x, B^-1 x B) with x commuting with B B*, when B is the invertible block of d. In example (b), (diag(1, 0), diag(1, 0)) has L = 0, so d leaves a residual U(1) x U(1) of U(2) x U(2) unbroken. The Standard Model finite triple has the same feature, through M_3(C).
2. **Spectrum.** The eigenvalues of D_F are +/- (lambda_i^2 + lambda_j^2)^{1/2}. In example (a) they are +/- sqrt(2) m with Connes distance 1/m, so the doubled spectrum differs from the masses in d, and the mass relations of Theorem 9.6 and lemma 11 would have to be redone.
3. **Parent.** The doubled parent is a choice. Its algebra sees one factor of X x X, its J covers the flip, which is outside the spin real structure of X x X, and generating A_F from O_F is an additional operation. We have no dynamical reason for it. For an ordinary compact spin X of dimension n, its Weyl exponent is 2 n with metric dimension n, which bears on the counting of 9, 10 and 11 and has not been analyzed.
4. **KO-6.** The Clifford point is an input, as F_0 is in Theorem D, and it fails finite orientability. In Theorem 16.7 the sign comes from the block structure of two sheets, which is itself an input. The documents checked here establish neither a construction of two sheets from one connected parent nor a no-go theorem. The grading-reversing structure of lemma 08 is a candidate we have not tested.
5. **Count.** For k = 4, dim H = 32, the value in the classification of Chamseddine-Connes, and k = 4 occurs in examples (b) and (c). For k = 4 in examples (b) and (c) alone this is a count and nothing more: the classification's algebra is M_2(H) + M_4(C), acting differently, and those examples do not produce it. Example 16.9 does, as abstract data with the listed inputs.

Consequence for T0 and T0b

T0 asks for finite real triples from commutative Kaluza-Klein data under specified notions of convergence. For one specified truncation procedure (compression, then generation, on the doubled parent with restricted algebra and flip J), this lemma gives a noncommutative finite algebra with an explicit bimodule, and a family with the tower decoupled (Example 16.6). It does not establish the target under every meaning of ordinary Kaluza-Klein data. For T0b, it gives examples with the requested signs and first-order relations; with two sheets it reaches the form of the classification, and in Example 16.9 the real algebra M_2(H) + M_4(C), bimodule and real structure of Chamseddine and Connes, with the grading added as abstract data. It leaves open an operator connecting the two pieces and hence the selection of A_F, a parent that yields the grading and nonzero mass data, and the beyond-first-order and twisted frameworks. See the background note.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random odd d and even a, b for dim V_+ = dim V_- = 2 with J = M K: J^2 = 1, J DD = DD J, J Gamma = Gamma J, DD Gamma = -Gamma DD, order zero and order one all hold to machine precision, with norm([a, b]) = 7.67; the KO-6 signs of Corollary 16.4 hold. For Theorem 16.7 and Example 16.9, with k = 4, W graded by Gamma_5 and V ungraded: J^2 = 1, J Gamma = -Gamma J, J D = D J, D Gamma = -Gamma D, D = Delta + J Delta J^-1, order zero and order one for random a, b, and Gamma_F = pi(z c) c° all hold to machine precision; j^2 = -1 commutes with the Gamma_a, and x_a(j w) = x_a(w); the real algebras generated by Gamma_a, by i Gamma_a and by all Hermitian matrices have real dimensions 16, 32 and 32; and sum_a x_a Gamma_a = 2 (P_w + P_{jw}) - 1. For example (b), by quadrature with 23 test functions: dim O_F = 4 and the generated algebra has dimension 8; the two displayed values Q(f_1) and Q(f_2) were reproduced by quadrature.

Part 22

The finite Dirac operator as a boundary condition on an interval

For STRATEGY T0, T0b, T10 and T13, following lemma 12 and lemma 20. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and one gap as first written, and amendments; all are incorporated below.

Outcome. Let (H_F, D_F, J_F, Gamma_F) be finite data of signs (+, +, -), and let l ||D_F|| < pi/2. Then D_F, with J_F and Gamma_F, is exactly the lowest spectral window of the pure derivative on an interval of length l, with no bulk mass and no Yukawa term, on the domain fixed by the boundary coupling tan(l D_F). So all the eigenvalues of D_F, Yukawa and Majorana blocks alike, are quantized internal momenta k with tan(k l) equal to a singular value of the coupling. The input is the finite triple: the content is that a consistent parent exists, with these properties:

- Preservation of the domain forces the observables at the wall into C_F, the commutant of D_F in A_F, while A_F acts freely in the bulk.
- The parent satisfies order one for any finite data with order zero.
- The compression is the exact average Q_l(a) = (2l)^-1 int_{-l}^{l} e^{ixD_F} a e^{-ixD_F} dx, so it tends to the finite representation as l -> 0, with defects of order l^2.
- The scalar inner fluctuations of the parent are pure gauge under its allowed endpoint gauge group. On the moving window, they act by the gauge orbit of D_F through the transport along the interval. This reproduces the quadratic term of Chamseddine-Connes-van Suijlekom on the orbit, and it leaves the singular values fixed, so the radial mode of the Higgs field is absent from this parent.

With the family X(l) = tan(l M) and l -> 0, the couplings of the light fermions vanish as m l at fixed mass, the tower decouples, and the broken gauge directions go to pi/(2l). The parent algebra is matrix-valued, so T0 for a commutative parent remains open.

Setting

Let H_F = H_+ (+) H_- be finite-dimensional, Gamma_F = diag(1, -1), let c : H_+ -> H_- be antiunitary, and put J_F(a, b) = (c^-1 b, c a), so that J_F^2 = 1 and J_F Gamma_F = -Gamma_F J_F. Let D_F = [[0, M*], [M, 0]] with M : H_+ -> H_-. Then J_F D_F = D_F J_F is equivalent to M = c M* c. By the Takagi factorization, in bases related by c, we may write M = V |M| with a unitary V satisfying V = c V* c.

Let l > 0 with l ||D_F|| < pi/2, and define X : H_- -> H_+ by

`tan(l D_F) = [[0, X], [X\*, 0]],  that is,  X = tan(l |M|) V\*.`

Since J_F commutes with real functions of D_F, c X = X* c^-1. Conversely, every X with c X = X* c^-1 arises from exactly one such D_F, namely D_F = l^-1 arctan of the block operator. In the limit, X = l M* + O(l^3).

The parent. On H_l = L^2([0, l]; C^2) (x) (H_+ (+) H_-), write psi = (Q_L, Q_R; U_L, U_R), with Q valued in H_+ and U in H_-. Let (D psi)_L = psi_R' and (D psi)_R = -psi_L' on both fields, the operator of lemma 12 with no bulk mass, on the domain

`Q_R(0) = 0,  U_L(0) = 0,  Q_R(l) = X U_R(l),  U_L(l) = -X\* Q_L(l).`

Call it D_X. Put Gamma^ psi = (Q_L, -Q_R; U_L, -U_R) and J^ psi = (c^-1 U_R, -c^-1 U_L; -c Q_R, c Q_L). Let (E psi)_L = psi_R and (E psi)_R = -psi_L, so that D_X = E d/dx on its domain.

In the Standard Model form of the classification, with E = Hom(V, W) and W = W_+ (+) W_-, one takes H_+ = E_+ (+) E_-* and H_- = E_- (+) E_+*, with c the adjoint map. At X = 0, the Q-fields have boundary type (a) and the U-fields type (b), in the terms of lemma 09; for X != 0 the modes continue from those zero modes, which generally lift. A single matrix X encodes the Yukawa blocks and the Majorana block together. Its entries need not keep the original block pattern, since the tangent brings in higher powers of D_F: for M = [[a, b], [b, 0]] real symmetric, X_22 = (l^3/3) a b^2 + O(l^5).

The window

Theorem 22.1. D_X is self-adjoint with compact resolvent, and J^^2 = 1, J^ D_X = D_X J^, J^ Gamma^ = -Gamma^ J^, Gamma^ D_X = -D_X Gamma^. Let R(x) = exp(x Gamma_F D_F), and define W : H_F -> H_l so that for phi = (a, b), the L-components (Q_L, U_L) of W phi at x are l^{-1/2} R(x)(a, 0), and the R-components (Q_R, U_R) are l^{-1/2} R(x)(0, b). Then W is an isometry and

```
D_X W = W D_F,    J^ W = W J_F,    Gamma^ W = W Gamma_F.
```

With sigma_k the singular values of M and theta_k = l sigma_k, the signed spectrum of D_X is (n pi +- theta_k)/l, n in Z, with multiplicities added when values coincide; a zero singular value gives a zero mode of multiplicity two. So the spectrum of |D_X| consists of the sigma_k and of (n pi +- theta_k)/l for n >= 1, and ran W is the spectral subspace of |D_X| <= ||D_F||, and the next level is at least (pi - l ||D_F||)/l.

Proof. Self-adjointness is Lemma 12.1. Since Gamma_F D_F is anti-Hermitian, R(x) is unitary, and with C(x) = cos(x |M|), C'(x) = cos(x |M*|), S(x) = V sin(x |M|),

```
R(x)(a, 0) = (C(x) a, -S(x) a),    R(x)(0, b) = (S(x)\* b, C'(x) b).
```

Isometry follows from the unitarity of R(x) and the orthogonality of the L- and R-components. The conditions at 0 hold since S(0) = 0. At l, using V* cos(l |M*|) = cos(l |M|) V*, we get X C'(l) = S(l)* and X* C(l) = S(l), which are the conditions at the wall. For the derivative, the L-part of D W phi is d/dx R(x)(0, b) = R(x) Gamma_F D_F (0, b) = R(x)(M* b, 0), and the R-part is -d/dx R(x)(a, 0) = R(x)(0, M a); together these are W D_F phi. The relations for Gamma^ are direct. For J^, use c |M| c^-1 = |M*| and V = c V* c, which give c^-1 C' = C c^-1 and c S(x)* = S(x) c^-1. The signs of the parent follow in the same way, using c X = X* c^-1 for the domain. For the spectrum, constant unitaries on Q and on U commute with E d/dx and bring X to diag(tan theta_k), which reduces to Example 12.5 in each channel. ■

Corollary 22.2 (conjugated form). D_X = R (D_0 + E Gamma_F D_F) R*, with R acting pointwise on the pair (Q, U) and D_0 the uncoupled operator, X = 0. So after this unitary change of domain, D_F is a constant matrix connection, odd for the grading, with relative endpoint rotation R(l). This is equation (21) of lemma 20, for general D_F.

Proof. D(R phi) = R(D phi + E R* R' phi), and R* R' = Gamma_F D_F. For phi in the uncoupled domain, the computation at the wall in the proof above shows that R phi satisfies the coupled conditions. ■

So in this family a fermion mass is a momentum: D_X has no mass term, and the light eigenvalue k = theta/l is fixed by tan(k l) = tan(theta), a singular value of X. In the finite description, the condition relating k to l is lost and the value appears as free data in D_F.

The parent as a real spectral triple

Let (A_F, H_F, D_F, J_F, Gamma_F) be finite data as above, with an even representation pi = pi_+ (+) pi_- of a unital finite-dimensional real or complex *-algebra A_F, faithful, unital and satisfying order zero, with pi°(b) = J_F pi(b)* J_F^-1. Order one is not assumed. Let

```
C_F = {a in A_F : [pi(a), D_F] = 0},    A_l = {f in C^infinity([0, l], A_F) : f(l) in C_F},
```

with f acting pointwise by pi_+(f(x)) on Q and pi_-(f(x)) on U.

Theorem 22.3.

1. A function f in C^infinity([0, l], A_F) preserves Dom D_X if and only if f(l) lies in C_F.
2. (A_l, H_l, D_X, J^, Gamma^) is a real even spectral triple, with [D_X, pi(f)] = E pi(f') and J^ pi(g)* J^^-1 = pi°(g) pointwise.
3. It satisfies order zero and order one, with signs (+, +, -).

So A_F acts freely in the bulk, and the boundary condition restricts the observables at the wall to the commutant of D_F. In the Standard Model data with nonzero Yukawa blocks, this is the algebra {(lambda, lambda, m)} of the photon and gluons; with vanishing Yukawa blocks the commutant is larger. Order one

holds in this parent whatever the finite data, for its pointwise algebra: the finite condition appears only on the truncation, and arbitrary compressions need not preserve either order condition.

Proof. Item 1: at l, the traces Q_L(l) and U_R(l) are free, and applying f to the conditions gives pi_+(f(l)) X = X pi_-(f(l)) and pi_-(f(l)) X* = X* pi_+(f(l)), as in Result 3 of lemma 20. That is [pi(f(l)), tan(l D_F)] = 0, equivalent to [pi(f(l)), D_F] = 0 because l ||D_F|| < pi/2 makes D_F a function of tan(l D_F). At 0 the conditions are preserved since pi is even. Item 2: the commutator is the Leibniz rule. Conjugation by J^ acts on (Q, U) as J_F does, and pi(g) acts identically on the two spin components, which gives the second formula. This preserves the domain since J_F commutes with D_F. Item 3: pi(f(x)) and pi°(g(x)) commute at each x by finite order zero, and E commutes with both; so [E pi(f'), pi°(g)] = E [pi(f'), pi°(g)] = 0. ■

This is consistent with Counterexample 4 of lemma 20, whose finite data A_F = C + C, D_F = m sigma_1 fail order one while the parent satisfies it.

The compression

Let Q_l(T) = W* T W for a bounded operator T on H_l. For T(x) a bounded function with values in the even operators on H_F, acting pointwise and identically on both spin components, we write T for the multiplication operator.

Theorem 22.4.

```
Q_l(T) = l^-1 int_0^l cos(x ad_{D_F})(T(x)) dx,      Q_l(E T) = -l^-1 int_0^l sin(x ad_{D_F})(T(x))
↪ dx.
```

For constant a in A_F,

```
Q_l(a) = sinc(l ad_{D_F})(pi(a)) = (2l)^-1 int_{-l}^{l} e^{ix D_F} pi(a) e^{-ix D_F} dx,
```

so in an eigenbasis of D_F with eigenvalues mu_i, the matrix elements are pi(a)_{ij} sin(l (mu_i - mu_j))/(l (mu_i - mu_j)). Moreover:

1. Q_l(A_l) = Q_l(C^infinity([0, l], A_F)), so the image of the parent algebra contains Q_l(a) for every a in A_F.
2. ||Q_l(a) - pi(a)|| <= (l^2/6) |||[D_F, [D_F, pi(a)]]|| <= (2/3) l^2 ||D_F||^2 ||a||.
3. Q_l([D_X, f]) = [D_F, Q_l(f)] for f in A_l.
4. With a' = Q_l(a) and b' = J_F Q_l(b)* J_F^-1, |||[a', b']|| <= (8/3) l^2 ||D_F||^2 ||a|| ||b||, and |||[[D_F, a'], b'] - [[D_F, pi(a)], pi°(b)]|| <= (16/3) l^2 ||D_F||^3 ||a|| ||b||.

So as l -> 0 with D_F fixed, D_F, J_F and Gamma_F stay exact, and the compressed observables converge to pi(A_F) with rate l^2. The finite order conditions hold in the limit exactly when they hold for the finite data.

Proof. From the definition of W, the expectation of T splits into the L- and R-parts, so Q_l(T) is l^-1 int of the Gamma_F-even part of R(x)* T(x) R(x). Put K = Gamma_F D_F. For even Z, ad_K Z = Gamma_F [D_F, Z], which is odd, and ad_K (Gamma_F [D_F, Z]) = -[D_F, [D_F, Z]]. So the even part of e^{-xK} T e^{xK} is cos(x ad_{D_F})(T). Likewise, E exchanges the L- and R-parts with a sign, so Q_l(E T) is l^-1 int of Gamma_F times the odd part of R* T R, and that odd part is -Gamma_F sin(x ad_{D_F})(T). The formula for constant a follows from cos(x ad_D)(Z) = (e^{ixD} Z e^{-ixD} + e^{-ixD} Z e^{ixD})/2.

Item 1: the map f -> Lambda(Q_l(f)), for a linear functional Lambda, is integration against a smooth density. If it vanishes on A_l, which contains all f of compact support in [0, l), the density vanishes on [0, l), hence everywhere by continuity. Item 2: by Taylor's formula with integral remainder, ||cos(x ad_D)(Z) - Z|| <= (x^2/2) ||ad_D^2 Z||, since cos(x ad_D) has norm at most 1 by the average formula; averaging over x gives l^2/6. Item 3: P commutes with D_X. It also follows from the two formulas by integration by parts, where the boundary term at l vanishes because f(l) is in C_F. Item 4: J_F commutes with D_F, so b' is the same average of pi°(b). Expand both commutators with the bound of item 2 and |||[D_F, Z]|| <= 2 ||D_F|| ||Z||. ■

The convergence is a statement about the maps Q_l, on the operator systems, in the setting of Connes and van Suijlekom (arXiv:2004.14115, sections 1 and 3). The generated algebra is not continuous at l = 0. An analytic example, from the referee: let M = [[0, 0, 1], [0, 0, 1], [1, 1, 0]], D = [[0, M], [M, 0]], with A_F = C^3 acting diagonally and equally on the two blocks. With P_1, P_2 the first two channel projections, [Q_l(P_1), Q_l(P_2)]_{12} = -l^2/3 + O(l^4), so the compressions of commuting projections generate a noncommutative algebra for small nonzero l. On the one-generation Standard Model data, a numerical closure with tolerance 10^-7 gives complex dimension 15 for the algebra generated by pi(B), with B = C + H + M_3(C), and 30 for Q_l(B) at l = 0.3; these are numerical values with no certified rank.

Scalar fluctuations

Theorem 22.5. Let h be smooth with values in the anti-Hermitian elements of A_F. The self-adjoint one-forms of the parent are exactly the E pi(h). Let u' = -h u, u(0) = 1, and let w = u(l), a unitary of A_F. Then

```
D_X + E pi(h) = pi(u) D_{X_w} pi(u)\*,      X_w = pi_+(w)\* X pi_-(w),
```

and the window of D_{X_w} carries pi(w)* D_F pi(w). With the real structure, for A = E pi(h) we have J^ A J^^-1 = E J_F pi(h) J_F^-1, and D_X + A + J^ A J^^-1 has the window U_w* D_F U_w, with U_w = pi(w) J_F pi(w) J_F^-1, and this commutes with J_F. Moreover, with u~(x) = u(x) w*, which lies in A_l since u~(l) = 1, and U~ = pi(u~) J_F pi(u~) J_F^-1,

```
D_X + A + J^ A J^^-1 = U~ D_X U~\*.
```

So every scalar connection in this model is gauge equivalent to the original parent under the allowed endpoint gauge group (referee).

Proof. A one-form is sum f_1 [D_X, f_2] = E pi(sum f_1 f_2'), and E pi(h) is self-adjoint exactly when h is anti-Hermitian, since E is anti-Hermitian and commutes with pi(h) and pi is faithful. Every smooth such h arises: with b(x) = int_l^x h, we have b(l) = 0, so b is in A_l, and [D_X, b] = E pi(h). Next, pi(u)* D pi(u) = D + E pi(u* u'), and pi(u)* E pi(h) pi(u) = E pi(u* h u), so the sum is D with no connection term. At 0 the conditions are unchanged. At l, the substitution psi = pi(u) phi gives the conditions with X_w. Then tan(l pi(w)* D_F pi(w)) has the block X_w. For general w, this operator need not commute with J_F, so we use the spectrum and the isometry of Theorem 22.1 without its real-structure part. For the real fluctuation, J^ commutes with E, u(x) is replaced by pi(u) J_F pi(u) J_F^-1, using finite order zero, and U_w commutes with J_F, so the real window follows. The last formula follows by differentiating, with the endpoint condition. ■

Corollary 22.6 (the quadratic term as transport). By Chamseddine-Connes-van Suijlekom (arXiv:1304.7583, section III), for unitary v in A_F, with U = v J v J^-1, one has U D U* = D + A_(1) + A~*(1)* + *A*(2), for the universal form v delta(v*). Taking v = w*, on the gauge orbit the fluctuation of the parent, which satisfies order one and has no quadratic term, gives on the moving window exactly the fluctuated operator of that framework (equations (10)-(11) of section III.2 there), quadratic term included, for finite data that fail order one. This answers the first open point of lemma 19 on these gauge orbits and with the moving window. For the two-point data with J = sigma_1 K and w = diag(e^{i alpha}, e^{i beta}), it gives the upper link m e^{-2 i (alpha - beta)}, which is m e^{2 i psi} of Corollary 19.2 with psi defined from v = w*. Theorem 22.5 and the covariant projection of lemma 14 share a transport principle. The fixed compression, as in lemma 13, is a different object: by Theorem 22.4, it gives Q_l(E pi(h)) = -[D_F, l^-1 int_0^l x pi(h) dx] + O(l^3 ||D_F||^3 ||h||) for h fixed, so that for constant h it is -(l/2)[D_F, h], while the first derivative of the transported operator is -l [D_F, h]. The difference is the contribution of the moving spectral projection.

No-go theorem 22.7 (the radial mode). *Hypotheses:* the parent above, on one interval, with fixed X, no bulk masses, and scalar inner fluctuations from the even algebra A_l, with the moving spectral window. *Conclusion:* the fluctuations leave all singular values of the finite Dirac operator fixed. In the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through the unit quaternions, a three-dimensional gauge orbit, with no radial mode. Degenerate data can have a smaller

orbit. A fixed compression can show changes in singular values at second order, so the window has to be named.

Proof. Theorem 22.5: D_F and pi(w)* D_F pi(w) are unitarily equivalent. The same holds with U_w. ■

Known escapes.

These lie outside the hypotheses, and each needs its own construction.

- The coupling X as a field on the wall, with its own kinetic term and potential.
- Connections odd for the grading, as in Corollary 22.2 and lemma 23: the fluctuation K -> K + delta K changes the singular values.
- Bulk masses that do not commute with the gauge action.
- Two or more internal dimensions, where a connection has curvature, although curvature alone does not prove a radial mode.
- An additional composite or other degree of freedom; a quantum effective action on the same configuration space does not by itself add a radial coordinate.
- A fluctuation of the finite data alone, as with the radial parameter of lemma 19, which is not the moving window.

Direction. On the interval, the even connection is pure gauge up to its boundary effect, so the finite orbit is all it can reach. The physical Higgs modulus must then be sought at the wall, in odd directions, or in more dimensions. At the 10 -> 9 endpoint this agrees with T10, where the Higgs is frozen.

The gauge sector and the 10 -> 9 family

Proposition 22.8. Consider the product of M^4 with this parent, with algebra of functions f(y, x) valued in A_F and f(y, l) in C_F. Then the gauge one-forms sum f_1 d_mu f_2 take values in C_F at x = l. Assume, in addition, a flat product, a positive five-dimensional Yang-Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Then the directions in the Lie algebra of C_F have masses n pi/l, n >= 0, and the orthogonal directions have masses (n - 1/2) pi/l, n >= 1. The physical gauge algebra comes from the skew-adjoint elements through the real gauge representation, with its kernel and any unimodularity condition.

Proof. The first statement is direct evaluation at x = l. The rest is the interval spectrum in the gauge A_5 = 0, with Neumann conditions at 0, and with Neumann or Dirichlet conditions at l. ■

In the Standard Model data with nonzero Yukawas, M_W = M_Z = pi/(2l). The Lie algebra of C_F has dimension 10 before unimodularity; with the unimodularity condition of Chamseddine-Connes-Marcolli (arXiv:hep-th/0610241, section 2.5), the massless modes are the photon and the gluons. These are the boundary conditions of Higgsless models (Csaki, Grojean, Murayama, Pilo and Terning, Phys. Rev. D 69 (2004) 055006), and the fermion formula tan(m L) = M_D L is equation (5.6) of Csaki, Grojean, Hubisz, Shirman and Terning (Phys. Rev. D 70 (2004) 015012); paper/LITERATURE.md, section C. Boundary spectral geometry for the Standard Model has been studied by Chamseddine and Connes (arXiv:1008.3980, section 5). What is added here, as far as our searches went, is the combination: the real structure, the finite triple as an exact window, the wall algebra C_F, the compression formula and the fluctuations. A wider claim of novelty would need a fuller comparison.

Corollary 22.9 (the 10 -> 9 family). Fix a block M_0 for the light fermions and a block Theta/l on an orthogonal channel, compatible with J_F and the grading, with 0 < Theta < pi/2, for the top and if wanted for the Majorana block. Let D_F(l) have M(l) = M_0 (+) Theta/l, and let l -> 0. Then:

1. X(l) has the light block tan(l |M_0|) V_0* = l M_0* + O(l^3) and a heavy block with singular values tan(Theta): the light couplings vanish as m l at fixed mass m.
2. The heavy channel has mass Theta/l, the broken gauge directions have mass pi/(2l), and the other levels diverge.
3. The massless gauge sector is C_F. Let P_L be the projection reducing D_F(l) to the fixed light block D_L. For constant a, P_L Q_l(a) P_L = sinc(l ad_{D_L})(P_L pi(a) P_L), with ||P_L Q_l(a) P_L - P_L pi(a) P_L|| <= (2/3) l^2 ||D_L||^2 ||pi(a)||, also when pi(a) mixes light and heavy channels.

The limiting light compression need not be a representation, since products can pass through the heavy subspace. The full estimate of Theorem 22.4 does not apply, because l ||D_F(l)|| tends to Theta. For example, with D_F(l) = sigma_1 (x) diag(m, Theta/l) and pi(a) = 1 (x) sigma_1, one has Q_l(a) = sinc(Theta - l m) pi(a), which tends to sinc(Theta) pi(a).

4. With l = pi/(2 M_W), we get X_q ~ (pi/2) m_q/M_W, which has the form of y_q = sqrt(2) m_q/v -> 0 at fixed mass, as the coupling vanishes while the mass stays. Identifying X_q with the four-dimensional Yukawa coupling is a hypothesis.
5. The ratio is m_t/M_W = 2 Theta/pi < 1, and M_W/M_Z = 1.

So the 10 -> 9 proposal is realized at this level as a limit l -> 0, with W, Z and top at the tower scale and the light fermions keeping explicit masses. Item 5 does not reach m_t/M_W = sqrt(8/3), which belongs to the interior. This is a statement in the stated family: lowest level, no bulk mass, strict wall algebra. Bulk masses for the top, and a changed gauge wall condition, are directions beyond it.

A numerical comparison for the interior (Python). A separate model, with a finite wall scalar and Neumann conditions at 0, lies outside Proposition 22.8 and No-go theorem 22.7. In it, the mass of a broken direction satisfies t tan t = s, with t = M l and s = g^2 v^2 l^2/4 (for the Z, s/cos^2 theta_0), which has the form of M_k tan(M_k pi R) = -V in Csaki et al. (arXiv:hep-ph/0305237, section on mixed boundary conditions, read in the LaTeX source), with the sign convention of V there. If g'^2/g^2 = 3/5 is taken at all scales, M_W^2/M_Z^2 goes from 5/8 as s -> 0 to 1 as s -> infinity. It equals the measured 0.777 at M_W l = 1.03, with 1/l = 78 GeV and a next W level at 280 GeV; whether collider searches allow this would need the couplings, branching fractions and an applicable limit, which we have not analyzed. The top needs M_W l < 0.96 for m_t/M_W = sqrt(8/3), or M_W l < 0.73 for the measured ratio, in the lowest level with no bulk mass. So in the flat tree-level toy without bulk masses or running, the top and gauge requirements are incompatible. It does show the interior interpolating between the spectral-action value 5/8 and the endpoint value 1. The directions are bulk masses, warping and running.

What is input and what is derived

The finite data (A_F, H_F, D_F, J_F, Gamma_F) are all inputs, so the existence of the window is close to a reparametrization: X = tan(l D_F). The derived statements are the wall algebra C_F for that input, the order conditions of the parent, the compression formula with its rate, the form of the fluctuations, and the gauge spectrum under the stated assumption. The window identifies D_F, J_F and Gamma_F exactly, while the represented algebra is obtained through compression and is not an exact representation of A_F at l > 0. In particular, this does not select A_F. T0 asks for a commutative parent: a natural next step is a two-stage parent, compressing functions on S^4 and CP^3 at level one to get A_F-valued functions on the interval, and then taking this window, where the difficulty is the same interface problem as after Result 7 of lemma 20.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^5, l ||M|| = 1.3: D u_+(a) = u_-(M a) and D u_-(b) = u_+(M* b) by finite differences, to about 10^-9; boundary conditions and J^ to about 10^-15; norm of W to 10^-13.
- Q_l(a) = sinc(l ad)(a), by quadrature, to 2 x 10^-8. The formula for Q_l(E T), with a nonconstant T, to 10^-14.
- One-generation Standard Model data with Yukawas and the rank-one Majorana block, with signs, order zero and order one exact on B = C + H + M_3(C), and commutant of D in B of real dimension 20. At l = 0.1 and 0.03, ||Q_l(a) - a + (l^2/6)[D, [D, a]]|| is 10^-5 and 4 x 10^-8, and the order-one defect of Q_l is 1.5 x 10^-4 and 1.2 x 10^-5, which scales as l^2.
- The dimensions of the generated algebras quoted above, 15 and 30, from a numerical closure with tolerance 10^-7, not retained and not certified; at smaller l the numerical rank depends on the tolerance.
- The roots of t tan t = s, and the values quoted in the comparison for the interior.

- On the two-point data with J = sigma_1 K: D + A_(1) + A~*(1)* + *A*(2) = U D U* to 4 x 10^-16, and the link of U_w* D U_w equal to m e^{-2 i psi}.
- The bounds of Theorem 22.4, items 2 and 4, on 30 random samples of the one-generation data: largest ratios to the bound 0.33, 10^-4 and 10^-3.

Part 23

Two gradings on an interval: masses as principal angles, the Higgs modulus, and exact W and Z masses

For STRATEGY T0b, T5b, T10 and T13, following lemma 22. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): four errors and one gap as first written, and amendments, all incorporated below. The simpler form of the mass formulas, and the proof of item 3 of Theorem 23.6, are by the referee.

Outcome. A self-adjoint boundary condition for the derivative on an interval is a grading at each wall. The whole finite Dirac operator is then the relative position of two gradings: its singular values are the principal angles between them, divided by l, and

```
D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0).
```

With a bulk prealgebra, this gives four results.

1. Aligned walls compress a prealgebra stable under the grading to its even part, with the grading inherited, which is the step that Example 16.9 had to add by hand.
2. Odd scalar fluctuations move the second grading, so they change the singular values and supply the radial mode that No-go theorem 22.7 excludes for even fluctuations. For M_2(H) the field space is S^4, and the Higgs modulus is half the geodesic distance between the points chosen by the two walls.
3. With a wall connector on one channel and a five-dimensional Yang-Mills action with common normalization, the masses are exact: sin^2(M_W l) = sin^2(2 theta)/2 and sin^2(M_Z l) = (4/5) sin^2(2 theta), for universal fermions of mass theta/l. At small theta this gives M_W^2 = 2 m^2 and M_Z^2 = (16/5) m^2, the constants in the lower bounds of the fully massive family of lemma 18, with M_W^2/M_Z^2 tending to 5/8 at both ends.
4. The heat-cutoff spectral action on the interval is a Jacobi theta function of theta, decreasing on [0, pi/2]. This family lies outside the hypotheses of the radial no-go of lemma 18: the radial direction is compact, with a stationary point at m = pi/(2l), the Connes-Lott value, where M_W = M_Z = 0.

Everything finite is still an input, and the parent algebra is matrix-valued. For a commutative parent, lemma 24 finds the window exactly, with spectral walls, while direct compression of scalar functions gives only R + R on the S^4 factor. The formulas for gauge masses depend on the assumed action. The relation to measured values fails in the flat toy, as stated at the end.

Two gradings

Let H be a finite-dimensional Hilbert space, and let Gamma_0, Gamma_1 be self-adjoint involutions on H. On L^2([0, l]; C^2) (x) H, with psi = (psi_L, psi_R), let D = E d/dx, where (E psi)_L = psi_R and (E psi)_R = -psi_L, on the domain

```
Gamma_0 psi_L(0) = psi_L(0),   Gamma_0 psi_R(0) = -psi_R(0),   Gamma_1 psi_L(l) = psi_L(l),
↪  Gamma_1 psi_R(l) = -psi_R(l).
```

Call it D(Gamma_0, Gamma_1), and let Gamma^ act as +1 on psi_L and -1 on psi_R.

Theorem 23.1.

1. D(Gamma_0, Gamma_1) is self-adjoint with compact resolvent, and it anticommutes with Gamma^.
2. Suppose Gamma_0 and Gamma_1 have eigenspaces of equal dimensions, and that -1 is not an eigenvalue of Gamma_1 Gamma_0. Let theta_k in [0, pi/2) be the principal angles between the +1 eigenspaces. Then K = (1/2l) log(Gamma_1 Gamma_0), with the principal branch, is anti-Hermitian,

Gamma_0 K Gamma_0 = -K, and D_F = Gamma_0 K is self-adjoint, odd for Gamma_0, with singular values theta_k/l on the +1 eigenspace. Moreover Gamma_1 = e^{lK} Gamma_0 e^{-lK}.

3. With H_+, H_- the eigenspaces of Gamma_0, and X the H_- -> H_+ block of tan(l D_F), as in lemma 22, D(Gamma_0, Gamma_1) = D_X, and the spectral part of Theorem 22.1 gives the spectrum and the window, with no real structure needed. Conversely, every D_F odd for Gamma_0 with l ||D_F|| < pi/2 arises from exactly one such Gamma_1.
4. Let an antiunitary J_1 satisfy J_1^2 = -1, J_1 Gamma_i J_1^-1 = -Gamma_i for i = 0, 1. Then (J^ psi)_L = J_1 psi_R, (J^ psi)_R = -J_1 psi_L defines a real structure of signs (+, +, -) for D(Gamma_0, Gamma_1). With J_F = Gamma_0 J_1 up to sign, J_F^2 = 1, J_F Gamma_0 = -Gamma_0 J_F and J_F D_F = D_F J_F, which are the signs of lemma 22.

So J_F itself does not anticommute with Gamma_1 in general: J_F Gamma_1 J_F^-1 = -Gamma_0 Gamma_1 Gamma_0, the reflection of Gamma_1 through Gamma_0.

Proof. Item 1: the boundary form is [<psi_R, phi_L> - <psi_L, phi_R>] evaluated between 0 and l, which vanishes on the domain because L-values and R-values lie in orthogonal eigenspaces at each wall, and these are half of the boundary data.

Item 2: U = Gamma_1 Gamma_0 is unitary and Gamma_0 U Gamma_0 = Gamma_0 Gamma_1 = U^-1, so K is anti-Hermitian and anticommutes with Gamma_0. The eigenvalues of U are e^{+-2 i theta_k}, by the standard two-projection decomposition, and K then has eigenvalues +-i theta_k/l. Since e^{lK} Gamma_0 e^{-lK} = e^{2lK} Gamma_0 = U Gamma_0 = Gamma_1, the last formula follows.

Item 3: in the notation of Theorem 22.1, R(l) = e^{lK} carries H_+ to the graph {(C a, -S a)} = graph(-X*), and H_- to graph(X). These are the +1 and -1 eigenspaces of Gamma_1, which is the condition at the wall. The converse uses the same formula, with uniqueness from injectivity of the exponential on the stated range.

Item 4: J^ commutes with E by direct computation, and J^^2 = -J_1^2 = 1. The domain is preserved since J_1 exchanges the eigenspaces of each Gamma_i. For the last statement, J_1 U J_1^-1 = U, so as J_1 is antilinear, J_1 K J_1^-1 = K in the sense of the principal branch, and J_F = Gamma_0 J_1 then satisfies J_F K J_F^-1 = -K and commutes with D_F = Gamma_0 K. ■

When theta_k = pi/2 the channel has type (c) of lemma 09, with no zero mode and lowest level pi/(2l), the Connes-Lott mass of Theorem A in the paper. There the principal logarithm and the isolated window cease to be unique.

The bulk prealgebra

Let P be a *-subalgebra of End(H), which need not commute with Gamma_0. Let

```
A(Gamma_0, Gamma_1) = {f in C^infinity([0, l], P) : [f(0), Gamma_0] = 0, [f(l), Gamma_1] = 0},
```

acting pointwise and identically on both spin components.

Theorem 23.2.

1. A function f in C^infinity([0, l], P) preserves the domain exactly when it lies in A(Gamma_0, Gamma_1). The commutators are [D, f] = E f', and Gamma^ commutes with every f, including those with values odd for Gamma_0.
2. Under the hypotheses of Theorem 23.1, item 2, with W the isometry of Theorem 22.1, Q(T) = l^-1 int_0^l E_0(R(x)* T(x) R(x)) dx, where E_0(Z) = (Z + Gamma_0 Z Gamma_0)/2 is the conditional expectation onto the even part.
3. *Aligned walls.* If Gamma_1 = Gamma_0, then D_F = 0, the window consists of constant L-values in H_+ and constant R-values in H_-, and Gamma^ restricts on it to Gamma_0. For an ambient constant multiplier a in P, Q(a) = E_0(a), and the image of the boundary algebra A(Gamma_0, Gamma_0) is E_0(P), by compactly supported profiles. If Gamma_0 P Gamma_0 = P, then E_0(P) is the even part of P, so the compressions of the prealgebra are exactly its even part, with the grading inherited from the even parent operator Gamma^. Without that hypothesis this can fail: for H = C^2,

Gamma_0 = sigma_z and P generated by sigma_x + sigma_z, E_0(P) = span{1, sigma_z}, while the even part of P is C.

4. *Order conditions.* Let J_1 be as in Theorem 23.1 and let P° = {J_1 b* J_1^-1 : b in P} commute with P. Then the parent has order zero and order one for A(Gamma_0, Gamma_1).

Proof. Item 1: as in Theorem 22.3, using that the conditions at a wall say that L- and R-values lie in the eigenspaces of Gamma_i. Item 2: as in the proof of Theorem 22.4. Item 3 is the case R = 1. Item 4: [D, f] = E f', and pointwise commutation gives both conditions. ■

Item 3 evades the inherited-grading obstruction of Result 11 of lemma 20, because hypothesis 4 there fails: for odd a, here Q(a) = 0, while there the coordinates are retained as Q(x_a) = Gamma_a/5. So at the level of a matrix-valued parent this addresses the question left in step 3 of the framework note: the prealgebra is the bulk algebra, the even algebra is what the window sees, and the grading is a boundary condition. The commutative-parent and retained-coordinate questions remain open, and Theorem 4 of lemma 24 shows a real difference: scalar functions compressed directly give R + R on the S^4 factor, and H + H needs the matrix algebra generated at the intermediate stage.

Remark 23.3 (the twist). On H = E (+) E*, with J(x, y) = (y*, x*), the operator J_1 = Gamma_0 J is the one that anticommutes with both gradings. For an odd anti-Hermitian h in P, J_1 h J_1^-1 is y -> +y h on E*, where J alone gives y -> -y h. So the opposite representation b° = J_1 b* J_1^-1 incorporates on the W-factor the grading automorphism sigma(b) = gamma_W b gamma_W. In this parent the twist comes from the requirement that J_1 preserve both boundary conditions. Twisted spectral triples for the grand symmetry (Devastato, Lizzi and Martinetti, JHEP 01 (2014) 042, arXiv:1304.0415; Devastato and Martinetti, Math. Phys. Anal. Geom. 20 (2017) 2, arXiv:1411.1320, Eqs. (1.8)-(1.11)) use a different automorphism: their rho exchanges left and right spinorial indices and algebra summands, while conjugation by gamma_W is inner on M_2(H) and fixes its centre. So this is a structural analogy between involutive automorphisms, and an identification needs a comparison of the representations, opposites and first-order conditions.

Odd fluctuations and the radial mode

Theorem 23.4. Let h be smooth with values in the anti-Hermitian elements of P, and let u' = -h u, u(0) = 1, w = u(l). Then

```
D(Gamma_0, Gamma_1) + E h = u D(Gamma_0, w\* Gamma_1 w) u\*.
```

So scalar fluctuations move the second grading along its orbit under the unitary group of P. If h has an odd part, the principal angles between Gamma_0 and w* Gamma_1 w change in general.

Proof. u* D u = D + E u* u', and u* E h u = E u* h u, so the connection cancels. Since u(0) = 1, the wall at 0 is unchanged. At l, psi = u phi gives the stated grading for phi. ■

This lies outside No-go theorem 22.7, whose hypotheses include the even algebra: an even connection gives a w commuting with Gamma_0, and the angles are kept, while an odd connection changes them. For a real fluctuation, add the conjugate one-form: D + E h + J^(E h)J^^-1, with generator of the form (L_F, R_F) in Example 23.5. The invariant is a double coset. With H_i = Stab(Gamma_i) in the connected unitary group G of P, a gauge transformation removing h has the form u(x) c, and it preserves both endpoints exactly when c is in H_0 and w c in H_1, that is, when w is in H_1 H_0. So the scalar moduli of the parent are the double cosets H_1 G / H_0, the relative position of the two gradings. This is with the moving window, as in lemma 22.

Example 23.5 (M_2(H) and S^4). Let W = H^2 = C^4, with the Clifford generators Gamma_a of lemma 16, and for a unit vector p in R^5 put Gamma_p = p_a Gamma_a. Each grading of W with one-dimensional quaternionic +1 and -1 eigenspaces is such a Gamma_p, so these gradings form S^4 = Sp(2)/(Sp(1) x Sp(1)), the quaternionic projective line. For two points, Gamma_q Gamma_p = p . q + (p wedge q), with eigenvalues e^{+-i d}, where d is the geodesic distance; so all the principal angles equal theta = d/2.

Take H = Hom(V, W) (+) Hom(W, V), J(x, y) = (y*, x*), Gamma_0(x, y) = (gamma x, -y gamma) with gamma = Gamma_{p_0}, and K(x, y) = (theta/l)(F x, y F), with F an odd anti-Hermitian element of

M_2(H), F^2 = -1. Then:

- D_F = Gamma_0 K has all singular values equal to theta/l: a universal Dirac mass, for all colours.
- The second grading is Gamma_1(x, y) = (Gamma_p x, -y Gamma_{p'}), where p is at distance 2 theta from p_0 and p' is its mirror image through p_0. J_1 = Gamma_0 J satisfies Theorem 23.1, item 4, while J alone does not.
- The coordinates (theta, q) on S^4, with F determined by the unit quaternion q, are the polar coordinates of a Higgs quaternion: theta is the radial coordinate, bounded by pi/2, and q gives three angular directions, undefined at the poles. At theta = pi/2 the finite logarithm and the lowest window are degenerate. Quotienting by the stabilizer leaves the radial interval. A canonically normalized field needs the scalar kinetic term, which we have not computed.

With bulk masses, by Theorem 12.6, the light masses are about (tan(theta)/l) Phi_Q Phi_U when small compared with the tower thresholds, with a universal matrix before flavour data; general finite-angle eigenvalues need the secular equation. This field space is the coset of the minimal composite Higgs model (Agashe, Contino and Pomarol, Nucl. Phys. B 719 (2005) 165, sections 2-3, with their h/f_pi corresponding to 2 theta). Their construction uses a warped interval and additional fermion and boundary data, so the shared coset does not identify the actions or potentials. The appearance here of the S^4 of the twistor window is an observation, and no identification between the two roles of S^4 is made.

The weak bosons

Take P = M_2(H) (+) M_4(C), acting on E = Hom(V, W) on the left and on the right. Assume the following.

- A five-dimensional Yang-Mills field valued in sp(2) (+) su(4), with the trace form Tr_W + Tr_V, so with one coupling (the normalization of the spectral action, g_3 = g_2 = sqrt(5/3) g_Y).
- At x = 0: Neumann for the stabilizer of p_0, Dirichlet for its complement. At x = l: Neumann for the stabilizer of p_l and of a connector, that is, the a with [a_W, Gamma_p] = 0, a_W w = lambda w and a_V v = lambda v, as in Results 2 and 5 of lemma 20, for a line w in the +1 eigenspace of Gamma_p and a line v in V; Dirichlet for the complement.
- The connector is prescribed gauge-breaking wall data here. If it is instead a term in the fermion boundary operator, as for a unit vector e in the +1 eigenspace of Gamma_1, f = J_1 e, K_c = |f><e| - |e><f| and Gamma_{1,c} = e^{eta K_c} Gamma_1 e^{-eta K_c}, then J_1 still anticommutes with Gamma_{1,c}, the parent order conditions are kept, and the affected fermion tower changes, so the universal fermion formula must be recalculated for that channel.

Theorem 23.6. With theta the angle above, the modes that are massless for all theta are exactly nine, the photon and the eight gluons, with three additional electroweak zero modes at theta = 0 and theta = pi/2. The lowest charged and neutral massive modes satisfy

```
sin^2(M_W l) = sin^2(2 theta)/2,
sin^2(M_Z l) = (4/5) sin^2(2 theta),
```

equivalently tan^2(M_W l) = sin^2(2 theta)/(1 + cos^2(2 theta)) and tan^2(M_Z l) = 4 sin^2(2 theta)/(1 + 4 cos^2(2 theta)).

The photon has 1/e^2 = (8/3)/g^2. Moreover:

1. As theta -> 0, with m = theta/l: M_W^2 -> 2 m^2 and M_Z^2 -> (16/5) m^2, so M_W^2/M_Z^2 -> 5/8.
2. As theta -> pi/2: M_W, M_Z -> 0, with M_W^2/M_Z^2 -> 5/8, while m -> pi/(2l).
3. At theta = pi/4: M_W l = pi/4, tan(M_Z l) = 2, and M_W^2/M_Z^2 = ((pi/4)/arctan 2)^2 = 0.5032..., the global minimum. So M_W^2/M_Z^2 stays in [0.5032, 0.625), with 5/8 as a limit at the ends, where the ratio itself is undefined.
4. m/M_W goes from 1/sqrt(2) at theta -> 0 to infinity at theta -> pi/2. It equals sqrt(8/3) at theta = 1.0589.

Proof. In the gauge A_5 = 0 the field is free, with cos(k x) on the Neumann directions at 0 and sin(k x) on the Dirichlet ones, and the condition at l holds in the frame rotated by Ad exp(theta F). The generators split according to the frequency of Ad F. Write L, R for su(2) on W_- and W_+, S = (L + R)/sqrt(2), A = (L - R)/sqrt(2), and O for the odd generators; Ad F has frequency 0 on S, and frequency 2 on the plane (A, O). So S = s cos kx, A = alpha cos kx, O = beta sin kx.

At l, in the rotated frame, A' = cos(2 theta) A + sin(2 theta) O and O' = -sin(2 theta) A + cos(2 theta) O, with O' = 0 at l, which gives beta = alpha tan(2 theta) cot(k l), and so A'(l) = alpha cos(k l)/cos(2 theta).

Charged directions. The connector forces R^+- to vanish at l, with L^+- Neumann. So S(l) = A'(l), and d/dx (S + A') = 0 at l. Substituting gives sin^2(kl) (1 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Neutral directions. Add B = b cos kx for the generator T_15 = diag(1, 1, 1, -3)/(2 sqrt 6) of su(4). With Y = sqrt(3/5) R + sqrt(2/5) B and Z' = -sqrt(2/5) R + sqrt(3/5) B, the connector forces Z' to vanish at l, with L and Y Neumann. The Neumann conditions give b = -sqrt(3) s, and Z' = 0 gives s = alpha/(4 cos(2 theta)). Substituting in the condition for L gives sin^2(kl) (1/4 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Massless modes and limits. At k = 0, a zero mode is a constant in the intersection of the two stabilizers. This consists of a_W = diag(q, q) on W_+ and W_-, identified by F, with q preserving the line of w, together with a_V = lambda (+) m, and with the trace condition this has dimension 1 + 8. In it, L = R and B/L = sqrt(2/3), which gives 1/e^2 = (1 + 1 + 2/3)/g^2. The limits follow by expanding. A second derivation, from the referee, uses the principal angles between the Neumann subspaces at the two walls: for each charged direction the squared sine of the angle is sin^2(2 theta)/2, and in the neutral sector the single nonzero value is (4/5) sin^2(2 theta); it also covers cos(2 theta) = 0, where the divisions above need a limit. For item 3, with a = sin(2 theta), the ratio is [arcsin(a/sqrt 2)/arcsin(2a/sqrt 5)]^2, and this decreases with a on (0, 1], since x/(sqrt(1 - x^2) arcsin x) increases. ■

The constants in item 1 are the values ||[D_F, a]||^2/||a||^2 in the Hilbert-Schmidt norm, for universal mass m, for a = L^+- and for the direction orthogonal to the photon and Z', which has A-component of squared length 4/5. These are the coefficients of the lower bounds for the poles of the fully massive family in Proposition 14 of lemma 18 (2 x < z_W, (16/5) x < z_Z); the small-x pole coefficients there are 3 and 24/5. The limit 5/8 is common to both, with different behaviour: the finite-action poles grow in the heavy limit of lemma 18, while the interval masses here tend to zero at the antipodal point. The spectral-action Taylor mass for universal m is M_W^2 = 3 m^2, which exceeds the value here by the factor 3/2 of Proposition F in the paper: there the mass matrix carries the relative heat coefficients of the action, and here we have assumed a five-dimensional Yang-Mills action, which gives the commutator norm alone. A direct matching claim needs the full parent spectral action, with its tower and boundary conditions.

Item 2 gives a geometric form of the behaviour that lemma 18 found: heavy fermions with light vectors. At antipodal points Gamma_p = -Gamma_{p_0}, so both walls have the same stabilizer in sp(2) again, and the fermion channels have type (c) with mass pi/(2l).

The spectral action on the interval

Proposition 23.7. For the heat cutoff and universal theta, with s = l Lambda, the trace over a channel is

```
Tr_channel exp(-D^2/Lambda^2) = 2 sum_{n in Z} exp(-(n pi + theta)^2/s^2).
```

This is proportional to the Jacobi function theta_3(theta, exp(-s^2)). It is strictly decreasing on [0, pi/2], with a maximum at theta = 0 and a minimum at theta = pi/2, where it is stationary, and it is symmetric under theta -> pi - theta. In the towerless regime s -> 0 at fixed x = theta^2/s^2, it tends to 2 e^{-x}, the form of the vacuum density in lemma 18 and Proposition 21.2.

Proof. The eigenvalues in a channel are +-(n pi + theta)/l, by Theorem 22.1, and the sum is over Z after combining the signs. By Poisson summation the sum equals (s/sqrt(pi)) sum_{k in Z} exp(-k^2 s^2) e^{2 i k theta}, and the Jacobi triple product writes this as a constant times the product over n >= 1 of (1 + 2 q^{2n-1} cos(2 theta) + q^{4n-2}), with q = exp(-s^2). Each factor is positive and strictly decreasing in theta on [0, pi/2]. ■

In the form of the Jacobi function, the trace is (2 s/sqrt(pi)) theta_3(theta, exp(-s^2)). This family lies outside the hypotheses of No-go theorem 9 of lemma 18, which concerns finite internal data and a free ray V = c V_*: here the operator family is different, the radial direction is a compact orbit of relative gradings, and the tower, which the finite description omits, supplies a second branch with a stationary point at theta = pi/2. In the present family the literal action has its minimum at antipodal gradings, m = pi/(2l), with M_W = M_Z = 0 by Theorem 23.6, where the lowest eigenspace is degenerate. This is a vacuum with massless vectors, so it is not the physical point. The sign of the term agrees with the fermion loop in the Hosotani mechanism, where gauge-boson loops enter with the opposite sign. So an interior minimum could arise from that competition, which the classical spectral action does not contain. We have not computed it.

The one-loop potential in the same family

From the exact spectra, the products over the towers are, relative to theta = 0 and in Euclidean momentum x = p l,

```
fermion channel:  1 + sin^2(theta)/sinh^2(x),
charged vectors:  1 + sin^2(2 theta)/(2 sinh^2(x)),
neutral vector:   1 + (4/5) sin^2(2 theta)/sinh^2(x).
```

So in the gauge-boson and fermion-loop form of the Hosotani mechanism, assuming the usual one-loop formula with three polarizations per vector, and with n_f = 2 x (number of physical Dirac species),

```
8 pi^2 l^4 V(theta) = int_0^infinity x^3 dx [ 3 log(1 + s^2/(2 sinh^2 x)) + (3/2) log(1 + (4/5)
↪  s^2/sinh^2 x) - n_f log(1 + sin^2(theta)/sinh^2 x) ],
```

with s = sin(2 theta). A Dirac top with three colours has n_f = 6, which needs a separately specified fermion sector, since the universal connection gives masses to all: one generation has n_f = 16 and three have n_f = 48. With I = int x^3/sinh^2 x dx = (3/2) zeta(3) and I' = int x^3/cosh^2 x dx = (9/8) zeta(3), near theta = 0 the bracket is (10.8 - n_f) I theta^2 + o(theta^2), and with epsilon = pi/2 - theta, near theta = pi/2 it is V(pi/2) + (10.8 I + n_f I') epsilon^2 + o(epsilon^2), where we have written V for the left side.

Proposition 23.8 (referee). In the massless flat family with this content, for every n_f > 0, theta = pi/2 is the unique global minimum on [0, pi/2], since the gauge logarithms are nonnegative and vanish there, and the negative fermion logarithm is minimized at sin^2(theta) = 1. For n_f >= 10.8, V is strictly decreasing on the interior. Indeed, with t = sin^2(theta) and J(z) = int_0^infinity x^3 dx/(sinh^2 x + z), the derivative with respect to t is, up to a positive factor,

```
(1 - 2t) [6 J(2t(1 - t)) + (24/5) J((16/5) t (1 - t))] - n_f J(t).
```

For t <= 1/2, both arguments in the bracket are at least t, and J decreases; for t > 1/2 both contributions have the required sign.

For 0 < n_f < 10.8, on grids of theta (Python), for n_f in {1, 2, 3, 4, 6, 8, 10}, V has one interior maximum, at about 0.33 for n_f = 8, and no interior minimum. Theorem 9 of lemma 24, by Astra 6, proves this for all n_f in that range: exactly one interior stationary point, a strict maximum, using a log-concavity property of the integral.

These are negative results in an exact scope: one interval, no bulk masses, fermions of frequency 1 under F, this gauge content, one loop, this condition at the connector wall. As a formal leading approximation, not uniform at x = 0, V has the form B sin^2(2 theta) - A sin^2(theta) with A, B > 0, whose interior stationary point cos(2 theta) = A/(4B) is a maximum. *Bulk masses.* With bulk masses, the fermion product becomes cos^2(theta) + sin^2(theta) G(x), with G = (w_Q coth w_Q - c_Q)(w_U coth w_U - c_U)/x^2 and w = (c^2 + x^2)^{1/2} in the secular form of lemma 12. For a = c_Q + c_U != 0, G = 1 - a/x + O(x^-2), so the potential has angle-dependent ultraviolet divergences, and a finite value depends on a regulator and subtraction: as the referee showed, an earlier scan of general pairs here did not establish a potential, and it is withdrawn. The exceptional case c_U = -c_Q = c has a common bulk mass on the two mixed components and G = 1 + w^2/(x^2 sinh^2 w), which decays exponentially. Keeping the flat gauge products as a separate tree-level assumption, a scan for c in {0, 1/2, 1, 3/2, 2, 3, 5}, n_f in {1, 3, 6, 8, 10, 16, 48}, on a grid of 721

values of theta, finds no interior minimum (Python). General bulk masses need a gauge-invariant completion, a regulator and wall counterterms.

So an interior minimum needs another change: a negative coefficient of sin^2(2 theta), as from fermions of frequency 2 under F, which the bimodule E = Hom(V, W) does not contain; general bulk masses or other gauge form factors, as with warping; a wall scalar with its own potential; or tree-level terms at the walls. With h/f_pi = 2 theta, the leading flat form has the angular form alpha cos(h/f_pi) - beta sin^2(h/f_pi) of Agashe, Contino and Pomarol (section 2.1, Eqs. (19)-(23)) with spinorial SO(5) fermions, with alpha = A/2 and beta = -B, up to a constant. Their form factors can give the sign and competition needed for an interior minimum, so the flat results here do not exclude that construction. This points away from a pure flat Wilson line, toward those changed form factors or a wall scalar with its own potential, for the physical interior.

Comparison with measured values in the flat toy

These are numerical comparisons (Python), with no running and no bulk masses, in lowest levels, using the Particle Data Group 2026 values quoted by the referee, m_t = 172.60 GeV as a direct-measurement mass, M_W = 80.3625 GeV and M_Z = 91.1879 GeV, so m_t/M_W = 2.1478 and M_W^2/M_Z^2 = 0.7767. The ratio m_t/M_W reaches sqrt(8/3) at theta = 1.0589 and the measured value at theta = 1.1626, where M_W l = 0.541, so 1/l is about 148.5 GeV. There, M_W^2/M_Z^2 = 0.581. The wall-scalar model of lemma 22 covers [0.625, 1], and this family covers [0.5032, 0.625); in the wall-scalar model, fitting the measured vector ratio gives M_W l = 1.028, where the lowest no-bulk-mass fermion channel has m/M_W $<$ 1.528. So neither flat family fits both ratios on its own, and the flat one-loop potential above does not select an interior point. Whether a tower at 1/l of this size is allowed would need the couplings and an applicable experimental limit, which we have not analyzed. The directions are bulk masses for the light fermions, warping, running of the couplings, and families combining a Wilson line with a wall scalar. These negative statements hold in the flat tree-level toy only.

Open questions

1. The interior vacuum: the effective potential with a wall scalar, or with warping, or with general bulk masses, with a gauge-invariant completion, a regulator and wall counterterms. The flat cases above have none on the grids.
2. The relation between the two roles of S^4: the base of the twistor window, and the field space of the Higgs quaternion.
3. A commutative parent: functions on S^4 x I and CP^3 x I, with the gradings as local boundary conditions.
4. The factor 3/2 between the five-dimensional normalization and the spectral action: whether the spectral action on the parent itself, with its tower, reproduces the formula of Theorem 23.6.
5. The comparison with twisted spectral triples, Remark 23.3.
6. The contribution of a real connector to the fermion tower, the degenerate antipodal window, and the scalar kinetic normalization needed for a canonically normalized Higgs field.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^4 with l ||M|| = 1.25: Gamma_1 is an involution, Gamma_1 = exp(2 l K) Gamma_0; D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0) to 3 x 10^-15; the principal angles over l equal the singular values of M to five digits.
- Sp(2) x SU(4), of dimension 25, with the Neumann and Dirichlet subspaces built from the stated stabilizers: the two-point boundary problem was solved as a linear system on a grid in k l. For theta = 0.2, pi/4, 1.06 and 1.3, the two lowest nonzero roots are (0.279, 0.356), (0.785, 1.107), (0.647, 0.868), (0.373, 0.479), equal to the formulas of Theorem 23.6 to three digits, with other roots at pi/2 and above. The intersection of the two stabilizers has dimension 9 at each theta tested.
- Small theta: M_W^2/m^2 = 2.000 and M_Z^2/m^2 = 3.200.

- The universal connection on E (+) E*: J D = D J, Gamma_0 D = -D Gamma_0, all |eigenvalues| equal to theta/l, Gamma_1 = (left Gamma_p, right mirror), J_1^2 = -1, J_1 Gamma_1 = -Gamma_1 J_1, while J alone fails; J_1 h J_1^-1 is right multiplication by +h on E*, for odd h.
- The theta function: monotone on a grid for s = 0.3, 1, 2, with the towerless value at s = 0.3 equal to e^{-x} to five digits; Poisson summation and the triple product to 10^-16.
- The ratio M_W^2/M_Z^2 on a grid: range [0.5032, 0.6250], symmetric under theta -> pi/2 - theta to 10^-15, minimum at pi/4. At theta = 1.1625, M_W l = 0.541, and at theta = 1.0589, M_W l = 0.648.
- The one-loop integrals by adaptive quadrature (scipy), and by Gauss-Legendre quadrature on 400 nodes for the scan with c_U = -c_Q. The referee recomputed the derivative with scipy, found one sign change for each n_f in {1, 2, 3, 4, 6, 8}, with the stationary angle about 0.3382 at n_f = 8, and checked I = (3/2) zeta(3).

Part 24

Commutative cylinders, spectral walls, and the two-stage compression

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/24_commutative_parent.md. Reproduced as delivered, with links made relative. The section "Numerical check" at the end is by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): no error in the displayed equations, and one gap in the general statement on compression maps. The record above the section "Referee amendments" stays as delivered, and those amendments apply to it.

A commutative cylinder admits an exact realization of the **operator window** of lemmas 22 and 23 when the two finite gradings are prescribed through nonlocal spectral boundary conditions. The higher K-modes then decouple as the radius of K tends to zero. Two qualifications remain essential: these boundary conditions restrict scalar functions at the walls, and compression of scalar functions through both stages differs from generating a matrix algebra between the stages. In particular, aligned walls give $\mathbb{R} \oplus \mathbb{R}$ on the quaternionic factor under direct scalar compression, whereas the intervening matrix-algebra construction gives $\mathbb{H} \oplus \mathbb{H}$. Local boundary gradings preserve all scalar functions, but their reduction to the unchanged four-section window has a precise restriction proved below. The radial no-go of lemma 22 survives rechecking in its stated scope. For the massless one-loop potential of lemma 23, an analytic argument strengthens the numerical statement to all $n_f > 0$.

1. Theorem — An exact spectral boundary construction.

Hypotheses and conventions. Let

$$K = \mathbb{CP}^3, \qquad \mathcal{H}_K = L^2\big(K, \Lambda^{0,*}T^*K \otimes \mathcal{O}(1)\big)\,,$$

and let

$$A = \sqrt{2}(\bar\partial + \bar\partial^*), \qquad \gamma_K = (-1)^{\deg}.$$

Use the Fubini–Study metric at a fixed reference scale. Write

$$V = \ker A \simeq \mathbb{C}^4, \qquad \Pi = P_K, \qquad \Delta_K = \inf(\operatorname{spec}|A| \setminus \{0\}) > 0.$$

Here $\gamma_K|_V = 1$. The radius parameter $r > 0$ replaces A by A/r, after the usual unitary identification of the L^2-spaces.

These are the analytic data of Example 16.9. Its S^4 option uses functions pulled back to this same twistor space. An actual operator on S^4 with the required kernel and compression map would need to be specified separately.

Let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

On $L^2(I_\ell; \mathbb{C}^2 \otimes \mathcal{H}_K)$, set

$$\mathcal{D}_r = E\partial_x + Z\,A/r, \qquad \hat{\gamma} = Z\gamma_K.$$

All unspecified tensor factors are identities.

Choose self-adjoint involutions Γ_0, Γ_1 on V, each with two-dimensional positive and negative eigenspaces, and assume

$$-1 \notin \operatorname{spec}(\Gamma_1\Gamma_0).$$

Extend Γ_i by zero on $V^\perp$, writing $C_i = \Pi\Gamma_i\Pi$. Define $\operatorname{sgn} A = 0$ on V, and impose

$$\chi_0\psi(0) = \psi(0), \qquad \chi_\ell\psi(\ell) = \psi(\ell),$$

where

$$\chi_0 = ZC_0 - S\operatorname{sgn} A, \qquad \chi_\ell = ZC_1 + S\operatorname{sgn} A. \tag{1}$$

Then:

1. The H^1 realization of $\mathcal{D}_r$ is self-adjoint with compact resolvent and anticommutes with $\hat{\gamma}$.
2. $\Pi \otimes 1$ reduces the operator, including its domain.
3. On this reducing subspace, the operator is exactly $D(\Gamma_0, \Gamma_1)$ of lemma 23.
4. Every eigenvalue on the complementary subspace satisfies

$$|\lambda| \geq \Delta_K/r. \tag{2}$$

Proof. Put $B = SA/r$. Since $ES = Z$,

$$\mathcal{D}_r = E(\partial_x + B).$$

The operators χ_i are self-adjoint involutions and anticommute with E. Their positive eigenspaces are therefore maximal isotropic for the boundary form

$$[\langle\psi, E\phi\rangle]_0^\ell.$$

On $V^\perp$, condition (1) selects the negative spectral subspace of B at 0 and its positive spectral subspace at ℓ. These are APS conditions with the inward-normal sign reversed between the two walls. On $\ker B = \mathbb{C}^2 \otimes V$, condition (1) supplies a Lagrangian subspace:

$$\Gamma_i\psi_L = \psi_L, \qquad \Gamma_i\psi_R = -\psi_R.$$

For q in the complementary domain, integration by parts gives

$$\|\mathcal{D}_r q\|^2 = \|q'\|^2 + \|Aq/r\|^2 + [\langle q, Bq\rangle]_0^\ell \geq \|q'\|^2 + \frac{\Delta_K^2}{r^2}\|q\|^2. \tag{3}$$

The boundary term is nonnegative by the spectral choices.

An eigenbasis of A decomposes the complementary problem into finite-dimensional interval problems with maximal isotropic endpoint conditions. Each is self-adjoint. Estimate (3) identifies the domain of their self-adjoint direct sum with the stated H^1 domain and gives compactness of its resolvent. The finite-dimensional tangential kernel contributes an ordinary self-adjoint interval problem. This proves self-adjointness, compactness, and (2).

Every boundary projector commutes with Π, proving reduction. Finally, $\{\gamma_K, A\} = 0$ implies $\{\hat{\gamma}, \mathcal{D}_r\} = 0$, and $\hat{\gamma}$ preserves both boundary subspaces. ■

Define

$$K_F = \frac{1}{2\ell}\log(\Gamma_1\Gamma_0), \qquad D_F = \Gamma_0 K_F, \qquad R(x) = e^{xK_F},$$

with the principal logarithm. Let $P_\pm = (1 \pm \Gamma_0)/2$. The finite-window embedding is

$$W_\ell v(x) = \ell^{-1/2} \begin{pmatrix} R(x)P_+ v \\ R(x)P_- v \end{pmatrix}. \tag{4}$$

Direct differentiation gives

$$\mathcal{D}_r W_\ell = W_\ell D_F, \qquad \hat{\gamma} W_\ell = W_\ell \Gamma_0.$$

Thus the prescribed finite operator and grading are exact, provided $\Delta_K / r > \|D_F\|$. The next eigenvalue in absolute value is at least

$$\min \left\{ \frac{\Delta_K}{r}, \frac{\pi - \ell \|D_F\|}{\ell} \right\}. \tag{5}$$

---

2. Proposition — Local walls and spectral walls have different scalar domain algebras.

Hypotheses. Use the product differential expression of Result 1.

For the local option, let $G_i(y)$ be smooth self-adjoint bundle involutions and impose

$$G_i \psi_L = \psi_L, \qquad G_i \psi_R = -\psi_R \quad \text{at } x = i.$$

Assume

$$[G_i, \gamma_K] = 0, \qquad [G_i, \sigma_A(y, \xi)] = 0 \quad (\xi \in T_y^* K), \tag{6}$$

where σ_A is the tangential principal symbol.

Then these local conditions give a self-adjoint elliptic realization with compact resolvent, and **every** smooth scalar function on $K \times I_\ell$ preserves its domain.

For the spectral option (1), the maximal smooth scalar $*$-algebra preserving the domain is

$$\mathcal{A}_{\mathrm{APS}} = \{ f \in C^\infty(K \times I_\ell) : f(\,\cdot\,, 0), f(\,\cdot\,, \ell) \text{ are constant on } K \}. \tag{7}$$

The same conclusion holds for the subalgebra pulled back from $S^4 \times I_\ell$.

Proof. For the local option, the boundary involution is ZG_i. It anticommutes with the normal matrix E. Condition (6) makes it anticommute with the principal symbol of the adapted tangential operator SA/r. Hence its positive eigenspace is an elliptic Lagrangian boundary condition. The corresponding local ellipticity and adjoint-domain criteria are described in Bär–Ballmann, Corollary 4.18 and Example 4.20. Primary reference

Scalar multiplication commutes pointwise with G_i, so it preserves the endpoint conditions and the H^1 domain. Moreover,

$$[\mathcal{D}_r, f] = E\, \partial_x f + Z[A, f]/r$$

is bounded for each $r > 0$.

For spectral walls, preservation by a scalar $*$-algebra is equivalent to

$$[M_{f_i}, \chi_i] = 0.$$

Independence of the spin matrices Z, S gives

$$[M_{f_i}, C_i] = 0, \qquad [M_{f_i}, \operatorname{sgn} A] = 0.$$

Since $C_i^2 = \Pi$, the first equation implies

$$[M_{f_i}, \Pi] = 0. \tag{8}$$

Consequently multiplication by f_i preserves $H^0(K, \mathcal{O}(1))$. If s is such a section, then

$$0 = \bar{\partial}(f_i s) = (\bar{\partial} f_i) s.$$

The sections of $\mathcal{O}(1)$ have no common zero, so $\bar{\partial} f_i = 0$ everywhere. A holomorphic function on compact connected $\mathbb{CP}^3$ is constant. Constants plainly preserve (1), proving (7). ■

The hypotheses in (6) are a sufficient local ellipticity condition. An arbitrary bundle grading supplies a pointwise isotropic condition, but self-adjointness on an H^1 domain and compact resolvent still require an ellipticity check. Likewise, prescribing a grading only on V leaves the boundary problem incomplete until a condition on $V^\perp$ is supplied.

---

3. Proposition — Compression of a local grading and the unchanged level-one window.

Hypotheses. Let G be a self-adjoint bundle involution acting by multiplication, and let

$$g = \Pi G \Pi|_V, \qquad Q_K(f) = \Pi M_f \Pi|_V.$$

Keep the level-one space and scalar representation of Example 16.9 unchanged.

Then

$$1_V - g^2 = \big((1-\Pi)G\Pi\big)^*\big((1-\Pi)G\Pi\big). \tag{9}$$

In particular, g is a grading exactly when $[G, \Pi] = 0$. In that event,

$$[g, Q_K(f)] = 0 \quad \text{for every scalar } f. \tag{10}$$

For either the full $\mathbb{CP}^3$ scalar algebra or the S^4 pullback algebra, this forces

$$g = 1_V \quad \text{or} \quad g = -1_V. \tag{11}$$

Proof. Expand $\Pi G^2 \Pi = \Pi$ using $1 = \Pi + (1 - \Pi)$; this gives (9). Vanishing of its positive right-hand side implies $(1-\Pi)G\Pi = 0$, and self-adjointness gives the other off-diagonal block.

Since scalar multiplication commutes with G, reduction then gives (10). The compressed $\mathbb{CP}^3$ functions generate $M_4(\mathbb{C})$. The compressed real S^4 functions generate $M_2(\mathbb{H})$, whose complex-linear commutant on $\mathbb{C}^4$ is also $\mathbb{C}1$. Thus a self-adjoint involution satisfying (10) is ± 1. ■

For **aligned local walls** with $[G, A] = 0$, the finite zero-mode window is explicitly

$$v \longmapsto \ell^{-1/2} \begin{pmatrix} P_+(g)v \\ P_-(g)v \end{pmatrix}.$$

The induced grading is g, and scalar compression is

$$P_+(g) Q_K(f) P_+(g) + P_-(g) Q_K(f) P_-(g) = Q_K(f).$$

By (11), the unchanged four-section window has trivial grading. Its generated algebra remains $M_2(\mathbb{H})$ or $M_4(\mathbb{C})$, according to the scalar algebra chosen. The globally constant choices $G = \pm 1$ give concrete self-adjoint examples with this window and a separated higher spectrum.

There is also an explicit local field illustrating failure of reduction. Add a trivial auxiliary $\mathbb{C}^4$ bundle and put

$$G(y) = \sum_{a=1}^{5} y_a \Gamma_a^{\text{aux}},$$

where $y \in S^4$ is the twistor projection and the Γ_a satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$. This is a local involution satisfying (6), but now the tangential kernel has dimension 16, and

$$g = \frac{1}{5} \sum_{a=1}^{5} \Gamma_a^{\text{section}} \otimes \Gamma_a^{\text{aux}}.$$

Using $\mathrm{tr}(\Gamma_a\Gamma_b) = 4\delta_{ab}$,

$$\mathrm{tr}(g^2) = \frac{16}{5} \neq 16.$$

Thus g is not a grading. Equation (9) even gives

$$\|(1-\Pi)G\Pi\| \geq \frac{2}{\sqrt{5}},$$

independently of r. This is a bound on boundary-trace mixing; it does not identify the actual low spectral window of the coupled local problem.

A local $G(y)$ specifies a **field of gradings**. It does not select a marked point p. In the tautological quaternionic line bundle,

$$\Pi_y = \frac{1 + y_a\Gamma_a}{2},$$

the natural field $2\Pi_y - 1$ acts as $+1$ on the fibre $\mathrm{ran}\,\Pi_y$. By contrast, the finite grading

$$\Gamma_p = p_a\Gamma_a = 5Q_K(p \cdot y)$$

uses a separately chosen point p. Its action on global section coefficients is nonlocal.

The restriction proved here concerns the unchanged level-one space, a local multiplication grading, and its compression as an actual involution. A different coupled window, additional fibres, or spectral boundary conditions remain available directions.

---

4. Theorem — The composed compression map and the aligned-wall algebra.

Hypotheses. Use the exact spectral construction of Result 1. For a scalar multiplier $f(y, x)$, let

$$T_f(x) = Q_K(f(\,\cdot\,, x)), \qquad \mathcal{E}_0(T) = \frac{1}{2}(T + \Gamma_0 T\Gamma_0).$$

Use real-valued functions and real algebra generation when discussing $M_2(\mathbb{H})$.

Then the compression directly from the scalar cylinder to the finite window is

$$\boxed{\Phi_\ell(f) = \frac{1}{\ell}\int_0^\ell \mathcal{E}_0\big(R(x)^* T_f(x) R(x)\big)\, dx.} \tag{12}$$

It is independent of r.

For aligned walls $\Gamma_1 = \Gamma_0 = \Gamma_p$, direct scalar compression on the S^4 factor generates

$$\mathbb{R} \oplus \mathbb{R}. \tag{13}$$

If instead one first generates $M_2(\mathbb{H})$ and then applies the interval compression, the result is

$$\mathbb{H} \oplus \mathbb{H}. \tag{14}$$

Proof. Insert (4) on both sides of the multiplication operator. Orthogonality of the two interval spin components gives exactly (12).

For real S^4 functions, Example 16.9 implies

$$Q_K(C^\infty(S^4, \mathbb{R})) = \mathrm{span}_{\mathbb{R}}\{1, \Gamma_1, \ldots, \Gamma_5\}. \tag{15}$$

Indeed all compressed functions are quaternionic Hermitian matrices; that space has real dimension six, and the displayed six matrices belong to the image and are independent.

For aligned walls $R = 1$. Clifford multiplication gives

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p.$$

Therefore the image of direct scalar compression is

$$\mathrm{span}_{\mathbb{R}}\{1, \Gamma_p\}.$$

This space is already the algebra $\mathbb{R} \oplus \mathbb{R}$, proving (13).

On the other hand,

$$\mathcal{E}_p(M_2(\mathbb{H})) = \{a \in M_2(\mathbb{H}) : [a, \Gamma_p] = 0\} = \mathbb{H} \oplus \mathbb{H},$$

proving (14). The noncommuting products needed to generate the quaternionic diagonal blocks have been inserted between the two compressions in this second procedure. ■

On the single $\mathbb{CP}^3$ factor, real scalar compression is onto the Hermitian 4×4 matrices. A chosen $2+2$ aligned grading therefore gives all Hermitian block-diagonal matrices and generates

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

For the **chosen two-sheet coarse correspondence** of Example 16.9, with

$$H_0 = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V), \qquad \Gamma_0(\xi, \eta) = (\Gamma_p \xi, -\eta \Gamma_p),$$

the same calculation gives

$$\boxed{\mathrm{alg}_{\mathbb{R}}(\text{direct aligned scalar compression}) = \mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).} \tag{16}$$

The V-side matrices act on the left of η and commute with its right grading, so that factor survives unchanged.

Endpoint restrictions do not change these finite-dimensional images. If a linear functional vanishes on compressions of functions supported in $0 < x < \ell$, its smooth matrix-entry density vanishes in the interior and hence everywhere. Finite-dimensional separation then gives equality with the unrestricted image, exactly as in Result 7 of lemma 20.

At the intermediate stage, however, spectral walls still give scalar endpoint values. Even after algebra generation, those values remain scalar. Enlarging them to the full stabilizers of Γ_0, Γ_1, as in lemma 23, is another change of the parent algebra. Equality of finite compression images does not identify the parent gauge groups or one-forms.

---

5. Theorem — Radius collapse, interval collapse, and controlled mixing.

Hypotheses. First use the exact construction of Result 1. Identify the varying Hilbert spaces by metric rescaling. For resolvent formulas, fix units in which the nonreal spectral parameter is i.

Then, at fixed ℓ,

$$\left\| (\mathcal{D}_r - i)^{-1} - \iota_\Pi (D(\Gamma_0, \Gamma_1) - i)^{-1} \iota_\Pi^* \right\| \le \frac{r}{\Delta_K}. \tag{17}$$

Thus radius collapse gives the entire matrix interval operator in generalized norm-resolvent sense.

If D_F is fixed, $\ell \to 0$, and

$$\Gamma_1(\ell) = e^{\ell \Gamma_0 D_F} \Gamma_0 e^{-\ell \Gamma_0 D_F},$$

then $r \to 0$ and $\ell \to 0$ decouple both towers and leave the finite operator D_F. No relation between the two rates is required in this exactly reducing construction.

Proof. The resolvent is block diagonal with respect to Π. Estimate (2) bounds its complementary block by r/Δ_K, proving (17). For the finite-window limit, (5) bounds the complementary resolvent by

$$\max\left\{\frac{r}{\Delta_K}, \frac{\ell}{\pi - \ell\|D_F\|}\right\},$$

which tends to zero. ■

A useful mixing statement requires additional hypotheses. Suppose a transformed problem has a common domain and the form

$$\widetilde{\mathcal{D}}_r = \mathcal{D}_r + V_r, \qquad V_r = V_r^*, \qquad \|V_r\| \le b_r.$$

Write

$$v_r = \|(1-\Pi)V_r\Pi\|.$$

For an eigenvector $\psi = p + q$, with $p = \Pi\psi$, $q = (1-\Pi)\psi$, and $|\lambda| \le L$, assume

$$g_r = \Delta_K/r - b_r - L > 0.$$

Then

$$\boxed{\|q\| \le \frac{v_r}{g_r}\|p\|, \qquad \|\text{Schur correction at } \lambda\| \le \frac{v_r^2}{g_r}.} \tag{18}$$

Proof. The complementary eigenvalue equation gives

$$q = -(\mathcal{D}_{r,Q} + V_{QQ} - \lambda)^{-1} V_{QP} p.$$

The inverse has norm at most g_r^{-1}. Substituting this formula into the p-equation gives the second bound. ■

For bounded b_r, v_r, both errors are $O(r/\Delta_K)$. If a common-domain construction instead produces terms of order $1/\ell$, sufficient conditions include

$$\frac{r v_r}{\Delta_K} \to 0, \qquad \frac{r v_r^2}{\Delta_K} \to 0,$$

together with convergence of the Π-diagonal perturbation.

An arbitrary local $G_i(y)$ generally fails $[G_i, \Pi] = 0$. Although the differential expression commutes with Π, its domain then does not. Such a boundary change is not automatically a bounded perturbation satisfying (18). The explicit local field in Result 3 has nonvanishing boundary mixing at every radius. No convergence to the prescribed two-gradings operator is established here for that nonreducing local problem.

At the level of observables, radius collapse leaves (12) unchanged. For an x-independent $T = Q_K(f)$,

$$\|\Phi_\ell(f) - \mathcal{E}_0(T)\| \le \ell\|D_F\|\,\|T\|. \tag{19}$$

If T is even for Γ_0, the stronger formula of Theorem 22.4 applies:

$$\Phi_\ell(f) = \operatorname{sinc}(\ell\ \mathrm{ad}_{D_F})(T),$$

$$\|\Phi_\ell(f) - T\| \le \frac{\ell^2}{6}\|[D_F,[D_F,T]]\|. \tag{20}$$

Equation (19) follows by integrating $\|R^*TR - T\| \le 2x\|D_F\|\|T\|$; equation (20) follows by taking the even part of the conjugation series.

Consequently there is convergence of the **compression maps on their scalar operator systems**. The limit is $\mathcal{E}_0 Q_K$. It does not turn that image into every element of $\mathcal{E}_0(P)$; Result 4 computes the difference explicitly. Algebra generation remains a separate operation.

---

6. Proposition — The second sheet, reality, even algebra, and connector remain prescribed data.

Hypotheses. Consider the scalar cylinder construction above and, separately, its extension using the two-sheet coarse correspondence of Example 16.9.

The resulting distinctions are:

Structure	What follows under the stated choices
Second sheet and the two Hom spaces	They are inputs in the coarse correspondence. A single four-section cylinder window does not supply them.
$J_1^2 = -1,\ J_1\Gamma_i = -\Gamma_i J_1$	These are additional compatibility conditions on the prescribed gradings and antiunitary.
$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$	This is the derived even part after choosing the quaternionic real form, the grading, and intermediate algebra generation. Direct aligned scalar compression gives (16).
Connector and $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	The connector's lines and coupling are chosen. Their intertwining algebra is then derived.

Proof. The one-factor construction starts with $V = \mathbb{C}^4$, and (4) has domain V. The two-sheet construction instead starts with

$$(W \otimes \overline{V}) \oplus (V \otimes \overline{W}).$$

Geometrically, its analytic realization uses product factors and a restricted scalar algebra, as Theorem 16.7 specifies. It is additional correspondence data over the scalar spaces.

For reality, suppose an antiunitary $\mathcal{J}_1$ on the tangential Hilbert space satisfies

$$\mathcal{J}_1^2 = -1, \quad \mathcal{J}_1 A = -A\mathcal{J}_1, \quad [\mathcal{J}_1, \gamma_K] = 0,$$

and its restriction J_1 to V anticommutes with both Γ_i. Then

$$\widehat{J} = E\mathcal{J}_1$$

preserves (1) and obeys

$$\widehat{J}^2 = 1, \qquad [\widehat{J}, \mathcal{D}_r] = 0, \qquad \widehat{J}\hat{\gamma} = -\hat{\gamma}\widehat{J}.$$

On the window,

$$J_F = \Gamma_0 J_1.$$

These identities follow directly from the stated commutation relations. They specify sufficient conditions; the natural $\mathcal{O}(1)$ cylinder has not supplied such an extension or its order-zero property here.

In particular, the quaternionic antiunitary j used to recognize $M_2(\mathbb{H})$ commutes with the Γ_a. It does not itself satisfy the required anticommutation with Γ_p.

The even-algebra assertion is the centralizer calculation in Result 4 after intermediate generation.

Finally, choose

$$w \in W_+, \quad v \in V, \quad \|w\| = \|v\| = 1, \quad \kappa \neq 0,$$

and prescribe

$$T_\kappa \xi = \kappa(w^* \xi v)\, v w^*.$$

Its intertwining conditions require the lines $\mathbb{C}w, \mathbb{C}v$ to be reducing, with the same scalar action. A quaternion preserving $\mathbb{C}w$ lies in its chosen copy of $\mathbb{C}$; $v^\perp$ carries arbitrary $M_3(\mathbb{C})$, and W_- carries arbitrary $\mathbb{H}$. Hence

$$\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m)\} \simeq \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$$

as in Result 2 of lemma 20. ■

A marked $p \in S^4$ fixes a quaternionic line, but choosing w also requires a complex line within it. The second line v, the coupling, and an interface realizing the connector remain necessary. Scalar agreement at marked points has unchanged compression image by Result 7 of lemma 20, so it supplies none of this selection by itself.

---

7. No-go theorem — Recheck of the radial statement in lemma 22.

Hypotheses. Let the parent be exactly the one-interval model of lemma 22: fixed wall coupling X, zero bulk mass, the even algebra $\mathcal{A}_\ell$, scalar inner fluctuations Eh with $h^* = -h$, and the **moving spectral window**. For real fluctuations, include the conjugate one-form and assume the stated order-zero representation.

Then every fluctuated finite operator has the same singular values as D_F. This is an exact finite-amplitude spectral statement; it uses no action or loop expansion.

Proof. Solve

$$u' = -hu, \qquad u(0) = 1, \qquad w = u(\ell),$$

and set $v(x) = u(x)w^*$. Then

$$v(\ell) = 1, \qquad v(0) = w^*.$$

Since h is even, w preserves the boundary splitting at 0; the identity at ℓ preserves the fixed coupled wall. Thus v preserves the original domain, and

$$vD_X v^* = D_X + Eh.$$

For the real fluctuation, the analogous implementing unitary is

$$vJ_F vJ_F^{-1};$$

order zero separates the two factors in its derivative.

The full operators are unitarily equivalent, so their moving spectral windows are unitarily equivalent. In the finite identifications of lemma 22, the operators are $w^* D_F w$, or $U_w^* D_F U_w$ in the real case. Their singular values coincide. ■

The moving-window hypothesis matters. For example, with $D_F = m\sigma_1$ and $h = i\sigma_3$, fixed compression gives

$$Q_\ell(Eh) = -\frac{1 - \cos(2m\ell)}{2m\ell}\sigma_2.$$

Adding a multiple of this operator to the fixed $m\sigma_1$ changes its singular values at second order.

The verdict agrees with the author's amended theorem and Terra's independent endpoint-preserving proof. In the usual nondegenerate Standard Model representation, the normalized shifted Higgs quaternion lies on the unit-quaternion orbit; its norm is fixed.

The known escapes are a variable wall coupling, odd connection directions, suitable bulk masses, additional internal directions with nontrivial connection data, or additional elementary or composite fields. A quantum effective action on the same configuration space does not add a radial coordinate. The proposed $K \times I_\ell$ parent has tangential directions and tangential one-forms, so this theorem does not decide its full fluctuation space.

---

8. Lemma — A logarithmic concavity property of the one-loop integral.

Hypotheses. Let

$$\mathcal{J}(z) = \int_0^\infty \frac{x^3}{\sinh^2 x + z}\, dx, \qquad z > 0,$$

and define

$$e(z) = -\frac{z\mathcal{J}'(z)}{\mathcal{J}(z)}.$$

Then $0 < e(z) < 1$, and e is nondecreasing.

Proof. Positivity and

$$0 < \frac{z}{\sinh^2 x + z} < 1$$

show that $e(z)$, expressed as a weighted average of this fraction, lies strictly between zero and one.

Put $x = e^v$, $z = e^u$. Then

$$\mathcal{J}(e^u) = \int_{\mathbb{R}} \exp\left(4v - \log(\sinh^2(e^v) + e^u)\right)\, dv. \tag{21}$$

The function

$$b(v) = 2\log\sinh(e^v)$$

is convex, because, with $x = e^v$,

$$b''(v) = 2x(\coth x - x\ \operatorname{csch}^2 x) > 0.$$

The last inequality is equivalent to $\sinh x \cosh x > x$.

The function $(u, v) \mapsto \log(e^{b(v)} + e^u)$ is therefore convex. The integrand in (21) is jointly log-concave. Marginal integration preserves log-concavity by Prékopa's theorem, so $u \mapsto \log \mathcal{J}(e^u)$ is concave. We use precisely Theorem 6, p. 342, of Prékopa's original paper.

Finally,

$$\frac{d}{du}\log \mathcal{J}(e^u) = -e(e^u).$$

Concavity makes this derivative nonincreasing, proving the assertion. ■

---

9. Theorem — The massless one-loop potential has no interior local minimum.

Hypotheses. Let the action be exactly the one-loop determinant displayed in lemma 23: a flat interval, zero bulk masses, frequency-one fermions, the stated charged and neutral gauge towers and multiplicities, and the prescribed connector wall condition. Assume that no fermion connector modifies the displayed fermion tower, and add no wall potential or other angle-dependent term.

Let $n_f > 0$, set $t = \sin^2\theta$, and write

$$\mathcal{V}(t) = 8\pi^2\ell^4 V(\theta).$$

Then:

1. $t = 1$, equivalently $\theta = \pi/2$, is the unique global minimum.
2. If $n_f \geq 54/5$, the potential is strictly decreasing on the interior.
3. If $0 < n_f < 54/5$, there is exactly one interior stationary point, and it is a strict maximum.

Proof. Define

$$F(z) = \int_0^\infty x^3 \log\left(1 + \frac{z}{\sinh^2 x}\right)\, dx, \qquad F'(z) = \mathcal{J}(z).$$

The exact potential is

$$\mathcal{V}(t) = 3F(2t(1-t)) + \frac{3}{2}F\left(\frac{16}{5}t(1-t)\right) - n_f F(t). \tag{22}$$

Its gauge terms are nonnegative and vanish at $t = 1$. Since F is strictly increasing, the fermion term is uniquely minimized there. This proves item 1.

Differentiate:

$$\mathcal{V}'(t) = (1-2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\left(\frac{16}{5}t(1-t)\right)\right] - n_f\mathcal{J}(t). \tag{23}$$

For $t \geq 1/2$, this is strictly negative.

For $0 < t < 1/2$ and $c \geq 2$, put

$$R_c(t) = (1-2t)\frac{\mathcal{J}(ct(1-t))}{\mathcal{J}(t)}.$$

Let $a = ct(1-t)$. Then $a \geq t$, and Result 8 gives $e(a) \geq e(t)$. Consequently,

$$\begin{aligned}\frac{d}{dt}\log R_c(t) &= -\frac{2}{1-2t} - \frac{1-2t}{t(1-t)}e(a) + \frac{e(t)}{t} \\ &\leq -\frac{2}{1-2t} + \frac{e(t)}{1-t} \\ &< -\frac{2}{1-2t} + \frac{1}{1-t} < 0.\end{aligned}$$

Thus every R_c is strictly decreasing. Since $\mathcal{J}(0)$ is finite and positive,

$$R_c(0+) = 1, \qquad R_c(1/2-) = 0.$$

Therefore

$$H(t) = 6R_2(t) + \frac{24}{5}R_{16/5}(t)$$

decreases strictly from 54/5 to zero. Equation (23) becomes

$$\mathcal{V}'(t) = \mathcal{J}(t)\,[H(t) - n_f] \quad (0 < t < 1/2).$$

For $n_f \geq 54/5$, its sign is negative. For $0 < n_f < 54/5$, it changes sign exactly once, from positive to negative. Since $t = \sin^2\theta$ is strictly increasing on the interior, the same stationary-point classification holds in θ. ■

This proves the massless statement for the entire parameter range, including the previously numerical range. No numerical computation is used in this proof.

The recorded author and Terra referee calculations agree on the sampled massless cases. The logarithmic-concavity proof above is an additional single-agent derivation; it has not received a fresh independent model review in this turn. The recorded agreement therefore should not be described as a multi-model review of this new proof.

The escapes remain changes to the stated determinant: frequency-two fermions, modified form factors through warping or other geometry, suitable bulk-mass sectors, wall potentials, and wall kinetic or interaction terms. The full effective action of the proposed $K \times I_\ell$ parent, including its K-tower, has not been computed here.

---

10. Proposition — Scope of the bulk-mass scans.

Hypotheses. Retain the flat gauge terms of (22) as an independent assumption. Replace the fermion factor by the expression in lemma 23:

$$1 + t(G(x) - 1), \qquad G(x) = \frac{(w_Q \coth w_Q - c_Q)(w_U \coth w_U - c_U)}{x^2}, \quad w_i = \sqrt{x^2 + c_i^2}.$$

If $c_U = -c_Q = c$, then the potential is finite and $t = 1$ is its unique global minimum for every $n_f > 0$. The reported absence of additional interior **local** minima at nonzero c remains a numerical statement here.

If $a = c_Q + c_U \neq 0$, the displayed unrenormalized integral has angle-dependent ultraviolet divergences.

Proof. For opposite masses,

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w}, \qquad w = \sqrt{x^2 + c^2}.$$

The added fraction is positive and decays exponentially at infinity. The negative fermion logarithm is strictly decreasing in t. The gauge logarithms are nonnegative and vanish at $t = 1$, proving the global-minimum assertion.

For general masses,

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

and hence

$$\log(1 + t(G - 1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by the integration measure $x^3 dx$ produces angle-dependent divergences. A finite potential therefore requires a specified completion, regulator, and counterterms. ■

The opposite-mass scan in lemma 23 samples finitely many masses, multiplicities, and angles. It cannot exclude unsampled local minima. General bulk masses and altered gauge form factors remain directions for constructing an interior vacuum; these conclusions leave that question open.

Consequence for T0 and T0b

For T0, a commutative scalar representation with prescribed spectral walls gives an exact interval operator window and controlled tower decoupling, while local walls preserve the full scalar algebra under the ellipticity hypotheses above. The derivation of a nontrivial local grading and the desired algebra from one coupled scalar window remains open, and intermediate algebra generation must be stated explicitly. For T0b, the second sheet, compatible reality, quaternionic grading choice, and connector remain inputs; their algebraic consequences can be derived once those choices are supplied. Neither the radial theorem nor the specified one-loop potential closes the commutative-parent problem.

Referee amendments

From Terra 6.1. These apply to the text above.

- **Result 2.** Write G_0 at x = 0 and G_l at x = l. Equation (6) supplies the principal-symbol condition of Bar-Ballmann, Corollary 4.18 and Theorem 4.11; their Example 4.20 assumes more. On the unaugmented Dolbeault bundle, the Clifford representation is irreducible, so a fibre endomorphism commuting with every sigma_A is scalar, and a smooth involution of this form is +1 or -1. Additional fibres and other local elliptic projectors lie outside this sufficient criterion.
- **Result 3.** The statement on marked points concerns the natural field displayed there. A local field can encode a chosen point: on the auxiliary bundle, G_p(y) = p_a Gamma_a^aux is a local constant field whose coefficients contain p.
- **Result 4.** Equation (16) has the two Hom spaces and the two scalar algebras of Example 16.9 as inputs. The endpoint argument proves equality of the finite-dimensional linear images, and does not identify the parent algebras, norms, gauge groups or one-forms.
- **Result 5, gap in the final assertion.** Convergence of compression maps needs a common domain and a collapse map. With aligned walls, D_F = 0 and f_l(y, x) = 6 (x/l)(1 - x/l), both endpoint values vanish while Phi_l(f_l) = 1. A correct form: for F(y, s) fixed on K x [0, 1] and f_l(y, x) = F(y, x/l), Phi_l(f_l) tends to E_0 Q_K(int_0^1 F(., s) ds). For x-independent ambient multipliers the limit is E_0 Q_K, by (19). The rate discussion after (18) also needs control of the denominator: a sufficient addition is r (b_r + L) <= (1 - epsilon) Delta_K. The case b_r = O(1/l) follows from the stronger estimate |D_{r,Q}| >= (Delta_K^2/r^2 + pi^2/(4 l^2))^{1/2}, from the one-ended Poincare inequality on the positive and negative B-spectral components.
- **Result 6.** At the finite level, if the gradings are Gamma_p, Gamma_q, a unit n orthogonal to p and q gives J_1 = Gamma_n j, with J_1^2 = -1 and the required anticommutation. Extending this choice to the parent remains a separate task.
- **Lemma 8 and Theorem 9.** An independent check of the key monotonicity, without log-concavity or Prekopa's theorem: with rho(s) = (arsinh sqrt(s))^3/(2 sqrt(s (1 + s))), J(z) = int_0^infinity rho(s)/(s + z) ds; with L(v) = log rho(e^v), L'(v) = 3 tanh(x)/(2x) - (1 + tanh^2 x)/2 for v = 2

log sinh x, which is strictly decreasing. For z = e^u, with the probability density in y proportional to e^y rho(e^{u+y})/(1 + e^y), e(e^u) is the mean of 1/(1 + e^y), and its derivative in u is Cov(1/(1 + e^y), L'(u + y)) > 0, as both are strictly decreasing. Theorem 9 is for theta in [0, pi/2]. At theta = pi/2 the determinant has its continuous value, while the principal-logarithm window is degenerate. Its hypotheses are the flat massless one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no other angle-dependent term.

- **Result 10.** The divergence is stated for nonzero fermion multiplicity, with no cancellation by additional sectors. The opposite-mass scan in lemma 23 is a numerical result on its sampled parameters.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept: E_p(Gamma_a) = p_a Gamma_p to 10^-15, which gives equation (13); tr(g^2) = 16/5 for the auxiliary field of Result 3; on grids, e(z) lies in (0, 1) and is nondecreasing for z in [10^-3, 50]; H(t) is strictly decreasing on (0, 1/2), with H near 54/5 at small t and near 0 at t = 1/2; for n_f in {1, 3, 6, 8, 10}, H - n_f has one sign change, with the stationary angle 0.34 at n_f = 8, which agrees with the referee of lemma 23. Results 1, 2, 5 and 6 were read and not separately computed.

Part 25

The weak-boson sum rule and the effective dimension of the interval

For STRATEGY T5b, T13 and T1, following lemma 18, lemma 22 and lemma 23. Proved by Claude Opus 5.5, with checks in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and two gaps as first written, and amendments, all incorporated below.

Outcome. Unfolded, the interval parent is a circle of length 2l, and the parent operator is -i d/dx (x) 1 + 1 (x) D_F: the tower is D_F shifted by the momenta k_n = n pi/l. So the exact quadratic kernel of lemma 18 applies termwise over the tower. For small masses, at leading order, for the heat cutoff, the Taylor mass of a vector with constant profile in the transported frame is

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2 / ||T||^2,      d_eff(s) = d log S / d log s,      S(s) =
↪  sum_{n in Z} exp(-n^2 pi^2/s^2),   s = l Lambda,
```

in Hilbert-Schmidt norms, where d_eff is the spectral dimension at the scale of the cutoff of the unfolded circle, equivalently of the paired Neumann and Dirichlet interval: 0 for l Lambda « 1 and 1 for l Lambda » 1. For the Standard Model representation, at the same order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,       m_t/M_W = (8/(3 - d_eff))^{1/2} for the top alone,
```

with M_W^2/M_Z^2 = 5/8 for all s. So the relation 8 M_W^2 = sum N_f m_f^2 of the asymptotic spectral action is the zero-dimensional value, with m_t/M_W = sqrt(8/3). A resolved interval gives 12 M_W^2 = sum N_f m_f^2, with m_t = 2 M_W, as five-dimensional covariance requires for this operator on the circle. The endpoint m_t = 2 M_W is known in flat gauge-Higgs models with suitable representations (references below); in the inspected records the referee did not find the coefficient (3 - d_eff)/24 or its crossover, and priority remains open pending a wider search. The factor 3/2 in Proposition F of the paper, the finite-action coefficient relative to the Yang-Mills commutator quotient, is the ratio of these two endpoints. The crossover is at l Lambda about 2.

Unfolding

Let D_X = E d/dx be the parent of lemma 22, and let D_0 + E Gamma_F D_F be its form after the transport of Corollary 22.2, where D_0 is the operator with X = 0.

Proposition 25.1. On the circle x in [-l, l] with periodic conditions, for psi in the domain of D_0, put

```
Psi_Q(x) = Q_L(|x|) + i sgn(x) Q_R(|x|),      Psi_U(x) = U_R(|x|) - i sgn(x) U_L(|x|).
```

After multiplying by 1/sqrt(2), this is a unitary from the parent Hilbert space onto L^2(circle) (x) (H_+ (+) H_-), which maps the H^1 domain of D_0 onto the periodic H^1 domain, and it carries D_0 + E Gamma_F D_F to

```
-i d/dx (x) 1 + 1 (x) D_F.
```

So the signed spectrum is k_n + mu_i, with k_n = n pi/l for n in Z and mu_i the eigenvalues of D_F. An even operator T = T_Q (+) T_U on H_F, constant in x in this transported frame and acting identically on both spin components, is carried to 1 (x) T.

Proof. Since Q_R(0) = Q_R(l) = 0 and U_L(0) = U_L(l) = 0, the functions Psi are continuous and periodic, and every smooth function on the circle arises from its even and odd parts. For x > 0, -i d/dx Psi_Q = Q_R' - i Q_L' = (D psi)_L + i (D psi)_R on Q, and -i d/dx Psi_U = -i U_R' - U_L' = (D psi)_R - i (D psi)_L on U, which are the transforms of D_0 psi; x < 0 is the same with the odd parts. For the connection term, with M* the block of D_F, the term E Gamma_F D_F sends Psi_Q to M* Psi_U and Psi_U to M Psi_Q. ■

In this form D_F is a constant connection on the circle, with holonomy exp(2 i l D_F) up to the convention for parallel transport, and the two walls are the fixed points of the reflection that carries D_F to -D_F. With U = exp(-i pi Gamma_0/4), the identity of lemma 23 reads Gamma_1 Gamma_0 = exp(2 l Gamma_0 D_F) = U exp(2 i l D_F) U*. This form of the interval as a circle with a reflection is the usual orbifold description. It also gives a one-line proof of the spectrum in Theorem 22.1.

The frame. By Corollary 22.2, D_X = R (D_0 + E Gamma_F D_F) R*, with R(x) = exp(x Gamma_F D_F). A generator T constant after transport is R(x) T R(x)* in the original frame of D_X, which generally varies with x and has an odd bulk component. Conversely, a constant generator in the original frame need not preserve the coupled wall, if it is broken. So the field used here requires an admissible enlarged gauge algebra, as in the prealgebra P of lemma 23, and it is generally not a field of the even algebra of lemma 22.

The kernel over the tower

Let A_mu(y) = a_mu(y) T on M^4 = R^4, with T even and constant along x, and consider D = -i gamma^mu (d_mu + i A_mu) + gamma_5 V with V = -i d/dx (x) 1 + 1 (x) D_F. Theorem 1 of lemma 18 has this form, with V_0 = V, whose eigenvalues are k_n + mu_i, and T_{n i, n' j} = delta_{n n'} T_{ij}.

Theorem 25.2. With d_ij = mu_i - mu_j, for the heat cutoff t = 1/Lambda^2, the kernel is

```
C_t(q^2) = (1/(8 pi^2)) sum_{n in Z} sum_{ij} |T_ij|^2 [ (2/t + d_ij^2 + q^2) I_t((k_n + mu_i)^2,
↪  (k_n + mu_j)^2; q^2) - (e^{-t (k_n + mu_i)^2} + e^{-t (k_n + mu_j)^2})/t ],
```

with I_t as in lemma 18. The sum converges, and it holds without expansion in D_F. Expanding in D_F, with C_t = C_0 + q^2 C_1 + …, and with t = 1, so with k_n = n pi/(l Lambda) dimensionless,

```
C_0 = (1/(8 pi^2)) ||[D_F, T]||^2 sum_n e^{-k_n^2} (1 - (2/3) k_n^2) + O(||D_F||^4),
C_1 = (1/(8 pi^2)) (2/3) ||T||^2 sum_n e^{-k_n^2} + O(||D_F||^2).
```

So, with <k^2> = sum k_n^2 e^{-k_n^2} / sum e^{-k_n^2}, at fixed s and for fixed nonzero T,

```
M^2 = C_0/C_1 = (3/2 - <k^2>) ||[D_F, T]||^2/||T||^2 + O(||D_F||^4 / Lambda^2).
```

Proof. V and T split into finite-dimensional blocks indexed by n, so equation (1) of lemma 18 applies to symmetric finite sets of blocks, and the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify the Taylor expansion under the sum. Terms odd in k cancel between n and -n: exactly, the bracket at q = 0 is d_ij^2 int_0^1 E(u) [1 - u(1-u)(2k + mu_i + mu_j)^2] du, with E(u) = exp(-(1-u)(k + mu_i)^2 - u (k + mu_j)^2), whose next term is d_ij^2 e^{-k^2} ((2k^3 - 5k)/3)(mu_i + mu_j), and the bracket at order q^2 begins e^{-k^2} (2/3 - (2k/3)(mu_i + mu_j)). For the leading term of the expansion, put a_i = (k + mu_i)^2 = k^2 + p mu_i + mu_i^2 with p = 2k. To second order in mu,

```
I(a_i, a_j) = e^{-k^2} [1 - (p/2)(mu_i + mu_j) - (mu_i^2 + mu_j^2)/2 + (p^2/6)(mu_i^2 + mu_j^2 +
↪  mu_i mu_j)],
```

and e^{-a_i} = e^{-k^2} [1 - p mu_i - mu_i^2 + p^2 mu_i^2/2]. Therefore

2 I(a_i, a_j) - e^{-a_i} - e^{-a_j} = -(p^2/6) d_ij^2 e^{-k^2},

and adding d_ij^2 I gives d_ij^2 e^{-k^2} (1 - p^2/6) = d_ij^2 e^{-k^2} (1 - (2/3) k^2). Finally, sum_ij |T_ij|^2 d_ij^2 = |||[D_F, T]||^2. For C_1, at D_F = 0 the bracket in equation (3) of lemma 18 has d_ij = 0 and integrates to 2/3. ■

Differentiating directly, <k^2> = (1/2) d log S/d log s = (s/2) d log S/ds, with S(s) = sum_n exp(-n^2 pi^2/s^2). This S is the heat trace of the Dirac operator of the unshifted circle of length 2l at t = 1/Lambda^2, equivalently the paired Neumann and Dirichlet trace on the interval, with S_N = (S + 1)/2 and S_D = (S - 1)/2 separately. So d_eff = d log S/d log s = -2 d log Tr e^{-t D^2}/d log t is its spectral dimension at that scale. For nonzero D_F, the parent trace has e^{-t (k_n + mu_i)^2}, and S is the unperturbed trace used in the leading expansion. Hence, at leading order,

M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2/||T||^2.

The two limits. We first take the coefficient at small D_F/Lambda at fixed s, and then the limit in s. Explicitly, d_eff ~ (4 pi^2/s^2) exp(-pi^2/s^2) as s -> 0, and d_eff = 1 - 4 s^2 exp(-s^2) + O(s^2 exp(-2 s^2)) as s -> infinity.

1. For s « 1, only n = 0 contributes. Then d_eff = 0, the factor is 3/2, and this is the towerless regime of lemma 18 and Proposition F of the paper.
2. For s » 1, the sum is a Gaussian integral, with <k^2> = 1/2. Then d_eff = 1, and the factor is 1. This is forced by covariance on the unfolded circle: V is the fifth component of a five-dimensional Dirac operator with constant A_5 = D_F, and the local five-dimensional heat coefficient contains F_{MN} F^{MN}, with F_{5 mu} = i [D_F, A_mu] up to convention and with the same coefficient as F_{mu nu}^2. So the mass is the Hilbert-Schmidt commutator norm alone, as at small angle for the Yang-Mills parent of Theorem 23.6. On a general space with boundary, the bulk coefficient alone does not exclude boundary contributions; here the operator with these transported fields is unitarily equivalent to the circle problem, which is the reason they are absent.

In table form, at small masses:

s = l Lambda	0.3	1	1.5	2	2.5	3	5
(3 - d_eff)/2	1.5000	1.4990	1.3935	1.1414	1.0240	1.0022	1.0000

Corollary 25.3 (the sum rule). For the three-generation Standard Model representation, with the literal particle and antiparticle trace, with Dirac masses small compared with Lambda and with 1/l, and with T a charged generator of SU(2)_L, at leading order,

M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,

which at d_eff = 0 is 8 M_W^2 = sum N_f m_f^2, and at d_eff = 1 is 12 M_W^2 = sum N_f m_f^2. For the top alone,

m_t/M_W = (8/(3 - d_eff))^{1/2},

from sqrt(8/3) = 1.633 at d_eff = 0 to 2 at d_eff = 1. The neutral direction scales with the same factor, so M_W^2/M_Z^2 = 5/8 at all s to this order, and the massless directions keep zero mass coefficient, while their kinetic coefficients can change. These are leading-order statements: already in the towerless top-only family, M_W^2 = (3/8) m_t^2 - (23/128) m_t^4/Lambda^2 + … (equation (19) of lemma 18).

Proof. The trace gives ||T||^2 proportional to Q_11 = 12 and |||[D_F, T]||^2 proportional to K_11 = sum N_f m_f^2, which gives the coefficient. At d_eff = 0 this is the CCM relation (hep-th/0610241, Eq. (4.16)), as recovered in Proposition 4 and Remark 16 of lemma 18 at small x. The factor is common to all generators in Theorem 25.2. ■

In terms of Yukawa couplings, with m_t = y_t v/sqrt(2) and M_W = g v/2, this reads y_t^2 = 4 g^2/(3 - d_eff), so y_t = 2 g/sqrt(3) at d_eff = 0, the top-dominance value of CCM (Eq. (5.27) there), and y_t = sqrt(2) g at d_eff = 1.

Remark (positive Laplace cutoffs). For f(z) = int e^{-tz} dnu(t) as in equation (4) of lemma 18, with t measured in units of 1/Lambda^2, the same computation gives at leading order

```
M^2 = ((3 - d_nu)/2) ||[D_F, T]||^2/||T||^2,     S_nu(s) = int S(s/sqrt(t)) dnu(t),     d_nu = d
↪ log S_nu/d log s = 2 <t k^2>_nu,
```

where the average has weights dnu(t) e^{-t k_n^2}. Under the moment hypotheses there, dominated convergence gives the same endpoints, and the crossover and its corrections depend on nu. A general smooth cutoff, such as one flat near zero, need not belong to this class.

Scope

- This is the Taylor quotient of a prescribed constant-profile vector, at leading order in D_F, about a prescribed background and not a demonstrated vacuum. Taylor and pole masses differ in general. Calling it the leading term of a lowest-mode mass also needs an admissible gauge field, a positive kinetic form and an isolated lowest branch; on the circle, translation invariance separates the zero Fourier sector, within which the generator matrix must still be diagonalized. For finite angles in the Yang-Mills parent, the exact masses are in Theorem 23.6, which expands as M_W^2 = 2 m^2 - (4/3) m^4 l^2 + … and M_Z^2 = (16/5) m^2 - (64/75) m^4 l^2 + …, in agreement with d_eff = 1 at leading order. That theorem is for its own Yang-Mills boundary problem, and it does not establish the full finite-cutoff gauge spectrum of the spectral action.
- T must be even and constant in the transported frame, so it preserves both walls there, and it needs the enlarged gauge algebra noted above. Vectors that are Dirichlet at a strict wall, as in Proposition 22.8, are a different boundary problem.
- All fields have flat profiles on one interval, with no bulk masses. In particular, for the top-only statement, the kinetic term counts all twelve doublets of three generations on the same interval, with the same S. With bulk masses to localize light fields, the spectra and overlaps change, and the kinetic coefficient must be recomputed from all eigenstates and overlaps.
- The spectral action is taken as a function of the fermion operator at a matching scale, with no running.
- For a boundary with reflection, the walls contribute heat-kernel boundary terms in general. Here the unfolding removes them for this operator and these generators.
- Other priority: the endpoint m_t = 2 M_W appears at tree level in Cacciapaglia, Csaki and Park, Fully radiative electroweak symmetry breaking, arXiv:hep-ph/0510366, section 5, for a suitable rank-four representation, and a limit approaching 2 M_W in Adachi and Maru, arXiv:1804.06012, Eq. (3.23). These are representation-dependent mechanisms, and they do not give the spectral-action crossover. Universal propagation alone does not impose it in ordinary universal extra dimensions (Cordero-Cid et al., arXiv:1108.2926, section 2).

Reading

This connects three items of the project.

1. *The emergence of a dimension.* The heat dimension of the interval, which distinguishes the labels 10 and 11, enters the mass relation directly. As the cutoff resolves the interval, the coefficient moves from the finite-space value to the five-dimensional value.
2. *Proposition F and lemma 13.* The factor 3/2 is the finite-action coefficient relative to the Yang-Mills commutator quotient, and in this constant-profile calculation the two are the two ends of one function. Proposition F also contains a bounded-fermion against tower-scale obstruction and overlap conditions, which keep their separate hypotheses.
3. *The measured ratio.* With the Particle Data Group 2026 listings, m_t = 172.60 +- 0.27 GeV as a direct-measurement Monte Carlo mass, and M_W = 80.3625 +- 0.0077 GeV, the ratio is 2.148, which would formally correspond to d_eff = 1.27, outside the range. With a running mass of about 163 GeV (Yoon and Peskin, arXiv:1810.12352, section 3.3 and Eq. (23), for a pole mass of 173 GeV, with

about 147 GeV at 2 TeV), it is 2.03, which corresponds to d_eff = 1.06. Dividing a running mass by an on-shell W mass is only a formal orientation and not a matching test at a common scale, and no running from the matching scale has been computed, so we draw no conclusion from this comparison. The known spectral-action value y_t = 2 g/sqrt(3) is the d_eff = 0 boundary condition. A tower near the weak scale, as in the separate unrefereed estimate of the steer note, item 9, would use the d_eff = 1 boundary condition y_t = sqrt(2) g, at a matching scale of that order, in a specified effective theory with thresholds and a renormalization prescription. In a one-loop orientation with universal bulk fields (the steer note, item 11, no referee), the running approaches an infrared quasi-fixed point near 211 to 232 GeV, nearly independently of the boundary value, so that family does not support the closeness of 2 M_W to the running top mass.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. The kernel of lemma 18, with quadrature in u, was summed over the tower, with explicit 4 x 4 level blocks for the operator on the interval, independently of the unfolding.

- Random complex K_1 and Hermitian T_Q, T_U on C^3, with ||K|| = 10^-3 min(s, 1): the ratio of C_0/C_1 to the Hilbert-Schmidt quotient agrees with 3/2 - <k^2> to six digits, for s in {0.3, 1, 1.5, 2, 2.5, 3, 5, 10}.
- Three generations, top only, with T = sigma_x/2 on the U-fields: m_t/M_W agrees with (8/(3 - d_eff))^{1/2} to six digits on the same grid, from 1.63299 at s = 0.3 to 2.00000 at s = 5.
- The neutral direction, with weights for lepton and colour: M_Z^2/m^2 goes from 4.801 to 3.200 for universal mass, and M_W^2/M_Z^2 stays at 0.625 to four digits.
- Finite mass in the top-only family: M^2/m^2 at s = 5 is 0.2500, 0.2498, 0.2477, 0.2409, 0.2241, 0.1478 for m/Lambda = 0.02, 0.1, 0.3, 0.6, 1, 2, and at s = 0.5 it is 0.3749, 0.3732, 0.3591, 0.3145, 0.2277, 0.0555, the latter matching lemma 18.

Part 28

Two sheets as the two ends of one connected commutative parent

For STRATEGY T0 and T0b, following lemma 16 (Example 16.9), lemma 24 (Proposition 6) and lemma 09. Propositions 28.1 to 28.4 and 28.13 are by Claude Opus 5.5. Results 28.8 to 28.12 and 28.14 to 28.21 are by Astra 6, job 28 (report, with the full proofs; prompt), checked by Claude Opus 5.5 by hand and in Python. Proposition 28.13 and the numerics are refereed by Terra 6.1 (report), with its corrections applied. The window is a conjecture with numerical support; the numbers come from a radial solve in Python (numpy and scipy), described in the section Numerical check; the script is not kept.

Correction, 3 Nov 2026. The first version of Result 28.5 and of the numbers in Remark 28.7 (commits dd1b14c, 530fa06) used the wrong sign of the spin connection in the Python slice operator. That changed the singular values and the boundary fibre parts, so those numbers are withdrawn, including the next singular value 1.04, the value 1.35 on the representation 16, and the boundary fibre parts -0.011 to -0.853. The exponents at the origin and the multiplicities are unaffected. Astra 6 did not reproduce the old boundary values and its proofs do not use them. We found the error when that operator gave a core zero mode in the representation 20, against the exact count of Proposition 28.11. The singular values, fibre parts and densities below are recomputed, and the script tests the operator against exact statements: {A_0, A_1} = 0, A_0^2 equal to the Casimir formula on S^4, D^2 on (0, 0)-forms with values in O(1) at the Fubini-Study radius equal to j(j + 4), and A_2 equal to its curvature contraction (the last test proposed by Terra 6.1).

Update, 4 Nov 2026. Lemma 29 proves Conjecture 28.6 in case (b) for the metric class of its Setting 29.0, and in case (a) at the Fubini-Study radius, refereed by Terra 6.1. Item 3 below and the Status of Proposition 28.13 describe the position before that lemma.

Outcome.

1. *The mechanism is proved at the level of compression maps.* A smooth function on the disc bundle X has independent values on the zero section S^4 and on the boundary CP^3, and at the zero section it is a function on S^4 alone (Proposition 28.8). For prescribed windows over this X, real functions generate exactly M_2(H) + M_4(C) at every localization scale (Theorem 28.21). In Example 16.9 the second sheet and the restriction to S^4 were inputs, so this is a connected source for them.
2. *The two ends of an actual local operator are understood.* There is a self-adjoint operator with a local wall (Theorem 28.9). The core carries exactly four states in the representation 4 (Proposition 28.11). At the wall, large mass keeps a tangential mixing (Lemma 28.12). For the pull-back connection the boundary mode then has the fibre part -w(rho) in closed form, with w(rho) = 2(rho^2+2)^2/(9+2(rho^2+2)^2), which reaches the level-one value of Example 16.9 only as rho -> infinity (Proposition 28.13). This is a statement on the finite wall model, and its convergence from the full operator is numerical. A connection preserving the line summands in a collar, with the Fubini-Study metric at the wall, gives the level-one value in that model.
3. *The isolated window of the whole operator stays a conjecture* (Conjecture 28.6): what is missing is the normal reductions with their errors, the matching, and a coercive estimate uniform in the mass and in all representations of Sp(2) (Lemma 28.14).
4. *The algebra M_2(H) + M_4(C) is the algebra of the limit map.* Generated algebras need not converge when compression maps do (Counterexample 28.17). At the sample points resolved numerically, the core mode has a nonzero density breaking the quaternionic structure, empirically of order lambda^2; if it is nonzero in the continuum, Proposition 28.16 gives M_4(C) for the core block at those masses. This is numerical evidence and does not establish the exact algebra at any finite mass.

Setting

Let G = Sp(2), K = Sp(1)_a x Sp(1)_b and H = Sp(1)_a x U(1)_b, so that G/K = S^4, K/H = S^2 and G/H = CP^3. Let S be the bundle G x_K (1, 2), so that CP^3 = P(S). The normal bundle of the zero section is G x_K (1, 3), and away from it X = (0, l] x CP^3, with radial coordinate x and S^2 fibres of radius rho(x), rho(x) ~ x at x = 0. Charges of U(1)_b are normalized so that the doublet of Sp(1)_b has +-1. The isotropy representation is (2)_{+-1} (+) (1)_{+-2}. The spinors of X restrict on a slice to Sigma_6 = Sigma^+ (+) Sigma^-, with c(d_x) = Z the chirality of Sigma_6, where

```
Sigma^+ = (1)_2 (+) (1)_0 (+) (2)_{-1},   Sigma^- = (2)_1 (+) (1)_0 (+) (1)_{-2}.
```

The representation 4 of G restricts to (2)_0 (+) (1)_1 (+) (1)_{-1}, that is (2, 1) (+) (1, 2) of K. Let Gamma_a be the Clifford generators on 4, with Gamma_5 commuting with K and equal to +-1 on the two summands, and let x_a(g) be defined by lambda(g) Gamma_5 lambda(g)^-1 = x_a(g) Gamma_a.

Proposition 28.1 (the twist).

1. For k != 0, O(k) on CP^3 does not extend to a line bundle on X, since H^2(X) = H^2(S^4) = 0 while O(k) has degree k on the fibre S^2.
2. The pullback of S to X restricts on CP^3 to O(1) (+) O(-1), as the tautological splitting. So E = S^3(S), which is pulled back from S^4 and has trivial connection along the fibres, restricts to O(3) (+) O(1) (+) O(-1) (+) O(-3). The summand O(3) gives the Dolbeault operator on O(1) of Example 16.9, since Sigma (x) O(3) = Lambda^{0,*} (x) O(1).
3. On E the section Phi(v) = phi(|v|) (v-hat . T), with T the spin-3/2 generators and phi odd and smooth, is smooth and G-equivariant, with zero set exactly S^4. In the frame of the slice it is phi(x) tau, with tau diagonal in the summands, with eigenvalues +-3/2, +-1/2.

So we take the Callias-type operator T = D_X(E) + i Phi on spinors of X tensored with E, and the self-adjoint operator D-hat = [[0, T^*], [T, 0]], with the grading Gamma-hat = diag(1, -1). At x = l we take the local condition Gamma-hat Z = -1 on the doubled spinor, that is u_even(l) = 0 for T and v_odd(l) = 0 for T^*. It is a chirality condition, elliptic and local, and it preserves the domain under all smooth functions on X.

Proposition 28.2 (multiplicities). On a slice, Hom_H(4, Sigma^+ (x) E) and Hom_H(4, Sigma^- (x) E) each have dimension 5. For the summand O(3), the dimensions are 1 and 0. So on 4, the operator

on the summand O(3) with its U(1) connection vanishes for every G-invariant metric on the slice, and its kernel contains H^0(CP^3, O(1)) = 4 for every fibre radius. This concerns the separate line connection. The pull-back connection of E does not preserve the summands, and its slice operator is invertible for small fibres (Proposition 28.10).

Proof. Tensoring the lists above with the charges of E gives the multiplicities. The operator is odd under Z and maps the multiplicity space of 4 into itself. ■

Proposition 28.3 (the zero section). The fibre operator c_N . nabla + i Phi on R^3 with values in E has zero modes (1, 3) (+) (1, 1) of K, by the Callias count for the positive weights 3/2 and 1/2, with one sign of Phi. On S^4 this gives the Dirac operator twisted by (1, 3) (+) (1, 1). Then Hom_K(4, S^- (x) (1, 3)) has dimension 1, through (1, 2), and Hom_K(4, S^+ (x) (1, 3)) = 0, so the twisted operator vanishes on that copy of 4, in agreement with its index 4. The untwisted part (1, 1) has eigenvalue 2 on the representation 4. Every map in Hom_K(4, S^- (x) (1, 3)) annihilates (2, 1) in 4. These are the twistor spinors of S^4, as sections of S.

Proof. The representation count is as in Proposition 28.2, and the Callias count is a reference, with the exact count in Proposition 28.11. For the last sentence, the first-order operator from (1, 2) to (2, 3) in the slice limit is the twistor operator, and its kernel is the 4. ■

On a representation V the twisted operator has D^2 = Cas_G(V) + 3/2 - Cas_K(tau), for the twist tau, with Cas_G(4) = 5/2, Cas_G(16) = 15/2, Cas_G(20) = 21/2, Cas_G(40) = 29/2, Cas_K(j_a, j_b) = 2 j_a(j_a+1) + 2 j_b(j_b+1). With the twist (1, 3) this gives 0, 5, 8, 12, and with the twist (1, 1) on the representation 4 it gives 4. The slice operator reproduces these (Result 28.5).

Proposition 28.4 (the compression). Let s_v(g) = F lambda(g)^-1 v be a section in the multiplicity space of 4 on a slice, with B = F^* F, which commutes with lambda(H). Write t-hat = 2 i t_3 of Sp(1)_b. Then, with normalization by the norm of s_v,

```
Q(x_a) = (tr(B Gamma_5)/tr B) Gamma_a/5,
```

and for the functions y_T(g) = tr(T Ad(g) t-hat) with T in the Lie algebra sp(2), Q(y_T) is a multiple of T, proportional to tr(B t-hat)/tr B.

Proof. End_H(4) is spanned by 1, Gamma_5 and t-hat. Since int lambda(g) Gamma_5 lambda(g)^-1 x_a = int x_a x_b Gamma_b = Gamma_a/5, the part with Gamma_5 gives the first formula. Equivariant maps from 5 to End(4) are multiples of Gamma_a, and tr(Gamma_a Ad(g) t-hat) = 0, so 1 and t-hat contribute nothing. The second formula is Schur's lemma on the representation 10. ■

Consequently, a mode with F supported on (1, 2) and K-symmetric has tr(B Gamma_5)/tr B = +-1 and tr(B t-hat) = 0. That is the zero section: Q(x_a) = +-Gamma_a/5, and the part of a function that depends on the fibre gives zero. On the boundary with the pure level-one mode, B is the projection on (1)_{-1}, so that tr(B t-hat)/tr B = +-1. Example 16.9 has these values.

The operator and its two ends

Proposition 28.8 (endpoint restrictions). Restriction C^infinity(X, R) -> C^infinity(S^4, R) + C^infinity(CP^3, R) is surjective, and in normal coordinates a(y, v) = a(y, 0) + O(|v|), uniformly on bounded subsets of C^1(X). (Report, Proposition 1.)

Theorem 28.9 (a self-adjoint realization). Take the connection metric g = dr^2 + h(r)^2 pi^* g_{S^4} + rho(r)^2 g_{S^2} with h smooth, even and positive at 0, rho odd and smooth with rho(r) = r + O(r^3), and a product near r = l. Let Phi_M = M b(|v|^2) v . J with b smooth and positive, T_M = D_X^E + i Phi_M, and D-hat_M the doubled operator. At r = l set Z = -i c(dr), chi = Gamma-hat Z, and impose chi psi = -psi. Then D-hat_M on this H^1 domain is self-adjoint, elliptic and odd, with compact resolvent, and every smooth scalar function preserves the domain. No boundary condition enters at the zero section: after the outer condition is fixed, removing S^4 creates no further self-adjoint extension. (Report, Theorem 2, using Bär-Ballmann, Corollary 4.18.)

Proposition 28.10 (three slice operators). For the standard complex structure on CP^3: (1) the kernel of the Dolbeault-Dirac operator on O(1) is C^4 for every Hermitian metric; (2) the Levi-Civita Dirac operator twisted by the separate line bundle O(3) with its invariant connection vanishes on the 4 for every invariant squashing, which gives a kernel of dimension at least four, with any further kernel in other representations at a locally discrete set of squashings; (3) the Levi-Civita Dirac operator twisted by the full pull-back connection on E is invertible for small fibres, since (D^E_rho)^2 >= 1/(2 rho^2) - C_0 - C_1 rho^2. (Report, Proposition 3.) In Python, on the representation 4, the smallest |eigenvalue| of that operator is 3.235, 1.842, 1.200, 0.722, 0.148 at rho = 0.3, 0.5, 0.707, 1, 2. So the window comes from the competition between the normal operator and the mass.

Proposition 28.11 (normal model). Let Q = sigma . d_y + y . J on L^2(R^3; C^2 (x) C^4), with J Hermitian of spin 3/2. Then ker Q = 0 and ker Q^* = 3 (+) 1 under the normal Sp(1), with a positive gap above the zero modes; changing the sign of the mass exchanges the two kernels. The proof uses Q^* Q = -Laplacian + (y . J)^2 - sigma . J >= 0, with equality excluded since sum_i J_i^2 = 15/4, and the Callias index for the positive eigenbundles of n . J, of degrees 3 and 1. The induced bundle over S^4 is S^2(S) (+) 1, the operator on Sigma (x) S^2(S) has exactly four zero modes, in one chirality, and on the parallel inclusion S -> S (x) S^2(S) the twisted equation is a multiple of the twistor equation. So the kernel consists of harmonic spinors of the twisted operator, which are twistor spinors as sections of S. Rescaling gives the localization scale r = O(M^{-1/2}) at the core and O(M^{-1}) at the wall. (Report, Proposition 4.) In Python, in an oscillator basis: Q^* Q starts at 1.02, QQ^* has four near-zero values, with total spin 0 and three values near spin 1, then 1.02.

Lemma 28.12 (tangential mixing). On a product collar, normalized stable profiles for mass eigenvalues M q_i overlap by 2 sqrt(q_i q_j)/(q_i + q_j), independently of M, which is sqrt(3)/2 for q = 1/2, 3/2. So large mass does not remove the coupling between the stable channels O(3) and O(1), which in the pull-back connection has order 1/rho. (Report, Lemma 5.)

Proposition 28.13 (the boundary operator on the representation 4). Take the sign of Phi for which the right singular vector of T is at the zero section. Its left singular vector v is at the wall, in the domain Z v = +v of Theorem 28.9 for T^*. A wall profile exp(q(x - l)), q > 0, for T^* = i Z d/dx + D_6 + i phi tau, with phi > 0, has q = -phi Z tau at leading order. So the stable channels of v are (Z, tau) = (+, -3/2) and (+, -1/2) twice, and they map to the channels (-, -1/2) twice. In unitary bases of these, the compression of D_6(rho) with the overlaps of Lemma 28.12 is

```
A(rho) = ( 3/(2 rho)   1/rho   rho + 1/rho )
         (     0       sqrt 2     -sqrt 2  )
```

with the matrix elements computed in Python, and reproduced by Terra 6.1 in exact symbolic arithmetic from the bracket and Clifford data. Its kernel is spanned by c = (-(2/3)(rho^2 + 2), 1, 1). The channel tau = -3/2 and one channel tau = -1/2 are supported on (1)_{-1} of the 4; the other channel tau = -1/2 is supported on (1)_{+1}. All three lie in Gamma_5 = -1, and their target spaces are mutually orthogonal, so the cross terms vanish in the traces of Proposition 28.4. Hence for the mode c

```
tr(B Gamma_5)/tr B = -1,    tr(B t-hat)/tr B = -w(rho),    w(rho) = 2(rho^2 + 2)^2 / (9 + 2(rho^2 +
↪ 2)^2),
```

for every rho, with horizontal scale 1; for horizontal scale h, replace rho^2 by rho^2/h^2. So w = 8/17 as rho -> 0, w = 2/3 at the Fubini-Study radius rho = 1 (the nearly Kähler radius is 1/sqrt 2), and w -> 1 as rho -> infinity, the level-one value. The nonzero singular values of A are the next wall values on the 4: 1.863 and 2.789 at rho = 1, with the core value 2 in between. With the opposite sign of Phi, the right singular vector is at the wall, in the channels Z = -1, tau > 0, with fibre part +w(rho), and its left singular vector at the zero section.

For the connection preserving the line summands, the first row is replaced by (0, 1/rho, rho + 1/rho), the kernel is (1, 0, 0), so w = 1 for every rho, and the block on tau = -1/2 has singular values with sum of squares rho^2 + 2/rho^2 + 6 and product sqrt 2 (rho^2 + 2)/rho, so it is invertible for every rho > 0, with D^2 = 3, 6 at rho = 1. This is the collar repair of the report: an invariant connection preserving the four line summands in a collar extends into X with a cutoff, with the pull-back connection near the zero section.

At the Fubini-Study metric the stable boundary model is then a sum of line-twisted Dirac operators, with four zero modes on O(3) and none on O(1), since H^q(CP^3, O(-1)) = 0 for all q.

Status. The kernel and the traces are exact for this finite block, which is the leading formal wall operator; the next correction is O(1/M) in the fixed-block expansion and does not change the limiting kernel, since the kernel is separated from the other singular values. That the boundary mode of T_M converges to it as M -> infinity is a numerical statement here (Result 28.5), since a uniform global reduction is unproved: the difference falls as phi_0 grows, from 0.037 to 0.009 at wall radius 0.5 and from 0.008 to 0.002 at wall radius near 1, between phi_0 = 16 and 32. The value 2/3 tests the factor sqrt 3/2, since without it w(1) would be 3/5.

Lemma 28.14 (a sufficient spectral criterion). Let U_M : C^8 -> Dom D-hat_M be an isometry with ||D-hat_M U_M|| <= eps_M -> 0 and ||D-hat_M psi|| >= delta ||psi|| for psi orthogonal to the range, with delta > 0 independent of M. Then for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), and the spectral space is within O(eps_M/delta) of the range. (Report, Lemma 6.) The missing steps for Conjecture 28.6 are the normal reduction with its errors, and the coercive estimate on every representation of Sp(2).

Conjecture 28.6 (the window). Take Theorem 28.9 with the sign of Proposition 28.13 and with (a) the pull-back connection and wall radius rho, or (b) the collar connection of Proposition 28.13 and the Fubini-Study metric at the wall. For large M the hypotheses of Lemma 28.14 hold, with the eight states being one copy of the 4 at the zero section in one Gamma-hat degree and one at the boundary in the other. For a family of metrics and mass profiles with a uniform positive decay rate, the off-diagonal block is exponentially small in l. The compression of a smooth function tends to (Q_B(a|*{S^4}), Q_Y(a/*{CP^3})), with Q_B and Q_Y(x_a) = -Gamma_a/5, and Q_Y(y_T) a multiple of T, equal to the level-one value of Example 16.9 times w(rho) in case (a), and equal to it in case (b). This concerns compression maps, in line with Counterexample 28.17.

Result 28.5 (Python, the radial system). We solve T on the multiplicity spaces, with Hermitian Clifford generators, so that c(dx) = i Z in the skew convention of Theorem 28.9. For horizontal scale h, after absorbing the half-density h^2 rho, T = i Z d/dx + A_0/h + A_1/rho + (rho/h^2) A_2 + i phi tau, with -s(x) A_E/rho added for a collar connection. We take h = 1, so that T = i Z d/dx + D_6(rho(x)) + i phi(x) tau, with D_6 built from the structure constants of sp(2), rho(x) = rho_inf tanh(x/rho_inf), phi(x) = phi_0 tanh(x/w_0), w_0 = 0.3, in units with S^4 of radius 1, by the quadratic form of T^* T with P1 elements on a graded grid. This rho is not exactly a product near the wall. Rayleigh quotients of the exact form bound the values from above, and with three-point quadrature they approximate those bounds. Small entries have a mesh floor that depends on the parameters: at (phi_0, l) = (6, 4) the lowest value is 0.00548, 0.00401, 0.00355 for 2500, 5000, 10000 elements, so such entries do not establish a zero continuum value. At x = 0 the exponents are the values for R^3: +-1, +-2 (three times), +-3 on the 4.

phi_0	l = 2	l = 3	l = 4	next singular value, l = 4
3	0.406	0.159	0.066	1.293
4	0.288	0.088	0.028	1.452
5	0.200	0.045	0.011	1.665
6	0.132	0.021	floor	1.818
8	0.049	floor	floor	1.852

(rho_inf = 1, representation 4; the next value at larger phi_0 is 1.86 to 1.87.) The lowest value falls with the exponent of the constant-coefficient system, the smallest |Re kappa| of i Z (D_6(rho) - i phi tau): measured between l = 3 and 4, 0.877, 1.155, 1.464 at phi_0 = 3, 4, 5, against 0.856, 1.151, 1.476, as numerical evidence for these profiles, without a general tunnelling theorem. At large phi, kappa = phi/2 - 2/rho + O(1/phi) (phi = 64: 30.07 at rho = 1, 31.07 at rho = 2). The formula 2 mu exp(-mu l) of lemma 09 belongs to a separable interval and does not carry over.

On the two modes, at rho_inf = 1, l = 3, with the sign of Proposition 28.13, the boundary mode being the left singular vector; the core fibre parts at phi_0 = 16, 40 are below what the computation resolves:

phi_0	zero section: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B	boundary: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B, Z, tau
10	0.39, -1.0000, -3e-7	2.93, -1.0000, -0.645, 0.944, -1.117
16	0.28, -1.0000, 1e-11	2.96, -1.0000, -0.658, 0.976, -1.146
40	0.16, -1.0000, 1e-11	2.99, -1.0000, -0.663, 0.996, -1.162

Against Proposition 28.13, at l = 3:

rho_inf	wall radius	-w(rho)	phi_0 = 16	phi_0 = 32	next value, phi_0 = 16	second singular value of A
0.5	0.500	-0.529	-0.493	-0.521	1.992	1.992
0.7	0.700	-0.579	-0.562	-0.575	1.966	1.966
1.0	0.995	-0.665	-0.658	-0.663	1.864	1.865
1.5	1.446	-0.788	-0.785	-0.787	1.660	1.661
2.0	1.810	-0.861	-0.858	-0.860	1.557	1.556
3.0	2.285	-0.921	-0.918	-0.919	1.487	1.486

Collar connection. With rho(l) = 1 and s(x) = (1 + tanh((x - l + 1)/0.15))/2 multiplying the vertical connection term to be removed, at phi_0 = 8, 16, 32: the boundary mode has tr(B t-hat)/tr B = -1.0000 and Q(x_a)/(Gamma_a/5) = -1.0000, the next value is 1.733, that is sqrt 3, and the core mode is unchanged. The switch is nonzero, though tiny, at x = 0, whereas the smooth extension needs a cutoff vanishing near the zero section, so these rounded values support the limiting model and do not establish exact finite-mass equality.

Other representations. The multiplicity is zero on 1, 5, 10, 14, 35. At phi_0 = 16, rho_inf = 1, l = 3, the lowest singular values are 2.188 (wall) and 2.235 (core) on the 16, 1.920 (wall) and 2.828 (core) on the 20, and 3.381 (wall) and 3.463 (core) on the 40, the core values being sqrt 5, sqrt 8 and sqrt 12 up to the mass. The multiplicity zero follows from the central character. The finite wall operator A_V(rho) has no zero at 1500 sampled radii in [0.1, 20] on the 16, 20, 40, 64, a sampled statement; its smallest singular value tends to the core value as rho -> 0 and falls slowly at large rho on the 16 and the 40 (0.33 and 0.67 at rho = 20). Higher representations were not computed, and no estimate in the representation is proved.

Core block at finite mass. The density of Proposition 28.16 is nonzero at these points, stable in the mesh to five digits:

phi_0, l	lambda	tr(B t-hat)/tr B, integrated	max of abs tr(B t-hat) over max of tr B	sup of the pointwise ratio	integrated value over lambda^2
4, 3	0.0882	-5.71e-3	6.48e-3	0.849	-0.73
5, 3	0.0453	-1.84e-3	1.97e-3	0.856	-0.89
6, 3	0.0212	-4.55e-4	4.73e-4	0.858	-1.01

The pointwise ratio reaches its maximum at the outer wall, where the core density is small, so the breaking comes from the tail of the core mode at the wall. Over phi_0 = 3 to 6, l = 2 to 4, the integrated value over lambda^2 lies between -0.4 and -1.0, an empirical observation with no O(lambda^2) asymptotic proved. These values support the hypothesis of Proposition 28.16 at the sample points, conditional on the density being nonzero in the continuum.

The compression and the algebra

Theorem 28.15 (compression, conditional on localization). Assume an eight-dimensional spectral window splitting under Gamma-hat as W_M (+) V_M with dimensions 4, 4. Then scalar compression is exactly block diagonal. If the isometries differ by at most eta_M from model embeddings with angular compressions Q_B, Q_Y and normal measures with first moments at most C M^{-1/2} at the core and C M^{-1} at the wall, then

```
|| Q_M(a) - (Q_B(a|_B) (+) Q_Y(a|_Y)) || <= C ||a||_{C^1} M^{-1/2} + 4 ||a||_infinity eta_M.
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(Report, Theorem 7.) For a spectral wall as in lemma 24, the domain has constant boundary values, and if a$|Y = c$ *then any boundary-localized family has* $//Q${Y,M}(a) - c|| <= C M^{-1} ||d_r a||_infinity. So localization just inside a spectral wall does not recover arbitrary boundary functions for a fixed smooth a, and the limit needs a local wall or an explicitly different comparison algebra.

Proposition 28.16 (the exact symmetry test). For an equivariant copy of 4, B(r) = F(r)^* F(r) = diag(b_a(r) I_2, b_+(r), b_-(r)). All real scalar functions compress to operators commuting with the quaternionic antiunitary j if and only if b_+(r) = b_-(r) for almost every r. If in addition the 5 component is nonzero, the block generates M_2(H). If both the 5 and the 10 components are nonzero, it generates M_4(C). A vanishing integral of tr(B(r) t-hat) alone does not decide this. If a radial h has Q_M(h) = alpha I_W + beta I_V with alpha != beta, then the block projections lie in the generated algebra. (Report, Proposition 8.) At the limit, the real linear image is Herm_2(H) + Herm_4(C), of real dimension 6 + 16 = 22, and the generated real algebra has dimension 16 + 32 = 48.

Counterexample 28.17 (generation is discontinuous under convergence of compression maps). Let R_0 = Q_B E_fib, with E_fib the fibre average on CP^3, R_1 = Q_Y, and R_eps = (1 - eps) R_0 + eps R_1. These are unital completely positive, ||R_eps - R_0|| <= 2 eps, and real functions generate M_4(C) under R_eps for every eps > 0, and M_2(H) at eps = 0. So the correct limit statement concerns specified compression maps, and it gives no convergence of the generated algebras. (Report, Counterexample 9.)

Theorem 28.21 (prescribed windows on the same X). Let P_y = (1 + y_a Gamma_a)/2 and p_z <= P_{pi(z)} the tautological rank-one projector. With normalized radial profiles chi_0, chi_l supported in disjoint eps-neighbourhoods of the two ends, U_0 v = sqrt 2 chi_0 P_y v and U_l v = 2 chi_l p_z v are isometries into L^2(X; C^4) (+) L^2(X; C^4). Real functions compressed by them generate M_2(H) + M_4(C) at every eps, and the compression differs from Q_B (+) Q_Y by at most C eps ||a||_{C^1}. (Report, Theorem 13.) This proves the connected-space mechanism without Conjecture 28.6, with the window as input. Allowing nonlocal smoothing perturbations, (1 - P) D (1 - P) + P d P makes such a window reducing, which enlarges the operator class and keeps the window as input.

The finite data

Proposition 28.18 (charge conjugation and the wall). With Hermitian generators J_i, C = C_7 (x) j_E has C^2 = -1 and commutes with T_M, D-hat_M and Gamma-hat as a differential expression, and CZ = -ZC. So it exchanges the two local wall conditions and does not preserve the chosen domain. If a pointwise parent antiunitary preserves a window and commutes with real multiplication, every real compressed multiplier commutes with its restriction, which cannot hold on a generated full M_4(C). (Report, Proposition 10.) The complex boundary block therefore requires this loss of the inherited symmetry, or a different real correspondence.

Remark 28.7 (why one block is quaternionic and the other complex). Proposition 28.18 proves the signs argument of the first version, with one correction of wording: j_E commutes with the anti-Hermitian Lie algebra action and reverses the Hermitian generators, so that C commutes with i Phi. Numerically (Python, phi_0 = 16, l = 3, rho_inf = 1), exchanging the wall condition exchanges (Z, tau, tr(B t-hat)/tr B) = (0.976, -1.146, -0.658) and (-0.976, 1.146, 0.658) on the boundary mode, with the same singular values, and leaves the zero-section mode unchanged. The formal parent symmetry does not by itself prove that the finite-mass core space is preserved, since C changes the wall domain, and the breaking computed in Result 28.5 shows the size of the effect at the sample points.

Proposition 28.19 (tunnelling and the grading). If the window consists of one copy of the 4 in each Gamma-hat degree, then d_M = [[0, conj(lambda_M) I_4], [lambda_M I_4, 0]] after equivariant unitary identifications, by Schur's lemma, and lambda_M = (1/4) Tr(U_{Y,M}^* T_M U_{B,M}). The inherited grading is I_W (+) (-I_V), the auxiliary grading of the doubling, which labels the two ends under localization. The radial Clifford involution Z does not extend across the zero section, so it does not serve as an interval chirality. This grading leaves all four core states in one degree, so W = H (+) H, for the even algebra H + H + M_4(C), still needs a chosen quaternionic grading, such as a marked Gamma_p. (Report, Proposition 11.)

Proposition 28.20 (the doubled bimodule). With H = W (+) V, Theorem 16.2 gives K = H (x) conj(H), of dimension 64, with DD = d (x) 1 + Gamma-hat (x) d^c. Its odd part is exactly K^- = Hom(V, W) (+) Hom(W, V), with the desired bimodule actions, and order zero and order one hold exactly on the full K, for its actual even algebra. If lambda != 0, then DD maps K^- into K^+, P_- DD P_- = 0 and DD^2 = 2 lambda^2. So K^- is not a reducing subspace. On K^- the inherited tensor grading is constant; the sheet-sign grading of Theorem 16.7 gives J(x, y) = (y^*, x^*), J^2 = 1, J Gamma_6 = -Gamma_6 J, with zero Dirac operator. The tunnelling map W -> V does not provide the map Hom(V, W) -> Hom(W, V) of lemma 20. (Report, Proposition 12.)

Balance against Proposition 6 of lemma 24

(Report, Proposition 14, with the numerical statements updated.)

Structure	What follows
Connected scalar parent	Supplied by the chosen disc bundle X.
Independent endpoint functions	Derived, by restriction and extension.
Base-function restriction at the collapsed end	Derived geometrically; quantitative for localized states.
Four core states	Proved for the linear normal model and the resulting twisted operator on S^4.
Four boundary states	Proved for the separated collar model; for the pull-back connection, the finite block of Proposition 28.13, with the convergence numerical.
Isolated 4 + 4 window	Conjecture 28.6.
Limiting quaternionic core compression	Derived once convergence to the core model is established.
Exact finite-mass quaternionic block	Requires b_+ = b_-; numerically not satisfied at the sample points resolved.
Exact level-one boundary compression	Requires rho -> infinity or the separated collar connection.
End grading	Inherited from the chosen self-adjoint doubling.
Quaternionic grading axis	Input.
The two Hom spaces	The odd part of the coarse correspondence; the selection is additional.
KO-6 reality and order conditions	Exact after the specified correspondence and grading choices.
Nonzero connector of lemma 20	Input.
Mass value and dependence on l	Radial matching; equivariance gives only lambda I_4; the exponent is computed numerically in Result 28.5.

Choices that stay inputs: X, E = S^3(S), the hedgehog, the wall condition, the collar connection in case (b), and the doubling of Theorem 16.2. A double with two identical collapsed zero sections removes the outer boundary condition, but the same local core construction gives two quaternionic endpoint maps, so

obtaining a complex block needs different structure at one end. T0 stays open, and this is a candidate for its first half.

Direction. The next analytic step is the coercive estimate of Lemma 28.14, in case (b) first, where both end models have exact kernels. Since Phi commutes with the Clifford symbol, D + iM Phi is a concentrating pair in the sense of Maridakis (arXiv:1510.07004, Definition 2.1 and Lemma 2.2), with [D, Phi] of order zero; the symbol condition goes back to Prokhorenkov and Richardson (arXiv:math/0307251). Two things are missing there. The spectral separation theorem of arXiv:2110.11654 (section 2, Assumption 3) needs a leading endomorphism independent of the normal direction, and this fails here, since (n . J)^2 depends on n for spin 3/2; so a separate core analysis or an anisotropic extension is needed. And the wall adds a boundary layer: on a half-line ||(d_t + M) f||^2 = ||f'||^2 + M^2 ||f||^2 - M |f(0)|^2, and exp(-Mt) is a zero mode away from the zero set of the mass. (Read by Terra 6.1 in arXiv HTML; the LaTeX sources were not available to it.)

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a script that ran the tests below as assertions and printed the tables of Result 28.5; by the rule of the project the script is not kept (removed on 4 Nov 2026; Terra 6.1 had reproduced its matrix of Proposition 28.13 in exact arithmetic). The slice operator on each multiplicity space is Hermitian, odd under Z, and closes on Hom_H. The structure constants reproduce [t_1, t_2] = t_3 in Sp(1)_a and Sp(1)_b. The decomposition D_6(rho) = A_0 + A_1/rho + A_2 rho holds. The form uses three-point Gauss quadrature on each element. The exact tests are {A_0, A_1} = 0, A_0^2 = 0, 4 on the 4, 5, 9 on the 16, 0, 8 on the 20, 12, 16 on the 40 (the value 0 on the 20 has the twist (1, 5) and gives no core zero mode), A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha, and at rho = 1 with the line connection the values D^2 = 0, 5, 5, 12 on the summand O(3) of the 4, 16, 20, 40, against j(j + 4) for (0, 0)-forms in O(1); the whole summand also has the value 10 on the 20. The closed form of Proposition 28.13 is tested to ten digits, and the core mode density in the mesh.

Part 29

The eight-state window on the disc bundle

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T0, without an Astra job. It continues lemma 28 and answers the prompt 29_window_coercive_estimate.md. Refereed by Terra 6.1 (report): case (b) stands with amendments, and the two errors it found in case (a), the definition of A_eff and the sign of the fibre part, are corrected below. The referee ran no computation. A fresh reading of the corrected case (a) in the referee report on lemma 30 found one more sign and one missing step in Proposition 29.10; these are corrected, and the corrections have not been read again.

Outcome

1. **Conjecture 28.6 holds in case (b)** for the class of metrics and mass profiles of Setting 29.0: for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), for every delta < sqrt 3 (Theorem 29.7).
2. **Conjecture 28.6 holds in case (a) at the Fubini-Study radius** rho = 1, with fibre part -2/3 (Theorem 29.11). At another wall radius, case (a) follows from one kernel condition K(rho), which is a finite check at each radius.
3. **The off-diagonal block is exponentially small**, |lambda_M| <= C exp(-(1 - eta)(M/2) int f dr), by an exact tunnelling identity and Agmon estimates (Proposition 29.8).
4. **The compression converges** to (Q_B(a|*{S^4}), Q_Y(a|*{CP^3})) with error O(M^{-1/2}) (Theorem 29.9). The statement concerns compression maps, in line with Counterexample 28.17.

The proof is short because Sp(2)-equivariance gives exact sums of squares at the core and at the wall. Each estimate holds on all of L^2 of a fixed manifold, with the boundary operators elliptic on CP^3 or S^4, so uniformity over the representations of Sp(2) needs no separate argument. The spectral separation hypothesis

of Maridakis, which fails for this mass (historical/reviews/2026-11-03_terra_check_28.md, item 6), is never used. General forms of two identities are in the literature: the square with its zero-order cross term in the bulk (Maridakis, arXiv:1510.07004, Lemma 2.2), and a horizontal and vertical factorization with a curvature contraction (Kaad and van Suijlekom, arXiv:1710.03182, Proposition 18, under other hypotheses).

Scope. These are spectral statements for a chosen metric family, with flat fibres near the zero section. They give the finite window from the operator, for the sign of Proposition 28.13. They leave open: other metrics in the class of Theorem 28.9, including the Bryant-Salamon metric; the connector of Proposition 28.20; the quaternionic grading of Proposition 28.19; any dynamics selecting M, l or rho. Finite-mass generated algebras stay outside the statement (Counterexample 28.17).

Setting

Setting 29.0. Let X be the disc bundle of lemma 28, G = Sp(2), K = Sp(1)_a x Sp(1)_b, H = Sp(1)_a x U(1)_b, E = S^3(S), Phi_M = eps M f(r) n . J, and T_M = D_X^E + i Phi_M, with D-hat_M the doubled operator and the wall condition of Theorem 28.9: Z u = -u for T_M, and Z v = +v for T_M^*. Assume:

- (H1) g = dr^2 + pi^* g_{S^4} + rho(r)^2 g_{S^2}, with S^4 of radius 1, rho(r) = r on [0, 2e], and rho constant, equal to rho_l, on the collar [l - 2a, l]. Case (b) and Theorem 29.11 take rho_l = 1, the Fubini-Study metric.
- (H2) f(r) = r on [0, 2e], f = f_0 > 0 on the collar, and f > 0 on (0, l]. This is M b(|v|^2) v . J with b positive.
- (H3) The connection on E is the pull-back connection on r <= 2e. In case (b) it preserves the four line summands on the collar, with any invariant interpolation in between. In case (a) it is the pull-back connection throughout. We use radial parallel gauge: nabla_{d/dr} = d/dr. An invariant dr-component a(r) q could otherwise be added to a line-preserving connection; it is removed by radial parallel transport, which changes the displayed formulas by a phase on each line.
- (H4) eps = -gamma_b, where gamma_b is the chirality of Sigma_4 whose K-type is (1, 2). This is the sign of Proposition 28.13: its right singular vector is at the zero section. The opposite sign exchanges T_M and T_M^* throughout. In the convention of the radial computation of lemma 28, gamma_b = +1.

On the collar put q = eps n . J, with eigenvalues +-3/2, +-1/2 on the line summands, and m = M f_0. In general q = -gamma_b n . J. In the notation of Proposition 28.13 and of the radial computation of lemma 28, q = -tau, and T_M = i Z d/dx + A + i m q on the collar, with x increasing towards the wall. The line q = +3/2 is the summand O(3), and q > 0 are the stable channels of Proposition 28.13.

Let C_R = ||A_2||, the norm on Sigma_6 (x) E of the curvature contraction A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha; it does not depend on V, and C_R = 1 in Python. Let mu_1 > 0 be the smallest positive eigenvalue of Q^* Q for Q = sigma . d_y + y . J of Proposition 28.11. It is positive since the resolvent is compact, because (y . J)^2 >= |y|^2 / 4, and ker Q = 0. Python gives 1.02.

The zero section

Proposition 29.1 (exact splitting). Identify sections over the tube r <= 2e with K-equivariant maps s : G -> F, F = Sigma_4 (x) L^2(R^3; C^2 (x) C^4), with c(e_a) = gamma_a (x) 1 and c(f_alpha) = gamma_5 (x) sigma_alpha. Let D_H = i sum gamma_a e_a, with e_a acting as left-invariant derivatives on G, and Q^ = gamma_5 sigma . d_y + eps M y . J. Then on this tube T_M = T_0 + R, with T_0 = D_H + i Q^ and R Hermitian of order zero with |R| <= C_R |y|, and

```
T_0^* T_0 = D_H^2 + Q^^* Q^,    T_0 T_0^* = D_H^2 + Q^ Q^^*,
```

with D_H^2 commuting with Q^^* Q^ and Q^ Q^^*.

Proof. The fibres are flat and totally geodesic by (H1), so the Levi-Civita connection of X differs from the connection associated with the principal K-bundle G -> S^4 only by the O'Neill tensor. Its component is the curvature of N evaluated at v. This gives the zero-order term R, which equals r A_2 in the radial form

D_6 = A_0 + A_1/rho + A_2 rho, of norm at most C_R |y|. For the pull-back connection, the remaining operator is D_H + i gamma_5 sigma . d_y + i Phi_M. The operator Q^ acts on the values of s, and it is K-equivariant, since sigma . d_y and y . J are invariant under the simultaneous rotation of y, of C^2 and of C^4. So Q^ maps equivariant maps to equivariant maps, and it commutes with each derivative e_a, because it does not depend on the point of G. Since gamma_a anticommutes with gamma_5 and commutes with y . J, D_H Q^ = Q^~* D_H, where Q^~* = -gamma_5 sigma . d_y + eps M y . J. Therefore

```
T_0^* T_0 = (D_H - i Q^~*)(D_H + i Q^) = D_H^2 + Q^~* Q^ + i (D_H Q^ - Q^~* D_H) = D_H^2 + Q^~* Q^,
```

and likewise for T_0 T_0^*. Finally D_H Q^~* Q^ = Q^ Q^~* D_H, so D_H^2 commutes with both. ■

In radial form. T_0 = i Z d/dx + A_0 + A_1/rho + i phi q, and the cross terms vanish by {A_0, Z} = 0, [A_0, tau] = 0 and {A_0, A_1} = 0. these three identities were asserted on the representations 4, 16, 20, 40 (Python, script not kept).

Proposition 29.2 (the model spectrum). On the total space of N, with f = r and T_0 as above, let Pi be the projection onto ker T_0. Then:

1. ker T_0 is one copy of the 4 of G; ker T_0^* = 0.
2. ||T_0 u||^2 >= min(4, mu_1 M) ||u||^2 for u orthogonal to ker T_0, and ||T_0^* v||^2 >= min(4, mu_1 M) ||v||^2 for all v in the domain.

Proof. Write Q^_gamma for Q^ on chirality gamma. By Proposition 28.11, Q^_gamma has kernel exactly when gamma = -eps; this is the rank-four K-representation W_0 = (1, 3) + (1, 1), and on the orthogonal complement Q^~* Q^ >= mu_1 M, after the rescaling y -> M^{-1/2} y. The same bound holds on the other chirality, and for Q^ Q^~* with the chiralities exchanged, since Q^* Q and Q Q^* have the same nonzero spectrum. The fibrewise projection P_W onto W_0, on both chiralities of Sigma_4, commutes with D_H, by Proposition 29.1. (The projection onto the fibre kernel of Q^ also selects gamma = -eps, so it does not commute with D_H, which is odd; it commutes with D_H^2.) On chirality -eps, Q^ vanishes on the range of P_W and is bounded below by sqrt(mu_1 M) on its complement; on chirality +eps, it is bounded below by sqrt(mu_1 M) throughout. On Sigma^gamma (x) W_0, D_H is the Dirac operator of S^4 twisted by the homogeneous bundle of W_0, so by Parthasarathy's formula D_H^2 = Cas_G(V) + 3/2 - Cas_K(tau), with Cas_K = 2 j (j + 1) on Sp(1)_b, that is 0 and 4 on W_0. Only V odd under -1 occur, with Cas_G(V) >= 5/2 and Cas_G(V) >= 15/2 for V not the 4. So D_H^2 >= 4 except for D_H^2 = 0 at V = 4, twist j = 1. This kernel needs Hom_K(4, Sigma^gamma (x) (1, 3)) != 0, which holds exactly for Sigma^gamma = (1, 2), where (1, 2) (x) (1, 3) contains (1, 2) once. By (H4), gamma_b = -eps, so T_0 sees this chirality and T_0^* the other. Decomposing u by chirality and by P_W, and using the sum of squares, gives both bounds. The computation holds on compactly supported smooth sections, and extends by closure, since the doubled model is essentially self-adjoint on that core; no boundary condition enters at infinity. ■

Lemma 29.3 (core estimate). Let u be in H^1 with support in r < 2e. Then

```
|| |y| u || <= (2/M) ||T_0 u|| + sqrt(10/M) ||u||,
```

and the same holds with T_0^*. So ||T_M u|| >= (1 - 2 C_R/M) ||T_0 u|| - C_R sqrt(10/M) ||u||, and likewise for T_M^* with T_0^*. In particular, for such u with Pi u = 0, and for all such v,

```
||T_M u|| >= (2 - o(1)) ||u||,    ||T_M^* v|| >= (2 - o(1)) ||v||,    M -> infinity.
```

Proof. Q^~* Q^ = -Laplacian_y + M^2 (y . J)^2 - eps M gamma_5 sigma . J. The eigenvalues of sigma . J are 3/2 and -5/2, and (y . J)^2 >= |y|^2/4. Hence M^2 || |y| u ||^2 / 4 <= ||Q^ u||^2 + (5/2) M ||u||^2 <= ||T_0 u||^2 + (5/2) M ||u||^2, by Proposition 29.1. The rest is the triangle inequality with |R| <= C_R |y|, and Proposition 29.2. ■

The bulk

Lemma 29.4 (bulk estimate). Let w be in H^1 with support in a compact subset of 0 < r < l, and let f_1 be the minimum of f on that support. Then

```
||T_M w||^2, ||T_M^* w||^2 >= (M^2 f_1^2 / 4 - C_Phi M) ||w||^2,
```

with C_Phi the supremum of the operator norm of the Clifford contraction sum_i c(e_i) nabla_{e_i} (f n . J). With 0 < r_1 < r_2 < l and a radial cutoff psi equal to 1 on [r_1, r_2], with compact support in 0 < r < l, we have for u in Dom T_M,

```
||u||^2_{L^2(r_1 <= r <= r_2)} <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

Proof. ||(D + i Phi) w||^2 = ||D w||^2 + ||Phi w||^2 + <w, i [D, Phi] w>, with [D, Phi] of order zero and pointwise bounded independently of V, and Phi^2 >= M^2 f^2 / 4. For the second statement, T_M (psi u) = psi T_M u + [D, psi] u. ■

The interpolation of the connection in (H3) takes place in this region, where it enters only through C_Phi.

The wall

Lemma 29.5 (line-twisted operators at the Fubini-Study metric). Let D_k be the Dirac operator on CP^3 twisted by O(k) with its invariant connection, k odd, at rho = 1. Then D_k^2 = C_2(L) + 3/2 - (3/8) k^2 on the SU(4)-type L, with C_2 normalized by C_2(4) = 15/8. Consequently:

1. On O(3): ker D_3 is one copy of the 4 of Sp(2), in Z = +1; and D_3^2 >= 5 on the V-isotypic part for V not the 4.
2. On O(1): ker D_1 = 0; D_1^2 >= 3 on the 4 and D_1^2 >= 6 on the V-isotypic part for V not the 4.

So the smallest nonzero |eigenvalue| on O(3) + O(1), over all V, is kappa_w = sqrt 3, attained on the 4; the lowest wall value on the 16 is sqrt 5.

Proof. The form of D_k^2 is Parthasarathy's formula for SU(4)/U(3), since the twist is a character of U(3). The Sp(2)-invariant connection on a line summand is unique, because the isotropy representation has no trivial summand, so it is the Chern connection. The normalization: j(j + m + 3) is the eigenvalue of 2 box on (0,0)-forms in O(m) on the type [m + j, 0, j], with C_2([m + j, 0, j]) - C_2([m, 0, 0]) = j (j + m + 3), which fixes the coefficient of C_2; scalar curvature 12 gives the constant 3/2; and the term in k^2 is fixed by the kernel of D_3 on the 4. Two checks: D_0^2 = 4 on the type 6, the Kirchberg limit R/3 at R = 12 (Ann. Global Anal. Geom. 4 (1986) 291-325); and the values in the Python table below. The kernel statements are H^q(CP^3, O(1)) = C^4 at q = 0 only, and H^q(CP^3, O(-1)) = 0, with K^{1/2} = O(-2). The types of odd 4-ality with C_2 < 55/8 are the 4, its conjugate, the 20 and its conjugate (enumerated in Python). The 4 occurs on O(3) once, by the multiplicity on the slice, and this copy is the kernel; since each 20 restricts to Sp(2) as 4 + 16, neither 20 occurs on O(3). So V != 4 has C_2 >= 55/8 on O(3), giving D_3^2 >= 5. On O(1), the 4 and its conjugate restrict only to the 4 of Sp(2), so V != 4 has C_2 >= 39/8, giving D_1^2 >= 6; and the slice block on the 4 has D_1^2 = 3, 6. The chirality of ker D_3 is that of (0,0)-forms, and the slice block fixes it as Z = +1 for q = +3/2 (Python). ■

Lemma 29.6 (wall estimate in case (b)). Let w be in H^1 with support in the collar, vanishing near its inner end, and p(x) = sqrt(3 m) exp((3/2) m (x - l)), so that the kernel of T_M^* on the half-cylinder is p (x) ker D_3. Then:

1. If Z w = -w at r = l, then ||T_M w||^2 >= min(m^2/4, 3) ||w||^2.
2. If Z w = +w at r = l, and w is orthogonal to p (x) ker D_3, then ||T_M^* w||^2 >= min(m^2/4, 3) ||w||^2.

Proof. On the collar, A commutes with q and with d/dx, and anticommutes with Z. Writing w = w_+ + w_- by Z, integration by parts gives, for T_M,

```
||T_M w||^2 = ||(d + m q) w_+||^2 + ||A w_+||^2 + ||(d - m q) w_-||^2 + ||A w_-||^2,
```

where the cross terms are purely imaginary, using w_+(l) = 0. For T_M^* the same holds with q replaced by -q and with w_-(l) = 0. There are three one-dimensional cases on (-infinity, l]. (i) If g(l) = 0, then ||(d +- m q) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2. (ii) With g(l) free, ||(d + m |q|) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2 + m |q| |g(l)|^2. (iii) With g(l) free, ||(d - m |q|) g||^2 is the form of -d^2 + m^2 q^2 with a Robin condition at l, whose spectrum consists of 0, on the profile sqrt(2 m |q|) exp(m |q| (x - l)), and of [m^2 q^2, infinity). Cases (i) and (ii) give the lower bound m^2/4. Case (iii) arises exactly in the channels (Z, q) = (-,

q > 0) for T_M and (+, q > 0) for T_M^*. On those, expand in eigenvectors of A^2: the term ||A w||^2 is at least 3 ||w||^2, except on ker D_3 in Z = +1, which by Lemma 29.5 only arises for T_M^* at q = 3/2, where the Robin gap (9/4) m^2 applies. ■

The window

Theorem 29.7 (the window, case (b)). Under Setting 29.0 with rho_l = 1 and connection (b), let delta < sqrt 3. For M large, D-hat_M has exactly eight eigenvalues in (-delta, delta). The spectral subspace splits under Gamma-hat as W_M (+) V_M, each a copy of the 4 of Sp(2), with W_M within O(M^{-1/2}) of the cut-off core modes in the domain of T_M, and V_M within O(M^{-1/2}) of the cut-off wall modes p (x) ker D_3 in the domain of T_M^*.

Proof. We verify Lemma 28.14. Take radial functions chi_c, chi_b, chi_w with chi_c^2 + chi_b^2 + chi_w^2 = 1, chi_c = 1 on r <= e with support in r < 2e, chi_w = 1 on [l - a, l] with support in the collar. Let U_M span (chi_c u_i, 0) and (0, chi_w v_k), where u_i is a basis of ker T_0 and v_k of p (x) ker D_3, with Gram-Schmidt. Radial functions preserve the wall condition.

Trial states. T_M (chi_c u_i) = [D, chi_c] u_i + R chi_c u_i, of norm O(M^{-1/2}), since u_i has Gaussian decay at the scale M^{-1/2}; and T_M^* (chi_w v_k) = [D, chi_w] v_k, which is O(exp(-(3/2) m a)). So eps_M = O(M^{-1/2}).

Coercivity. Since [D, chi] = c(d chi) with c(d chi)^2 = |d chi|^2, and sum chi_i d chi_i = 0,

```
||T_M u||^2 = sum_i ||T_M chi_i u||^2 - sum_i || |d chi_i| u ||^2,
```

and likewise for T_M^*. The functions d chi_i vanish outside the bulk region e <= r <= l - a. So by Lemma 29.4,

```
(1 + C M^{-2}) ||T_M u||^2 >= ||T_M chi_c u||^2 + ||T_M chi_w u||^2 - C M^{-2} ||u||^2,    ||chi_b
↪  u||^2 <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

If u is orthogonal to every chi_c u_i, then Pi (chi_c u) = 0 exactly, since <u_i, chi_c u> = <chi_c u_i, u>, so Lemmas 29.3 and 29.6 (1) give ||T_M u||^2 >= (3 - o(1)) ||u||^2. If v is orthogonal to every chi_w v_k, then chi_w v is orthogonal to p (x) ker D_3, so Lemmas 29.3 and 29.6 (2) give the same for T_M^*. Since ||D-hat (u, v)||^2 = ||T_M u||^2 + ||T_M^* v||^2, Lemma 28.14 gives the count and the distance to the range. The window is G-invariant and close to the range of U_M, so as representations W_M and V_M are copies of the 4. ■

Remark. The gap is set by the wall at sqrt 3, with the core next at 2. On the 16, 20 and 40 the lowest values are sqrt 5, sqrt 5, sqrt 12.

Proposition 29.8 (tunnelling). Let T_M u = lambda v and T_M^* v = lambda u be a singular pair of the window, with unit vectors. By Proposition 28.19, the four singular values coincide. For a radial chi equal to 1 near r = 0 and 0 near r = l,

```
lambda (<u, chi u> - <v, chi v>) = <v, c(d chi) u>.
```

For every eta in (0, 1) there are M_eta and C_eta with

```
|lambda_M| <= C_eta exp(-(1 - eta)(M/2) int_e^{l - a} f dr),    M >= M_eta.
```

Proof. The function chi u lies in Dom T_M, since chi vanishes near the wall. So <T_M^* v, chi u> = <v, T_M chi u>, and T_M chi u = chi lambda v + c(d chi) u. This gives the identity.

For the decay of u, let xi = psi exp(g) be radial, with psi = 0 on r <= e/2 and psi = 1 on r >= e, with g = 0 on r <= e, g' = (1 - eta)(M/2) f on [e, r_*], and g constant on r >= r_*. Since xi^2 u lies in Dom T_M,

```
||T_M (xi u)||^2 = lambda^2 ||xi u||^2 + || |d xi| u ||^2.
```

As in the proof of Theorem 29.7, with no core piece, the bulk bound of Lemma 29.4 and Lemma 29.6 (1), or Proposition 29.10 in case (a), bound the left side from below by the integral of (M^2 f^2/4 - C M) |xi u|^2 on the bulk, plus (3 - o(1)) ||xi u||^2 on the collar, minus C M^{-2} ||xi u||^2. On the bulk the excess over (g')^2 is at least (eta/4)(M^2 f^2 / 4) for M large, with f >= f_1 > 0, so ||exp(g) u||*{L^2(r >= e)}* <=

C_eta. For v we use the same argument from the wall inward, with g-tilde = 0 on [l - a, l], with -g-tilde' = (1 - eta)(M/2) f, and with Lemma 29.3 replacing Lemma 29.6 on the core, since T_M^* has no low states there. Take chi with |d chi| <= 2/s, supported on [r_, r_ + s] in the bulk. By Theorem 29.7, <u, chi u> -> 1 and <v, chi v> -> 0, so the identity gives |lambda| <= (4/s) ||u||{[r_, r_ + s]} ||v||{[r_, r_ + s]}. Run the weights with eta' = eta/2, and take s so small that (1 - eta')(I - int{r_}^{r_ + s} f) >= (1 - eta) I, with I = int_e^{l - a} f. This gives the stated exponent. The absorption uses lambda -> 0, supplied by the trial states. The factor 1/2 holds on every channel, since (n . J)^2 >= 1/4. ■

For a family with l -> infinity, M >= M_eta, with uniform bounds on the end geometry, the cutoffs, the connection and the positive bulk mass, the constants are uniform, so lambda is exponentially small in l at the rate (1 - eta)(M/2) inf f. This is the uniform decay rate required by Conjecture 28.6, as amended by the referee.

Theorem 29.9 (compression). Under Theorem 29.7 or Theorem 29.11, for a in C^1(X),

```
|| Q_M(a) - (Q_B(a|_{S^4}) (+) Q_Y(a|_{CP^3})) || <= C ||a||_{C^1} M^{-1/2},
```

where Q_B(a) = int a(x) 2 P_x d mu(x), with P_x = (1 - x . Gamma)/2, and Q_Y(a) = int a(z) B(z) d mu(z). In case (b), B(z) = 4 p_z, the level-one map of Example 16.9. In case (a) at radius rho, B(z) = 2((1 + w) p_z + (1 - w) p_{jz}), with p_{jz} the quaternionic partner of p_z and w = w(rho); so Q_Y(x_a) = -Gamma_a/5, Q_Y(y_T) is w(rho) times the level-one value, and the signed fibre part is -w(rho). Restriction to the two ends is surjective (Proposition 28.8), so real functions on X generate M_2(H) + M_4(C) under the limit map, by lemma 10 and Example 16.9, for every w(rho) > 0.

Proof. Theorem 28.15, with eta_M = O(M^{-1/2}) from Theorems 29.7 and 29.11, after choosing equivariant unitary identifications of the window spaces with the model copies of the 4. The normal first moments are O(M^{-1/2}) at the core, by the Gaussian decay of the model, and O(M^{-1}) at the wall, by the profiles. At the core, F in Hom_K(4, (1, 2) (x) (1, 3)) kills (2, 1), so B = 2 P_x in the normalization Q_B(1) = I. The sign in P_x is the value tr(B Gamma_5)/tr B = -1 of Proposition 28.13. At the wall, the pointwise inner product of different line summands vanishes, so the density is a sum over channels with weights from Proposition 28.13. ■

Case (a)

On the collar in case (a), A = A_d + A_o, where A_d = D_line preserves the lines and A_o = A_E/rho has order zero, with [A_o, q] != 0. In Python, the block of A_E between q = 3/2 and q = 1/2 has norm sqrt 3 and ||A_E|| = 2, on the full fibre, hence for all V. Let P be the projection onto {sum_{q > 0} p_q g_q : Z g_q = -g_q}, with p_q = sqrt(2 m q) exp(m q (x - l)), and Y the same with Z = +1, with Pi_Y the projection onto the second space. Let F = Pi_Y (A_d + A_o) P, from the range of P to Y, and let A_eff be the operator on the sum of both ranges with blocks F and F^*. Identified with sections over CP^3 of the lines q > 0, it is, by Lemma 28.12,

```
A_eff = D_3(rho) (+) D_1(rho) + (sqrt 3 / 2) B(rho),
```

with B(rho) the block of A_o between the stable channels, of norm sqrt 3 / rho. This is self-adjoint and elliptic on the fixed manifold CP^3, so its nonzero spectrum has a gap kappa_a(rho) > 0, uniformly in V. On the 4, F^* is the matrix A(rho) of Proposition 28.13, from Y to the range of P.

Condition K(rho). The kernel of A_eff(rho) is exactly one copy of the 4, in Z = +1, so F is injective and ker F^* is that copy.

Proposition 29.10 (wall estimate in case (a)). Assume K(rho). For w as in Lemma 29.6, with the kernel p (x) ker D_3 replaced by the kernel of the half-cylinder operator T_M^*, the two statements of Lemma 29.6 hold with the constant 3 replaced by kappa_a(rho)^2 - O(1/m). The half-cylinder kernel is exactly one copy of the 4 for m large, with fibre part tending to -w(rho) at rate O(1/m).

Proof. T_d = i Z d/dx + A_d + i m q preserves the block structure: T_d P = Pi_Y T_d P, since T_d (p_q g) = p_q A_d g. Also T_d maps the complement of P to the complement of Y, because for y in Y the function T_d^* y = A_d y lies in P. On the complement of P, ||T_d w_2|| >= (m/2) ||w_2||, by the

sum of squares of Lemma 29.6, where the channels with Robin kernel are removed. So with w = w_1 + w_2 and w_1 = P w,

```
||T_M w|| >= ||(1 - Pi_Y) T_M w|| >= (m/2) ||w_2|| - ||A_o|| ||w||,
||T_M w|| >= ||Pi_Y T_M w|| >= ||F w_1|| - ||A_o|| ||w_2||.
```

Eliminating ||w_2||, one gets (1 + 2 ||A_o|| / m) ||T_M w|| >= ||F w_1|| - (2 ||A_o||^2 / m) ||w||. Under K(rho), F is bounded below by kappa_a(rho), and F^* is bounded below on the complement of its kernel. The same elimination holds for T_M^*, with P and Y and F and F^* exchanged. On the half-cylinder T_M is thus injective with closed range, so T_M^* is surjective; its kernel is the kernel of the Schur complement of the block T_d + (1 - Pi_Y) A_o (1 - P), whose inverse has norm at most (m/2 - ||A_o||)^-1. So this Schur complement differs from F^* by at most ||A_o||^2 / (m/2 - ||A_o||), and the graph of the Schur solution gives ||P_{ker T_M^*} - P_{ker F^*}|| = O(1/m). This projection estimate turns orthogonality to the exact half-cylinder kernel into orthogonality to ker F^* up to O(1/m), which gives the adjoint bound. Off the 4, this is invertible. On the 4 the block is a 2 x 3 matrix of rank 2, so its kernel has dimension exactly 1, within O(1/m) of (-(2/3)(rho^2 + 2), 1, 1). ■

Theorem 29.11 (the window in case (a)). Under Setting 29.0 with the pull-back connection, K(1) holds, and for delta < min(2, kappa_a(1)) and M large the conclusion of Theorem 29.7 holds at the Fubini-Study radius, with the wall modes replaced by the kernel of Proposition 29.10, and with fibre part tending to -w(1) = -2/3. Here kappa_a(1) >= sqrt 5 - 3/2 off the 4, and it equals 1.863 on the 4, the squared values there being (45 +- 3 sqrt 33)/8. At another wall radius the same conclusion holds under K(rho).

Proof. By Lemma 29.5, on V != 4 we have |D_3 (+) D_1| >= sqrt 5, while ||(sqrt 3/2) B(1)|| = 3/2. So A_eff is invertible there, with lower bound sqrt 5 - 3/2 > 0. On the 4, F^* is A(1), with kernel of dimension one in Z = +1, the vector (-2, 1, 1), and nonzero singular values 1.863 and 2.789, by Proposition 28.13. So K(1) holds. The proof of Theorem 29.7 then applies with Proposition 29.10 in place of Lemma 29.6. The trial wall states are cut-offs of the exact half-cylinder kernel, so their defect is exponentially small. ■

Other radii. For K(rho), only finitely many V can carry a kernel, those with |D_3(rho) (+) D_1(rho)| <= 3/(2 rho), by ellipticity. In Python, on the 16, 20, 40 and for rho in [0.2, 4], the compressed block has smallest singular value at least 1.30. A check of K(rho) on all relevant V at a given radius needs a lower bound for D_line(rho) on all V, which we have not written down away from rho = 1.

Remarks

Remark 29.12 (the index). The summands of E on CP^3 have indices 4, 0, 0, -4, of total zero, as equivariant cobordism invariance requires: E extends over X (Witten 1983, preprint pp. 43-44). The window has one copy of the 4 in each Gamma-hat degree, so its Sp(2)-index is 4 - 4 = 0. So index theory allows lambda != 0, and the eight states come from the estimates above. The sign in (H4) is consistent with this: with the opposite sign, the end-model modes at the core belong to T_M^* and those on O(-3) at the wall to T_M (Proposition 28.13), so the two ends never lie in the same Gamma-hat degree. At finite mass the full operator can have nonzero lambda, so these states need not lie in exact kernels. The relative sign of the two ends enters through the chirality of ker D_3. It follows from the slice weights, which contain the 4 once in Z = +1 and not in Z = -1, and the sign of the core is the chirality statement behind (H4). If the O(3) zero mode lay in Z = -1 with q > 0, it would be a wall zero mode of T_M, in conflict with Lemma 29.6 (1). As a check in Python, the mean position of the modes is 0.51 at the core for T_M and 2.96 at the wall for the adjoint problem, with l = 3.

Remark 29.13 (physical units). All statements are in shape units. Under the dilation g -> R^2 g, with D -> D/R and the potential divided by R, the operator is unitarily equivalent to D-hat_M / R. So the light mass is lambda_M / R, the gap delta / R, and their ratio is unchanged, while the compression maps stay the same; this observation is from the audit of 3 November 2026 by Astra 6, section 5.2, conditional there on the estimates proved here. A family with R -> infinity, with lambda_M / delta small, closes the physical gap and keeps the eight states separated. Whether a dynamics selects such a family is open. The limiting geometry of such a family is a separate question: nothing here identifies it with a complete Bryant-Salamon manifold.

Remark 29.14 (other metrics). In the class of Theorem 28.9, T_M - T_0 also contains first-order terms with coefficients O(r^2), and M (b(r^2) - b(0)) y . J. One route shrinks the core piece to r < M^{-1/2 + s}, where Lemma 29.4 controls the complement since M^2 r^2 dominates. We have not carried this out, and the horizontal first-order terms need a bound in terms of D_H that takes account of the unbounded curvature terms.

Numerical check

Python (numpy, scipy), scripts not kept by the rule of the project. The exact inputs were asserted: [A, q] = 0 and {A, Z} = 0 at rho = 1 with the line-preserving connection, the values of Lemma 29.5 against the formula, the norms used above, and the identities of Proposition 29.1. The direct solve is a test of the prediction: the limit values below were fixed by Proposition 29.2 and Lemma 29.5 before the run, and a failure would have refuted the argument. Setting: rho(x) = x for x <= 0.4, rho(x) = 1 for x >= 1.2, phi = M rho, l = 3, with the connection switched on [0.6, 1.2], and P1 elements at 2500 cells. Lowest singular values of T_M:

V	M = 8	M = 16	M = 32	predicted limit
4	small, 1.7321, 1.9861, 2.4495	small, 1.7321, 1.9963, 2.4495	small, 1.7323, 1.9991, 2.4497	0, sqrt 3, 2, sqrt 6
16	2.2245, 2.2361, 2.4495	2.2324, 2.2367, 2.4495	2.2351, 2.2409, 2.4497	sqrt 5 (core), sqrt 5 (wall), sqrt 6
20	2.2361, 2.8148, 2.8284	2.2367, 2.8248, 2.8284	2.2409, 2.8275, 2.8286	sqrt 5 (wall), sqrt 8 (core), sqrt 8 (wall)
40	3.4461, 3.4641, 3.4641	3.4585, 3.4641, 3.4645	3.4626, 3.4642, 3.4672	sqrt 12 three times

The wall values have a mesh error growing with M, of size 0.005 at M = 32 on the 16 and 20. The small value on the 4 is uninformative in this table: it halves when the mesh doubles, so it is a mesh floor, and Proposition 29.8 is proved without it. The wall mode of the adjoint problem has fibre part -1.00000 and Gamma_5 part -1.00000 at M = 8, 16, 32, as Theorem 29.9 says in case (b).

Part 30

The massless limit as a joint limit: a finite window and an opening tower

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T14. Theorem 30.1, Corollary 30.2, Proposition 30.3, Corollary 30.4 and Proposition 30.6 were proposed in an audit of commit b2ddf98 by Astra 6 (assessment in the review); the proofs below were redone against the source lemmas. Theorem 30.5 and Proposition 30.7 are new here and rest on lemma 29, whose referee is queued. Refereed by Terra 6.1, together with the note (report); its corrections are applied and have not been read again, since the Codex quota is spent.

Outcome

The claim that W and Z become massless as a dimension opens needs a family. Along a general path to M_W = M_Z = 0 nothing forces a tower to open: lemma 23 has theta -> 0 at fixed l, with the tower spacing fixed. The families below have the hierarchy

```
m_weak << E_eps << Delta_KK -> 0,
```

so a moving low-energy window keeps a fixed finite number of states. The hierarchy alone does not make the count at fixed energy diverge: the signed spectrum 0, +-eps^alpha, +-1, +-2, … has the hierarchy with a count bounded at fixed energy below 1. So the divergence at fixed positive energy is a separate property, proved below for each family.

1. On the exact interval this holds with an internal dimension of one (Theorem 30.1), also with localized end modes (Proposition 30.3).
2. On the cylinder of lemma 24 and on the disc bundle of lemma 29 it holds with seven internal dimensions. On the disc bundle the window comes from the operator, with one copy of the 4 at each end, and the count at fixed energy has seven-dimensional Weyl growth (Theorem 30.5).
3. On the disc bundle the core block of the window is quaternionic up to a defect that is O(M^{-1/2}) in general, and exponentially small when the connection preserves the quaternionic structure of E, for all bounded real observables (Propositions 30.6, 30.7).

Scope. These are statements about operators, compression maps and spectra along families given by hand. No dynamics selects them (Remark 30.8). The vector masses of Corollary 30.2 belong to the five-dimensional Yang-Mills toy of lemma 23. On the disc bundle only the fermion window is treated, so the relation between lambda and M_W is unproved there.

The interval

Theorem 30.1 (window and tower on the interval). Let (H_F, D_, *J_F, Gamma_F) satisfy the hypotheses of Theorem 22.1, with the algebra A_F represented by pi as in Theorem 22.4, r = dim H_F, and for 0 < alpha < 1, l_* > 0 and small eps > 0 let D_F(eps) = eps D_, *l_eps = l_* eps^{-alpha}, with D_eps the interval operator D_X of Theorem 22.1 for these data. Let Delta_eps be the smallest element of spec |D_eps| outside spec |eps D_*|. Then for eps small:

1. W_eps^* D_eps W_eps = eps D_*, with J_F and Gamma_F exact.
2. Delta_eps = (pi / l_) *eps^alpha - eps //D_||.*
3. ||Q_eps(a) - pi(a)|| <= (2/3) l_^2 *//D_||^2* eps^{2(1 - alpha)} ||a||.
4. For alpha < beta < 1, E_* > 0 and E_eps = E_* eps^beta, the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals r.
5. For fixed E > 0, N_eps(E) ~ (2 r l_* E / pi) eps^{-alpha}.

Proof. The condition l_eps ||D_F(eps)|| = l_* ||D_// *eps^{1 - alpha} < pi/2 holds for small eps. Theorem 22.1 gives item 1, and the next level is (pi - theta_max)/l = pi/l - //D_F||, which is item 2. Theorem 22.4 (2) gives item 3. For item 4, eps //D_||* « eps^beta « eps^alpha. For item 5, the signed spectrum is (n pi +- theta_k)/l_eps, which is r arithmetic progressions of spacing pi / l_eps, counted with multiplicity also when singular values coincide or vanish, each with (2 E l_eps / pi)(1 + o(1)) points in [-E, E]. ■

Theorem 22.1 starts from supplied finite data and a matrix-valued parent algebra, and the wall data change with eps by hand. A fixed Majorana block does not scale as eps D_*, so applying the theorem to an electroweak Dirac block needs a reducing sector or a treatment of the heavy neutral sector.

Corollary 30.2 (vector poles in the toy of lemma 23). For the universal channel of lemma 23, with m_eps = eps m_* and theta_eps = l_eps m_eps -> 0,

```
M_W = arcsin(sin(2 theta_eps)/sqrt 2) / l_eps,    M_Z = arcsin(2 sin(2 theta_eps)/sqrt 5) / l_eps,
```

so M_W^2 = 2 m_eps^2 (1 + O(theta_eps^2)) and M_Z^2 = (16/5) m_eps^2 (1 + O(theta_eps^2)). The vector poles follow the hierarchy. The constants 2 and 16/5 belong to this five-dimensional action. The small-mass pole coefficients of the fully massive finite spectral action are 3 and 24/5 (lemma 18, Proposition 14, and lemma 23), for the family with both members of every doublet at the same Dirac mass and no Majorana mass. Lemma 25 connects the two pairs at the level of the leading coefficient: its M^2 = ((3 - d_eff(s))/2) ||[D_F, T]||^2 / ||T||^2, with s = l Lambda, is the leading Taylor quotient of the heat-cutoff action for a generator constant in the transported frame, so the coefficient is 3 and 24/5 at d_eff = 0, and 2 and 16/5 at d_eff = 1. This does not identify the spectral-action pole problem with the Yang-Mills toy, and a joint-limit use needs a remainder uniform in s, since lemma 25 expands at fixed s. Along Lambda = E_eps we have s = l_* E_* eps^{beta - alpha} -> 0, the finite value, and at fixed Lambda > 0 we have s -> infinity, the interval value. Varying the cutoff of the action is an extra prescription: an observational cutoff alone does not change the action.

Proposition 30.3 (a localized family). Take M = -mu and the mixed conditions of lemma 9, with mu_*

> 0, 0 < alpha < 1, mu_eps = mu_* eps^alpha and the length l_eps^loc = c_eps / mu_eps, where c_eps = (1 - alpha) log(1/eps) + c_0. This length differs from the l_eps of Theorem 30.1. Let s = kappa l be the root of s = c tanh s, and m = mu sech s. Then

```
m_eps ~ 2 mu_* exp(-c_0) eps,    Delta_eps ~ mu_* eps^alpha,    (1/kappa) / l_eps = 1/s -> 0.
```

With the left profile L(x) proportional to exp(-kappa x) - r exp(kappa x), r = exp(-2 kappa l), the following identities are exact:

```
r = (m / (mu + kappa))^2,    mu - kappa = m^2 / (mu + kappa).
```

Proof. Lemma 9.5 gives m = 2 mu exp(-c)(1 + o(1)) and kappa/mu -> 1. The first oscillatory root has pi < k_1 l < 3 pi/2 and lambda_1^2 = mu^2 + k_1^2, so Delta/mu -> 1. From mu = kappa coth s and m = kappa / sinh s we get mu + kappa = kappa exp(s) / sinh s, hence m / (mu + kappa) = exp(-s). The second identity is m^2 = mu^2 - kappa^2. ■

The localization width and the length both grow, with a shrinking ratio. The finite two-point distance 1/m and the length l are both divergent, with (1/m)/l ~ exp(c)/(2c), so a metric convergence statement would need its own comparison maps.

Corollary 30.4 (the cylinder of lemma 24). Use the operator D_r = E d_x + Z A / r on the interval times CP^3, with A as in lemma 24, finite data on V = ker A = C^4 with gradings of type (2, 2) as in lemma 24, so that r = 4, the scaling of Theorem 30.1, and r_eps = r_* eps^{-gamma}, 0 < gamma < 1. Other values of r need an enlarged tangential bundle, which we do not construct. Then for max(alpha, gamma) < beta < 1, the cutoff E_* eps^beta isolates the finite window, since the next level is at least min((Delta_K / r_) *eps^gamma, (pi / l_)* eps^alpha - eps ||D_*||), by lemma 24, (5). At fixed E > 0 the count is proportional to l_eps r_eps^6 E^7, the seven-dimensional Weyl growth, by Dirichlet and Neumann bracketing of the squared product operator as l_eps and r_eps grow. For gamma = alpha the geometry expands homothetically in all seven directions.

The window here is prescribed by the spectral walls of lemma 24.

The disc bundle

Theorem 30.5 (joint limit on the disc bundle). Fix a shape as in Setting 29.0, with connection (b), or with connection (a) at rho = 1. Let delta > 0 be as in Theorem 29.7 or 29.11, G_0 = int_e^{l - a} f dr, 0 < alpha < beta < 1, E_* > 0, eta in (0, 1), and

```
M_eps = (2 (1 - alpha) / ((1 - eta) G_0)) log(1/eps),    R_eps = eps^{-alpha},    D_eps =
↪  D-hat_{M_eps} / R_eps.
```

By the dilation of Remark 29.13, which is exact under the canonical spinor identification and the L^2 unitary factor R^{-7/2}, D_eps is the operator of the metric R_eps^2 g with potential (M_eps / R_eps) f(s / R_eps) n . J in the physical radial coordinate s = R_eps r, of size M_eps / R_eps -> 0. For eps small:

1. D_eps has exactly eight eigenvalues in (-delta eps^alpha, delta eps^alpha), and they are +-m_eps with multiplicity 4 (or zero with multiplicity 8), where m_eps <= C eps. The next level is at most C log(1/eps) eps^alpha, so it tends to zero.
2. For E_eps = E_* eps^beta the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals 8.
3. The compression converges to the endpoint maps, with error C ||a||_{C^1} (log 1/eps)^{-1/2} in shape coordinates.
4. For fixed E > 0, N_eps(E) ~ C_X E^7 eps^{-7 alpha}, with C_X = 64 omega_7 Vol_g(X) / (2 pi)^7 the Weyl constant of the elliptic operator D-hat_0 on X, where omega_7 is the volume of the unit ball and 64 = 8 x 4 x 2 counts the spinors, E and the doubling.

Proof. Item 1: Theorem 29.7 or 29.11 applies since M_eps -> infinity, Proposition 28.19 gives the multiplicity, and Proposition 29.8 gives lambda <= C exp(-(1 - eta)(M_eps/2) G_0) = C eps^{1 - alpha}, hence m_eps = lambda / R_eps <= C eps. The bound on the next level: the shape-unit value is at most s_9(D-hat_0) + c M_eps, with the comparison below. Item 2: eps « eps^beta « eps^alpha. Item 3: Theorem 29.9, with M_eps^{-1/2} of the stated order. Item 4: D-hat_M = D-hat_0 + M V on the same domain, with V

bounded by c = (3/2) sup f, so by min-max N(D-hat_0, Lambda - c M) <= N(D-hat_M, Lambda) <= N(D-hat_0, Lambda + c M). Take Lambda = E R_eps, so that M_eps / Lambda -> 0, and use the Weyl law for the fixed self-adjoint elliptic operator D-hat_0. ■

So the eight finite states stay separated at the moving scale, and at fixed physical energy the spectrum has the growth of a seven-dimensional internal space, for eleven dimensions in M^4 x X. Item 1 is an upper bound on m_eps, since a lower bound on lambda is unproved. A family with l_eps growing at fixed M has the same form, with the rate of Proposition 29.8. The constants M_eta, C_eta of Proposition 29.8 are fixed once the shape and eta are fixed, so no uniformity in a varying shape is used. The limit at fixed energy resolves a neighbourhood whose shape is fixed and whose size grows, so the pointed limit is flat at a fixed interior point and a half-space at a fixed boundary point. Nothing here identifies it with a complete Bryant-Salamon manifold, or with a solution of field equations.

The quaternionic defect

For the 4-copy of the window at the core, let B(r) = F(r)^* F(r) = diag(b_a I_2, b_+, b_-) as in Proposition 28.16, let j be the quaternionic antiunitary commuting with Sp(2), and P_H(A) = (A + j A j^-1)/2. Put

```
eta_t = int |b_+(r) - b_-(r)| dr / int tr B(r) dr.
```

Proposition 30.6 (all bounded real observables). For the normalized compression Q of this mode, ||Q(f) - P_H Q(f)|| <= 2 eta_t for every real f with ||f||_infinity <= 1. If X_1, …, X_k are such compressions, then ||X_1 … X_k - P_H(X_1 … X_k)|| <= 2 k eta_t.

Proof. With angular Haar measure of total mass 1, Schur's lemma gives the angular average (tr B / 4) I, so Q(f) = (4 / int tr B) int int f(r, g) lambda(g)^-1 B(r) lambda(g) dg dr. Since j commutes with lambda(g), Q(f) - P_H Q(f) is the same integral with B - P_H B. Here j exchanges b_+ and b_- and fixes the block b_a, so ||B - P_H B|| = |b_+ - b_-|/2. The triangle inequality gives the first bound. For the second, write X-tilde = j X j^-1, so that X - P_H X = (X - X-tilde)/2 and ||X_i - X-tilde_i|| <= 4 eta_t. Telescoping, X_1 … X_k - X-tilde_1 … X-tilde_k is a sum of k terms, each a product of contractions with one factor X_i - X-tilde_i, so its norm is at most 4 k eta_t; dividing by 2 gives 2 k eta_t. ■

So bounded expressions become quaternionic when eta_t -> 0. An unrestricted generated algebra may use coefficients of size 1/eta_t and retain the other directions, as in Counterexample 28.17.

Proposition 30.7 (the size of the defect). Under Theorem 29.7 or 29.11, eta_t = O(M^{-1/2}). Assume moreover that the quaternionic structure j_E of E is parallel on all of X. This holds for the pull-back connection and for the interpolation (1 - s(r)) nabla^pull + s(r) nabla^line with real s, and it can fail for other invariant interpolations allowed by (H3). Then for every eta in (0, 1) there are M_eta and C_eta with

```
eta_t <= C_eta exp(-(1 - eta) M int_{2e}^{l - a} f dr),    M >= M_eta.
```

This is the exponent of the bound on lambda in Proposition 29.8 doubled, with the integral starting at 2e. A bound in terms of lambda itself would need a lower bound on lambda, which is open.

Proof. For the first bound, fix a normalized equivariant model F_0 at the core with C F_0 = F_0 for the local real structure below, so that j B_0 j^-1 = B_0. Theorem 29.7 gives ||u - chi_c F_0|| = O(M^{-1/2}) in L^2, and |b_+ - b_-| <= (|F| + |C F|) |F - C F| gives the bound, by the Cauchy-Schwarz inequality.

For the second, j_E is parallel and j_E (n . J) j_E^-1 = -n . J, so C = C_7 (x) j_E commutes with D_X^E and with i Phi_M, hence with T_M as a differential expression, with C^2 = -1, and it exchanges the wall conditions (Proposition 28.18). For the line-preserving connection, j_E P_q j_E^-1 = P_{-q} gives j_E nabla^line j_E^-1 = nabla^line, and real affine interpolation keeps this. On Hom_G(4, -) put (C F) = C F j^-1, which is antilinear with square +1. It commutes with D_H, with the full fibre term i Q^ and with R separately, since each part of the Clifford structure transforms in the same way, so it acts on the one-dimensional space Hom_G(4, ker T_0) as a real structure. Fix F_0 there with C F_0 = F_0, so that B_0 is j-invariant. Let u = F be the mode with T_M u = lambda v, T_M^* v = lambda u, with the phase

for which <chi_c F_0, u> is real, with chi_c as in Theorem 29.7. Put zeta = u - C u and nu = v - C v. In the interior,

```
T_M zeta = lambda nu,    Pi (chi_c zeta) = 0,
```

where the second holds since <chi_c F_0, C u> is the conjugate of <chi_c F_0, u>. Take the weights of the proof of Proposition 29.8 starting at r = 2e and meeting at r_* = l - a: g = 0 on r <= 2e, g' = (1 - eta)(M/2) f on [2e, l - a], g constant beyond; and g-tilde = G - g on r <= l - a, g-tilde = 0 beyond, with G = g(l - a). So g-tilde is constant on r <= 2e. That proof gives ||exp(g) u|| <= C_eta on X, and ||exp(g-tilde) v|| <= C_eta on r <= l - a/2, and lambda <= C_eta exp(-G). The same bounds hold for C u and C v, since C is pointwise isometric.

Let xi = psi-tilde exp(g-tilde), with psi-tilde = 1 on r <= l - a and 0 on r >= l - a/2. Then ||T_M (xi zeta)|| <= lambda ||xi nu|| + || |d xi| zeta ||. Since xi is constant on the support of chi_c, Pi (chi_c xi zeta) = 0, so Lemmas 29.3 and 29.4 and the partition of Theorem 29.7 bound ||T_M (xi zeta)|| from below by the bulk term with (M/2) f and by (2 - o(1)) on the core. On the bulk |d xi| = (1 - eta)(M/2) f xi, which is absorbed with the factor eta to spare, leaving

```
||xi zeta|| <= C (lambda + ||u||_{L^2(l - a <= r <= l - a/2)}) <= C exp(-G).
```

Finally B - j B j^-1 = F^* F - (C F)^* (C F), so |b_+ - b_-| <= (|F| + |C F|) |zeta| pointwise in r. On r <= l - a, the Cauchy-Schwarz inequality with the weight exp(g) gives

```
int (|F| + |C F|) |zeta| <= 2 ||exp(g) u|| ||exp(-g) zeta|| = 2 ||exp(g) u|| exp(-G) ||xi zeta|| <=
↪  C exp(-2 G),
```

since exp(-g) = exp(g-tilde - G). On r >= l - a, both |F|^2 and |C F|^2 integrate to at most C exp(-2 G). ■

The samples in section 6.1 of the audit (Python, in the pull-back connection, rows reproduced by Claude Sonnet 5.5) have eta_t / lambda^2 between 0.5 and 0.9. They vary the length at fixed mass, with profiles outside Setting 29.0, so they are a qualitative comparison and do not test this statement as M -> infinity. Under the added hypothesis the defect comes from the wall, which breaks C: for an operator with the same interior and no wall, on a complete manifold, Hom_G(4, ker T) carries a real structure, and if it is one-dimensional the core block is exactly quaternionic.

Remark 30.8 (what is missing, and the direction). The next steps are, in order:

1. The vector sector on the family of Theorem 30.5: the kinetic matrix and mass Hessian of the vector fluctuations, H_eps u = M^2 K_eps u, and then the full momentum-dependent kernel. The aim is M_W, M_Z ~ c lambda / R with finite nonzero couplings. The tunnelling eigenvalue alone does not establish this.
2. The gauge normalization. For a bulk Yang-Mills field with fixed microscopic coupling and constant profile, g_4^-2 = Vol(X) / g_11^2, so plain dilation gives g_4 proportional to R^{-7/2}, which must be separated from restoration at finite coupling. Gauge fields from isometries have Killing overlap integrals instead.
3. The action in the radius and the electroweak order parameter. A toy V = a r^-p + b eps^2 r^q with a, b, p, q > 0 has its unique minimum at r proportional to eps^{-2/(p+q)}, so alpha = 2/(p+q) lies in (0, 1) if p + q > 2. The repository has not derived such a potential, and adding it by hand would derive nothing.
4. The connector of Proposition 28.20, and the origin of the quaternionic grading of Proposition 28.19.

Numerical check

No statement here needs a computation. The audit tabulates the localized family in Python (mu_* = 1, alpha = 1/2, c_0 = 2), and Claude Sonnet 5.5 recomputed the rows; for eps = 10^-2, c = 4.3026 and m = 2.710 x 10^-3, against the asymptotic 2.707 x 10^-3.

Part 31

The interval of H as an eleventh dimension

Lemma 31. Written as file 01 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line of research (README). Propositions 1.1, 1.4 and 1.5 are proved here; Proposal 1.3 is a dictionary to be tested. The joint-limit family of the section of that name is from the Astra 6 audit of 3 Nov 2026 (historical/reviews/2026-11-03_astra_audit_d11.md, §3 and §4), restated here in circle language and checked. Attempted by Claude (Anthropic), 3 Nov 2026; checks in Python, checks_11d.py, sections F, F2 and H. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (historical/reviews/2026-11-04_terra_check_11d.md, historical/reviews/2026-11-04_sonnet_second_check_11d.md).

Referee note (GPT-6.1 Terra, 4 Nov 2026). Proposition 1.1 and the joint family's Weyl constant are independently checked; the unfolded circle is scalar, and even extensions of arbitrary smooth interval functions need not be smooth at the fixed points. See the referee report. Proposition 1.5 is amended at its statement below.

Referee note on Proposition 1.4 (GPT-6.1 Terra). The endpoint value, minimum and small-angle coefficient are independently checked under the stated one-loop hypotheses; see the report.

Correction (3 Nov 2026). There was an error in the first version of Proposition 1.4: it read the potential of lemma 24 as the whole energy of the walls, but that potential is taken relative to theta = 0 and omits the theta-independent Casimir energy. The corrected statement is below; the reading "the walls attract" is withdrawn. The first version of Proposal 1.3 also placed the W at the bottom of the tower, M_W ~ 1/(2R), along every path to v = 0; after the audit (its §2.2), that holds only on the scale-free path of Proposition 1.5.

Setting

Hypothesis H of STRATEGY.md reads the two sheets of the finite geometry as the minimal truncation of an interval I_l with opposite chiral boundary conditions, with l ~ 1/M_W. Its three regimes carry the labels 9, 10 and 11. These are KO labels. At the 11 endpoint (v -> 0, l -> infinity) the heat dimension is 4 + 1 = 5, since the finite factor F keeps heat dimension 0 for every v (canonical §2.1). To read 11 as a metric dimension, the KO-6 factor must itself be resolved into a six-dimensional space, which gives the count

```
11 = 4 (spacetime) + 1 (the interval) + 6 (a space resolving F).
```

The files of this folder take the six-dimensional space to be B_6 = CP^2 x S^2, the base of Witten's spaces (02), and study two ways of completing 4 + 6 to eleven dimensions (03). This file treats the "1", and the path along which v -> 0 opens it.

Proposition 1.1 (the chiral interval is the antiperiodic circle)

Statement. Let D_l act on H_l = L^2([0, l], C^2) as in lemma 05, eq. (8): D_l(L, R) = (R', -L'), with R(0) = 0 and L(l) = 0, grading Gamma = diag(1, -1) and the scalar functions C^infinity([0, l]). Let S_R be the circle of length 2l, that is of radius R = l/pi, with coordinate x in [-l, l] and antiperiodic functions u(x + 2l) = -u(x). The map

```
(W(L, R))(x) = (L(|x|) + i sgn(x) R(|x|)) / sqrt 2
```

is a unitary H_l -> L^2_antiperiodic(S_R) with

1. W D_l W^* = -i d/dx on its antiperiodic domain;
2. W Gamma W^* = P, the reflection (Pu)(x) = u(-x);
3. W f W^* = multiplication by f(|x|), so the algebra goes to the reflection-even functions;

4. the walls x = 0 and x = l go to the two fixed points x = 0 and x = l ~ -l of the reflection.

Hence spec D_l = {(k + 1/2)/R : k in Z}, each eigenvalue simple, which is (9) of lemma 05, and the Connes-Lott mass of the minimal truncation is m = pi/(2l) = 1/(2R).

Proof. The inverse map is L(x) = (u(x) + u(-x))/sqrt 2, R(x) = (u(x) - u(-x))/(i sqrt 2) for x in [0, l], so W is a bijection, and |u(x)|^2 + |u(-x)|^2 = |L(x)|^2 + |R(x)|^2 gives unitarity. For H^1 data, u is continuous at 0 exactly when R(0) = 0, and u(l) + u(-l) = sqrt 2 L(l), so the antiperiodic matching u(-l) = -u(l) is the condition L(l) = 0. On x > 0, -i (L + iR)' = R' + i(-L'), the image of (R', -L') = D_l(L, R). On x < 0, u(x) = (L(-x) - i R(-x))/sqrt 2 and -i u'(x) = (R'(-x) - i(-L'(-x)))/sqrt 2, which is again the image of (R', -L'). Gamma(L, R) = (L, -R) maps to L(|x|) - i sgn(x) R(|x|) = u(-x). Items 3 and 4 are immediate. The antiperiodic spectrum of -i d/dx on a circle of length 2l is pi(2k + 1)/(2l) = (k + 1/2)/R. ■

So the two sheets of H are the two fixed points of a reflection on a circle whose spinors are antiperiodic, and the grading of the finite geometry is that reflection. With finite data D_F supplied as in lemma 22, the parent unfolds to -i d/dx + D_F on the periodic circle of length 2l (lemma 25, Proposition 25.1), and the angle theta interpolates between that circle (theta = 0) and the antiperiodic circle of Proposition 1.1 (theta = pi/2): D_F is a constant connection, a Wilson line, and the wall angles theta_k = l mu_k (mu_k the eigenvalues of D_F) are its holonomy phases.

The joint family: weak masses below a descending tower

At fixed l, the vector masses of lemma 23 vanish as theta -> 0 and also as theta -> pi/2, while the tower spacing pi/l stays fixed; vanishing masses alone open no dimension (audit §2.2). The audit gives a family in which they do.

Proposition (Astra 6, audit §3; restated). Let D_F(epsilon) = epsilon D_* and l_epsilon = l_* epsilon^{-alpha}, 0 < alpha < 1, with wall data X = tan(l D_F) as in lemma 22. Then the exact window is W^* D W = epsilon D_, *the rest of the spectrum lies at |lambda| >= pi/l_epsilon - epsilon ||D_||*, the compression error is at most (2/3) l_^2 ||D_||^2 epsilon^{2(1-alpha)} ||a|| (Theorem 22.4, item 2), and for alpha < beta < 1 a cutoff E_epsilon = E_* epsilon^beta contains exactly the r = dim H_F finite states, while at every fixed E > 0 the number of states N_epsilon(E) grows like l_epsilon E, which is proportional to epsilon^{-alpha}.

In the circle language of lemma 25 (the periodic circle, on which D_F is a Wilson line; the window is its n = 0 mode) this reads: the radius R_epsilon = l_epsilon/pi goes to infinity, and the Wilson-line angles theta_epsilon = l_epsilon epsilon mu_k go to zero like epsilon^{1-alpha}. The hierarchy is

```
m_weak ~ epsilon  <<  E_epsilon ~ epsilon^beta  <<  1/R_epsilon ~ epsilon^alpha  ->  0.
```

In the five-dimensional Yang-Mills toy of lemma 23 the vector poles along this family are M_W^2 = 2 m^2 (1 + O(theta^2)) and M_Z^2 = (16/5) m^2 (1 + O(theta^2)), so M_W^2/M_Z^2 -> 5/8 (audit §3.1, from Theorem 23.6).

Check (Claude). The parent space of lemma 22 is L^2([0, l]; C^2) ⊗ H_F, of fibre dimension 2r. By Theorem 22.1 its signed spectrum is (n pi +- theta_k)/l, n in Z, two progressions of spacing pi/l for each of the r/2 singular values, so N(E) = 2 r l E/pi + O(r), the constant of the audit (the scalar circle of Proposition 1.1 gives 2 E l/pi; with fibre H_F the count is 2 r l E/pi). The window, gap and compression statements follow from Theorems 22.1 and 22.4 as stated. (Python, section H: exact spectra for random singular values.)

The audit also gives a localized family (its §4): a bulk mass mu_epsilon = mu_* epsilon^alpha and mu l = (1 - alpha) log(1/epsilon) + c_0, with light mass m ~ 2 mu_* e^{-c_0} epsilon, gap ~ mu_* epsilon^alpha, and the exact tail identity e^{-2 kappa l} = (m/(mu + kappa))^2. Its table of roots is reproduced in section H (Python, bisection on tanh s = s/c and tan y = y/c): the four light masses and next masses agree with the audit to the digits printed there.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The localized asymptotics and tail identity are independently checked analytically; the length includes a logarithmic factor. The numerical table was not recomputed. See the referee report.

Proposal 1.3 (the circle of H as the M-theory circle)

Take the circle of Proposition 1.1 to be the eleventh direction of M-theory over a ten-dimensional type IIA background, so R = g_s l_s. Then, along the joint family:

H	circle of Proposition 1.1	M-theory / string side (recalled, to check)
v -> 0 along the joint family, label 11	R -> infinity, Wilson-line angles -> 0, m_weak R -> 0	strong coupling: D0-brane bound states of mass n/(g_s l_s) become light and are the momentum modes of the eleventh dimension (Witten 1995); the weak masses lie below this tower
v finite, label 10, two sheets	R finite, fixed points of a reflection	M-theory on an interval with two walls (Hořava-Witten); here with walls of opposite chirality
v -> infinity, label 9, sheets merge	R -> 0, antiperiodic spinors	a circle with antiperiodic fermions shrinking: bulk fermions get masses >= 1/(2R); compare type 0 strings, which have no bulk spacetime fermions (Bergman-Gaberdiel 1999), and the walls of opposite chirality of Fabinger-Hořava 2000

Each row matches a statement already in the corpus: the 11 endpoint needs the tower (canonical §2.1, lemma 05 Part 4), and at the 9 endpoint the bare interval keeps no light fermion, so light fermions must come from another sector (STRATEGY, T1 and T3). On the string side the same happens: the light fermions of a type 0 theory come from open strings on branes. The proposal therefore places the light fermions of the 9 endpoint on branes or walls, and 03 develops a brane origin for them.

Tests.

1. Along the joint family m_weak/M_KK ~ epsilon^{1 - alpha} -> 0, so the weak scale is far below 1/(g_s l_s); the scale-free path of Proposition 1.5, with M_W R fixed, is the other possibility and keeps no finite window.
2. Gauge couplings. A five-dimensional gauge field with constant profile on the interval has g_4^{-2} = l/g_5^2 = pi R/g_5^2 (audit §7.2 for the general volume formula), so at fixed g_5 the four-dimensional couplings vanish as R -> infinity, and M_W = g v/2 could vanish through g. The limit sought keeps g finite, which needs g_5^2 ~ R, and this must be shown for the vectors in question before the masses are read as restoration of the symmetry.
3. The path must be selected by an action: Proposition 1.5 shows what the flat interval alone selects.

Proposition 1.4 (the angle-dependent one-loop term at the vacuum angle)

Statement. Under the hypotheses of Theorem 9 of lemma 24 (flat interval, zero bulk masses, frequency-one fermions, the stated gauge towers and multiplicities, no wall potential), the angle-dependent part of the one-loop potential, taken relative to theta = 0 as in lemma 23, has its minimum at theta = pi/2 with value

```
V(pi/2) - V(0) = - 93 zeta(5) n_f / (512 pi^2 l^4),
```

and near theta = 0 it is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) theta^2/l^4 + o(theta^2). At theta = pi/2 the exact vector poles of lemma 23 give M_W = M_Z = 0, with the fermion at the tower scale m = pi/(2l).

Proof. In the notation of lemma 24, V = $\mathcal{V}$/(8 pi^2 l^4), with F(0) = 0, and $\mathcal{V}$(1) = -n_f F(1) since the gauge terms vanish at t = 1. With 1 + 1/sinh^2 x = coth^2 x and log coth x = 2 sum over odd k of

e^{-2kx}/k,

```
F(1) = 2 int_0^infinity x^3 log coth x dx = 4 sum_{k odd} (1/k) 3!/(2k)^4 = (3/2) sum_{k odd}
↪  k^{-5} = (3/2)(31/32) zeta(5) = 93 zeta(5)/64.
```

Near t = 0, $\mathcal{V}$'(0) = (6 + 24/5 - n_f) J(0) with J(0) = int x^3/sinh^2 x dx = (3/2) zeta(3), the coefficient already given in lemma 23. Python: F(1) = 1.506786, J(0) = 1.803085, series against quadrature to 10^-6. ■

Reading. The θ-independent Casimir energy of the towers, a_0/l^4, is not part of this potential and is not computed in lemmas 23 or 24; the direction in which l moves depends on the sign of a_0 - 93 zeta(5) n_f/(512 pi^2). The minimum over the angle is the point with massless W and Z at fixed l, which opens no dimension (lemma 23 already notes that it is not the physical point).

Proposition 1.5 (a scale-free potential with an isolated stationary angle fixes M_W l)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The derivative identity is correct, and an isolated stationary angle gives $l \propto 1/m$. The universal conclusion has a flat-direction counterexample $G(\theta) = \theta^4$; for the displayed one-loop determinant the small-angle obstruction is recovered when $n_f \neq 54/5$. See the referee report.

Statement. Let V(l, theta) = l^{-4} G(theta) with G of class C^1, the form of every one-loop potential of a flat interval whose only scales are l and the wall angles (lemmas 23 and 24, with a_0 included in G).

1. If the angle is free, V is monotone in l at every fixed theta with G(theta) != 0. At the minimizing angle theta_min the length runs away: to l -> infinity if G(theta_min) > 0, to l -> 0 if G(theta_min) < 0. Stationary points exist only on the set G = G' = 0.
2. If the angle is tied to a fixed mass, theta = l m (as for X = tan(l M) in lemma 22), the stationary lengths satisfy theta G'(theta) = 4 G(theta) at theta = l m. So l = theta_*/m *for a root theta_* that does not depend on m, and M_W l, M_Z l (functions of theta alone by Theorem 23.6) are the same at every m.

Hence, when the roots of theta G' - 4 G are isolated, a stationary branch has l = theta_*/m: the path l ~ 1/M_W of H's original statement, alpha = 1 in the scaling of the joint family (Remark 10.7 of 10 for a small fixed ratio). A root interval, as for G = theta^4 near 0, makes every alpha < 1 stationary and selects none (referee amendment). The joint family with alpha < 1 needs a second scale in the action: a bulk mass (the localized family, G(l m, l mu)), the cutoff of the spectral action (lemma 25, G(theta, l Lambda)), curvature, flux or a wall term.

Proof. d/dl [l^{-4} G(l m)] = l^{-5} (theta G'(theta) - 4 G(theta)) at theta = l m. For item 1, d/dl V = -4 l^{-5} G(theta) and d/dtheta V = l^{-4} G'(theta). ■

Amendment (4 Nov 2026). The window can separate on the scale-free path, with a fixed small ratio, if the light mass is exponentially small in a stabilized shape parameter, and a_0 is now computed. See 10, Remark 10.7.

Consequence for the modulus target of the audit (§7.1). The audit proposes V = V_*(a r^{-p} + b epsilon^2 r^q) with a, b, p, q > 0, whose minimum r ~ epsilon^{-2/(p+q)} gives alpha = 2/(p + q) < 1 when p + q > 2. In the flat one-loop family the epsilon-dependent term near theta = 0 is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) m^2/l^2, so q = -2, and with the Casimir term p = 4 the stationary point has l ~ 1/m, alpha = 1, in agreement with item 2. A term growing with l at fixed v, as the ansatz requires, is absent from the flat family and must come from one of the second scales listed above.

What remains open here

- Whether the identification of Proposal 1.3 survives the gauge sector: lemmas 13, 14 and 25 tie M_W to the tower with model-dependent coefficients, which here become a relation between M_W, g_s and l_s, and test 2 above requires g_5^2 ~ R.

- The theta-independent Casimir coefficient a_0 of the flat family, which decides the runaway direction of Proposition 1.5, item 1.
- A potential with a second scale that selects alpha < 1: the localized family of the audit is the first candidate, with G(l m, l mu) computable from the exact secular equations of lemma 09.
- A fermion with both a fibre momentum and a base twist: on a nontrivial circle bundle (Witten spaces, file 02) the antiperiodic shift corresponds to a spin structure whose fibre circle does not lift the circle action. Which Witten spaces carry such a structure is an open item for the next job.

Referee proofs (GPT-6.1 Terra, 4 November 2026)

The arguments below are those of the referee report on the 11D line, written out in the notation of this lemma; the report in historical/reviews/ stays as the provenance record.

Independent proof of Proposition 1.4. *From the referee report on the 11D line, section 'Proposition 1.4: the one-loop endpoint and small-angle coefficient', transferred verbatim in substance on 4 Nov 2026.*

Hypotheses: those of Theorem 9 of lemma 24, including n_f > 0 and its normalization. The result is relative to theta $= 0$ and leaves the theta-independent Casimir term open.

Let

```
F(t) = int_0^infinity x^3 log(1 + t/sinh^2 x) dx,   t = sin^2 theta,
```

so that, by equation (22) of lemma 24, $\mathcal{V}(t) = 8$ pi^2 l^4 V(theta) = 3 F(2t(1 - t)) + (3/2) F((16/5) t(1 - t)) - n_f F(t). For every $x > 0$ the integrand increases strictly with t, so F is strictly increasing, and F(0) $= 0$. The two gauge terms are nonnegative and vanish at $t = 1$, where their arguments vanish; the fermion term -n_f F(t) is strictly decreasing for n_f > 0. Hence $t = 1$ is the unique global minimum.

Endpoint value. At $t = 1$, $1 + 1/\sinh^2 x = \coth^2 x$. With $q = e^{-2x}$, $\coth x = (1 + q)/(1 - q)$, so

```
log coth x = log(1 + q) - log(1 - q) = 2 sum_{k >= 1, k odd} e^{-2kx}/k,     int_0^infinity x^3
↪  e^{-2kx} dx = 3!/(2k)^4.
```

All terms are positive, so the series may be integrated term by term, and

```
F(1) = 2 int_0^infinity x^3 log coth x dx = 4 sum_{k odd} (1/k) 3!/(2k)^4 = (3/2) sum_{k odd}
↪  k^{-5} = (3/2)(1 - 2^{-5}) zeta(5) = 93 zeta(5)/64.
```

Since $\mathcal{V}(0) = 0$ and $\mathcal{V}(1) =$ -n_f F(1), division by 8 pi^2 l^4 gives the stated endpoint value V(pi/2) - V(0) $=$ -93 zeta(5) n_f/(512 pi^2 l^4).

Small-angle coefficient. Differentiation under the integral at $t = 0$ is allowed because $x^3/\sinh^2 x$ is integrable on (0, infinity). With $1/\sinh^2 x = 4 e^{-2x}/(1 - e^{-2x})^2 = 4$ sum_{n >= 1} n e^{-2nx},

```
F'(0) = int_0^infinity x^3/sinh^2 x dx = 4 sum_{n >= 1} n 3!/(2n)^4 = (3/2) sum_{n >= 1} n^{-3} =
↪  (3/2) zeta(3),
```

which is J(0) in the proof of Proposition 1.4. The two gauge arguments 2t(1 - t) and (16/5) t(1 - t) have derivatives 2 and 16/5 at $t = 0$, with prefactors 3 and 3/2, so

```
V'(0) = (3 * 2 + (3/2)(16/5) - n_f) F'(0) = (54/5 - n_f)(3/2) zeta(3).
```

With $t = \sin^2 \text{theta} = \text{theta}^2 + o(\text{theta}^2)$, V(theta) - V(0) $= \mathcal{V}'(0)$ theta^2/(8 pi^2 l^4) + o(theta^2) $=$ (3 zeta(3)/(16 pi^2))(54/5 - n_f) theta^2/l^4 + o(theta^2), the claimed coefficient. The exact vector formulas of lemma 23 vanish at theta $=$ pi/2, where the lowest fermion mass is pi/(2l).

Counterexample (referee) to the universal conclusion of Proposition 1.5. *From the referee report on the 11D line, section 'Proposition 1.5: scale-free potentials and the joint path', transferred verbatim in substance on 4 Nov 2026.*

Verdict of the referee: an error in the universal conclusion, and correct with an amendment for isolated stationary angles. The derivative identities in the proof hold. The universal conclusion, that every stationary family has constant angle and that a second scale is necessary for the existence of a joint family, fails by the counterexample below.

At fixed m > 0 put theta = l m. Then

```
d/dl V(l, l m) = l^{-5} (theta G'(theta) - 4 G(theta)).
```

If a continuous stationary branch chooses an isolated positive root theta_, *then theta_* is constant along the branch and l = theta_*/m. A root interval gives a flat direction instead.

Counterexample. Let G(theta) = theta^4 on a neighbourhood 0 <= theta < delta, with a C^1 extension to the angle domain. There

```
V(l, l m) = l^{-4} (l m)^4 = m^4
```

is independent of l. For any 0 < alpha < 1, the family m = epsilon, l = epsilon^{-alpha} has theta = l m = epsilon^{1 - alpha} < delta for all sufficiently small epsilon, so it is stationary, and l m -> 0. The vector mass formula of Theorem 23.6 gives M_W l ~ sqrt 2 l m -> 0. The potential leaves this path undetermined and does not dynamically select it. A smooth extension can also be chosen with G(theta) >= theta^4 on the rest of the fundamental interval; then V(l, l m) >= m^4, with equality on l m < delta, and the example is a flat family of minima.

The specified flat one-loop determinant. By Proposition 1.4, with a_0 the theta-independent Casimir coefficient,

```
G(theta) = a_0 + c theta^2 + o(theta^2),    theta G'(theta) - 4 G(theta) = -4 a_0 - 2 c theta^2 +
↪  o(theta^2),
```

where c = (3 zeta(3)/(16 pi^2))(54/5 - n_f) is proportional to 54/5 - n_f (the expansion of G' follows from that of $\mathcal{V}$', since $\mathcal{V}$ is C^1 at t = 0). If c != 0, there is no sequence of positive stationary angles tending to zero, whether or not a_0 = 0: for a_0 != 0 the right side tends to -4 a_0, and for a_0 = 0 it equals -2 c theta^2 (1 + o(1)). In particular this covers every integer n_f under the displayed determinant. The tuned case c = 0 requires its higher-order term to be checked. The comparison with the audit's ansatz with q > 0, in the paragraph on the modulus target, is correct: the displayed quadratic contribution is l^{-4} c (l m)^2 = c m^2/l^2.

Free angle. For a free angle, d/dl V = -4 l^{-5} G(theta); a nonzero minimized G produces the stated runaway, and a zero minimum leaves the length flat. At an endpoint of a constrained angle interval, the stationarity condition on G' must be understood one-sided.

Scope and direction. The corrected fixed-angle result concerns isolated stationary roots of a scale-free C^1 potential. A flat modulus, a tuned higher-order potential, or additional dimensionful terms leave other families possible. A mechanism that selects a joint path with finite gauge couplings remains to be constructed.

Independent proof of the asymptotics of the localized family (audit, its §4). *From the referee report on the 11D line, section 'Audit section 4 and file 01: localized joint family', transferred verbatim in substance on 4 Nov 2026.*

The asymptotic light mass, the gap, the localization ratio and the exact tail identity follow analytically from the interval equations. The localized length includes a logarithmic factor, and so differs from the pure-power length of the boundary-angle family above. The numerical table of section H was not recomputed by the referee.

Let c = mu l, s = kappa l, and s = c tanh s, the light-mode condition kappa = mu tanh(kappa l). For c > 1 there is a positive solution, and as c -> infinity it has s -> infinity and s/c = tanh s -> 1. Moreover

```
0 < c - s = c (1 - tanh s) = 2c e^{-2s}/(1 + e^{-2s}) <= 2c e^{-2s} -> 0.
```

Therefore

```
m = mu sech s = 2 mu e^{-s}/(1 + e^{-2s}) ~ 2 mu e^{-s} ~ 2 mu e^{-c}.
```

With mu = mu_* epsilon^alpha and c = (1 - alpha) log(1/epsilon) + c_0, e^{-c} = e^{-c_0} epsilon^{1 - alpha}, which gives m ~ 2 mu_* e^{-c_0} epsilon and l = c/(mu_* epsilon^alpha). The localization width is 1/kappa, so its ratio to l is 1/s ~ 1/c.

The first oscillatory root y_1 = k_1 l of tan y = y/c satisfies pi < y_1 < 3 pi/2, since y/c > 0 is small for large c. The gap above the light pair is therefore

```
Delta = sqrt(mu^2 + k_1^2) = mu sqrt(1 + (k_1 l/c)^2) ~ mu = mu_* epsilon^alpha.
```

Finally, s = c tanh s reads mu = kappa coth s, and with it m = mu sech s = kappa/sinh s. Then

```
m/(mu + kappa) = 1/(sinh s (coth s + 1)) = 1/(cosh s + sinh s) = e^{-s},    m^2 = kappa^2/sinh^2 s
  = kappa^2 (coth^2 s - 1) = (mu - kappa)(mu + kappa),
```

and the first identity squared is e^{-2 kappa l} = (m/(mu + kappa))^2. This proves both displayed tail identities.

Part 32

Witten spaces as the hypercharge quotient of the five-dimensional representation

Lemma 32. Written as file 02 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line of research (README). Theorem 2.1 and Propositions 2.2 to 2.4 and 2.6 are proved here, Theorem 2.1 and Proposition 2.6 conditionally as the referee notes say; Proposition 2.5 is a negative result in a stated scope, refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5. Attempted by Claude (Anthropic), 3 Nov 2026, with a remark from the steer of the same day on the mixing of u and d. Checks: checks_11d.py, sections A to D, Python with exact fractions. Referee pending.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The exclusion in Proposition 2.5 has an independent analytic proof. The required-bidegrees lemma and the general assertion that u^c, d^c must have the same flux image require a common ten-dimensional chirality; the exclusion itself also holds without that extra assumption. See the referee report; other reviewed items are noted separately.

Referee note on Theorem 2.1 and Propositions 2.2, 2.4 (GPT-6.1 Terra). The quotient and tensor-centre calculations are independently checked. The cited M^{111} identification is conditional on the recalled fibration; the original quotient metric needs rescaling to be Sasaki-Einstein, and the action on holomorphic functions is dual to the action on U. See the report.

Referee note on Proposition 2.3 (GPT-6.1 Terra). The Gysin, Smith normal form and spin calculations are independently checked; the cited Kreck-Stolz paper was unavailable. See the report.

Setting

Witten's spaces M^{pqr} = G/H, with G = SU(3) x SU(2) x U(1) and H = SU(2) x U(1) x U(1), are the seven-dimensional spaces of least dimension with the isometry group of the Standard Model, so 4 + 7 = 11 (Witten 1981; paper/LITERATURE.md §F). Castellani, D'Auria and Fré gave them Einstein metrics; Killing spinors exist only for p = q, where the AdS4 vacuum has N = 2 and the space is M^{111} (Fabbri, Fré, Gualtieri, Termonia, hep-th/9903036). Fabbri et al. (hep-th/9907219, §4) write the cone over M^{111} as the Kähler quotient of C^5 by a circle acting with weights 2 on a triplet U and -3 on a doublet V, with D-term 2|U|^2 = 3|V|^2, and M^{111} as a circle bundle over P^2 x P^1. In later notation it is M^{3,2}.

These weights are six times the hypercharges of the 5-bar of SU(5). The rest of the file develops that observation.

Theorem 2.1 (M^{111} is the hypercharge quotient of the 5-bar)

Statement. Let 5bar = C^3 + C^2 carry the Standard Model action on the left-handed fields d^c in (3bar, 1)_{1/3} and L in (1, 2)_{-1/2}; write U for d^c and V for L, so 6Y = 2 on U and -3 on V. Let mu_Y =

(2|U|^2 - 3|V|^2)/6 be the moment map of hypercharge, and

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Y_{3,2} = (mu_Y^{-1}(0) ∩ S^9) / U(1)_Y.
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1. Y_{3,2} is a smooth, simply connected seven-manifold, the circle bundle of O(3, 2) over CP^2 x CP^1 (up to orientation), and its cone is the Kähler quotient C^5 // U(1)_Y of Fabbri et al.
2. The cone is Calabi-Yau (Gorenstein, sum of weights 0) because tr Y = 0 on the 5-bar, that is, because U(1)_Y lies in SU(5).
3. The group acting effectively on Y_{3,2} through the quotient is (U(3) x U(2))/U(1)_Y. Its subgroup S(U(3) x U(2))/U(1)_Y is PSU(3) x SO(3), and the overall phase of U(5) supplies one further circle, the Reeb (R-symmetry) direction.
4. The functions on Y_{3,2} that are polynomial in U, V and their conjugates have U(1)_Y-charge zero; the holomorphic ones are spanned by the monomials of degree (3k, 2k) in (U, V), whose span as tensors is Sym^{3k}(3bar) x Sym^{2k}(2) and as functions its dual Sym^{3k}(3) x Sym^{2k}(2): for k = 1 the functions form the (10, 3) of Fabbri et al.

Proof. On mu_Y^{-1}(0) ∩ S^9 we have |U|^2 = 3/5 and |V|^2 = 2/5, a product of round spheres S^5 x S^3. The circle acts by (lambda^2 U, lambda^{-3} V); a fixed point would need lambda^2 = lambda^3 = 1, so lambda = 1 and the action is free. The quotient of S^5 x S^3 by the full torus T^2 of phases of U and V is CP^2 x CP^1, and Y_{3,2} -> CP^2 x CP^1 is a principal bundle for T^2/U(1)_Y. A character lambda_U^m lambda_V^n of T^2 is trivial on U(1)_Y exactly when 2m - 3n = 0, so the characters of T^2/U(1)_Y are generated by (m, n) = (3, 2), whose line bundle is O(3) ⊠ O(2) up to the sign convention; this gives item 1 and item 4. Simple connectivity follows from the exact homotopy sequence of U(1) -> S^5 x S^3 -> Y_{3,2}. Item 2: the sum of weights is 3 * 2 + 2 * (-3) = 0. Item 3: an element (g, h) of U(3) x U(2) acts trivially on the quotient exactly when (g, h) = (lambda^2, lambda^{-3}), because mu_Y^{-1}(0) spans C^5. The intersection of SU(3) x SU(2) with U(1)_Y consists of lambda^6 = 1, generated by (e^{2 pi i/3} I_3, -I_2), the centre Z_3 x Z_2. ■

So the isometry group of the Witten space with Killing spinors, SU(3) x SU(2) x U(1) (Fabbri et al.), is the centralizer of hypercharge in U(5), divided by hypercharge. The SU(3) x SU(2) part is the Standard Model one; the remaining circle is the R-symmetry, and hypercharge itself has been quotiented. Witten assigned the U(1) of G to hypercharge; in this realization hypercharge is a different U(1) (03, Proposal 3.1).

Proposition 2.2 (uniqueness of the count 7)

Statement. (a) Among the quotients Y_{a,b} = (S^5 x S^3)/U(1), with weights b on C^3 and -a on C^2 and gcd(a, b) = 1, a, b > 0, the cone is Calabi-Yau only for (a, b) = (3, 2). (b) For a complex representation of dimension n and a free circle action, (S^{2n-1} ∩ mu^{-1}(0))/U(1) has dimension 2n - 3; this is 7 only for n = 5. Among representations of SU(5), with hypercharge as the circle, only the 5 and the 5-bar give a seven-dimensional quotient.

Proof. (a) The Calabi-Yau condition is 3b - 2a = 0. (b) Dimension count: 2n - 1 for the sphere, minus one for the moment map, minus one for the circle. The irreducible representations of SU(5) of dimension 5 are the 5 and the 5-bar. ■

Witten's count reads: 11 = 4 + (2 * 5 - 3). The seven internal dimensions come from the smallest representation of SU(5), with hypercharge as the circle. For the 10 the quotient has dimension 17, and for the 5 + 10 it has 27.

Proposition 2.3 (topology of Y_{a,b})

Statement. Let Y_{a,b} -> B = CP^2 x CP^1 be the circle bundle with Euler class a h_1 + b h_2, a != 0. Then H^2(Y) = Z + Z/gcd(a, b), b_3(Y) = 0, H^4(Y) is a group of order a^2, cyclic when gcd(a, b) = 1, and Y is spin exactly when a is odd and b is even. For (a, b) = (3, 2): H^2 = Z, H^4 = Z/9, spin. The pullback H^2(B) = Z^2 -> H^2(Y) = Z is (x, y) -> 2x - 3y up to sign: the two divisor classes h_1 = [{l(U) = 0}] and h_2 = [{m(V) = 0}] go to 2 and -3, the values of 6Y on d^c and L.

Proof. The Gysin sequence H^{k-2}(B) -> H^k(B) -> H^k(Y) -> H^{k-1}(B) -> H^{k+1}(B), with the first map the cup product with c = a h_1 + b h_2 and H^*(B) = Z[h_1, h_2]/(h_1^3, h_2^2). In degree 2 the cokernel of 1 -> (a, b) is Z^2/(a, b). In degree 4 the map H^2(B) -> H^4(B) has matrix [[a, 0], [b, a]] in the bases (h_1, h_2), (h_1^2, h_1 h_2), with determinant a^2 and Smith invariants (gcd(a, b), a^2/gcd(a, b)); since H^3(B) = 0 this gives H^4(Y), and the same matrix is injective, so H^3(Y) = 0. For spin: w_2(B) = h_1 mod 2, and the kernel of H^2(B; Z/2) -> H^2(Y; Z/2) is spanned by c mod 2; so w_2(Y) = 0 exactly when (1, 0) = (a, b) mod 2. For (3, 2), Z^2/(3, 2) -> Z by (x, y) -> 2x - 3y is an isomorphism. ■ (Python: Smith invariants and the spin condition, section C.)

The product member CP^2 x S^2 x S^1 of Witten's family (a = b = 0) is not spin.

Proposition 2.4 (the Z_6 of the Standard Model)

Statement. (a) Every monomial in U and V, and every state of the exterior algebra Λ^*(5bar), satisfies 2t + 3d + 6Y = 0 mod 6, with t the triality (1 for 3, -1 for 3bar) and d = 2j mod 2. (b) The U(1)_Y-neutral monomials have t = 0 and d = 0. (c) Every tensor field on Y_{3,2} carries a representation of (U(3) x U(2))/U(1)_Y, on which SU(3) x SU(2) acts through PSU(3) x SO(3). So no tensor Kaluza-Klein mode on Y_{3,2}, on a background invariant under the displayed action, is a colour triplet or a weak doublet; spinors are open.

Proof. (a) holds for U (t = -1, d = 0, 6Y = 2) and V (t = 0, d = 1, 6Y = -3) and is additive. (b) U^alpha V^beta is neutral when 2 alpha = 3 beta, so 3 | alpha and 2 | beta. (c) The effective group of Theorem 2.1 acts on the frame bundle and on every tensor bundle, and its intersection with SU(3) x SU(2) is the centre Z_6 (Theorem 2.1, item 3). ■

(a) is the statement that the Standard Model group is (SU(3) x SU(2) x U(1))/Z_6 = S(U(3) x U(2)). In the quotient realization the Z_6 is the kernel of SU(3) x SU(2) -> Isom(Y_{3,2}). States with fractional hypercharge are those charged under the quotient circle, and they cannot be functions or tensors on Y_{3,2}. For spinors the isometry group may lift to a double cover; whether fermionic Kaluza-Klein modes on Y_{3,2} can carry half-integer weak isospin or triality is open (next job, item 2 of the README).

Chiral zero modes on the base with abelian fluxes

Witten's argument makes Kaluza-Klein fermions on an odd-dimensional space vectorlike, for gauge fields acting by isometries. The even base B_6 = CP^2 x CP^1 escapes it: abelian fluxes on B_6 give chiral four-dimensional fermions through the index of the twisted Dirac operator. This section asks which fluxes give one generation.

Setting. Ten-dimensional Weyl fermions on M^4 x CP^2 x CP^1, each with definite charges under some U(1) factors with fluxes on B_6. Since CP^2 is not spin, a fermion is a spin^c spinor, Λ^{0,*} ⊗ K^{1/2} ⊗ N with N the flux line bundle of its charges, and the four-dimensional chiral content is the virtual representation H^*(O(alpha, beta)) with (alpha, beta) = (-3/2, -1) + pairing of the charges with the fluxes. The cohomology is H^0(O(alpha)) = Sym^alpha(3bar) for alpha >= 0, H^2(O(alpha)) = Sym^{-alpha-3}(3) for alpha <= -3, and zero otherwise on CP^2; on CP^1, H^0(O(beta)) = Sym^beta(2) for beta >= 0 and H^1(O(beta)) = Sym^{-beta-2}(2) for beta <= -2. Each Standard Model Weyl field must come from one ten-dimensional field, from the whole cohomology of one line bundle.

Lemma (required bidegrees). Writing P_s = (alpha + 3/2, beta + 1) for the pairing of the charges of field s (up to the common sign of charge and chirality):

field	P_s	note
Q (3, 2)	(-5/2, 2)	H^2(O(-4)) x H^0(O(1))
u^c, d^c (3bar, 1)	(5/2, 1)	H^0(O(1)) x H^0(O(0))
L (1, 2)	(+-3/2, 2)	
e^c, nu^c (1, 1)	(+-3/2, 1)	

The second entry is 2j + 1 and the first is +-5/2 on colour triplets and +-3/2 on singlets. (Python, section D, by enumeration of the line bundles with these cohomologies and the Riemann-Roch check.)

Proposition 2.5 (no generation from fluxes along hypercharge and B - L) *Scope note (4 Nov 2026, after the referee).* The statement on equal images and the linearity of 2j + 1 assume a common ten-dimensional chirality for the fields. The conclusion holds without it: in the span of 6Y and 3(B - L) the charges of L are -3 times those of Q, so their second pairings are P_2(L) = -3 P_2(Q), against |P_2| = 2 on both doublets.

Referee note on the third-charge search (GPT-6.1 Terra, 4 Nov 2026). The bounded search below has an independent analytic proof. The same argument extends to any real third charge and independent source chiralities; this stronger extension was checked by Claude Sonnet 5.5 (second check: the vectors y, b, p, w have rank 4 for every sign pattern), so it stands as a no-go in the same scope with two model identifiers. See the referee report.

Statement. In the setting above, with the canonical spin^c root and fluxes only in U(1) directions spanned by Y and B - L (any flux quanta), no choice reproduces one generation, with or without nu^c. More generally, the pairing must give u^c and d^c the same image, and 2j + 1 = (2, 1, 1, 2, 1, 1) on (Q, u^c, d^c, L, e^c, nu^c) must be a linear function of the fluxed charges.

Proof. The pairing M is linear in the charges, and M c_s = P_s with c_s the charges of the left-handed field s. With only Y and B - L fluxed, c_u - c_d = (-6, 0) in the coordinates (6Y, 3(B - L)), so M vanishes on the hypercharge direction and P_s = 3(B - L)_s m for one vector m. The second entries 2 on Q and 1 on u^c would then need 3(B - L) to take values in the ratio 2 : 1 on (Q, u^c), against 1 : -1. In general, 2j + 1 is not in the span of 6Y = (1, -4, 2, -3, 6, 0) and 3(B - L) = (1, -1, -1, -3, 3, 3): from Q and u^c the coefficients are -1 and 3, and then d^c gives -5 against 1. ■ An exhaustive search adding a third integer charge z in [-3, 3]^6 to Y and B - L finds no solution either (Python, section D).

Scope and escapes. Abelian fluxes on CP^2 x CP^1 with the Fubini-Study complex structure; one ten-dimensional Weyl field per Standard Model field; canonical spin^c root; Dolbeault index as the chiral content. Known escapes: nonabelian bundles (an SU(2)_R doublet bundle), several ten-dimensional fields per four-dimensional field, a changed common twist, orbifolds and walls, and non-Kaluza-Klein (brane) states. The result points to the next proposition.

Proposition 2.6 (one generation from a universal charge, the grading and B - L) **Referee note (GPT-6.1 Terra, 4 Nov 2026).** The index calculation works with the stated cohomology convention. For the colour action of the earlier quotient, reverse the first-component coefficients to obtain the labelled colour representations; global flux quantization remains to be specified. See the referee report.

Statement. Let the fluxed charges be F (one on every field: a change of the common spin^c twist), chi (+1 on Q and L, -1 on u^c, d^c, e^c, nu^c) and B - L. Then one generation with nu^c is the exact index content for the pairing

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P^{(1)} = c_0 chi + c_1 (B - L),   P^{(2)} = (3F + chi)/2,
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with (c_0, c_1) = (-9/4, -3/4) or (-3/2, -3). The bidegrees are Q: (-4, 1); u^c, d^c: (1, 0); L: (-3, 1) or (0, 1); e^c, nu^c: (0, 0) or (-3, 0). Without chi, or without B - L, there is no solution.

Proof. Substitution, and the two failures by the span test (Python, section D2). ■

Reading.

1. u^c and d^c see the same line bundle O(1, 0), and so do e^c and nu^c. The flux treats them as SU(2)_R doublets with a common internal profile; hypercharge Y = T_3R + (B - L)/2 enters through T_3R, which carries no flux. This is the left-right pattern of Pati-Salam, produced here by the index.
2. chi is the finite grading gamma_F on the particle basis. The flux therefore couples to the grading, an axial abelian connection on B_6; in the language of the finite geometry this is a fluctuation proportional to gamma_F, the structure of the two gradings of lemma 23.

3. **Mass and charge bases.** Because u^c and d^c share one zero-mode profile, overlap integrals on B_6 cannot separate them; their masses come from data that break SU(2)_R (Higgs or wall couplings), and the basis that diagonalizes the masses is set by those data, independently of the basis in which the charges are diagonal. The steer of 3 Nov 2026 remarks that u and d should still mix, since mass eigenvalues are not charge eigenvalues, and that at the 9 label, with the left-handed interactions off, charge eigenvalues can coincide. Proposition 2.6 puts the coincidence in the flux charges: equal flux charges for u^c and d^c, unequal hypercharges through T_3R. At the 9 label the electroweak part of the algebra is removed (lemma 21, STRATEGY T2), and what distinguishes u^c from d^c in the light theory is then the vectorlike charge alone. Making this precise needs the SU(2)_R-breaking wall data and the 9D reduction of this index; listed as item 4 of the README.

What this file establishes, and the direction

Referee note (GPT-6.1 Terra, 4 Nov 2026). The M^{111} identification is conditional on the recalled fibration and metric normalization. The flux examples establish a particular construction and scoped obstructions; a general necessity for the grading flux remains open. See the wording audit.

The Witten space with Killing spinors is the hypercharge quotient of the 5-bar (Theorem 2.1), the count 7 is fixed by SU(5) (Proposition 2.2), and the Z_6 of the Standard Model is the kernel of the isometry action (Proposition 2.4). The price is that hypercharge is no longer an isometry, and no tensor Kaluza-Klein mode on an invariant background carries triality, weak isospin 1/2 or fractional hypercharge. Kaluza-Klein chirality on the base needs fluxes that see the grading and leave T_3R free (Propositions 2.5 and 2.6). Both results point to states outside the Kaluza-Klein spectrum of Y_{3,2} for the quarks and leptons: 03 takes them to be wrapped branes.

Referee proofs (GPT-6.1 Terra, 4 November 2026)

The arguments below are those of the referee report on the 11D line, written out in the notation of this lemma; the report in historical/reviews/ stays as the provenance record.

Independent proof of Proposition 2.5 (the exclusion). *From the referee report on the 11D line, section 'Proposition 2.5: hypercharge and B - L fluxes', transferred verbatim in substance on 4 Nov 2026.*

Hypotheses: one abelian line bundle per field, the canonical formal shift, fluxes in the span of Y and B - L. The exclusion holds even if the ten-dimensional Weyl fields have different chiralities.

Let M_2 be the component of the flux pairing M along CP^1, and use the charges c = (6Y, 3(B - L)). A weak doublet occurs for a line bundle O(beta) on CP^1 only when beta = 1 or beta = -3, which give H^0(O(1)) = 2 and H^1(O(-3)) = 2 respectively. The formal K^{1/2} shift on CP^1 is -1, so M_2(c) = beta + 1 and |M_2(c)| = 2 for each doublet. This remains true if a right-handed mode is conjugated to the left-handed convention, since conjugation reverses the sign of the pairing.

For the left-handed doublets,

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c_Q = (1, 1),   c_L = (-3, -3) = -3 c_Q.
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Linearity gives M_2(c_L) = -3 M_2(c_Q), so |M_2(c_L)| = 3 |M_2(c_Q)|, whereas both absolute values must be 2. This contradiction excludes a generation with or without nu^c, for arbitrary flux quanta in the two specified directions. It uses neither the colour component nor a choice of signs for the individual doublets.

Amendment to the required-bidegrees lemma. On CP^2 x CP^1, K^{1/2} is a formal notation: K = O(-3, -2) has no line-bundle square root. A genuine spin^c structure and its charged twists must give integral final bidegrees. With a common ten-dimensional chirality, Serre duality relates a right-handed conjugate to the sign-reversed pairing, and the convention P^{(2)} = 2j + 1 of the table is consistent. If the ten-dimensional chiralities can be chosen independently, the table is incomplete. For example, O(-4, -3) has H^3 = H^2(O(-4)) ⊗ H^1(O(-3)) = (3, 2) and pairing (-5/2, -2); a suitable ten-dimensional chirality makes this a left-handed Q, and (-5/2, -2) is neither the table entry (-5/2, 2) nor its overall negative.

Likewise, with independent chiralities and independent flux directions, u^c can arise from H^0(O(1, 0)) = 3bar, while d^c arises by conjugating a right-handed H^2(O(-4, 0)) = 3 with opposite abelian charges. Their pairings in the left-handed charge convention are (5/2, 1) and (5/2, -1). The equality of their images, and the unsigned linear formula P^{(2)} = 2j + 1, are therefore conditional on the common-chirality convention; the direct doublet obstruction above uses neither.

Scope and direction. This concerns the Dolbeault cohomology of a single abelian line bundle for each field on CP^2 x CP^1, with the canonical formal shift and fluxes in the span of Y and B - L. Nonabelian bundles, additional charges or a common twist, multiple source fields, walls, orbifolds and brane states remain possible directions. This is an independent check of the scoped obstruction and makes no claim about those alternatives.

Independent proof of the third-charge exclusion in Proposition 2.5. *From the referee report on the 11D line, section 'Proposition 2.5: adding a third charge', transferred verbatim in substance on 4 Nov 2026.*

Status as the referee stated it: the argument proves the original bounded search, a third integer charge in [-3, 3]^6 under the common-chirality convention. The same argument covers an arbitrary real third charge and independently chosen ten-dimensional chiralities; the referee stated that extension as a conjecture awaiting an independent check by another model, and the referee note at Proposition 2.5 records that check by Claude Sonnet 5.5.

Write, on (Q, u^c, d^c, L, e^c, nu^c),

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y = 6Y = (1, -4, 2, -3, 6, 0),    b = 3(B - L) = (1, -1, -1, -3, 3, 3).
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(Here b is a charge vector, unrelated to the Euler-class coefficient b of Proposition 2.3.) For the stated colour convention the necessary CP^2 pairings are

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p = (-5/2, 5/2, 5/2, p_L, p_e, p_nu),   |p_L| = 3/2,
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and the CP^1 pairings have the form

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w = (2 s_Q, s_u, s_d, 2 s_L, s_e, s_nu),   s_i in {+1, -1}.
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The signs allow independently chosen source chiralities. Reversing the colour convention reverses p and leaves the argument unchanged.

Suppose that p and w both lie in the span of y, b and z, for some third charge z. The Q and L entries of y and of b have ratio -3, so every vector of span(y, b) has L entry equal to -3 times its Q entry, while |w_L/w_Q| = 1; hence w lies outside span(y, b). Then span(y, b, w) has dimension three and lies in span(y, b, z), so the two spans coincide, and

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p = a y + b_0 b + t w
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for real a, b_0, t. Apply the combination 2Q + u^c + d^c, the linear form v -> 2 v_Q + v_u + v_d. It vanishes on y (2 - 4 + 2 = 0), on b (2 - 1 - 1 = 0) and on p (-5 + 5/2 + 5/2 = 0), and takes the value 4 s_Q + s_u + s_d on w. So

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0 = t (4 s_Q + s_u + s_d).
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The factor in parentheses is nonzero, since |4 s_Q| = 4 > 2 >= |s_u + s_d|; hence t = 0. Equality of the u^c and d^c entries of p = a y + b_0 b, -4a - b_0 = 2a - b_0, gives a = 0, and their common value -b_0 = 5/2 gives b_0 = -5/2. The resulting L entry is -3 b_0 = 15/2, contradicting |p_L| = 3/2. The same four fields suffice without nu^c.

Scope and direction. This concerns the one-line-bundle-per-field Dolbeault construction, the specified Y and B - L charges and the canonical formal shift. A fourth charge direction, a changed common twist, nonabelian bundles, multiple source fields, walls, orbifolds and brane states remain directions to investigate.

Independent proof of Proposition 2.3. *From the referee report on the 11D line, section 'Proposition 2.3: topology of the circle bundle', transferred verbatim in substance on 4 Nov 2026.*

The calculation is independent of the specific M^{111} identification. The cited Kreck-Stolz paper was unavailable to the referee, so no direct comparison with that paper is claimed.

Let B = CP^2 x CP^1, H^*(B; Z) = Z[h_1, h_2]/(h_1^3, h_2^2), and let the Euler class be e = a h_1 + b h_2, with a != 0. The Gysin sequence gives

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H^2(Y) = Z^2/<(a, b)> ~= Z ⊕ Z/g,   g = gcd(a, b).
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The map H^2(B) -> H^4(B), cup product with e, sends h_1 to a h_1^2 + b h_1 h_2 and h_2 to a h_1 h_2 (since h_2^2 = 0); in the bases (h_1, h_2) and (h_1^2, h_1 h_2) its matrix is

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[[a, 0], [b, a]].
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Its determinant is a^2 != 0, so its kernel is zero, and since H^3(B) = 0 the Gysin sequence gives H^3(Y) = 0. The greatest common divisor of its entries is g, so its Smith invariants are g and a^2/g, and H^4(Y), its cokernel, is Z/g ⊕ Z/(a^2/g): a group of order a^2, cyclic when g = 1.

Spin. The vertical tangent line is trivial, spanned by the generator of the circle action, so w_2(Y) = pi^* w_2(B) = pi^*(h_1 mod 2), using w_2(CP^2) = h_1 mod 2 and w_2(CP^1) = 0. The mod-two Gysin sequence H^0(B; Z/2) -> H^2(B; Z/2) -> H^2(Y; Z/2), whose first map is the cup product with e mod 2, shows that the kernel of the pullback on H^2 is the span of e mod 2. Thus Y is spin exactly when h_1 ≡ e mod 2, that is, when a is odd and b is even. For (a, b) = (3, 2), the primitive homomorphism (x, y) -> 2x - 3y is onto Z and has kernel generated by (3, 2), so it identifies H^2(Y) with Z; and H^4(Y) ≅ Z/9.

Independent proofs for Theorem 2.1 and Propositions 2.2 and 2.4: freeness, the neutral-monomial congruence, the dual action and the Sasaki-Einstein normalization. *From the referee report on the 11D line, section 'Theorem 2.1 and Propositions 2.2 and 2.4: quotient and tensor action', transferred verbatim in substance on 4 Nov 2026.*

Hypotheses: the identification with M^{111} is conditional on the recalled description of M^{111} as the circle bundle of class 3h_1 + 2h_2; the cited paper could not be accessed by the referee.

Freeness and the bundle. On the moment level in the unit sphere, |U|^2 = 3/5 and |V|^2 = 2/5. Both factors are nonzero, so the stabilizer of a point under (lambda^2 U, lambda^{-3} V) has lambda^2 = lambda^3 = 1, hence lambda = 1, and the action is free. The quotient has dimension 9 - 1 - 1 = 7 and is simply connected by the homotopy sequence of U(1) -> S^5 x S^3 -> Y_{3,2}. The residual character of the two Hopf phases has primitive weight (3, 2); its circle bundle over CP^2 x CP^1 therefore has Euler class ±(3h_1 + 2h_2). Principal circle bundles over this base are classified by their Euler classes, which gives the conditional identification with M^{111}. The zero sum 3(2) + 2(-3) = 0 gives the Gorenstein holomorphic cone with trivial canonical bundle; a Ricci-flat metric needs a separate argument.

The congruence of Proposition 2.4. The congruence 2t + 3d + 6Y ≡ 0 (mod 6) holds on U, where (t, d, 6Y) = (-1, 0, 2), and on V, where (t, d, 6Y) = (0, 1, -3), and extends additively to tensor products and to conjugates, which reverse all three quantities. For a monomial in U, V and their conjugates, let Delta_U be the number of factors U minus the number of factors Ubar, and Delta_V likewise for V. Neutrality under U(1)_Y reads

```
2 Delta_U - 3 Delta_V = 0,
```

so 3 | Delta_U and 2 | Delta_V. The triality of the monomial is -Delta_U ≡ 0 (mod 3) and d ≡ Delta_V ≡ 0 (mod 2): zero triality and integer weak isospin. This extends item (b) of Proposition 2.4 from the holomorphic monomials U^alpha V^beta to all monomials in U, V and their conjugates.

Dual action. If U is the vector representation 3bar and projective space parametrizes its lines, the conventional pullback action on holomorphic coordinate functions is the dual 3: functions of degree 3k transform as Sym^{3k}(3). Treating U as formal field components gives Sym^{3k}(3bar) instead. The convention must be fixed consistently when identifying the 10 versus the 10bar and the later cohomology table.

Sasaki-Einstein normalization. With the standard unit-sphere quotient metric, the two transverse Fubini-Study factors have relative scales 3/5 and 2/5. CP^2 is the base of the Hopf circle on the sphere of radius sqrt(3/5), with Ricci constant 6/(3/5) = 10, and CP^1 is the base of the Hopf circle on the sphere of

radius sqrt(2/5), a round sphere of radius sqrt(2/5)/2 with Ricci constant 4/(2/5) = 10; so each has Ricci constant 10. A seven-dimensional Sasaki-Einstein metric needs transverse Ricci constant 8. The Sasaki metric obtained by the standard transverse homothety of factor 5/4 has the same underlying circle bundle and the required Einstein normalization. Thus the existence of the Sasaki-Einstein metric on that manifold is compatible with the quotient, and the initial round-sphere quotient metric is identified with it only after the rescaling.

Independent proof of Proposition 2.6, with the colour convention. *From the referee report on the 11D line, section 'Proposition 2.6: flux assignments for one generation', transferred verbatim in substance on 4 Nov 2026.*

Verdict of the referee: correct with amendments, conditional on global flux quantization. The bidegrees and the two linear solutions are correct with the cohomology convention of the file. With the colour action on the quotient fixed as in Theorem 2.1, that convention is conjugated and the first-component coefficients must be reversed.

The pairings. The charges of the left-handed fields are F = 1 on all; chi = +1 on Q, L and -1 on u^c, d^c, e^c, nu^c; and B - L = 1/3, -1/3, -1/3, -1, 1, 1 on Q, u^c, d^c, L, e^c, nu^c. For (c_0, c_1) = (-9/4, -3/4), direct substitution in P^{(1)} = c_0 chi + c_1 (B - L) and P^{(2)} = (3F + chi)/2 gives

```
P_Q = (-5/2, 2),   P_{u^c, d^c} = (5/2, 1),   P_L = (-3/2, 2),   P_{e^c, nu^c} = (3/2, 1);
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for instance P_Q^{(1)} = -9/4 - 1/4 and P_L^{(1)} = -9/4 + 3/4. For (c_0, c_1) = (-3/2, -3), the quark pairings are unchanged (-3/2 - 1 = -5/2 and 3/2 + 1 = 5/2), and the first components for L and for e^c, nu^c are +3/2 and -3/2 respectively. Subtracting the formal shift (3/2, 1) yields exactly the listed integral bidegrees. On CP^2, the nonzero cohomology of a line bundle O(k) is H^0(O(k)) for k >= 0, or H^2(O(k)) ≅ H^0(O(-k - 3))^* for k <= -3; CP^1 has the analogous H^0/H^1 formula. So O(-4, 1) gives (3, 2), O(1, 0) gives 3bar, O(-3, 1) and O(0, 1) give (1, 2), and O(0, 0) and O(-3, 0) give a singlet, in the convention of the Setting. All listed choices have even total Dolbeault degree (0 or 2), so the common-chirality convention gives the indicated chiral index.

The colour convention. If the base is P(3bar), as in the quotient of Theorem 2.1, then H^0(O(1)) = 3 and H^2(O(-4)) = 3bar, and the listed quark bundles carry the conjugates of the labelled colour representations. Replacing (c_0, c_1) by (9/4, 3/4) or (3/2, 3), with the second component unchanged, reverses the first pairings and exchanges O(1) with O(-4) for the quarks; this gives the labelled Q and u^c, d^c with the colour action of the quotient. Alternatively, the original table is valid for P(3), with the corresponding conjugate colour convention.

Without chi. The required weak-doublet pairings have absolute value 2. Write P^{(2)} = alpha F + beta (B - L), so P_Q^{(2)} = alpha + beta/3 and P_L^{(2)} = alpha - beta. If the two signs for Q and L agree, then beta = 0 and |alpha| = 2, and u^c gets |P^{(2)}| = 2, the wrong absolute value for a weak singlet. If they disagree, say alpha + beta/3 = 2 and alpha - beta = -2, then beta = 3, alpha = 1, and u^c gets P^{(2)} = 1 - 1 = 0.

Without B - L. Q and L have identical charges F = 1, chi = +1, so their pairings coincide, while they require colour pairings of absolute values 5/2 and 3/2 respectively.

These obstructions use only absolute values, so they also allow independent source chiralities.

Global data. The final line-bundle degrees are integral, but K^{1/2} itself is formal and the displayed coefficients include fractional flux components. An actual background requires a specified global gauge group, charge lattice, spin^c determinant and compatible flux quantization. The index calculation alone establishes neither those data nor the physical interpretation of the grading as a gauge connection.

Part 40

Dualities between nine and eleven dimensions, adapted to the spectral triple

Lemma 40. Written as file 07 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 7.1 to 7.3 are proved here by hand; the string-theory statements of section 1 are recalled without rereading; the readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (historical/reviews/2026-11-04_terra_check_11d.md, historical/reviews/2026-11-04_sonnet_second_check_11d.md).

1. The dualities (recalled)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The duality descriptions are conditional on the recalled sources. The M2 wrapping number and the auxiliary torus flux are separately specified; a map identifying them remains open. See the referee report.

1. **M-theory on T^2 and type IIB on S^1** (Schwarz, hep-th/9508143; Aspinwall, hep-th/9508154). The modulus tau of the torus is the IIB coupling and SL(2, Z) is IIB S-duality; the IIB radius scales as a negative power of the area A of the torus. M2-branes wrapped on T^2 are IIB momentum states, and momenta on T^2 are (p, q)-strings wound on S^1. Eleven dimensions at A -> infinity, ten (IIB) at A -> 0, nine in between.
2. **Sixteen supercharges, rank 17** (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209). The two heterotic strings on S^1 are T-dual through Wilson lines; their type I' form is an interval with two orientifold planes O8 and sixteen D8-branes whose positions are the Wilson-line eigenvalues; at strong coupling, M-theory on an interval times a circle, with E8 walls.
3. **Sixteen supercharges, lower rank.** The CHL string (rank 9) and M-theory on the Möbius strip; a rank-1 component related to the Klein bottle (Aharony, Komargodski and Patir, hep-th/0702195); further components with discrete theta angles (Montero and Parra de Freitas, arXiv:2209.03361).
4. **Light-cone M-theory.** On a light-like circle with momentum N, M-theory is the U(N) quantum mechanics of nine Hermitian matrices (Banks, Fischler, Shenker and Susskind, hep-th/9610043; Susskind). On T^2 with a light-like component of C_3, the torus becomes noncommutative and SL(2, Z) acts by Morita equivalence (Connes, Douglas and Schwarz, hep-th/9711162).

2. What each one offers the finite geometry

duality	9	11	the finite triple
1	the nine-dimensional theory	large torus	the lowest Landau level of a torus with flux p: algebra M_p(C), J from charge conjugation with doubling (Proposition 7.3); towers = Landau levels; p = wrapped M2-branes = IIB momentum

duality	9	11	the finite triple
2	type I' interval, or heterotic on S^1	Hořava-Witten on S^1	two sheets = two orientifold walls; D_F = Wilson lines = D8 positions (Propositions 6.1, 7.1, 7.2); A_F from Chan-Paton algebras; J from the orientifold
3	rank-reduced nine-dimensional theories	M-theory on non-orientable surfaces	J as an orientation reversal; the reflection with antiperiodicity of Proposition 1.1 is the Möbius-type ingredient
4	nine transverse matrices, Spin(9)	large N	A_F as the commutant of block-diagonal matrices: stacks of 1, 2, 3 give C + M_2(C) + M_3(C); H_F in off-diagonal blocks; D_F = Γ^i ad(X^i), masses = separations

Duality 2 is the one with an exact meeting point with the corpus, through the angle of the top channel.

3. Proposition 7.1 (the angle of a channel lives on an interval)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The spectral quotient and endpoint multiplicities are independently checked. Theorem 22.1 uses the strict interior; the antiperiodic endpoint is a limit, with a degenerate window. Equality of one channel spectrum does not by itself identify full physical backgrounds. See the referee report.

Statement. In the parent of lemma 22, a channel with singular value σ of the Yukawa block and angle θ = l σ has the signed spectrum {(n pi + θ)/l, (n pi − θ)/l : n in Z} (Theorem 22.1). As a set, this depends only on the class of θ in R modulo the group generated by θ -> −θ and θ -> θ + pi, whose fundamental domain is [0, pi/2]. At the endpoints:

1. θ = 0: the spectrum is n pi/l with a zero mode of multiplicity two (periodic);
2. θ = pi/2: the spectrum is (n + 1/2) pi/l, each value twice (antiperiodic), two copies of the spectrum of the chiral interval of Proposition 1.1; it is the limit X -> infinity of the wall data X = tan(θ), where the lowest eigenspace is degenerate, and in lemma 23 the point of antipodal gradings, where M_W = M_Z = 0 at fixed l with the fermion at pi/(2l).

Proof. The set is invariant under θ -> −θ (it exchanges the two progressions) and θ -> θ + pi (it shifts n). The orbit of θ meets [0, pi/2] in one point. The endpoint values are substitutions; for θ -> pi/2 the wall conditions Q_R(l) = X U_R(l), U_L(l) = −X^* Q_L(l) tend to U_R(l) = Q_L(l) = 0. The statements at θ = pi/2 are those of lemma 23 (Theorem 23.6 and the reading after Proposition 23.7). ■

So the parameter space of one channel of the finite Dirac operator, for a given parent, is an interval with two distinguished ends, and lemma 22's restriction l ||D_F|| < pi/2 is the choice of the fundamental domain.

4. Proposition 7.2 (the far end is the displaced D8-brane on the other wall)

Referee note (GPT-6.1 Terra, 4 Nov 2026). At $a = 1/2$ the full $SO(32)$ centralizer is $S(O(30) \times O(2))$, whose identity component is $SO(30) \times SO(2)$. The stated one-loop minimum uses the hypotheses of lemma 24 and $n_f > 0$; the D8 dictionary remains a proposal. (Second check.) In Spin(32)/Z_2 the component of

determinant -1 sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a + 1 holds for SO(32) matrices, and the lift has period 2. See the referee report.

Statement. Put a = θ/pi in [0, 1/2]. By Proposition 6.1 the channel spectrum is that of the states (30)_{±1} of SO(32) on a circle of radius R = 1/pi with the Wilson line W_a = 1_30 ⊕ R(2 pi a). The interval [0, 1/2] is the moduli interval of one Wilson-line eigenvalue modulo a -> −a and a -> a + 1; the unbroken group is SO(32) at a = 0 and SO(30) x SO(2) for 0 < a <= 1/2; at a = 1/2 the holonomy is −1 on the plane of SO(2), W_{1/2} = diag(1_30, −1_2), and the (30)_{±1} tower is (n + 1/2)/R, the antiperiodic tower of Proposition 1.1. The one-loop minimum of lemma 24, Theorem 9, θ = pi/2, is the point a = 1/2.

Proof. Proposition 6.1 and Proposition 7.1 with θ = pi a; the commutant of W_a in SO(32) for 0 < a < 1/2 is SO(30) x SO(2), and at a = 1/2 its identity component is still SO(30) x SO(2), since only two eigenvalues are −1. Lemma 24, Theorem 9, item 1, places the minimum at t = sin^2 θ = 1. ■

Reading (proposal). In type I' the Wilson line W_a is fifteen D8-branes, with images, on one orientifold plane and the sixteenth at distance proportional to a from it (proportional only at constant dilaton; see 10, Proposition 10.1); a = 1/2 puts it on the other orientifold plane (recalled: hep-th/9711098, hep-th/0108135). The dictionary of file 06 then reads: the top angle is the position of the top's D8-brane on the type I' interval, the two ends of the fundamental domain are the two orientifold walls, and the one-loop vacuum of the flat family of lemmas 23 and 24 carries the top brane to the far wall, where the tower of the stack-brane strings is antiperiodic. In supersymmetric type I' the D8 positions are flat directions; the potential of lemma 24 belongs to the non-supersymmetric flat family, the setting in which a force on the brane is expected.

Under the strong-coupling duality of item 2 of section 1, the D8 position is a Wilson-line eigenvalue of the heterotic string on S^1, and in eleven dimensions an E8 holonomy around the circle on a Hořava-Witten wall. So the same datum, the top angle, is a brane position at 9 and a holonomy at 11.

5. Proposition 7.3 (a torus with flux, doubled, has the colour form)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The signs, kernel dimensions and generated matrix corner check out, with a spin-line twist for nontrivial spin structures. The displayed action is degenerate and nonunital on the doubled Hilbert space, so the claimed unital finite triple needs an extra scalar summand. See the referee report.

Statement. Let T^2 be a flat torus of area A with a spin structure, L a line bundle of degree p >= 1 with a connection of constant curvature, and Lbar = L^{-1} with the conjugate connection. On H = L^2(S ⊗ L) ⊕ L^2(S ⊗ Lbar) put D = D_L ⊕ D_Lbar, gamma = gamma_S ⊕ gamma_S, and J(ψ, χ) = (Cχ, −Cψ), where C is the charge conjugation of the spin structure (antiunitary, C^2 = −1, C D = D C, C gamma_S = −gamma_S C, mapping S ⊗ Lbar to S ⊗ L). Let C^infinity(T^2) act by multiplication on the first summand only. Then:

1. J^2 = 1, J D = D J and J gamma = −gamma J, the signs of KO-dimension six, and the algebra satisfies order zero;
2. ker D = H^0(L) ⊕ H^1(Lbar) ≅ C^p ⊕ conj(C^p), with gamma = +1 on the first summand and −1 on the second; the other eigenvalues are the Landau levels, of order (p/A)^{1/2}, which go to infinity as A -> 0 at fixed p;
3. the compressions of the functions to the kernel generate M_p(C) on C^p ⊕ 0, so the window is the finite triple (M_p(C), C^p ⊕ conj(C^p), 0, J, gamma), with KO-6 signs and order zero: M_p(C) acts on the first summand and, through J, by its opposite on the second.

Proof. (1) J^2(ψ, χ) = (C(−Cψ), −C(Cχ)) = (ψ, χ) since C^2 = −1; JD = DJ and J gamma = −gamma J follow from the properties of C; order zero holds because the algebra acts on the first summand and J b J^{-1} on the second. (2) On a torus the spinor bundle is Λ^{0,*} up to a flat factor, so ker D_L = H^0(L) ⊕ H^1(L) = C^p ⊕ 0 by Riemann-Roch in genus one, and ker D_Lbar = 0 ⊕ H^1(Lbar) ≅ conj(C^p) by Serre duality; the Landau levels of a constant field of total flux p on area A scale as (p/A)^{1/2}. (3) The compression of a character e_{m,n} to H^0(L) is a nonzero multiple of a clock-and-shift Weyl operator U^m V^n with UV = e^{2 pi i/p} VU (Berezin-Toeplitz quantization of the torus; recalled; in the theta-function

basis the compression is a Gaussian factor times the Weyl word, checked for p = 2 to 5 in the second check), and these operators span M_p(C). ■

The construction is the torus analogue of lemma 16 (a doubled parent with restricted algebra and a J covering the flip; Example 16.9 for the twistor case), and its window at p = 3 has the form of the colour sector of the Standard Model triple: M_3(C) on the quarks and its opposite on the antiquarks. Berezin-Toeplitz quantization of the torus is known; the KO-6 doubling is the corpus device.

Reading for duality 1 (proposal). The area A -> 0 at fixed flux keeps the window and sends the tower to infinity, the towerless limit; A -> infinity brings the Landau levels down. In M-theory on T^2, flux and wrapping number of M2-branes on T^2 are IIB momentum states, so the dimension p of the window counts IIB momentum on the nine-dimensional circle, and the SL(2, Z) of the torus acts on the family by Morita equivalence of the noncommutative tori of Connes, Douglas and Schwarz when a C_3 background is on. Whether p = 3 for colour has a reason in this picture is open.

6. A remark on duality 4 (known)

For U(N) matrices X^i (i = 1, …, 9) that are block-diagonal, X^i = $\oplus_a$ *x_a^i 1*{n_a} with distinct points x_a in R^9, the commutant of all X^i is $\oplus_a$ *M*{n_a}(C), and the fermion operator Γ^i ad(X^i) on the adjoint, with the sixteen-component spinor of Spin(9), vanishes on the diagonal blocks and has eigenvalues ±|x_a − x_b|, each with multiplicity 8 n_a n_b per ordered block (a, b), that is 16 n_a n_b per unordered pair. With stacks of sizes 1, 2, 3 the commutant is C + M_2(C) + M_3(C), the complex form of A_F, the off-diagonal blocks are bifundamentals as in the Krajewski diagrams of the finite geometry, and the masses are the distances between stacks; coincident stacks restore U(N), the analogue of massless W and Z, and large N is the eleven-dimensional limit. A quaternionic factor H needs an orientifold projection on the two-stack, and chirality needs more than the nine Hermitian matrices.

6b. The three labels in type I' variables (proposal)

Correction (4 Nov 2026). The length l of H and the type I' length are T-dual, and the limit at label 11 below ends at ten-dimensional SO(32). See 10, Proposition 10.3.

Hypothesis H has two parameters for the top channel: the interval length l and the angle θ = l m_t in [0, pi/2] (Proposition 7.1). In the type I' reading of Proposition 7.2 they are the length of the interval and the position of the top's D8-brane as a fraction of it. Then:

- **label 9:** θ = pi/2 at fixed l, the brane on the far orientifold wall: the top at the tower scale pi/(2l) (lemma 22, Corollary 22.9: W, Z and top at the tower scale in the 10 -> 9 family), the antiperiodic tower, and the one-loop minimum of the flat family (lemma 24);
- **label 10:** 0 < θ < pi/2, the brane between the walls, a window below the tower;
- **label 11:** θ -> 0 together with l -> infinity, the joint family of 01 (θ_epsilon ~ epsilon^{1 − alpha}): the brane returns to the stack, SO(32) is restored on that channel, and the interval opens.

In this reading the 9 and 11 endpoints are the two ends of the brane's path, the far wall and the stack, and the second scale required by Proposition 1.5 must fix how l grows as the brane returns. The flat one-loop potential pushes the brane to the far wall (label 9); reaching label 11 needs a term that pulls it back to the stack while the interval grows.

7. What this adds, and what remains open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and E_8-holonomy readings below are proposals. Proposition 7.3's displayed algebra action is nonunital; the corrected finite datum is in the referee report. Proposition 1.5 also permits flat directions.

Added: the angle of the top channel lives on an interval whose ends are the two orientifold walls of type I' (7.1, 7.2); the corpus's one-loop vacuum puts the top brane on the far wall, the SO(32) Wilson line diag(1_30, −1_2); the same datum is an E8 holonomy in the eleven-dimensional frame; a torus with flux gives the colour form of the finite triple with KO-6 signs (7.3).

Open: - A nine-dimensional spectral triple with sixteen D8 positions as the eigenvalues of D_F on the type I' interval, and its strong-coupling limit with E8 walls; the configuration 15 + 1 lies at the SO(32) end, while E8 needs seven D8-branes at an orientifold plane with divergent coupling. - The quaternionic factor H from a torus with even flux and a real structure of quaternionic type, and the full A_F from several tori or from a rank-three bundle with flux. - The lower-rank nine-dimensional components (duality 3), whose J-like orientation reversal may supply the real structure of the finite triple.

Referee proofs (GPT-6.1 Terra, 4 November 2026)

The arguments below are those of the referee report on the 11D line, written out in the notation of this lemma; the report in historical/reviews/ stays as the provenance record.

Independent proof of Proposition 7.1. *From the referee report on the 11D line, section 'Proposition 7.1: channel-angle fundamental domain', transferred verbatim in substance on 4 Nov 2026.*

The stated fundamental domain classifies the signed channel spectrum, including its endpoint multiplicities. The spectrum modulo pi/l consists of the two residues $\pm\theta/l$. Equality of two such multisets is equivalent to $\theta' \equiv \pm\theta \pmod{\pi}$: if $\theta' = \pm\theta + k\pi$ the progressions are relabelled, and conversely the residue sets $\{\pm\theta'\}$ and $\{\pm\theta\}$ modulo pi must agree. This gives the unique representative in [0, pi/2]. In the interior the two progressions are disjoint, since $n\pi + \theta = m\pi - \theta$ would need $2\theta \in \pi Z$. At zero they coincide on $\pi Z/l$; at pi/2 they coincide on $(Z + 1/2)\pi/l$. Thus every endpoint eigenvalue has multiplicity two, including the zero at the first endpoint.

The channel has two copies of the scalar antiperiodic circle spectrum at the second endpoint. Theorem 22.1 assumes the strict window $l\,||D_F|| < \pi/2$; $\theta = \pi/2$ is the corresponding boundary-condition limit, and its lowest level is degenerate. The massless electroweak vectors at that endpoint come from the specified two-grading model of lemma 23 and do not follow from the spectrum of an isolated channel. The endpoints require separate domains or limits, and the quotient of one spectrum need not identify complete physical backgrounds.

Independent proof of Proposition 7.2, with the full centralizer at a = 1/2. *From the referee report on the 11D line, section 'Proposition 7.2: Wilson-line endpoint and one-loop minimum', transferred verbatim in substance on 4 Nov 2026.*

For $0 < a < 1/2$, the nonreal conjugate eigenvalues $e^{\pm 2\pi i a}$ of the rotation force a commuting orthogonal transformation to preserve the two-plane of $R(2\pi a)$ and its complement, and to act on the plane by an element of O(2) commuting with a rotation by an angle other than 0 and pi, that is, by a rotation. Its centralizer in SO(32) is therefore SO(30) x SO(2). At $a = 1/2$ the holonomy is $\mathrm{diag}(1_{30}, -1_2)$; a commuting orthogonal transformation preserves both eigenspaces and is otherwise arbitrary on them, so the full centralizer is

$$S(O(30) \times O(2)) = \{(A, B) \in O(30) \times O(2) : \det A \det B = 1\},$$

with identity component SO(30) x SO(2) and a second component of pairs with $\det A = \det B = -1$. At $a = 0$ the centralizer is SO(32). Conjugation by a reflection in each subspace, an element of SO(32), sends a to -a, and $a \to a + 1$ leaves this SO(32) holonomy unchanged.

Minimum. For the determinant of lemma 24, write $t = \sin^2\theta$. Its two gauge terms are nonnegative and vanish at $t = 1$; the fermion term $-n_f F(t)$ is strictly decreasing for $n_f > 0$. Thus the unique global minimum is $t = 1$, that is $a = 1/2$. This conclusion assumes the flat massless determinant, its stated field content and wall conditions, and no additional angle-dependent term. It does not derive a force on a supersymmetric type I' brane, and the D8 interpretation is conditional on the proposed dictionary.

Independent proof of Proposition 7.3, items 1 and 2, and of the generation step in item 3. *From the referee report on the 11D line, section 'Proposition 7.3: doubled torus and the finite algebra', transferred verbatim in substance on 4 Nov 2026.*

Zero modes. Let S_0 be the flat spin line of the chosen spin structure, with $S_0^2 \cong O$, so that $S \otimes L = \Lambda^{\{0,*\}}(L \otimes S_0)$. The two zero-mode spaces are

H^0(L ⊗ S_0) and H^1(L^{-1} ⊗ S_0).

The first has dimension p: a negative-degree line bundle has no holomorphic sections, so by Serre duality (K trivial on the torus) H^1(L ⊗ S_0) = 0, and Riemann-Roch on a genus-one curve gives χ(L ⊗ S_0) = p. For the same reason H^0(L^{-1} ⊗ S_0) = 0, and Serre duality identifies H^1(L^{-1} ⊗ S_0) with the dual of H^0(L ⊗ S_0^{-1}) = H^0(L ⊗ S_0), hence with its complex conjugate after choosing the inner product. With the stated orientation these have opposite gradings. The untwisted formula of item 2, H^0(L) ⊕ H^1(Lbar), is literal for the trivial spin structure; the dimensions hold for all four.

J and the levels. The assumed charge conjugation C intertwines the two coupled Dirac operators and reverses the grading. Consequently the displayed J satisfies J^2 = 1, JD = DJ and J gamma = -gamma J. In a constant field B = 2 pi p/A, the nonzero Dirac levels have magnitudes sqrt(2|B| n), n >= 1, in the standard Dirac normalization.

Compression. For a reciprocal lattice vector k, split the position operator as x = R + , with guiding-centre part R and oscillator part , which commute with each other and satisfy [_1, _2] = -[R_1, R_2], of modulus 1/|B|. Then e^{ik · x} = e^{ik · R} e^{ik · }, and the oscillator-vacuum matrix element of e^{ik · } is e^{-|k|^2/(4|B|)}. So projecting multiplication by e^{ik · x} onto the lowest Landau level gives

P e^{ik·x} P = e^{-|k|^2/(4|B|)} e^{ik·R} P,

the nonzero Gaussian factor times a guiding-centre translation. For primitive reciprocal vectors k_1, k_2, with |k_1 x k_2| = (2 pi)^2/A and 1/|B| = A/(2 pi p), the two translations U = e^{ik_1 · R} P and V = e^{ik_2 · R} P are unitary on the lowest Landau level and form a clock-and-shift pair, UV = e^{±2 pi i/p} VU. Their p^2 products U^m V^n, 0 <= m, n < p, span M_p(C): the commutation relation forces tr(U^m V^n) = 0 unless m ≡ n ≡ 0 (mod p), so the p^2 products are orthogonal for the trace inner product. Flat spin and connection holonomies change only phases, and leave the nonzero Gaussian factor. Compression generates this algebra, and the compression map itself is not multiplicative on the function algebra.

Counterexample (referee) to the unital finite triple of Proposition 7.3, item 3, and the unital repair. *Same section of the referee report, transferred verbatim in substance on 4 Nov 2026.*

On V ⊕ Vbar, with V = H^0(L ⊗ S_0) ≅ C^p, the generated representation is (a) = a ⊕ 0. Thus (1) = 1_V ⊕ 0, and both the parent and the finite representations are degenerate on the second summand. They do not have the unital representation required in the finite-data setting of lemma 22. In particular, for p = 3, no complex-linear unital M_3(C) representation on C^3 ⊕ conj(C^3) can satisfy order zero with a unital opposite action: the two commuting unital actions would make this six-dimensional space a module for M_3(C) ⊗ M_3(C)^op ≅ M_9(C), whose nonzero modules have dimensions divisible by nine.

One repair is to unitize the represented corner:

A = M_p(C) ⊕ C, (a,) = a ⊕ I_{Vbar}.

In conjugate bases J (b,)^* J^{-1} = I_V ⊕ b^T, which commutes with (a,). This yields a unital finite KO-6 datum with D = 0 and order zero. It has an additional scalar summand; deriving the full Standard Model algebra and its representation remains open.

Second-check derivation (Claude Sonnet 5.5, 4 November 2026)

Spin lift of W_{1/2} and connectedness of the heterotic centralizer (Proposition 7.2). *From the second check on the 11D line (historical/reviews/2026-11-04_sonnet_second_check_11d.md, batch 4, item B2), adopted in the note at Proposition 7.2 and written out on 4 Nov 2026.*

Let G_1, ..., G_32 generate the Clifford algebra of R^32, with G_i G_j + G_j G_i = 2 _ij, and let pi: Spin(32) -> SO(32) be the covering map, under which g~ acts on vectors by v -> g~ v g~^{-1}. The rotation by angle 2 pi a in the plane of the coordinates 31 and 32 lifts to

W~_a = exp(pi a G_31 G_32) = cos(pi a) + sin(pi a) G_31 G_32,

using (G_31 G_32)^2 = -1. At a = 1/2 the lift of W_{1/2} = diag(1_30, -1_2) is W~ = G_31 G_32, up to the sign of the covering. The product of the reflections in e_1 and in e_31 lies in the component of

S(O(30) x O(2)) with det A = det B = -1, and lifts to g~ = G_1 G_31. Since G_1 anticommutes with G_31 and with G_32, it commutes with their product, and

```
g~ W~ g~^{-1} = G_1 G_31 (G_31 G_32) G_31 G_1 = G_1 G_32 G_31 G_1 = G_32 G_31 = -W~.
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Every other element of that component is the product of this one with an element of SO(30) x SO(2), whose lifts are products of exp(φ G_i G_j/2) with i, j <= 30 or {i, j} = {31, 32}, all commuting with G_31 G_32. So every lift of the determinant -1 component conjugates W~ to -W~. The computation involves only G_1, G_31 and G_32 and holds in every even dimension.

Let = G_1 G_2 … G_32 be the volume element. It is central in Spin(32), ^2 = 1, != ±1, and pi( ) = -1. The heterotic group is the quotient of Spin(32) by {1, } (or by {1, - } for the other chirality convention; the argument is the same). The class of g~ centralizes the class of W~ exactly when g~ W~ g~^{-1} ∈ {W~, W~}. The case W~ would give pi(g~) W pi(g~)^{-1} = -W, which is impossible, since W and -W have the eigenvalue -1 with multiplicities 2 and 30. So pi(g~) lies in S(O(30) x O(2)). On the determinant -1 component g~ W~ g~^{-1} = -W~, and -1 differs from 1 and from . Hence the centralizer of the class of W~ in Spin(32)/Z_2 is the image of the preimage of SO(30) x SO(2) in Spin(32). That preimage is connected, since it contains the path exp(φ G_1 G_2/2), φ ∈ [0, 2 pi], from 1 to -1. So the heterotic centralizer is connected. Finally W~_{a+1} = -W~_a: the SO(32) holonomy has period 1 in a, and its lift, in Spin(32) and in Spin(32)/Z_2, has period 2.

Part 35

The sign of lemma 06 as an orientifold type, and E8 walls from the configuration 15 + 1

Lemma 35. Written as file 08 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line of research (README, PLAN, task C, next step of 07). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 8.1 is proved (it adds a commutant statement to lemma 06); Proposition 8.2 is proved from Proposition 7.1; Remark 8.3 rests on recalled string-theory rules and on the dictionary of 06; readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (historical/reviews/2026-11-04_terra_check_11d.md, historical/reviews/2026-11-04_sonnet_second_check_11d.md).

1. Setting

In type I' the two ends of the interval are orientifold planes. A stack of n Chan-Paton labels (k D8-branes with their images, n = 2k) on an orientifold plane of type O^- carries SO(n), and on one of type O^+ carries USp(n) (recalled; the physical O8 planes of the SO(32) web are O8^-). Lemma 06 factors a KO-6 finite triple into an odd one through a grading-reversing unitary involution U, commuting with the algebra and with D, with U J = s J U; s = −1 gives KO-5 (label 9) and s = +1 gives KO-7 (label 11), and the antiunitary j induced on H_+ satisfies j^2 = s and commutes with d^2 = D^2|{H+} (lemma 06, proof of Result 2 and the check of the electrodynamics data). STRATEGY T2c: with a realistic seesaw sector only s = +1 occurs.

2. Proposition 8.1 (the factorization sign fixes orthogonal or symplectic symmetry)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The antiunitary commutants are independently checked. They need not preserve *d* or the remaining data, and the Chan-Paton identification is a proposal. See the referee report.

Statement. Let E be an eigenspace of d^2 in H_+, of dimension n. The unitaries of E that commute with j form O(n) if s = +1, and USp(n) if s = −1, in which case n is even.

Proof. j preserves E because it commutes with d^2. An antiunitary with j^2 = +1 is a real structure: E = E_R ⊗ C with E_R its fixed points, and a unitary commutes with j exactly when it preserves E_R, which gives O(n). An antiunitary with j^2 = −1 is a quaternionic structure: E is a right H-module of quaternionic dimension n/2, and the unitaries commuting with j form USp(n). ■

The even dimension for s = −1 is lemma 06's own statement (Kramers degeneracy of the singular values). The proposition adds the symmetry group.

Reading (proposal). The groups are those of Chan-Paton labels on an orientifold plane: s = +1 with O^-(SO), s = −1 with O^+ (USp). So the two odd factorizations of the modern triple, which carry the labels 11 and 9, have the form of the two kinds of orientifold planes. The seesaw sector selects s = +1 (T2c), which is the O^- type of the SO(32) web used in 06 and 07: the physical neutrino sector and the type I' walls of the dictionary agree in type. Two refinements from the second check: the symmetries of the pair (j, d) are smaller, O(n_+) x O(n_-) for s = +1 and U(n/2) for s = -1, since for s = -1 the antiunitary j swaps the +-mu eigenspaces of d; and there is a tension of labels, since here label 9 goes with O^+ while section 6b of 07 puts label 9 on a far O8^- wall (the two are the KO sign and the position of the top's brane, see 10).

3. Proposition 8.2 (wall channels come with doubled multiplicity)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated count is for one signed eigenvalue. The lowest $|D|$ multiplicity is $2k$ in the interior and at zero, and $4k$ at $\pi/2$; interpreting the signed count as Chan-Paton labels requires further data. See the referee report.

Statement. In the parent of lemma 22, k channels at a common angle θ contribute, at the eigenvalue θ/l of the window, an eigenspace of dimension k if 0 < θ < pi/2 and of dimension 2k at the ends θ = 0 and θ = pi/2.

Proof. By Proposition 7.1 each channel has the spectrum {(n pi ± θ)/l}; the value θ/l occurs once (n = 0, sign +) for 0 < θ < pi/2, and twice at the ends: at θ = 0 as the zero mode of multiplicity two, at θ = pi/2 as (0 + pi/2)/l = (pi − pi/2)/l. ■

Reading (proposal). k D8-branes in the bulk carry k Chan-Paton labels and k D8-branes on an orientifold plane carry 2k, with their images; the parent reproduces this count, with the ends of the angle interval as the walls. With Proposition 8.1, a stack of channels at a wall has a symmetry of type SO(2k) or USp(2k), as a stack of D8-branes on an O8^- or O8^+ plane. The rank-reduced nine-dimensional theory with one O8^- and one O8^+ (duality 3 of file 07) would then be a parent whose two walls carry opposite signs s: one sheet of type 11 and the other of type 9.

4. Remark 8.3 (E8 walls from 15 + 1; conditional)

Correction (4 Nov 2026). The far-wall group is reached at the string-scale radius R = R_*, with the tower fixed, and only its charged states are D0-branes. See 10.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and slope arithmetic is independently checked conditional on the recalled rule. Wall enhancement requires tuning $z = 0$; for $7 + 7$ the two bulk positions must sum to L. The E_2 identification with the physical electroweak sector needs further data. The cited primary sources were unavailable to the referee; the slope rule and the E_{n+1} table were confirmed against Polchinski-Witten and Bergman-Gaberdiel-Lifschytz in the second check. See the referee report.

Conditions: the dictionary of 06 (fifteen D8-branes for the five light quarks in three colours, the sixteenth for the top) and the recalled rule that an O8^- plane with n <= 7 D8-branes on it, at infinite coupling, carries E_{n+1} (E_4 = SU(5), E_5 = SO(10), E_6, E_7, E_8), while n >= 8 admits no such point (Seiberg, hep-th/9608111; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; recalled).

1. Dilaton bookkeeping: the dilaton is e^{phi} = z^{−5/6} with z piecewise linear along the interval (conformal coordinate; Polchinski and Witten, hep-th/9510169) and slope proportional to 8 − n_left, where n_left counts the D8-branes to the left (O8^- carrying −8 units), so the coupling grows toward the wall with fewer branes; this reproduces the constant dilaton of 8 + 8, the SU(2) at the empty wall of the SO(32) point (the self-dual radius of the heterotic string) and 7 + 7 for E8 x E8 (extrapolated

from the single-wall rule; neither Polchinski-Witten nor Bergman-Gaberdiel-Lifschytz states it). In the configuration 15 + 1 the wall with fifteen branes has no E point (15 > 7), and the coupling can diverge only at the far wall. With the top's brane on the far wall (the label-9 configuration of Proposition 7.2, n = 1) the far wall enhances to E_2 = SU(2) x U(1), the electroweak group; with the top's brane in the bulk it is empty and enhances to E_1 = SU(2). The light-quark wall keeps SO(30).

2. If branes move in colour triples (a light flavour is three D8-branes), a wall holds 3j branes, or 3j + 1 with the top's brane. The reachable wall groups are E_1, E_2 (the top's brane alone), E_4 = SU(5) (one flavour), E_5 = SO(10) (one flavour and the top's brane), E_7 (two flavours) and E_8 (two flavours and the top's brane); E_6 (five branes) is not reachable.
3. Hence an E8 wall needs the top's brane together with two light flavours, and two E8 walls are impossible with one top brane: E8 x E8 requires breaking a colour triple.

In this reading the top is what completes E8 at a wall (7 = 2 x 3 + 1), and with colour triples the proposed E8 x E7 arrangement has seven branes at one wall, six at the other and a bulk triple at x_b, while two E8 walls require relaxing the colour-triple rule (referee replacement): with that arrangement, the slopes are proportional to +1 on (0, x_b) and −2 on (x_b, L), so z vanishes at both walls when x_b = 2L/3. The E_n series also contains E_3 = SU(3) x SU(2) (two branes at a wall), which colour triples do not reach.

Reading (proposal). At strong coupling the wall that holds the top's brane in the label-9 configuration carries SU(2) x U(1), and the wall of the five light quarks in three colours carries SO(30). The electroweak group then appears at the top's wall, where the M-theory circle opens locally, and the quarks' flavour group at the other wall.

5. What remains open

- The bulk symmetry: k D8-branes in the bulk carry U(k), while Proposition 8.1 gives O or USp on every eigenspace of d^2; the relation between the window's degenerate levels in the bulk and U(k) needs the parent's own structures in addition to j.
- The dilaton: a spectral-triple counterpart of the type I' dilaton profile (a coupling varying along the interval, with slope set by the brane charge to one side) and the infinite-coupling points of Remark 8.3. Proposition 9.1 of 09 writes the profile as a trace of a function of |D_F|, only implicitly after 10, Proposition 10.1, and the gradient is not the second scale in the scope of Propositions 10.5 and 10.6 there.
- The physical status of E8 x E7 with colour triples, and whether the top's brane at an E8 wall has a counterpart in the finite geometry.

Referee proofs (GPT-6.1 Terra, 4 November 2026)

The arguments below are those of the referee report on the 11D line, written out in the notation of this lemma; the report in historical/reviews/ stays as the provenance record.

Independent proof of Proposition 8.1, with its caution. *From the referee report on the 11D line, section 'Proposition 8.1: antiunitary commutants', transferred verbatim in substance on 4 Nov 2026.*

On each finite-dimensional eigenspace E of d^2, the commutant of j in the unitary group of E is O(n) for j^2 = 1, and USp(n) for j^2 = -1, with even complex dimension n in the latter case.

For j^2 = 1, the fixed points of j form a real subspace E_R with E = E_R ⊕ i E_R, and the inner product is real on E_R, since <jv, jw> = conj<v, w>. A real orthonormal basis of E_R is an orthonormal basis of E and identifies j with complex conjugation; the unitaries commuting with complex conjugation are the real orthogonal matrices, O(n). For j^2 = -1, a unit vector v and jv are orthogonal: antiunitarity gives <jv, j^2 v> = conj<v, jv> = <jv, v>, and j^2 v = -v gives <jv, j^2 v> = -<jv, v>, so <jv, v> = 0. The span of v and jv is j-invariant, and so is its orthogonal complement; induction gives a quaternionic orthonormal basis v_1, jv_1, ..., v_m, jv_m, with n = 2m. The unitaries commuting with j are then the compact symplectic group USp(n).

Caution. This commutant is defined using j and the d^2-eigenspace alone. It need not commute with d, with the represented algebra, or with other physical structures. For example, on C^2 let j be complex conjugation

and d = diag(1, -1): the commutant on the d^2 = 1 space is O(2), and a generic rotation does not commute with d. Moreover O(n) contains disconnected transformations, while SO(n) is its identity component for n >= 2. The orientifold comparison therefore requires an explicit Chan-Paton representation and compatibility with the rest of the data.

Independent proof of Proposition 8.2, with the multiplicities of |D|. *From the referee report on the 11D line, section 'Proposition 8.2: endpoint channel multiplicities', transferred verbatim in substance on 4 Nov 2026.*

For k independent channels at the same angle, the signed eigenvalue theta/l has multiplicity k in the interior and 2k at either endpoint. For 0 < theta < pi/2, only n = 0 in the positive progression gives +theta/l: (n pi + theta)/l = theta/l forces n = 0, and (n pi - theta)/l = theta/l forces theta = n pi/2, outside the interior. At zero both progressions have n = 0. At pi/2 the positive n = 0 and negative n = 1 branches coincide. This proves the claimed signed count.

The multiplicity of the corresponding lowest eigenvalue of |D| is 2k in the interior (the values ±theta/l, k each), 2k at zero, and 4k at pi/2 (the values ±pi/(2l), 2k each). At zero the two interior signed eigenvalues simply coalesce; at pi/2 the next level also meets the window. Thus the signed count alone does not establish a doubling of independent physical Chan-Paton labels or an orientifold projection.

Independent proof of the arithmetic of Remark 8.3. *From the referee report on the 11D line, section 'Remark 8.3: wall bookkeeping and enhancements', transferred verbatim in substance on 4 Nov 2026.*

The slope and brane-count arithmetic below is conditional on the recalled type I' rules; the referee could not access the primary sources, and the second check later confirmed the slope rule and the E_{n+1} table against Polchinski-Witten and Bergman-Gaberdiel-Lifschytz (note at the remark). An enhancement requires tuning the coupling to a zero of z, and identifying the resulting E_2 factor with the physical electroweak sector requires additional data.

Assume z' = c (8 - N), with c > 0, N(x) the number of D8-branes to the left of x, and z > 0 in the open interval.

- *8 + 8.* N = 8 in the interior, so the interior slope is zero.
- *16 + 0.* z' = -8c, and the empty far wall reaches z = 0 only when z(0) = 8cL.
- *7 + 7.* With the other two branes at x_1 < x_2 in (0, L), the slopes are c, 0 and -c on (0, x_1), (x_1, x_2) and (x_2, L). Both walls have z = 0 precisely when

  ```
  z(0) = 0,    x_1 + x_2 = L,
  ```

  since z(L) = z(0) + c x_1 - c (L - x_2). Then z is positive in the interior. Thus "7 + 7 for E8 x E8" includes a coupling and a position condition.
- *Isolated zeros.* With n branes at a wall where z = 0, the slope next to the wall is c (8 - n) at x = 0 and c (n - 8) at x = L, so z > 0 inside requires n <= 7: an isolated zero with a positive interior requires at most seven branes at that wall under this slope rule.
- *15 + 1.* With the single brane at L, the slope is c (8 - 15) = -7c, so the far-wall zero is at z(0) = 7cL. Assuming the recalled E_{n+1} rule gives E_2, with Lie algebra su(2) ⊕ u(1), at that tuned wall. The rule alone does not determine the Standard Model charges, couplings, global group or embedding.

Colour triples. If every light flavour moves as a triple and there is a single top brane, a wall holds 3j or 3j + 1 branes, so the allowed wall counts up to seven are 0, 1, 3, 4, 6, 7. Only a wall containing the top can have seven (7 = 2 x 3 + 1); two such walls require relaxing the triple constraint or the single-top assumption. For seven branes at 0, six at L, and the remaining triple at x_b, N = 7 on (0, x_b) and N = 10 on (x_b, L), so the slopes are +c and -2c. With z(0) = z(L) = 0, c x_b - 2c (L - x_b) = 0 gives x_b = 2L/3, and

```
z(x) = c x            for 0 <= x <= 2L/3,
z(x) = 2c (L - x)     for 2L/3 <= x <= L.
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The resulting E8 x E7 interpretation is conditional on the recalled enhancement rule and the proposed brane dictionary.

Part 36

The type I' dilaton as a spectral function, the SO(32) of the EPJC paper, and two more dualities

Lemma 36. Written as file 09 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026, after the steer of the same day (the coincident-brane route is overused; the type I' dilaton and the colour-flavour idea are worth pursuing; compare with the SO(32) of the EPJC paper). Proposition 9.1 is proved from the recalled type I' slope rule; Proposition 9.2 is proved from representation theory and the recalled orientifold projections; Proposition 9.3 is proved (linear algebra and Connes's sign table); the comparison of section 1 is read from the arXiv text of the paper through a summarizing fetch; the rest are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (historical/reviews/2026-11-04_terra_check_11d.md, historical/reviews/2026-11-04_sonnet_second_check_11d.md).

Correction (same day): the first version stated the slope rule for e^{−phi}. The piecewise-linear function is z, with e^{phi} = z^{−5/6}; Proposition 9.1 holds as stated with z in place of e^{−phi}, and infinite coupling is z = 0.

1. The SO(32) of A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The branching algebra is independently checked. The archived original abstract also states an additional chiral charge-+4/3 quark per generation. The full paper was unavailable for direct comparison, so its detailed attribution remains pending. See the referee report.

That paper uses the adjoint of SO(32) to classify the scalars of a supersymmetric Standard Model with three generations, as composites: pairs of quarks as colour triplets and quark-antiquark pairs as singlets and octets, with three extra scalars of charge +4/3. Its pattern is SO(2n) ⊃ SO(n_1) x SU(n_2) x SU(3) x U(1) with 2n = n_1 + 6 n_2, which for SO(32) and n_2 = 5 gives SO(2) x SU(5) x SU(3) x U(1): the 30 = 6 n_2 are the five light quarks u, c, d, s, b ("turtles") in three colours with conjugates, and the top is the non-participating "elephant". Its explicit route also passes through SO(32) ⊃ SU(16) x U(1), with 496 = 1 + 120 + 120bar + 255 (quarks from the 120s, leptons from the 255), and it mentions open strings, orientifolds and an SU(9) for M2 and M5.

Reading of the LaTeX source (Claude Sonnet 5.5, 4 Nov 2026; the earlier comparison used a summarizing fetch). The paper uses colour singlets for mesons and colour antitriplets for diquarks, and keeps only (1, 24, 1), (1, 15, 3bar) and (1, 15bar, 3) of its decomposition; colour octets and sextets lie outside it. The top is the favoured non-binding quark ("we have not a concrete argument"), and the postulates force the non-binding quark to be up-type. The chain is SU(16) x U(1) to SU(15) to SU(5) x SU(3), with SO(30) and SO(32) reached by "adding singlets". The identification of the top with the SO(2) factor, the O8^- plane and the Chan-Paton reading belong to these files, and "orientifold" in the paper means planar orientifold QCD. The footnote on the 84 + 12 split and SU(9) for M2 and M5 anticipates Proposition 21.3 and 04. The abstract specifies a chiral +4/3 quark per generation; the body has three scalars of charge +4/3 and leaves their fermion partner open. The paper gives (15, 3bar) as the antisymmetric square of one fundamental without stating chirality, so the question of Proposition 9.3 stays open; its remark on odd-parity diquarks is consistent with that proposition. The paper's selection of an up-type quark by its postulates and lemma 21's by the spectral-action values are independent arguments with the same outcome.

Coincidence. The chain SO(32) ⊃ SO(2) x SO(30) ⊃ SO(2) x SU(15) x U(1) ⊃ SO(2) x SU(5) x SU(3) x U(1), the 30 as five light quarks in three colours, and the top as the factor SO(2), are the same as lemma 21 and the dictionary of 06. Lemma 21 records the branching 496 -> (435, 1) + (1, 1) + (30, 2) and leaves SO(32) unexplored; the EPJC paper explores it through composites, and the files 06 to 09 through type I' branes. What the 11D files add: the type I' realization of the SO(2) as the top's D8-brane and of the 30 as the Chan-Paton labels of fifteen D8-branes on an O8^- wall (06); the top angle as the brane's position and the one-loop vacuum on the far wall (Propositions 7.1, 7.2); the orientifold type O^- matched by the seesaw sign (Proposition 8.1); the far wall's enhancement to SU(2) x U(1) at infinite coupling (Remark 8.3); the dilaton (Proposition 9.1); the diquark counts as strings between the stack and its image, which need the O^- type (Proposition 9.2); and the real-structure condition under which these counts arise, as connection components along the interval, outside the finite inner fluctuations (Proposition 9.3). In type I' language the paper's other chain, SU(16) x U(1), is the configuration with all sixteen D8-branes coincident in the bulk, and the chain with SO(2) x SO(30) is the configuration with fifteen on a wall and one apart; the second is the one the corpus selects. The SU(9) of M2 and M5 in the paper is the grading of e8 used in 04.

2. Proposition 9.1 (the type I' coupling function is a trace of a function of the finite Dirac operator)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The integration is independently checked conditional on the recalled slope rule. The trace is over sixteen selected channels, and the full physical D_F Hilbert space enters only through a specified projection onto them; direct comparison with the cited source was unavailable to the referee. (Second check, against Polchinski-Witten.) The formula z(L) = z(0) - c (8L - sum x_k) is exact in the positions; the dictionary x_k = (2L/pi) l m_k holds only for constant z, and in general theta_k = (pi/2) W(x_k)/W(L) with W = int z^{-1/3} (10, Proposition 10.1). In the profile 15 + 1 the endpoint values 7cL and 8cL stand, and theta_t = pi/4 gives 7.358 cL in place of 7.5 cL. See the referee report.

Setting. Type I' on an interval [0, L], in the conformal coordinate x, with sixteen D8-branes at x_1, ..., x_16 and O8^- planes at the ends. Recalled rule (Polchinski and Witten, hep-th/9510169): the dilaton is e^{phi} = z^{−5/6} for a function z that is continuous and piecewise linear in x, with slope c (8 − N(x)), where N(x) is the number of D8-branes to the left of x and c > 0; the coupling diverges where z vanishes. Dictionary (06, 07): x_k = (2L/pi) θ_k with θ_k = l m_k the angles of sixteen channels of the finite Dirac operator (fifteen light-quark channels with colour, and the top).

Statement.

1. z(x) = z(0) + c Σ_k min(x/2, x_k − x/2) for x in [0, L], so z is z(0) plus c Tr g_x(l |D_F|) over the sixteen channels, with g_x(θ) = min(x/2, (2L/pi) θ − x/2): a trace of a function of the finite Dirac operator, as the spectral action is.
2. At the far wall, z(L) = z(0) − c (8L − Σ_k x_k).
3. Infinite coupling at the far wall (z(L) = 0) holds exactly when z(0) = (2cL/pi)(4 pi − l Σ_k m_k): the coupling at the light-quark wall is fixed by the sum of the sixteen channel masses, the trace of |D_F| on these channels.

Proof. *∫0^x N(y) dy = Σ_k (x − x_k)+*, so z(x) = z(0) + c (8x − Σ_k (x − x_k)+) = *z(0) + c Σ_k (x/2 − (x − x_k)+)*, and x/2 − (x − x_k)_+ = min(x/2, x_k − x/2). Item 2 is x = L with Σ_k (L − x_k) = 16L − Σ x_k. Item 3 sets z(L) = 0 and substitutes x_k = (2L/pi) l m_k. ■

Correction (4 Nov 2026). The dictionary x_k = (2L/pi) theta_k is exact only for constant z; the map from angles to positions is nonlinear. See 10, Proposition 10.1.

Reading (proposal). With the light quarks near θ = 0, Σ_k m_k ≈ m_t and the condition is z(0) ≈ (2cL/pi)(4 pi − θ_t): 7cL at label 9 (θ_t = pi/2) and 8cL at label 11 (θ_t -> 0). The coupling therefore varies along the interval on a scale set by the spectrum of D_F, which is the kind of second scale that Proposition 1.5 requires to select a joint family; whether the type I' coupling can play that role in the corpus's flat family is the next question. Since the slope decreases with x, z is concave, so z > 0 on [0, L) once z(0) > 0 and z(L) = 0; positivity of z(0) gives l Σ_k m_k < 4 pi, a bound on the trace of |D_F| in units of the tower scale, met when only the top is heavy (θ_t <= pi/2).

3. Proposition 9.2 (diquarks as strings between the stack and its image)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The representation and charge counts are independently checked, conditional on the recalled projections. The shifts describe the specified gauge-field channels; Λ^2 of a line excludes that component only, and leaves the other open-string states and the strings to other branes. See the referee report.

Setting. V = C^5 ⊗ C^3, five light flavours in three colours, with Q = 2/3 on u, c and Q = −1/3 on d, s, b; the 30 of lemma 21 is V ⊕ Vbar. Fifteen D8-branes on an O8 plane carry the Chan-Paton space 30 (branes and images), with group SO(30) on an O8^- and USp(30) on an O8^+, and U(15) when the stack is moved off the plane (recalled).

Statement.

1. Under U(15), the adjoint of SO(30) is Λ²V ⊕ Λ²Vbar ⊕ (V ⊗ Vbar) and the adjoint of USp(30) is S²V ⊕ S²Vbar ⊕ (V ⊗ Vbar). The summands Λ²V or S²V and their conjugates are the strings between the stack and its image; V ⊗ Vbar are the strings with both ends on the stack.
2. Λ²V = (S²5, 3bar) ⊕ (Λ²5, 6) and S²V = (S²5, 6) ⊕ (Λ²5, 3bar).
3. In the colour antitriplet, Λ²V has the charge counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21, and S²V has (−2/3: 3, +1/3: 6, +4/3: 1). So the antisquark counts of three generations recorded in lemma 21 (6 of charge −2/3 and 6 of charge +1/3 in the 3bar) occur among the stack-image strings of an O8^- plane, while those of an O8^+ plane have the counts (3, 6, 1).
4. With the angles θ_k = l m_k of 06, a string from channel i to the image of channel j has the spectrum (n pi ± (θ_i + θ_j))/l, and a string from channel i to channel j has (n pi ± (θ_i − θ_j))/l, n in Z.
5. A single brane has no string to its own image on an O8^- plane (Λ² of a line vanishes) and one on an O8^+ plane.

Proof. 1: 30 = V ⊕ Vbar as a U(15) module, Λ²(V ⊕ Vbar) = Λ²V ⊕ Λ²Vbar ⊕ (V ⊗ Vbar) and likewise for S²; the orientifold projection keeps the antisymmetric part for SO and the symmetric part for USp, and the mixed summand V ⊗ Vbar is the U(15) adjoint. 2: Λ²(A ⊗ B) = (S²A ⊗ Λ²B) ⊕ (Λ²A ⊗ S²B) and S²(A ⊗ B) = (S²A ⊗ S²B) ⊕ (Λ²A ⊗ Λ²B), with Λ²3 = 3bar. 3: split 5 = 2 ⊕ 3 by charge; S²5 = S²2 ⊕ (2 ⊗ 3) ⊕ S²3 has dimensions 3, 6, 6 with charges 4/3, 1/3, −2/3, and Λ²5 = Λ²2 ⊕ (2 ⊗ 3) ⊕ Λ²3 has dimensions 1, 6, 3 with the same charges. 4: Proposition 6.1 with charges (+1, +1) and (+1, −1) under U(1)_i x U(1)_j, whose holonomy is e^{2 pi i (a_i ± a_j)}, and θ = pi a. 5: dimensions. ■

Reading (proposal). (11, Theorem 11.3 and Proposition 11.4, qualify the selection below: the window sees the counts (3, 6, 1) also at s = +1, through the commutator with the masses, and parent scalars with the counts (6, 6, 3) need M_15(C) at both walls.) The diquark counts of lemma 21 and of the EPJC paper are the stack-image strings of an O8^- plane, the type that Proposition 8.1 associates with the factorization sign s = +1 selected by the seesaw sector (T2c); the O8^+ alternative gives the wrong number of up-type antisquarks. So the seesaw sign and the scalar counts select the same orientifold type. Two features follow from the projection. Diquark masses go with the sum θ_i + θ_j and meson masses with the difference θ_i − θ_j, so when the stack leaves the wall the diquarks become massive and the mesons stay massless: the diquarks are tied to the orientifold. And the top's single brane has no string to its own image, while its strings to the stack, the (30, 2), have masses near m_t (Proposition 6.1); in the EPJC paper the top's exclusion is an input, and here the O8^- projection with one brane for the top removes the top-top channel only (item 5); the top-to-stack strings (30, 2) remain, with masses near m_t, so the rest of the exclusion is a mass statement, as in the paper.

4. Proposition 9.3 (the real structure decides between the two diquark counts)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated sign constraint excludes a nonzero antisymmetric block in the three KO labels. The signs permit all of S^2 V (second check: B ranges over exactly S^2 V for eps eps' = +1, with the diagonal block free); the algebra, grading and order conditions of a triple then select a subspace. Distinct-multiplet blocks remain related to their transposes. See the referee report.

Setting. E = V ⊕ Vbar with V as in Proposition 9.2, an antiunitary J on E that maps V onto Vbar and Vbar onto V (as particles and antiparticles are exchanged), J² = ε, and a Hermitian X on E with JX = ε'

XJ, the condition that inner fluctuations obey in a KO-dimension with signs (ε, ε'). Let B be the block of X from Vbar to V, an element of V ⊗ V.

Statement.

1. B is symmetric (B in S^2V) if εε' = +1 and antisymmetric (B in Λ^2V) if εε' = −1.
2. In the KO-dimensions 5, 6, 7 of the finite part for the labels 9, 10, 11, (ε, ε') = (−1, −1), (+1, +1), (+1, +1), so εε' = +1: the diquark block of every J-compatible fluctuation lies in S^2V, whose colour antitriplet ($\Lambda^2$5, 3bar) has the counts (−2/3: 3, +1/3: 6, +4/3: 1).
3. For a connection on a circle whose real structure is complex conjugation (KO-dimension 1, (ε, ε') = (+1, −1)), the block lies in Λ^2V, whose colour antitriplet ($S^2$5, 3bar) has the counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21. The SO(32) Wilson line of Proposition 6.1 is of this type.

Proof. Choose a basis of V and let K: V -> Vbar be coordinatewise conjugation; J(v, w) = (ε K^{-1} w, K v) has $J^2 = \varepsilon$. Write X = [[A, B], [B*, D]]. The component of JX = ε' XJ that maps V to V gives ε K^{-1} B* = ε' B K, so B = εε' K^{-1} B* K^{-1} = εε' B^T in coordinates. Under the U(15)-equivariant identification of Hom(Vbar, V) with V ⊗ V, symmetric matrices form S^2V and antisymmetric ones Λ^2V. Items 2 and 3 use Connes's sign table, ε = (+, +, −, −, −, −, +, +) and ε' = (+, −, +, +, +, −, +, +) for KO-dimensions 0 to 7, and the counts of Proposition 9.2. On the circle, D = −i d/dθ and J = conjugation give JD = −DJ, and a Hermitian connection A with JA = −AJ is i times a real antisymmetric matrix, an element of so(n), where the generator of R(2 pi a) lies. ■

Item 2 is a negative statement with the following scope: the blocks of J-compatible Hermitian operators between particles and antiparticles, for one multiplet V, in the three labels. It is the finite-geometry form of the symmetry of Majorana mass matrices, and of the Fermi-statistics constraint on a Lorentz-scalar bilinear ψ_i ψ_j of one Weyl multiplet without derivatives (recalled), under which the (15, 3bar) vanishes. Known escapes: item 3 (connection components along a circle or an interval), blocks between two distinct multiplets, which J does not constrain, and derivative couplings.

Reading (proposal). The counts of lemma 21 and of the EPJC paper, (6, 6, 3), belong to the antisymmetric block. In the corpus's setting this block arises as the component of an SO(30) connection along the circle or interval: the Wilson-line moduli of the type I' stack, whose stack-image part is Proposition 9.2. The finite triple alone, in any of the three labels, gives the symmetric block and the counts (3, 6, 1). So the composite scalars with the counts of lemma 21 point to the interval parent of lemma 22 and its gauge field, and the selection of O8^- in Proposition 9.2 concerns these connection components. This narrows the spin gap of Propositions 6.1 and 9.2: the Λ^2V states are scalars as circle components of a nine-dimensional gauge field, and they come with that field's vector and gaugino partners.

5. Two more dualities between 9 and 11 (recalled; proposals)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The E_1 alternatives need correction: the canonical-orthogonal root lattices distinguish $\mathbb{P}^1 \times \mathbb{P}^1$ from $dP_1 = \mathbb{F}_1$, and the usual local engineering assigns the latter to $\widetilde{E}_1$. The dP_2 and brane duality claims remain conditional on the unavailable primary sources. The Weyl action is a gauge relabelling of a holonomy, subject to the full background data. See the report on the del Pezzo comparison and Weyl action.

1. **E-type walls and del Pezzo surfaces.** The E_{n+1} symmetry of an O8^- plane with n D8-branes at infinite coupling is the symmetry of the five-dimensional theory on a D4-brane probe, which M-theory realizes on a Calabi-Yau threefold with a shrinking del Pezzo surface dP_{n+1} (Seiberg, hep-th/9608111; Morrison and Seiberg, hep-th/9609070; Intriligator, Morrison and Seiberg, hep-th/9702198). So the far wall with the top's brane (n = 1, E_2 = SU(2) x U(1)) has an eleven-dimensional description by a local dP_2, and the empty far wall (E_1 = SU(2)) by dP_1 or P^1 x P^1; the light-quark wall (n = 15) has none. The Witten base CP^2 x CP^1 of 02 is built from the same pieces (CP^2 = dP_0, and P^1 factors).
2. **Colour-flavour exchange as a T-duality.** In the heterotic string on S^1, the Weyl group of the gauge group acts on the Wilson lines as part of the T-duality group O(17, 1; Z). The colour-family transposition M of 05 is an element of the Weyl group of E8, so it is a T-duality of the nine-dimensional

E8 x E8 string, and through Hořava-Witten a symmetry of the E8 holonomy on a wall in eleven dimensions. As a Weyl element it acts as a global gauge transformation, so by itself it relabels gauge-equivalent vacua; it exchanges colour and family physically only relative to a structure that tells them apart, such as the placement of H_F in 05. The four-family requirement of Propositions 5.1 and 5.4 remains.

6. Open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The finite KO signs restrict the symmetric block without producing the full representation. Light interval scalar modes, their boundary conditions and the physical string dictionary remain to be constructed; the detailed EPJC comparison is pending.

- Whether the type I' coupling, as a spectral function of D_F (Proposition 9.1), supplies the second scale of Proposition 1.5 in the flat family of lemmas 23 and 24.
- The local dP_2 of the top's wall and its relation to the Witten base CP^2 x CP^1.
- Spin: in type I' the strings of Proposition 9.2 belong to nine-dimensional vector multiplets, with one real scalar each (the relative position), the same gap as in Proposition 6.1. By Proposition 9.3 a finite triple in the labels 9 to 11 gives S^2V, so the Λ^2V scalars must come from the connection along the interval; the next computation is the inner fluctuations of the parent of lemma 22 with an SO(30) algebra, and whether its interval component carries Λ^2V with the counts of lemma 21.
- Whether the (15, 3bar) of the EPJC paper could be read as bilinears of two distinct multiplets (the paper itself gives one fundamental with itself) (for example left-handed quarks and conjugates of right-handed ones arranged into one 30), which Proposition 9.3 does not constrain.

Referee proofs (GPT-6.1 Terra, 4 November 2026)

The arguments below are those of the referee report on the 11D line, written out in the notation of this lemma; the report in historical/reviews/ stays as the provenance record.

Independent proof of Proposition 9.1, with the scope of its trace. *From the referee report on the 11D line, section 'Proposition 9.1: integrated coupling profile', transferred verbatim in substance on 4 Nov 2026.*

Hypothesis: the recalled slope rule. The Polchinski-Witten source could not be accessed by the referee; the comparison with it is the second-check note at the proposition.

Let N(x) = sum_k 1_{x_k < x}. Integrating z' = c (8 - N) almost everywhere, with z continuous, gives

```
z(x) = z(0) + c (8x - sum_{k=1}^{16} (x - x_k)_+) = z(0) + c sum_{k=1}^{16} min(x/2, x_k - x/2),
```

since int_0^x N(y) dy = sum_k (x - x_k)+, *and x/2 - (x - x_k)+* = min(x/2, x_k - x/2) (both sides equal x/2 for x <= x_k and x_k - x/2 for x > x_k). At x = L, with every x_k <= L, this becomes

```
z(L) = z(0) - c (8L - sum_k x_k).
```

With x_k = 2 l L m_k/pi, the dictionary of the Setting (exact only for constant z, by the correction after the proof), the zero condition z(L) = 0 is z(0) = (2cL/pi)(4 pi - l sum_k m_k), as stated. The slopes c (8 - N(x)) are nonincreasing, so z is concave; if z(0) > 0 and z(L) = 0, then z(x) >= (1 - x/L) z(0) > 0 on [0, L).

Scope of the trace. The trace is over the sixteen selected channel values. The equality with a spectral trace assumes a chosen sixteen-dimensional channel space with eigenvalues m_k, or an explicitly normalized projection selecting those values. The physical D_F has additional chirality, conjugation, colour and generation multiplicities, which cannot be dropped from its full trace without specifying that projection. The assignment of the sixteen brane positions to those channels is also part of the proposal. This profile alone does not derive a stabilizing term for the interval modulus. The slope, coordinate and e^phi-to-z conventions are those of the recalled rule.

Independent proof of Proposition 9.2, items 2 and 3, with the scope of items 4 and 5. *From the referee report on the 11D line, section 'Proposition 9.2: diquark representations and string channels', transferred verbatim in substance on 4 Nov 2026.*

Hypothesis: the recalled orientifold projections. For V = C^5 ⊗ C^3, the swap of V ⊗ V is the product of the swaps of the two tensor factors, so its eigenspaces combine the symmetric and alternating projectors on the factors:

```
Λ²V = (S²C^5 ⊗ 3bar) ⊕ (Λ²C^5 ⊗ 6),
S²V = (S²C^5 ⊗ 6) ⊕ (Λ²C^5 ⊗ 3bar),
```

with $\Lambda^2 3 = 3\text{bar}$ and $S^2 3 = 6$. Their dimensions are 15 * 3 + 10 * 6 = 45 + 60 = 105 and 15 * 6 + 10 * 3 = 90 + 30 = 120, which are C(15, 2) and C(16, 2). Splitting the flavour space into two charge-2/3 and three charge-(-1/3) directions, $S^2C^5 = S^2 2 \oplus (2 \otimes 3) \oplus S^2 3$ and $\Lambda^2 C^5 = \Lambda^2 2 \oplus (2 \otimes 3) \oplus \Lambda^2 3$ give the antitriplet counts (6, 6, 3) in S^2C^5 and (3, 6, 1) in $\Lambda^2 C^5$, ordered by charges (-2/3, 1/3, 4/3). The full adjoints have dimensions 2(105) + 225 = 435 = dim so(30) and 2(120) + 225 = 465 = dim usp(30).

Scope of items 4 and 5. For the assumed Wilson-line action, the antisymmetric or symmetric block has weights a_i + a_j and the mixed block has weights a_i - a_j; hence their field-theory Kaluza-Klein momenta have the displayed shifts (n pi ± (θ_i ± θ_j))/l, with θ = pi a. This calculation leaves out the rest of the open-string spectrum, which also includes oscillator states and their orientifold projections. The sum-weight component with i = j is absent in Λ^2 and present in S^2; neutral diagonal modes remain. The claim of item 5, that a single brane has no string to its own image, holds in this sector. Absence of the same-line component leaves the strings between the top brane and other branes. Moving all stack branes by the same angle makes the mixed-block differences vanish; different individual positions can give nonzero differences. Identifying the resulting Chan-Paton modes with the proposed composites is an additional physical construction. The O8^-/O8^+ projection and global tadpole conditions were not independently verified, since the primary sources were unavailable to the referee.

Independent proof of Proposition 9.3, item 1, with the scope of item 2. *From the referee report on the 11D line, section 'Proposition 9.3: real-structure constraint on diquark blocks', transferred verbatim in substance on 4 Nov 2026.*

Choose an antiunitary identification K: V -> Vbar and write

```
J(v, w) = (eps K^{-1} w, K v),    X = [[A, B], [B*, D]].
```

Then JX(v, w) = (ε K^{-1}(B* v + D w), K(A v + B w)) and XJ(v, w) = (ε A K^{-1} w + B K v, ε B* K^{-1} w + D K v). The V -> V component of JX = ε' XJ, the coefficient of v in the first entry, gives

```
eps K^{-1} B* = eps' B K,   hence   B = epseps' K^{-1} B* K^{-1} = epseps' B^T,
```

the last equality in coordinates in which K is coordinatewise conjugation. Thus εε' = +1 forces B ∈ S^2V, and εε' = -1 forces B ∈ Λ^2V. For the displayed KO-5, KO-6 and KO-7 conventions the products are all +1. The zero block belongs to both subspaces, so "symmetric if εε' = +1" states the symmetry forced by the sign. For V = C^5 ⊗ C^3 the decomposition of S^2V and the counts are those of the proof of Proposition 9.2: $\Lambda^2 C^5$ has multiplicities 3, 6, 1 at charges -2/3, 1/3, 4/3, and S^2C^5, hence the colour-3bar part of Λ^2V, has 6, 6, 3.

Scope of item 2. A J-compatible finite fluctuation can occupy only a subspace of S^2V; the algebra, grading and order conditions determine which blocks actually occur. For example, with D = 0 every inner one-form vanishes, despite the symmetry condition. The KO signs alone therefore exclude a nonzero antisymmetric block and leave the full symmetric scalar spectrum to be established. For a circle operator D = -i d/dx with ordinary conjugation J, JD = -DJ. A Hermitian connection satisfying the same sign is purely imaginary and antisymmetric in a real basis; the complexified so(30) representation contains Λ^2V and its conjugate. Whether these components survive the particular interval boundary conditions as light scalars requires a separate construction. For distinct multiplets, an off-diagonal block B_12 may be chosen freely, while the full constraint still relates it to B_21 = εε' B_12^T.

Scope and direction. The exclusion concerns a Hermitian J-compatible operator between one multiplet and its conjugate, with the stated signs. Distinct multiplets, other real structures or signs, derivative couplings, and circle or interval connection components remain possible directions; each requires its own field and boundary data.

Independent proof of the branching in section 1. *From the referee report on the 11D line, section 'File 09, section 1: the SO(32) comparison', transferred verbatim in substance on 4 Nov 2026.*

The alternative chain follows from 32_C = 16_{+1} ⊕ 16bar_{-1} under U(16) ⊂ SO(32): so(32)_C = $\Lambda^2$32 = $\Lambda^2$16 ⊕ $\Lambda^2$16bar ⊕ (16 ⊗ 16bar), so

```
496 = 120_{+2} ⊕ 120bar_{-2} ⊕ 255_0 ⊕ 1_0.
```

Under SU(15), 16 = 15 ⊕ 1 and 120 = Λ^2(15 ⊕ 1) = 105 ⊕ 15. With 15 = 5 ⊗ 3, the plethysm of Proposition 9.2 gives

```
105 = Λ²(5 ⊗ 3) = (15, 3bar) ⊕ (10, 6),
```

with $S^2$5 = 15 and $\Lambda^2$5 = 10, and 15 ⊗ 15bar = (5 ⊗ 5bar, 3 ⊗ 3bar) = (24, 8) ⊕ (24, 1) ⊕ (1, 8) ⊕ (1, 1). The antitriplet charge multiplicities are those checked in Proposition 9.2. These group products are Lie-algebra shorthand; global embeddings involve finite central quotients.

Scope. The SO(2) plane is a colour singlet. Assigning it to a top brane is a label in the proposed dictionary and does not by itself construct a colour-triplet top field. Chan-Paton bilinearity and matching representation counts likewise do not by themselves establish dynamical quark composites. The referee could not access the paper itself; the comparison with its text is the reading of the LaTeX source in section 1.

Counterexample (referee) to the unqualified E_1 surface identification in section 5, item 1. *From the referee report on the 11D line, section 'File 09, section 5: del Pezzo wall correspondence', transferred verbatim in substance on 4 Nov 2026.*

The two degree-eight surfaces have different root lattices orthogonal to the canonical class. On dP_1, let H^2 = 1, E^2 = -1, H · E = 0 and K = -3H + E. If D = aH + bE is orthogonal to K, then D · K = -3a - b = 0, so b = -3a and D^2 = a^2 - b^2 = -8a^2. There is no integral class with D^2 = -2. On P^1 x P^1, with ruling classes F_1, F_2 (F_1^2 = F_2^2 = 0, F_1 · F_2 = 1) and K = -2F_1 - 2F_2, the difference F_1 - F_2 is orthogonal to K and has square -2. These surfaces therefore cannot be interchanged in the root-lattice comparison. In the usual local engineering description (recalled), P^1 x P^1 gives E_1, whereas dP_1 = F_1 gives the distinct $\tilde{E}$_1 theory.

For dP_2, with exceptional classes E_1, E_2 and K = -3H + E_1 + E_2, the class E_1 - E_2 is orthogonal to K and has square -2; the orthogonal lattice has rank two, consistent with an A_1 factor and an additional abelian direction. The dP_2 comparison is compatible with the expected A_1 root. This computation alone does not construct the corresponding five-dimensional field theory or the type I' duality, and the brane-to-geometry map remains conditional on the cited dualities.

Sources. The distinction between the two degree-eight engineering models is also stated in a secondary geometric-engineering discussion. The Seiberg and Morrison-Seiberg primary sources were unavailable to the referee for direct comparison. The occurrence of CP^2 and CP^1 in the separate six-dimensional base does not by itself identify that compactification with a local dP_2.

Independent proof of the Weyl action in section 5, item 2. *From the referee report on the 11D line, section 'File 09, section 5: Weyl action on a Wilson line', transferred verbatim in substance on 4 Nov 2026.*

A Weyl transformation of E8 acts by conjugation on an E8 holonomy and gives a lattice automorphism in the heterotic circle frame. Whether it acts as a symmetry of a fixed compactification depends on all of its background and boundary data.

Let T ⊂ E8 be a maximal torus. An element w ∈ W(E8) = N(T)/T has a representative g ∈ N(T), and the constant gauge transformation by g sends a holonomy W ∈ T to g W g^{-1} ∈ T. Thus the holonomy alone determines the same gauge orbit. The corresponding isometry w ⊕ 1 of E8 ⊕ E8 ⊕ II_{1,1} is an element of the automorphism group of the nine-dimensional Narain lattice. Proposition 5.2 places the transposition M in that Weyl group.

If a bundle, a defect, an algebra embedding or a distinguished set of zero modes is held fixed, the same g need not preserve it. Transforming the entire configuration produces a gauge-equivalent description when the transformation is allowed; an exchange relative to fixed colour and family structures needs an explicit

action on those structures. The root no-go applies to its stated three-slot configuration, while bundle zero modes remain an open escape.

Part 37

The type I' dilaton against the second scale of Proposition 1.5

Lemma 37. Written as file 10 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, after the merge of branch 11D into master, to answer the last item left open in 09: whether the dilaton can be the second scale that Proposition 1.5 requires. Sources read from the arXiv LaTeX: J. Polchinski and E. Witten, hep-th/9510169, section 3; O. Bergman, M. Gaberdiel and G. Lifschytz, hep-th/9711098, section 3. Numbers in Python (numpy, scipy, mpmath), scripts not kept by the rule of the project. The Codex quota is spent, so no Terra referee is available. An independent reading by Claude Sonnet 5.5, with the error and amendments below, is in the review. The amendments were rechecked by Claude Opus 5.5 before being applied, and the corrected text has not been read again.

Correction (4 Nov 2026). The first version counted 21 gauge generators with integer towers. Lemmas 23, 24 and 26 use sp(2) (+) su(4), with 25 generators: 16 with integer towers at theta = 0 and 9 with half-integer towers. The bracket for one generation is 41.3125, so the near-cancellation 64 - 63 stated in the first version came from the wrong count, and it is withdrawn. So is the small value theta_* = 0.202 that followed from it.

Outcome

1. **In the scope stated below, the dilaton gradient does not supply the second scale.** Along any path that opens the circle, its relative variation falls as 6/R^2 (Proposition 10.5). It enters as a power of alpha'/l^2, and that can matter only if the leading Casimir coefficient a_0 vanishes (Proposition 10.6).
2. **a_0 is now computed**, which was the open item of 01: 8 pi^2 a_0 = (3 zeta(5)/16)(4 n_D' - 3 (n_i - (15/16) n_h) - n_s - 5 n_g), with n_i and n_h the generators with integer and half-integer towers at theta = 0. For the corpus content it is positive, so at v = 0 the interval runs to large length, in the direction H wants.
3. **The dictionary of 06 to 09 needs three corrections** (Propositions 10.1 to 10.4). The map from angles to positions is nonlinear. The tower of Proposition 6.1 is a type I' winding state, so the length l of H and the type I' length are T-dual, and l -> infinity shrinks the type I' interval at fixed heterotic coupling. And the configuration 15 + 1 has a Romans mass in the bulk, which bounds length times coupling, so the classical profile has no eleven-dimensional regime.
4. **A scale-free action can still separate the window**, with a fixed small ratio (in lemma 30 the ratio tends to zero), if the light mass is exponentially small in a stabilized shape parameter (Remark 10.7), which amends the last sentence of Proposition 1.5.

The direction: the second scale must enter through a bulk energy depending on the electroweak order parameter, through a cancellation a_0 = 0, or through a fixed shape with an exponentially small ratio, as in lemma 29. These targets are stated in Proposition 10.6 and Remark 10.7. None is derived.

The dictionary against the source

Polchinski and Witten give, in conformal gauge, g = Omega^2(x) eta with Omega = C z^{-1/6}, e^phi = z^{-5/6}, and z = 3 C (B mu_8 - nu_0 x)/sqrt 2, with nu_0 = (N(x) - 8) mu_8 piecewise constant. This

confirms the slope rule recalled in 08 and 09. We write z for this function up to its constant, normalized so that its slope is exactly 8 - N(x), which fixes the normalization in the formula for R below, with x in [0, L].

Proposition 10.1 (the map from angles to positions). Put W(x) = int_0^x z^{-1/3} dy. By Polchinski-Witten, equations (mass) and (wilmap), the Wilson-line eigenvalue of a brane at x_k is lambda_k = W(x_k) / (2 W(L)), that is

```
theta_k = pi lambda_k = (pi/2) W(x_k) / W(L),
```

and with L = 2 pi, in their units alpha' = 2, R^2 = (3/4) int z^{1/3} dx / (int z^{-1/3} dx)^2. So the linear dictionary x_k = (2L/pi) theta_k of Proposition 9.1 holds exactly when z is constant (8 + 8), and asymptotically for large R.

Proof. The string stretched from the wall to x_k has mass (1/(2 pi alpha')) int_0^{x_k} Omega^2 dx, since its area element is Omega^2 dx^0 dx^9, and Omega^2 is proportional to z^{-1/3}. A string winding from 0 to L and back has the mass 1/R. The ratio gives lambda_k. The formula for R is their equation (rad), reduced as in their equation (dil). ■

Consequence for Proposition 9.1: its items 1 and 2 hold as statements about z in terms of the positions x_k. Reading them as a trace of a function of |D_F| needs the map above, which depends on the whole configuration, so the coupling function is a trace only implicitly. Item 3 changes accordingly.

Proposition 10.2 (15 + 1 with infinite coupling at the far wall). Take fifteen branes at x = 0 and one at x_t, with z(L) = 0, and put rho = (z(x_t)/z(0))^{2/3} in [0, 1]. Then, with alpha' = 2,

```
theta_t = (pi/2) 8 (1 - rho) / (8 - rho),        R_*^2 = 14 (8 - rho^2) / (8 - rho)^2.
```

So R_* runs from sqrt 2, for the top's brane on the stack, to sqrt 7 / 2, for the top's brane on the far wall, which are the values (1/2)|n - 8|^{1/2} of Polchinski and Witten at n = 16 and 15. In units of the string length, R_*/sqrt(alpha') lies in [0.935, 1]. In this chamber, R >= R_*.*

Proof. Integrate z^{1/3} and z^{-1/3} on the two segments, with slopes -7 and -8, and use z(x_t) = 8 (L - x_t). ■ Python agrees at x_t/L = 0.5, with theta_t = 0.5859 and R^2 = 1.9652, against 0.7854 for the linear dictionary.

So infinite coupling sits at the *smallest* radius of the chamber, at the string scale, and large R is large B, the weakly coupled region. Polchinski and Witten say this directly: "When B is large, the coupling e^{phi(x^9)} is small everywhere and the Type I' theory is weakly coupled; in this range the heterotic radius R is large".

Proposition 10.3 (T-dual lengths). The tower of Proposition 6.1, n/R, is a type I' winding tower: "a heterotic Kaluza-Klein state which is dual to a Type I' winding state" (Polchinski and Witten). With constant dilaton, the proper length L_p of the type I' interval and the radius satisfy L_p R_I = pi alpha' on the type I side, with alpha' that of type I'. So with l = pi R_I, measured in the type I' metric, l L_p = pi^2 alpha'. For the heterotic radius in the heterotic metric, l L_p = pi^2 alpha' sqrt(lambda_h), by Bergman, Gaberdiel and Lifschytz, with lambda_h the heterotic coupling.

Hence the walls of H, at distance l, and the O8 planes, at distance L_p, coincide only at the self-dual length. In particular, the reading "the interval opens" in section 6b of 07 has to be restated: at fixed heterotic coupling, l -> infinity shrinks the type I' interval, and it decompactifies the SO(32) circle.

The endpoint of that limit is the ten-dimensional SO(32) theory (heterotic or type I), since at fixed lambda_h the type I' coupling falls as 1/(sqrt(lambda_h) R_h). *Recalled, and supported in the two sources only by remarks of Bergman, Gaberdiel and Lifschytz at the constant-dilaton point:* the eleven-dimensional limit of the rank-17 moduli space is M-theory on an interval times a circle, with E8 x E8 at the walls, which by Remark 8.3 is out of reach with colour-respecting moves. If so, label 11 of section 6b is not reached by theta -> 0, l -> infinity in the chamber 15 + 1.

Proposition 10.4 (Romans mass, length and coupling). On a segment with N branes to the left, the proper distance is linear in the inverse coupling:

```
L_p = (C_L l_s / |8 - N|) (e^{-phi_1} - e^{-phi_2}),     C_L = 8 pi / 5,
```

with C_L from the standard normalization of the Romans mass, F_0 = (8 - N)/(2 pi l_s). So in the configuration 15 + 1, g(0) L_p <= C_L l_s / 7 for the total length. Near an infinite-coupling wall the coupling is g = C_L l_s / (|8 - N| s) at proper distance s, and the region g >= g_0 has length (3/4) (C_L / |8 - N|) g_0^{-4/3} in eleven-dimensional Planck units, which is 0.54 g_0^{-4/3} for N = 15. Therefore no region of the classical 15 + 1 background has both strong coupling and a length resolved in eleven-dimensional units.

Proof. Omega dx = C z^{-1/6} dz / (a C (8 - N)) integrates to a multiple of z^{5/6} = e^{-phi}. ■ This agrees with Bergman, Gaberdiel and Lifschytz, who conclude from the D0-brane bound states that they "expect neither to be applicable to the N != 0 backgrounds", meaning an M-theory interpretation or a matrix theory, and that "only the combination of eight D8-branes and an orientifold plane is known to arise in M-theory". Here their N is 8 - N. It holds wherever N != 8. The scope is the classical profile, which is not controlled where g is of order one, so this is a statement about the extrapolated background.

The far-wall electroweak group. Remark 8.3 places SU(2) x U(1) at the far wall at infinite coupling. By Bergman, Gaberdiel and Lifschytz, the states that become massless there are D0-branes at the wall, heterotic winding states. Let n be the number of branes at the strong wall, here n = 1 with the top's brane there, placed at their x^9 = 0 by reflecting the interval. Combining their equations (windingmass) and (D0mass), with N' = 8 - n, A = (B + 2 pi N')^{2/3} and b = B^{2/3},

```
M_D0 R = (N'/2) b / (A - b) = R^2 - R_*^2    (alpha' = 2),
```

which vanishes at B = 0, where R = R_*, and grows like R^2. Only W^+ and W^- can be such states: a neutral Z is a Cartan combination, and the rank of a nine-dimensional theory with sixteen supercharges is fixed, so no massive neutral vector arises this way. So in the chamber 15 + 1 with the top's brane on the far wall, if the W^+- are these D0 states, they are massless at the string-scale radius, with the tower fixed, and they become heavier as R grows: M_W l rises with l. This is the fixed-tower path of the audit (its section 2.2), and away from it the W moves the wrong way for hypothesis H. So on one path, in this scope, the dictionary can place the electroweak group at the far wall, or the opening of H on the circle of Proposition 6.1, but not both. Escapes: other chambers; the W as a Wilson-line string between D8-branes, as in lemma 23; the top's brane in the bulk, with the far wall giving E_1.

The test

Proposition 10.5 (the gradient is a string-scale effect). With L = 2 pi and alpha' = 2, for large R, R^2 ~ 3 B / (8 pi), and the relative variation of z across the interval is at most 16 pi / B ~ 6 / R^2. So along a joint family, R_eps ~ eps^{-alpha}, this variation falls as eps^{2 alpha}, the map of Proposition 10.1 linearizes, and the dilaton is constant in the limit.

Proof. z is B plus a function of size at most 8 L = 16 pi. Insert in the formula for R. ■

So the corrections to any potential from the gradient are powers of alpha'/l^2. In the supersymmetric type I' background there is no potential at all: D8 positions, B and C are flat directions. A potential needs broken supersymmetry, as in the flat family of lemmas 23 and 24.

The coefficient a_0. Write 8 pi^2 l^4 V = A_0 + 3 F(2u) + (3/2) F((16/5) u) - n_f F(t), the potential of lemma 24 with its theta-independent part A_0 restored, where n_f counts twice the Dirac species at frequency one (lemma 26). The relative terms vanish at theta = 0, and each tower at theta = 0 is half of a circle tower, periodic or antiperiodic:

```
A_0 = (3 zeta(5) / 16) (4 n_D' - 3 (n_i - (15/16) n_h) - n_s - 5 n_g),
```

with n_D' = 2 N_D counting twice all Dirac species, of any frequency (for frequency zero the tower is the theta = 0 tower), n_i and n_h the five-dimensional gauge generators with integer and half-integer towers at theta = 0, n_s the real bulk scalars with integer towers, and n_g = 1 if the five-dimensional graviton is included. This assumes that the bulk vacuum energy and the wall tensions are tuned away, since they carry the terms of order Lambda^5 l and Lambda^4.

Proof. A real degree of freedom on a periodic circle of length 2 l has V = -3 zeta(5)/(4 pi^2 (2 l)^4), and the

interval tower n >= 0 gives half of this, -3 zeta(5)/(128 pi^2 l^4). In the normalization of lemma 24 this is -(3/16) zeta(5) per real degree of freedom. A half-integer tower has the factor -15/16 relative to an integer one, so a generator with mixed conditions contributes +3 (15/16) (3/16) zeta(5). For a fermion channel the sign is opposite, with four degrees of freedom per unit of n_f. The check: the shift to theta = pi/2 multiplies a tower by -15/16, so the difference is (31/16)(3/4) zeta(5) = 93 zeta(5)/64 = F(1), which is the value computed in Proposition 1.4. ■

In lemmas 23, 24 and 26 the gauge field is valued in sp(2) (+) su(4), with 25 generators. At theta = 0, sixteen have integer towers: 8 gluons, 3 W_L, 1 hypercharge combination with Neumann conditions at both walls, and the 4 odd generators O with Dirichlet conditions at both, whose interval components are the Higgs quaternion. Nine have mixed conditions and half-integer towers: 2 R^+-, 6 X and 1 orthogonal neutral direction. So n_i - (15/16) n_h = 121/16. For three generations, with 24 Dirac species, the bracket is 192 - 22.6875 = 169.3125 before gravity and bulk scalars, and A_0 = 32.92. So a_0 > 0: at v = 0 the length runs to infinity, in the direction of H, and the first half of the modulus target of the audit, a r^{-p} with a > 0 and p = 4, is present. For one generation the bracket is 41.3125.

Proposition 10.6 (when the gradient can be the second scale). Let V = l^-4 [G_0(theta) + (alpha'/l^2) G_1(theta) + …], theta = l m, with G_0(0) = a_0 and G_0 = a_0 + b theta^2 + … near 0.

1. If a_0 != 0, then at small m the leading balance is the scale-free one of Proposition 1.5: alpha = 1 or a runaway. The term in alpha'/l^2 changes neither.
2. If a_0 = 0, G_1(0) = a_1 > 0 and b < 0, there is a minimum at l^4 = 3 a_1 alpha' / (|b| m^2), so alpha = 1/2: 1/l is the geometric mean of m and the string scale, up to constants, and m « 1/l « 1/l_s.
3. If supersymmetry is exact at theta = 0, then V = 0 at theta = 0 at all orders, a_1 = 0, and V = m^2 l^-2 H(theta, alpha'/l^2). Then l is fixed by H independently of m, or runs away, and no joint family results, unless the stationarity condition has a root interval (a flat direction), as in the counterexample to the universal form of Proposition 1.5.

So in the scope of a potential analytic in alpha'/l^2 around the flat family, with corpus content, the gradient is not the second scale. Item 2 would need a Bose-Fermi balance at theta = 0 that is not a supersymmetry at theta = 0, with about 169 further real bosonic degrees of freedom for three generations.

Proof. Items 1 and 2: differentiate; in item 2, V = b m^2/l^2 + a_1 alpha'/l^6 has V'' = 8 |b| m^2/l^4 > 0 at the stationary point. Item 3: V is even in theta and vanishes at theta = 0 for every l, so V = theta^2 l^-4 H = m^2 l^-2 H. Stationarity is -2 H + theta H_theta - 2 s H_s = 0 with s = alpha'/l^2, which at small m reduces to an equation independent of m, or has no root. ■

The flat family with the angle tied. For the top alone, n_f = 6: b is proportional to 54/5 - 6 > 0, and G > 0 on [0, pi/2] (Python: minimum 23.88, with A_0 = 32.92). So V has no stationary point with theta < pi/2, and l runs until theta leaves the fundamental domain of Proposition 7.1, where the top joins the tower; the periodically continued potential has stationary points only far outside the window, near theta = 8.2. For a universal mass at three generations, n_f = 48, G changes sign and there is one stable stationary point, theta_* = 1.009, with M_W/m = 0.685. For a universal mass at one generation, n_f = 16, theta_* = 1.031, with M_W/m = 0.653 (Python). Top-only content therefore has no vacuum in the window in this scope, and adding a graviton does not change the classification. The stationary angles are of order one, so these vacua do not separate the window.

A localized channel. For a channel with bulk mass M = -mu and c = mu l, the spectrum of lemma 9 gives the one-loop term

```
8 pi^2 l^4 V_loc = -2 int_0^infinity x^3 log(1 + ((w + c)/(w - c)) exp(-2 w)) dx,    w = sqrt(x^2 +
↪  c^2),
```

from the secular function cosh(w) - (c/w) sinh(w). At c = 0 it is -(45/64) zeta(5) = -0.729, and for large c it is -4 c^3 exp(-2 c) up to corrections. Corrections at large c are of relative order 1/c. In between it has a minimum, of depth -1.3246, at c = 1 to numerical precision (Python, mpmath). This concerns the interaction term alone: the bulk term, -(4/15) c^5 in dimensional regularization, and the wall term are

dropped as renormalizations of the bulk energy and wall tension, and with the bulk term included the minimum disappears. So the interaction term of one fermion channel prefers c = 1, where the light pair merges with the tower (lemma 9.7). A large c_*, which is what a separated window needs, must come from other fields, such as the bosonic tower of the scalar that gives mu.

Other second scales

Remark 10.7 (amendment of Proposition 1.5, last sentence). On the scale-free path, m l is a fixed number. Proposition 1.5 says that on it the finite window never separates from the tower. That holds for theta_* of order one, which is what the flat family gives above. If m is exponentially small in a stabilized shape parameter, the ratio m/Delta is a small fixed number. On the disc bundle of lemma 29, fix M and let R grow: m = lambda_M / R and Delta = delta / R both tend to zero, with lambda_M / delta exponentially small, and the count at fixed energy has seven-dimensional growth. So a potential R^-4 G(M), with M stabilized at a large M_* and G(M_*) > 0, *gives the hierarchy of lemma 30 with alpha = 1 and the fixed ratio lambda_{M_}*/delta, with no second scale; in lemma 30 the ratio tends to zero. The cost is that R then has no vacuum at finite value. The physical point at finite v needs one more term, and the stabilization at large M_* is a target, since the interaction term of one fermion channel prefers c = 1.

Remark 10.8 (a bulk energy). The ansatz V = a r^-p + b eps^2 r^q, with a > 0, has a minimum exactly when p + q > 0 and q b > 0, and then alpha = 2/(p + q), so alpha < 1 needs p + q > 2: for p = 4, q > -2 with q != 0. On the flat family the angle is a holonomy (Corollary 22.2), which no local bulk operator sees: at large l its effect is the Casimir term, q = -2, the borderline case alpha = 1. So a term falling more slowly than m^2/l^2 must come from elsewhere, and the natural place is a local bulk energy in the electroweak order parameter: for q = 1, V = a/l^4 + b v^2 M^3 l gives l^5 proportional to 1/v^2, so alpha = 2/5, if the bulk energy vanishes at v = 0. That requires the order parameter to be a bulk field with a bulk vacuum energy, and it departs from D_F as a pure boundary condition. Whether the spectral action provides this is open. In the heat cutoff, the dependence of the potential on D_F is exponentially small at large l Lambda, by Poisson summation over the tower, of relative size 4 exp(-(l Lambda)^2), so that action has no such term at this order. The factor (3 - d_eff)/2 of lemma 25 multiplies the term quadratic in the gauge field, a different term.

Corrections to the earlier files

- 09, Proposition 9.1 and its reading: subject to Proposition 10.1.
- 07, sections 4 and 6b: Proposition 7.2 stands as a statement on the circle; the reading "at distance proportional to a" in section 4 is corrected by Proposition 10.1, and the reading "the interval opens" at label 11 by Proposition 10.3.
- 08, Remark 8.3 and its reading: the far-wall group is reached at R = R_*, with the tower at the string scale, by Propositions 10.2 and 10.4 and the section on the far wall. There is also a tension of labels: Proposition 8.1 pairs label 9 with O^+, and section 6b of 07 puts label 9 on a far O8^- wall. These are two different meanings of the label, the KO sign and the position of the top's brane.
- 06, section 2: "9, 10, 11" mixes KO labels with metric dimensions. In the string web, nine dimensions is finite R, and R -> 0 is T-dual to R -> infinity, whereas in H the finite interior is labelled 10 and l -> 0 is labelled 9.
- 05, Proposition 5.4: the conclusion is correct on rerun (Python, exact integers, all 192 configurations). The sentence "In every case S - N has d = 19 and d = 20 only, while S - T has d = 16 on the neutrino roots of one slot and d = 22 on those of the two others" is wrong: S - N has only these values in 48 of the 192 configurations, and the other 144 show {13, 17, 19, 20}; the statement on S - T fails in 96 configurations, with the pattern 16, 16, 22. The values of d differ in every configuration, and that is what the proof needs.

Scope and escapes

Proposition 10.6 assumes a potential with a regular expansion in alpha'/l^2, around the flat family with its towers. It concerns the dilaton *gradient*. The following lie outside it: a dilaton potential from string

tree level, in non-supersymmetric strings or with tadpoles, for which a term in positive powers of 1/g could grow with l at fixed v through the law of Proposition 10.4; a different frame relation between l and the four-dimensional Planck mass; wall-localized dynamics; and the other chambers of the moduli space, where 8 + 8 removes the Romans mass and E8 x E8 is available. The string-theory statements are read from the two sources for type I' and recalled for the moduli-space limits; only the first have been checked against the text. By AGENTS.md, a negative result needs an independent check by another model. This one has had a reading by Claude Sonnet 5.5, which found the generator count above, and no further reading after the corrections.

Part 38

Inner fluctuations of the interval parent with an SO(30) algebra

Lemma 38. Written as file 11 of the eleven-dimensional line (branch 11D, by another Claude session) and moved here on 4 November 2026 with its text unchanged; its proposals keep their labels, and the plan and the unfinished files of that line are in historical/11D/. Internal references to "file 0n" of that line mean the lemma with the same content: 01 is lemma 31, 02 is 32, 03 is 39, 04 is 33, 05 is 34, 06 is in historical/11D, 07 is 40, 08 is 35, 09 is 36, 10 is 37, 11 is 38.

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, to answer the question left in 09: whether the gauge field along the interval of lemma 22 carries the antisymmetric block with the counts of lemma 21, given that Proposition 9.3 excludes it from the finite triple alone. Numbers in Python (numpy), scripts not kept by the rule of the project. The Codex quota is spent, so no Terra referee is available. An independent reading by Claude Sonnet 5.5, whose amendments are applied below, is in the review. The corrected text has not been read again.

Outcome

1. **Yes, under a condition on the walls.** With a real algebra M_30(R) on the light block, the fluctuation along the interval is an so(30) connection, and its block between particles and antiparticles lies in Λ^2 V, with the counts (6, 6, 3) in the colour antitriplet (Proposition 11.2). Massless scalars in exactly Λ^2 V + conjugate appear when both walls reduce the algebra to M_15(C), the commutant of quark number (Theorem 11.3).
2. **In lemma 22's own parent the massless scalars are absent.** Its algebra is free at x = 0, so the interval component has no zero mode, for any algebra (Theorem 11.3, item 2).
3. **A generator has a massless vector or a massless scalar, never both, on the interval**, in free five-dimensional Yang-Mills at trivial background. So the whole of Λ^2(30) of lemma 21 cannot be massless scalars with colour unbroken. The antisquark counts appear as scalars, while the slepton part (24, 1) + (1, 1) appears as vectors (Theorem 11.3, item 4).
4. **The finite window sees a symmetric shadow.** The connection acts on the window through its commutator with the masses, B_ij -> (m_i - m_j) B_ij, which maps the (S^2 5, 3bar) of Λ^2 V onto the (Λ^2 5, 3bar) of S^2 V, with counts (3, 6, 1) (Proposition 11.4). This reconciles Propositions 9.2 and 9.3. The kernel is the same-flavour diquarks.
5. **SO(30) has a precise meaning in the finite geometry:** it is the commutant of the real structure j = J U of lemma 06 on the massless light block, for the sign s = +1, and USp(30) for s = -1 (Proposition 11.1).

Setting

Let V = C^5 ⊗ C^3, with Q = 2/3 on u, c and Q = -1/3 on d, s, b. In the finite Hilbert space of the five light quarks, with colour, the chirality gamma = +1 subspace is

```
H_+ = {q_L} ⊕ {conj(q_R)} = V ⊕ Vbar,
```

of dimension 30: the left-handed Weyl fields of fifteen Dirac fermions. H_- = {q_R} ⊕ {conj(q_L)}, and J_F exchanges H_+ and H_-, with c = J_F restricted to H_+, as in lemma 22. Let U be a unitary,

grading-reversing and J-related as in lemma 06, with U q_L = q_R and U conj(q_R) = s conj(q_L), so U J_F = s J_F U. Then j = U^-1 c is an antiunitary on H_+, with j^2 = s, j q_L = s conj(q_R) and j conj(q_R) = q_L. Here U only defines j: the representation pi below does not commute with U, so this is not a doubling in the sense of lemma 06, in which U commutes with the algebra.

The interval parent is that of lemma 22: D_X = E d/dx on L^2([0, l]; C^2) ⊗ (H_+ ⊕ H_-), with Q in H_+, U-components in H_-, and the wall conditions Q_R(0) = U_L(0) = 0, Q_R(l) = X U_R(l), U_L(l) = -X^* Q_L(l).

The block

Proposition 11.1 (the commutant of j). Let s = +1. In the basis V ⊕ Vbar with j(v, w) = (K^-1 w, K v), the skew-Hermitian a with j a j^-1 = a form so(30), and

```
a = [[alpha, B], [conj(B), conj(alpha)]],    alpha^* = -alpha,   B^T = -B,
```

so so(30) = u(15) ⊕ Λ^2 V, with u(15) the commutant of the complex structure I = (i, -i), the quark number. The Hermitian generator H = i a satisfies j H j^-1 = -H. For s = -1 the same computation gives usp(30) = u(15) ⊕ S^2 V.

Proof. Linear algebra, with Proposition 8.1 for the groups. For s = +1, j a j^-1 = a gives the block form, and a^* = -a gives alpha^* = -alpha and B^* = -conj(B), that is B^T = -B. Then j (i a) j^-1 = -i a. ■

So the SO(30) of lemma 21 is the symmetry of the massless block for the sign s = +1, the sign that Proposition 8.1 and STRATEGY T2c associate with label 11 and with the seesaw; and the counts of lemma 21 select the same sign:

	colour antitriplet	its charges (-2/3, +1/3, +4/3)	colour sextet
s = +1, Λ^2 V	(S^2 5, 3bar)	(6, 6, 3)	(Λ^2 5, 6)
s = -1, S^2 V	(Λ^2 5, 3bar)	(3, 6, 1)	(S^2 5, 6)

Fluctuations along the interval

Take on the light block D_F = 0, so X = 0, and let C_0, C_l be subalgebras of A = M_30(R), the real elements of End H_+. Let

```
A_l = {f in C^infinity([0, l], A) : f(0) in C_0, f(l) in C_l},    pi(f) = f on Q, 0 on the H_-
↪ components.
```

Proposition 11.2. Assume 1 lies in C_0 and C_l.

1. pi(f) preserves the domain, for every choice of C_0, C_l. Order zero and order one hold on this block.
2. The self-adjoint one-forms sum pi(f_i) [D_X, pi(g_i)] are exactly E a_x on the Q components, with a_x in C^infinity([0, l], so(30)), with no condition at the walls. The fluctuated operator is

   ```
   D_A = E (d/dx + a_x ⊕ U a_x U^-1).
   ```

3. J-hat A J-hat^-1 = E (U a_x U^-1 on H_-), with sign +, as KO-dimension six requires. The block of a_x between Vbar and V is antisymmetric. The two statements agree since E is antisymmetric: the block of the Hermitian operator E a on C^2 ⊗ E is symmetric, as Proposition 9.3, item 2 requires, with B antisymmetric and the factor E carrying the other sign. So the interval component evades Proposition 9.3 through the escape in its item 3.

For D_F != 0 on this block, order one fails in the finite window: [[D_F, pi(a)], J pi(b) J^-1] has the blocks -U b U^-1 M a from H_+ to H_-, and -a M^* U b U^-1 from H_- to H_+, which vanish for all a, b only if M = 0. The parent still satisfies order one (Theorem 22.3), but its wall algebra shrinks to the commutant of D_F in pi, which is zero for pi = (a, 0) with M invertible. So this algebra describes the massless block, and the masses need a treatment without order one, as in lemma 19.

Proof. (1) The conditions Q_R = 0 and U_L = 0 at the walls are preserved by any operator acting on the H_+ index. [D_X, pi(g)] = E g' on Q, and J-hat pi(b) J-hat^-1 acts on the H_- components only, so both commutators vanish. (2) Given T, choose tau with support inside (0, l) and the same integral as T, which is a finite sum sum f_i g_i' of functions supported inside the interval. Take f = 1 and g = int_0^x (T - tau), which vanishes at both walls, so g lies in A_l and g' = T - tau. Self-adjointness of E T, with E skew, requires T skew. (3) J-hat E = E J-hat (direct from the formula for J-hat in Theorem 22.1), and c a c^-1 = U j a j^-1 U^-1 = U a U^-1. ■

Theorem 11.3 (zero modes of the interval component). Let k_0, k_l be the skew parts of C_0, C_l, and assume a five-dimensional Yang-Mills action for the fluctuation on a flat product with M^4, as in Proposition 22.8.

1. The gauge group is {g : I -> O(30), g(0) in K_0, g(l) in K_l}. The holonomy of a_x transforms as P -> g(l) P g(0)^-1, so the moduli are the double coset K_l SO(30) / K_0, and the massless four-dimensional scalars from a_x form

   ```
   the orthogonal complement of k_0 + k_l in so(30).
   ```

2. In lemma 22's algebra, C_0 = A, so k_0 = so(30), and there are no such scalars, for any algebra of this kind.

3. If C_0 = C_l = M_15(C), the massless scalars are exactly Λ^2 V ⊕ Λ^2 Vbar, with colour-antitriplet counts (6, 6, 3), and the massless vectors are u(15).

4. A generator in k_0 ∩ k_l has a massless vector and no massless scalar; a generator in the orthogonal complement of k_0 + k_l has a massless scalar and no massless vector; all others have neither. If the orthogonal projections onto k_0 and k_l commute (aligned walls), the towers are n pi/l in the first two cases, and (n - 1/2) pi/l for a generator in exactly one of k_0, k_l. For nonaligned walls the towers move: k_0 spanned by T_1 and k_l by cos(phi) T_1 + sin(phi) T_2 give k l = n pi +- phi, which is the mechanism of lemma 23.

5. With colour unbroken at both walls, the (1, 8) of Λ^2(30) is a vector, so Λ^2(30) cannot consist entirely of massless scalars. In item 3, the (24, 8) + (24, 1) + (1, 8) + (1, 1) of lemma 21 are vectors, and the scalars are (15, 3bar) + (10, 6) + conjugates.

Proof. (1) A gauge parameter xi(x) with xi(0) in k_0, xi(l) in k_l shifts int a_x by xi(l) - xi(0), so the linear moduli are so(30)/(k_0 + k_l), and a constant a_x in the orthogonal complement is a zero mode of the quadratic form. (2), (3): substitution, with Proposition 11.1. (4) A_mu has Neumann or Dirichlet conditions at each wall according to membership in k_wall, and a_x the opposite, so the mode functions are cos(n pi x/l), sin(n pi x/l) or their half-integer versions. (5) Item 4, and the branching of lemma 21. ■

Scope of item 4. Free five-dimensional Yang-Mills at trivial background. Escapes: a circle without walls, nontrivial holonomy, brane-localized terms, further internal dimensions.

Note that the choice in item 3 is an input: the domain does not force the wall algebra to M_15(C), since by Proposition 11.2, item 1, every f preserves it at D_F = 0. It says that the walls conserve quark number and the bulk does not. On a circle, without walls, every generator has both a massless vector and a massless scalar, and the whole of Λ^2(30) appears, which is the nine-dimensional adjoint scalar used in 09. A wall condition, or the reflection-even algebra of Proposition 1.1, removes half. So the two readings differ: for the stack off the O8 plane, with holonomy I, diquarks are massive and mesons have massless scalars; on the interval with parity Ad(I), diquarks have massless scalars and mesons have none.

The shadow in the finite window

Proposition 11.4. Let s = +1, M = U Lambda with Lambda real for j and symmetric, and let a_tot = a ⊕ U a U^-1 be constant. Then:

1. [D_F, a_tot] has the off-diagonal blocks U S and S U^-1, with S = [Lambda, a] real and symmetric, and [D_F, [D_F, a_tot]] = [Lambda, S] ⊕ U [Lambda, S] U^-1.

2. By Theorem 22.4, the compression of E a_tot to the window is -l^-1 int_0^l sin(x ad_{D_F})(a_tot) dx. For Lambda = diag(m_i) on flavours, on both V and Vbar, the block B_ij goes to

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   - ((1 - cos(l (m_i - m_j))) / (l (m_i - m_j))) B_ij,    which is -(l/2)(m_i - m_j) B_ij to
   ↪ leading order,
   ```

   as a change of Lambda. The result is symmetric, as Proposition 9.3 requires of a finite J-compatible fluctuation.

3. For distinct masses, B -> (m_i - m_j) B_ij maps (S^2 5, 3bar) in Λ^2 V onto (Λ^2 5, 3bar) in S^2 V, with kernel the five same-flavour diquarks, and maps (Λ^2 5, 6) into (S^2 5, 6). In charges, (6, 6, 3) - (3, 0, 2) = (3, 6, 1). If the wall condition is imposed on the gauge unitaries g ⊕ U g U^-1, so that k_l is the commutant of Lambda in so(30), then with k_0 = u(15) the massless scalars of Theorem 11.3 are the same off-diagonal blocks. If it is imposed on pi(f) itself, as in the algebra of lemma 22, then k_l = 0 for invertible M.

4. For a universal mass, Lambda = m, the compression vanishes: S = 0, and M = m U is invariant under SO(30). Fifteen Dirac fermions of equal mass are thirty Majorana fermions of equal mass.

Proof. (1) Direct multiplication, using a_tot on H_- equal to U a U^-1. S is real and symmetric since a is real and antisymmetric and Lambda real and symmetric. (2) Item 1 gives sin(x ad)(a_tot) = (sin(x delta)/delta) ad(a_tot) on B_ij, with delta = m_i - m_j; integrate. (3) In the colour antitriplet, B is symmetric in flavour, and (m_i - m_j) B_ij is antisymmetric in flavour, hence total symmetric. Colour sextets are analogous. (4) S = 0. ■ Checked in Python on a small model, with V = C^3: reality, symmetry, the blocks of item 1 and their squares. Claude Sonnet 5.5 also checked the compression by direct quadrature of W^*(E a_tot)W, with five flavours in three colours, against Theorem 22.4 and the closed block (Python, 10^-15).

Reading. The commutator [D_F, a] is the form of the Higgs field in Connes' finite geometry, and E a_x is a Kaluza-Klein component along the interval. Theorem 22.4 (3), Q_l([D_X, f]) = [D_F, Q_l(f)], says that the second compresses to the first. Indeed E a_tot = [D_X, f_tot] with f(x) = (x - l) a, and f(l) = 0 lies in C_F. So in this parent, to first order and for the fixed compression, a finite Higgs one-form is the shadow of a gauge field along the interval, and it inherits the symmetry type of the finite triple by the commutator with masses. The moving window of Corollary 22.6 gives -l [D_F, h] and a quadratic term (lemma 19). This shadow is tangent to the O(30) orbit of Lambda, so it gives no radial mode, in agreement with No-go theorem 22.7; the odd generators of lemma 23 are absent here since M_30(R) maps H_+ to H_+. A generator commuting with the masses has no shadow. This explains the difference between the counts (6, 6, 3) of the parent and (3, 6, 1) of the finite triple, and it replaces the alternative O8^- against O8^+ of Proposition 9.2, for these blocks, by a parent and its window.

What remains open

- Whether a boundary condition, for example a wall term that conserves quark number, can derive M_15(C) at the walls, where at present it is chosen.
- The treatment of nonzero masses, where order one fails on the block: the quadratic terms of lemma 19 for this algebra.
- The sleptons: on an interval the (24, 1) + (1, 1) are vectors. Scalars in these representations need components along further internal directions, such as the base of 02.
- The colour sextets (10, 6) come with the antitriplets, and nothing removes them in this construction.
- The relation to the top: the (30, 2) of SO(32), with the second factor as the top's SO(2).

Part 41

The axioms and the lift along the chain Pati-Salam, Standard Model, colour and electromagnetism

For STRATEGY T17, following lemma 06 (the representation of Result 5), lemma 17 (Corollary 17.4), lemma 20 (Theorem 20.2), lemma 21 and results/orientability_under_heavy_centralizer.md. Written by Claude Fable 5.1 on 4 November 2026. The three matrices, the orientation cycle and the traces were computed by hand and confirmed with numpy on the 32-state representation (one generation, Dirac neutrinos; script not kept, by the rule of the project). Refereed by Claude Opus 5.5 (effort max, 4 November 2026; report in historical/reviews/2026-11-04_opus_check_41_45.md): Lemma 41.1 correct; Theorem 41.2 and Proposition 41.4 correct with amendments; Remark 41.3 and Proposition 41.5 had errors, corrected below as the referee stated them (the generator of B, the state counts, the word "vectorlike", the e_R obstruction); the referee recomputed every matrix and trace.

Outcome. Along the chain of heavy operators, A_PS = H_R + H_L + M_4(C) -> A_SM = C + H + M_3(C) -> B = C + M_3(C), the finite data keep the KO-6 signs, order zero and the first-order condition of Theorem 20.2, and they lose the other two axioms of a finite real spectral triple: the trace-form Poincaré duality fails at every step, with intersection forms of rank 2, 2 and 0 (the last because every range rho(p_i) rho(p_j)° in the light block has graded dimension zero, the bimodule balance behind the KO-7 doubling of lemma 06), and orientability holds at the Pati-Salam step, for a sheet-diagonal D of the form of Corollary 16.8, and fails at the two later ones, through the pair nu_R, J nu_R and the pair e_R, J e_R at the Standard Model step and through the central mass blocks at the third. The unimodularity trace of the abelian generator changes along the chain, (4i, 12i) on the full space, (2i, 12i) with the Majorana pair removed, (-2i, 6i) with the up-type sector removed and (-4i, 6i) on the light block, so the inherited lift and a lift reimposed later impose different hypercharge constraints, and only the inherited one gives the photon its charges.

Setting

One generation with Dirac neutrinos, in the convention of lemma 06, Result 5: F = C + C^3 (lepton line and colour), H_p = (C^2 (x) F)_L + (C^2 (x) F)_R, H = H_p + H_pbar, dim H = 32. For a = (lambda, q, m) in A_SM, the left action is q (x) I_4 on the left-handed particles, Q_lambda (x) I_4 = diag(lambda, conj lambda) (x) I_4 on the right-handed ones, and I_2 (x) B_{lambda, m} = I_2 (x) diag(lambda, m) on the antiparticles; J(p, pbar) = (K pbar, K p); gamma = +1 on p_L and pbar_R, -1 on p_R and pbar_L. The opposite action is a° = J a* J^{-1}. The Pati-Salam parent acts by q_L (x) I_4 on p_L, q_R (x) I_4 on p_R and I_2 (x) m_4 on the antiparticles, and A_SM is its centralizer under the heavy Majorana block (Theorem 20.2, rank one, kappa != 0). The light block of the third step is the 18-dimensional space of d_L, d_R (three colours), e_L, e_R, nu_L and their conjugates, on which B = {(lambda, Q_lambda, m)} acts (lemma 06, Proposition 5), with the kept mass blocks commuting with B (lemma 17, Corollary 17.4).

Minimal projections, one per summand: p_R' = (1, 0, 0), p_L' = (0, 1, 0), p_4 = (0, 0, e_11) for A_PS; p_1 = (1, 0, 0), p_2 = (0, 1_H, 0), p_3 = (0, 0, e_11) for A_SM; for B the minimal projection of its summand C is (1, Q_1, 0) = (1, 1_H, 0) = p_1 + p_2, which is the identity on the nine light particles and on the light antileptons, and p_3 is that of M_3(C). The trace form is Q_ij = Tr(gamma rho(p_i) rho(p_j)°).

Poincaré duality

Lemma 41.1 (the trace form is skew). For a finite real triple with J^2 = +-1, J gamma = -gamma J, an algebra even for gamma and order zero, Q_ij = -Q_ji. *Proof.* For an antiunitary J, Tr(J A J^{-1}) = conj(Tr A). Then Q_ij = conj Tr(J gamma p_i p_j° J^{-1}) = -conj Tr(gamma p_i° p_j) = -conj Tr(gamma p_j p_i°) = -conj(Q_ji) = -Q_ji, using order zero to commute p_i° past p_j and the reality of Q. ■ (This is the observation of Ćaćić for KO-dimension 6; the corpus recorded it as rs_N7.)

Theorem 41.2 (the three forms). In the bases above,

```
Q_PS = [[0, 0, -2], [0, 0, 2], [2, -2, 0]],   Q_SM = [[0, -2, -2], [2, 0, 2], [2, -2, 0]],   Q_B =
↪  [[0, 0], [0, 0]].
```

Each has even rank: Q_PS and Q_SM have rank 2 with kernels spanned by p_R' + p_L' and by p_1 + p_2 - p_3, and Q_B vanishes. So the trace-form Poincaré duality fails at each step of the chain: at the first two because a skew form on an odd-rank K_0 is singular, at the third because every range rho(p_i) rho(p_j)° in the light block has graded dimension zero.

Proof. Each entry counts, with the sign of gamma, the states on which the left projection p_i and the right projection p_j° both act as the identity. For Q_SM: p_1 acts as the identity on all of p_R (eight states) and on the lepton line of pbar_L and pbar_R; p_2 on p_L; p_3 on one colour of pbar_L and pbar_R; and p_1°, p_2°, p_3° are the J-transports, so p_1° is the identity on pbar_R and on the lepton line of p_L, p_R, p_2° on pbar_L, p_3° on one colour of p_L, p_R. Then Q_12 counts the lepton line of pbar_L, two states of grading -1; Q_13 one colour of p_R, two states of grading -1; Q_23 one colour of p_L, two states of grading +1; the diagonal entries vanish by Lemma 41.1 (directly: Q_11 counts the lepton line of p_R and of pbar_R with opposite signs). Q_PS follows in the same way with p_R' the identity on p_R, p_L' on p_L and p_4 on one line of F on the antiparticles: Q_{R L} counts p_R against pbar_L, which is empty, and Q_{R 4}, Q_{L 4} count one line of p_R and of p_L. For Q_B, on the light block the projection (1, 1_H, 0) of the summand C is the identity on the nine light particles and on the three light antileptons, p_3 on one colour of dbar_L, dbar_R, and p_3° on one colour of d_L, d_R; Q_13 counts d_L and d_R of one colour, of opposite gradings, so it vanishes, and Q_11 vanishes by Lemma 41.1 (on the particles its range is nu_L, e_L, e_R with graded count +1, on the antiparticles the conjugates with -1). With the projection (1, 0, 0), which lies outside B, the form would be [[0, -1], [1, 0]]. The kernels are read off. ■

Remark 41.3 (the form is the bimodule balance). Q_ij is the graded dimension of the range of rho(p_i) rho(p_j)°, one colour of an isotypic component when p_j = e_11. It vanishes identically exactly when every such range has equal numbers of states of the two gradings, which is the bimodule balance condition behind the doubling of lemma 06 and is weaker than physical vectorlikeness: the light block of the third step keeps the unpaired nu_L, whose particle content restricted to SU(3) x U(1)_em has [V_L] - [V_R] = (1)_0, nonzero, while Q_B = 0, because the trace form sees the full real space, where the neutral class cancels against its conjugate and the characters lambda and conj(lambda) of the real summand C share one projection (referee; lemma 06 draws the same distinction in its no-go under [U, Q] = 0). On the full 32-state space with nu_R, where B acts and the Standard Model index restricts to zero, the form of B also vanishes, and there the two statements agree. At the first two steps the form has rank 2 because the quark doublets and the leptons carry the chiral asymmetry; the kernel vector p_1 + p_2 - p_3 of Q_SM is the direction along which the graded counts of the three summands cancel.

Orientability

Proposition 41.4. (i) The Pati-Salam data are orientable: gamma = rho(p_L' - p_R') rho(1_4)° - rho(1_4) rho(p_L' - p_R')°. (ii) The Standard Model data are not, with or without a right-handed neutrino: nu_R and J nu_R carry the same left character lambda and the same right character lambda and have opposite gradings, so no sum of left-right products separates them (Stephan 2006, hep-th/0610097, for the KO-6 models with neutrinos); and e_R carries (conj lambda, lambda) with gamma = -1 while J e_R carries (lambda, conj lambda) with gamma = +1, on which a real cycle takes complex-conjugate values and cannot be -1 on one and +1 on the other, an obstruction that persists when the nu_R pair is removed (referee, checked numerically: removing the nu_R and e_R pairs restores orientability). (iii) The light block of the third step is not: the kept mass blocks are nonzero and commute with B (lemma 17, Corollary 17.4), and a nonzero D commuting with both actions commutes with every orientation cycle while anticommuting with gamma; directly, d_L and d_R of one colour carry the same characters (conj lambda on the left, e_11 on the right) and opposite gradings.

Proof. (i) On the particles rho(1_4)° is the identity and rho(1_4) vanishes, so the right side is p_L' - p_R', which is +1 on p_L and -1 on p_R; on the antiparticles rho(p_L' - p_R') vanishes and rho(1_4) is the identity, so the right side is -(p_L' - p_R')°, which is -1 on pbar_L and +1 on pbar_R. This is gamma. (ii) and (iii): a cycle c = sum rho(a_i) rho(b_i)° takes the same value on two states with the

same pair of characters, while gamma takes opposite values. For the first sentence of (iii), if [D, rho(a)] = 0 = [D, rho(b)°] for all a, b then [D, c] = 0 for every cycle, and c = gamma would give D gamma = gamma D, which with D gamma = -gamma D forces D = 0; this is the corpus statement rs_N9 (historical/corpus/connes_kk_research_state.md, section 5.4), restated here with its proof. ■

The explicit model of results/orientability_under_heavy_centralizer.md shows the same loss under a heavy centralizer in a four-summand toy; here it happens in the physical chain, at the first heavy step when nu_R is present and at the second step in every case.

Unimodularity

Proposition 41.5. Let X_lambda be the generator of the abelian direction of the summand C, acting in each algebra as its own element: in A_SM the element (i, 0, 0), which acts as diag(i, -i) (x) I_4 on p_R and as i on the lepton line of the antiparticles; in B the element (i, Q_i, 0), which in addition acts as diag(i, -i) (x) I_4 on p_L, that is by i on nu_L and u_L and by -i on e_L and d_L. Let X_c be the common colour phase, i on the three colours of the antiparticles. The traces of the left action are

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full H, 32 states, A_SM:                          Tr X_lambda = 4i,   Tr X_c = 12i;
Majorana pair removed, 30 states, A_SM:            2i,                12i;
up-type sector removed, nu_R kept, 20 states, B:  -2i,                 6i;
light block of the third step, 18 states, B:      -4i,                 6i.
```

So the unimodularity condition Tr A = 0 of Chamseddine-Connes-Marcolli on the one-form (the left action; the adjoint action u J u J^{-1} has determinant one for every unitary, so a condition on it would be empty), imposed on the parent, gives theta_lambda + 3 theta_c = 0 between the two abelian phases, the infinitesimal form of lambda det m = 1, and reproduces lambda^2 (det m)^4 at the 30-state step recorded by the corpus; the same condition reimposed on the light block gives -4 theta_lambda + 6 theta_c = 0, a different embedding of the surviving abelian generator. Under the adjoint action, normalized so that the electron has charge -1, the inherited condition gives the charges (2/3, -1/3, -1, 0) to (u, d, e, nu) and the reimposed one gives (1/6, -5/6, -1, 0): only the inherited lift gives the photon (referee). This is the condition Proposition 22.8 of lemma 22 uses, where the ten generators of u(C_F) reduce to the nine of the photon and the gluons.

Proof. Count the states with their phases. Full space: Q_i is traceless on p_R and X_lambda vanishes on p_L, so only the antilepton line contributes, two weak components in two sectors, 4i; X_c has 12i on the antiquarks. Removing nu_R (phase +i) and J nu_R (antilepton line, +i) gives 2i. For B, (i, Q_i, 0) is traceless on p_L and on p_R by itself; removing the up-type sector removes u_L and u_R, three colours each with phase +i, and their conjugates, which carry no abelian phase, so the particle part gives -3i on p_L (nu_L at +i, d_L at -3i, e_L at -i) and -3i on p_R, and the antilepton line keeps 4i: -2i on 20 states; the antiquarks halve to 6i. Removing also nu_R and J nu_R takes i twice: -4i on 18 states. ■

The first version of this proposition used the element (i, 0, 0) on the spaces where only B acts and counted 26 states where there are 20; the referee corrected both (the element (i, 0, 0) gives the traces 4i, 2i, i, -i on the four spaces, the numbers of the first version).

Scope

One generation with Dirac neutrinos in the stated representation; the trace form on minimal projections as the test of Poincaré duality (a real Kasparov duality is a different question, as the corpus noted); the strict projections of lemma 06 for the third step; at the Pati-Salam step a sheet-diagonal D of the form of Corollary 16.8 (a Pati-Salam-symmetric Yukawa with no Majorana block), since with the Majorana block order one fails there, which is how Theorem 20.2 selects A_SM. With three generations the matrices multiply by three and nothing else changes. The direction the result points to: the finite triple of the main paper carries, along the chain, exactly the data that lemma 22 derives from the interval (signs, order zero, first order), and none that it does not; the two failed axioms are the ones a Kaluza-Klein parent would have to supply through its topology (orientability, Poincaré duality for the parent manifold), and the vanishing of the form at the third step is the bimodule balance of lemma 06, which coexists with the unpaired nu_L.

Part 42

The Higgs representation in the window: the tangent of the orbit of the marked lines

For STRATEGY T14, item (ii) of the corpus reading (CORPUS_SALVAGE item 13), following lemma 28 (the group data G = Sp(2), K = Sp(1)_a x Sp(1)_b, H = Sp(1)_a x U(1)_b, and Proposition 28.19), lemma 29 (the eight-state window) and lemma 20 (Theorem 20.2 and item 2 of its readings: the connector as a marked pair of points and a coupling). Written by Claude Fable 5.1 on 4 November 2026, by hand; the representation theory is standard and is checked against the isotropy representation stated in lemma 28. Refereed by Claude Opus 5.5 (effort max, 4 November 2026; report in historical/reviews/2026-11-04_opus_check_41_45.md): Proposition 42.2 correct (stabilizer dimensions and ad-eigenvalues recomputed), Proposition 42.1 correct with its escape added, Proposition 42.3 had an error in the count of the connector moduli, corrected below as the referee stated it.

Outcome. The window operator of lemma 29 carries no Higgs representation by itself: Sp(2)-equivariance makes its off-diagonal part a scalar (Proposition 28.19). The Higgs representation of the first-stage corpus note (section 6: T_p HP^1 = Hom_H(L, L^perp), the (2, 2) of Spin(4), one doublet of hypercharge 1/2 after marking a U(1)) is the tangent space of the orbit of the marked line that the connector of lemma 20 puts on the quaternionic sheet. That orbit is CP^3, and its tangent at the marked line is the isotropy representation (2)_{+-1} + (1)_{+-2} of lemma 28: a doublet of the weak Sp(1)_a with U(1)_b charge +-1, which is the Higgs doublet, and a singlet of charge +-2 along the twistor fibre, which is an SU(2)-singlet scalar direction of hypercharge 1. The moduli of the connector under Sp(2) are the Hermitian and the symplectic overlaps of the two marked lines, the modulus of the coupling, and the phase of kappa alpha^2 when the Hermitian overlap alpha is nonzero.

Setting

The window of lemma 29 is W + V, two copies of the 4 of G = Sp(2): the core copy W, on which the real functions of S^4 compress to M_2(H), and the wall copy V, on which the real functions of CP^3 compress to M_4(C) (Example 16.9, Theorem 28.21). The connector of Theorem 20.2 is the rank-one block T_kappa x = kappa K x K with K = v w*, that is, a complex line Cw in W, a complex line Cv in V and the coupling kappa; the centralizer of T_kappa is C + H + M_3(C) (lemma 20, Result 2). In the notation of lemma 28, K = Sp(1)_a x Sp(1)_b is the stabilizer of a quaternionic line L in W = H^2, with Sp(1)_b acting on L and Sp(1)_a on L^perp, and H = Sp(1)_a x U(1)_b is the stabilizer of a complex line l in L, so that G/K = S^4, G/H = CP^3 and K/H = S^2 is the twistor fibre. Charges of U(1)_b are normalized so that the doublet of Sp(1)_b has charges +-1. In the flag algebra of the corpus note (section 5), marking l reduces H_R = Sp(1)_b to C and leaves H_L = Sp(1)_a, the weak isospin, on L^perp.

Statements

Proposition 42.1 (no Higgs representation in the equivariant window). If the window consists of one copy of the 4 in each degree of the end grading, then after equivariant identifications d_M = [[0, conj(lambda_M) I_4], [lambda_M I_4, 0]], and a phase makes lambda_M >= 0 (Proposition 28.19, by Schur's lemma). So the off-diagonal part of the window operator is a multiple of the identity, invariant under G, and it transforms under the isotropy group of any marked point trivially. The question of item (ii) of CORPUS_SALVAGE, whether that part transforms as T_p S^4, has the answer no for the equivariant window, for this reason. The escape (referee): as a representation of G, Hom(W, V) = 4 (x) 4 = 1 + 5 + 10 contains T_p S^4 = (2, 2) twice after restriction to the isotropy group of p (once in the 5, once in the 10), so a G-breaking off-diagonal block at p, such as the abstract d = mu n^a Gamma_a of Example 16.9 with n orthogonal to the axis, carries the representation of item (ii); the answer no is a statement about the G-invariant block, and the result points to the G-breaking blocks, which are the variation of the marked data described below.

Proposition 42.2 (the tangent of the marked line). Let w be a unit vector of W = H^2 = C^4, l = Cw its complex line, L = l + jl its quaternionic line, and p the point of S^4 = HP^1 under L. The G-orbit

of l is CP^3 = G/H and the G-orbit of L is S^4 = G/K. The tangent spaces at l and L decompose under the stabilizers as

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T_L S^4 = (2, 2) of Sp(1)_a x Sp(1)_b = Hom_H(L, L^perp),
T_l CP^3 = (2)_{+-1} + (1)_{+-2} under H = Sp(1)_a x U(1)_b,
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where (2)_{+-1} is the restriction of T_L S^4 (the horizontal part, the doublet of the weak Sp(1)_a with U(1)_b charge +-1, one complex doublet of hypercharge 1/2 in the normalization Y = charge/2) and (1)_{+-2} is the tangent of the twistor fibre K/H = S^2 at l (one complex singlet of hypercharge 1, electric charge +1).

Proof. sp(2) = sp(1)_a + sp(1)_b + m with m = (2, 2) of real dimension 4, the isotropy representation of S^4; this is the decomposition of so(5) under so(4). Under H, sp(1)_b = u(1)_b + (1)_{+-2}, since the adjoint of Sp(1) restricted to its maximal torus has charges 0, +-2, and (2, 2) restricted to Sp(1)_a x U(1)_b is (2)_{+1} + (2)_{-1}, since the 2 of Sp(1)_b has charges +-1. The isotropy representation of G/H is sp(2)/h = (1)_{+-2} + (2)_{+-1}, as stated in lemma 28, with (1)_{+-2} tangent to the fibre K/H and (2)_{+-1} the lift of the tangent of G/K. The identification T_L HP^1 = Hom_H(L, L^perp) with the action t -> q_a t q_b^{-1} is the corpus note, section 6. ■

Proposition 42.3 (the moduli of the connector). Let w in W and v in V be the two marked unit vectors, with G acting on both copies of the 4 by the same representation, and let h and omega be the G-invariant Hermitian and symplectic forms on C^4 (omega(x, y) = h(jx, y)). The G-orbits of the pair of lines (Cw, Cv) are classified by the two real invariants

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a = |h(w, v)|,   b = |omega(w, v)|,   a^2 + b^2 <= 1;
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the orbit of the pair is determined by the position of Cv relative to the quaternionic line L of w: a^2 + b^2 is the squared norm of the component of v in L, a the overlap with l and b the overlap with jl. The G-orbits of the connector T_kappa itself are classified by a, b, |kappa| and, when alpha = h(w, v) != 0, the phase of kappa alpha^2: with the identity 1 in Hom(V, W) fixed by G, tr T_kappa(1) = kappa (w^* v)^2 = kappa alpha^2 is a G-invariant, unchanged by the rephasings that fix T_kappa (referee). So the connector has four real moduli when alpha != 0 and three when alpha = 0; a symmetry larger than the diagonal Sp(2), for instance an independent phase on V, removes the fourth.

Proof. G acts transitively on the unit sphere of H^2 with stabilizer Sp(1)_a (the stabilizer of w and jw), so the orbit of the pair is the Sp(1)_a-orbit of v. Write v = alpha w + beta jw + u with u in L^perp = H; Sp(1)_a acts on u by the quaternionic scalars, transitively on each sphere of L^perp, and fixes alpha = h(w, v), beta = h(jw, v) = conj(omega(w, v)) up to the convention. Rephasing w by e^{i phi} and v by e^{i theta} sends (alpha, beta) to (e^{i(theta - phi)} alpha, e^{i(theta + phi)} beta), which removes both phases. So the invariants of the pair of lines are |alpha|, |beta|, and |u|^2 = 1 - |alpha|^2 - |beta|^2 is determined; here beta = omega(w, v) exactly. For the connector, the rephasings w -> e^{i phi} w, v -> e^{i theta} v, kappa -> e^{-2i(theta - phi)} kappa leave T_kappa = kappa K x K fixed and multiply alpha^2 by e^{2i(theta - phi)}, so kappa alpha^2 is invariant under them, and it is invariant under G because 1 is; T_kappa and T_{-kappa} on the same pair of lines with alpha != 0 have the same a, b, |kappa| and opposite invariants, so they lie in different orbits. ■

Reading

The Higgs doublet of the first-stage note is recovered in the window as the tangent of the orbit of the marked core line, projected to S^4: it is the representation in which the point p of S^4 under the connector moves. The connector of lemma 20 is an input of the window (Theorem 28.21 prescribes the windows; the connector is not derived), so the proposition says where the doublet sits once the connector is a field: in the variation of the marked lines, which is a variation of the finite data, in the moduli a, b, |kappa| and arg(kappa alpha^2), and in G-breaking off-diagonal blocks, and never in the equivariant window operator d_M. Beside the doublet, the twistor fibre contributes a charged singlet direction of hypercharge 1; whether this direction is a dynamical scalar, a gauge direction of the U(1)_b, or frozen by the collar connection of lemma 29 (which preserves the line summands, hence the fibre structure) is open, and it is the same question as the grading axis of Proposition 28.19, which marks the quaternionic line L. The dictionary note keeps one

row from this: the Higgs doublet is the tangent of the base S^4 at the point marked by the connector, with the twistor fibre as a charged singlet beside it.

Scope

The window of lemma 29 with its Sp(2)-equivariance; the connector of Theorem 20.2 as a marked pair of lines and a coupling; the normalization Y = charge/2 of the corpus note. Nothing here makes the marked lines dynamical, computes a potential, or fixes the coupling; Proposition 42.3 counts moduli and does not identify them with a vacuum expectation value; Proposition 42.1 concerns the G-invariant block only. The direction it points to: a parent in which the connector's lines vary along S^4, that is, a Higgs field as a section over the base, for which the collar connection of lemma 29 and the engineered connections of T15 are the available tools.

Part 43

Which connections on a vertical kernel are Berry connections

For STRATEGY T15 (from CORPUS_SALVAGE items 1 and 2), following lemma 10 (Proposition 10.1 and the second-moment identity), lemma 29 (the collar connection of case (b)) and lemma 38. Written by Claude Fable 5.1 on 4 November 2026, by hand. Refereed by Claude Opus 5.5 (effort max, 4 November 2026; report in historical/reviews/2026-11-04_opus_check_41_45.md): Proposition 43.1 and Theorem 43.2 correct with amendments (the mean-curvature term, the transposed adjoint, the exclusion of vertical components), the two computations of Corollary 43.3 correct, and its last clause, which read lemma 29's collar connection as an instance, an error, withdrawn below. The construction was used in the corpus (historical/corpus/canonical_research.md, section 9, on N over S^5 with the flux (1, 1, 2)); this lemma supplies the general statement, and leaves the question of lemma 29's collar connection open.

Outcome. Let a fibrewise Dirac family on a bundle of compact fibres F_b have kernels K_b of constant rank, forming a bundle K over the base B, with the Berry connection induced from a unitary connection on the coefficient bundle E. Changing the coefficient connection by a horizontal one-form alpha changes the Berry connection by the fibrewise compression P alpha P. Every connection on K is a Berry connection of some coefficient connection exactly when the fibrewise compression Gamma(F_b, u(E)) -> u(K_b) is surjective at every point of B, and that holds exactly when the pairwise products s_i s_j^* of a frame of kernel sections are linearly independent as sections of End(E) over the fibre. For scalar changes of the connection the same criterion reads: the pointwise inner products <s_i, s_j> are linearly independent functions on the fibre. On CP^3 with the kernel H^0(O(1)) this holds (sixteen independent functions z-bar_i z_j/|z|^2), so every connection on the rank-four kernel bundle is a Berry connection of an abelian change; for functions pulled back from the base S^4 the compressed image has real dimension six out of sixteen (Example 16.9 with Proposition 10.1), so scalar changes along the base reach six of the sixteen directions. The collar connection of lemma 29 is outside this statement: it acts at the wall, on the CP^3 copy, and it changes the fibre operator and moves the kernel, the effect excluded below (referee).

Setting

Let pi: P -> B be a fibre bundle with compact fibres F_b and a connection (a horizontal distribution), E -> P a Hermitian vector bundle with a unitary connection nabla, and D = (D_b) a smooth family of fibrewise elliptic self-adjoint operators on L^2(F_b, E|*{F_b}) whose kernels K_b have constant dimension k; P_b denotes the orthogonal projection onto K_b. The Berry connection on the bundle K -> B is nabla^K_X s = P_b nabla*{X^h} s for X in T_b B, with X^h the horizontal lift and s a section of K read as a family of sections of E over the fibres; this is unitary for the fibrewise L^2 product when horizontal transport preserves the fibre measure (Riemannian submersions with totally geodesic fibres, as in the homogeneous examples), and in general the unitary connection adds (1/2) k(X) with k the mean-curvature form of the fibres (Bismut; Berline-Getzler-Vergne, chapter 9), a term common to all the connections compared here, so that the statements below hold with that term included or with volume-preserving transport assumed (referee). For a fibre F_b, the compression map is

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C_b : Gamma(F_b, End(E)) -> End(K_b),   C_b(X) = P_b X P_b,
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and in a unitary frame s_1, …, s_k of K_b its matrix elements are C_b(X)_{ij} = int_{F_b} <s_i(y), X(y) s_j(y)> dy. Its adjoint for the L^2 pairing on Gamma(F_b, End(E)) with the Hilbert-Schmidt pairing on End(K_b) is C_b^*(Y) = rho_Y = sum_{ij} Y_{ij} s_i s_j^*, a smooth section of End(E) over F_b (the density of Y; the first version had the indices transposed, which changes nothing below since the two families of products coincide as sets).

Statements

Proposition 43.1 (the change of the Berry connection). Let nabla' = nabla + alpha with alpha in Omega^1(P, u(E)) vanishing on vertical vectors. Then the Berry connection of nabla' is nabla^K + beta, where beta_b(X) = C_b(alpha(X^h)|_{F_b}): the compression of the horizontal component of alpha, fibre by fibre, exactly. A fibrewise twisted Dirac operator uses only the vertical part of the coefficient connection, so such an alpha leaves the family D_b and its kernels unchanged. A vertical component of alpha changes D_b and moves the kernels, and a first-order motion of the kernels changes the Berry connection at first order through the second fundamental form (for a kernel line n(b) in C^2 over S^2 moved along the flow of a vector field xi, the Berry curvature changes by the Lie derivative along xi); such components are excluded here, and this is the effect at work in the collar connection of lemma 29 (referee).

Proof. With the kernels unchanged, nabla'*{X^h}* s = *nabla*{X^h} s + alpha(X^h) s, and P_b of the second term is C_b(alpha(X^h)) s. ■

Theorem 43.2 (which connections are Berry connections). The following are equivalent for the family (P, E, D):

1. Every unitary connection on K -> B is the Berry connection of nabla + alpha for some alpha in Omega^1(P, u(E)) vanishing on vertical vectors.
2. C_b : Gamma(F_b, u(E)) -> u(K_b) is surjective for every b in B.
3. For every b and every (one, hence every) unitary frame s_1, …, s_k of K_b, the k^2 sections s_j s_i^* of End(E) over F_b are linearly independent.

When these hold, a preimage can be chosen smoothly: alpha(X^h) = C_b^*(C_b C_b^*)^{-1} beta_b(X) is a smooth u(E)-valued form on the horizontal distribution, extended by zero on vertical vectors.

Proof. (1) implies (2): a connection whose difference from nabla^K at b is a prescribed Y in u(K_b) must have Y in the image of C_b. (2) implies (3): C_b is surjective iff C_b^* is injective on u(K_b); C_b^* is complex linear with C_b^*(Y^*) = C_b^*(Y)^*, so its kernel in End(K_b) is closed under the adjoint and meets u(K_b) trivially exactly when it is zero, and C_b^*(Y) = sum Y_{ij} s_i s_j^* vanishes for a nonzero Y exactly when the sections s_i s_j^* are dependent. (3) implies (1): with C_b surjective, C_b C_b^* is a positive invertible operator on u(K_b) depending smoothly on b, so for a given beta in Omega^1(B, u(K)) the form alpha(X^h) = C_b^*(C_b C_b^*)^{-1} beta_b(X) is smooth, u(E)-valued, and C_b(alpha(X^h)) = beta_b(X); by Proposition 43.1 the Berry connection of nabla + alpha is nabla^K + beta. ■

Corollary 43.3 (scalar changes). Restricting alpha to scalar forms (alpha = i a (x) 1_E, a real), the attainable changes are the compressions of real functions, and every connection on K is attained by a scalar change iff the k^2 functions <s_i, s_j> on F_b are linearly independent. For the kernel H^0(CP^3, O(1)) of the Dolbeault-Dirac operator twisted by O(1), with s_i = z_i/|z| in the unitary frame, <s_i, s_j> = z-bar_i z_j/|z|^2, sixteen linearly independent functions, so the criterion holds and every connection on the rank-four kernel bundle is the Berry connection of an abelian change of the coefficient connection; in matrix form, Q(<w, B w>/<w, w>) = (B + tr B)/5 (lemma 10, the second-moment identity), a bijection of Hermitian matrices. For functions pulled back from the base S^4 of the twistor fibration, the compressed image is the real span of 1 and Gamma_1, …, Gamma_5, of dimension six (a real function from S^4 compresses to a Hermitian element commuting with j, Example 16.9, item 1, and Proposition 10.1 shows the span is attained), so scalar changes along the base attain only the connections whose difference from nabla^K lies in that span. Whether the complementary ten directions are reached by matrix-valued alpha of a restricted class, such as components between the summands of E, is a further application of criterion 3 to the core

modes that has not been made; the first version of this corollary asserted it and read the collar connection of lemma 29 as an instance, which it is not (that connection acts at the wall, on the CP^3 copy, and moves the kernel).

Proof. The first sentence is Theorem 43.2 with the scalar subalgebra in place of Gamma(u(E)). The CP^3 statement is the second-moment identity Q(w^* B w/w^* w) = (B + tr B)/5 (Example 16.9, item 2, from the fourth-moment formula in the proof of Proposition 10.1), which maps Herm(4) onto itself with eigenvalues 1/5 and 1; the S^4 statement is Example 16.9, item 1, with Proposition 10.1. ■

Remark 43.4 (the test case of the corpus). On the Berger family of S^5 with the kernel of flux (1, 1, 2) the corpus engineered a connection with a prescribed Berry connection and found that all three coefficient lines keep a circle zero mode exactly when eta^2 is an integer; that calculation rests on stated spin lifts and was checked for its phase arithmetic only (historical/corpus/canonical_research.md, section 9, and research_checks, section 4.7). Theorem 43.2 says when such engineering is possible; it does not redo that calculation, and the integrality statement stays a record.

Scope

Constant-rank kernels, compact fibres, unitary connections with volume-preserving horizontal transport or the mean-curvature term; the Berry connection as the compressed horizontal derivative; changes of the connection with no vertical component, which leave the kernels fixed. The theorem is linear algebra fibre by fibre and a smoothness remark; its content for the project is the criterion of item 3, which separates the two sheets of lemma 29 for scalar changes: the wall copy (functions on CP^3) attains every connection, the core copy (functions on S^4) six directions of sixteen. The collar connection of lemma 29 is a vertical change at the wall and moves the kernel, so the question of T15, whether it is a choice or a construction, stays open; a theorem that covers it must include the motion of the kernels (referee).